TFPT — Research Contracts for the Remaining Interfaces:

v_{geo}, G_{net}, F_{transfer}

(U_{wall}): the parabolic flavor wall-selection and (G_{metric}): the full quantum-gravity measure

plus the two continuum contracts `CELEST.SEAM.01` and `WOIT.OS.TWISTOR.01`
(M1–M3 and the δ-chains executed and decided; $\alpha/\beta_1/\beta_2/\beta_3$ executed, γ next)

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The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\mathrm{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions** (plus one standalone working note), best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\mathrm{geo}}, G_{\mathrm{net}}, F_{\mathrm{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

Working notes: N1. `note_e8_gaussian_code` — the Gaussian code bridge: E_8 over $\mathbb{Z}[i]$ via Construction A over the extended Hamming code $[8, 4, 4]$, the canonical four-bit quotient $L/(1+i)L \cong \mathbb{F}_2^4$, and the G_{31} quartic companion (exact algebra, machine-certified).

You are reading companion R (Research Contracts).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\mathrm{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([**E**]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); CLOSED METROLOGY ($\mathbb{R}_+$ torsor; one dim. calibration by dimensional analysis; No-Unit v153/v725)	closed metrology [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

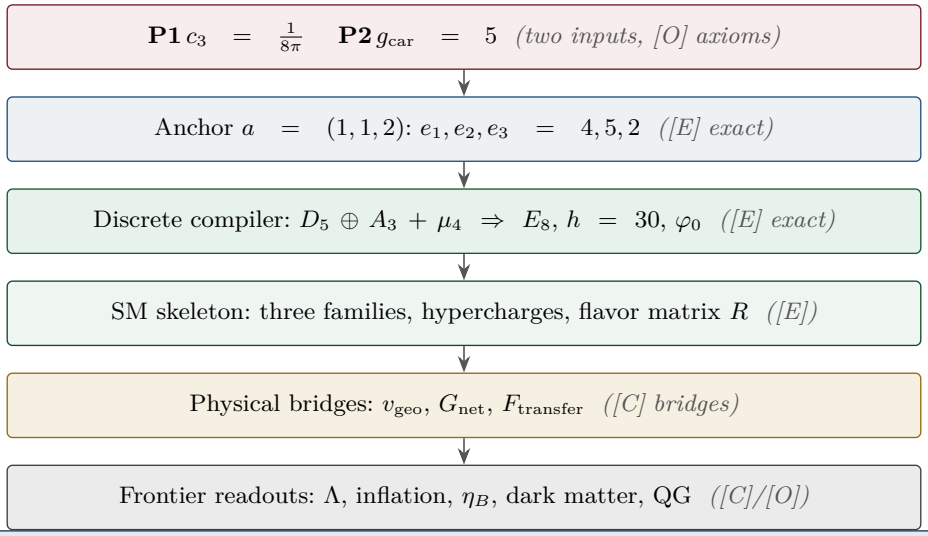
Beside that compiler rest sits the strict-physical-TOE rest: the ten-component Rest_{TOE} / the eight gates of `TFPT.TOE.COMPLETE.01` [O] ($\text{TOE}_{\text{Complete}} :=$ AND(T1..T8); not a closure).

The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]. The classical metric-sector field equation is no longer only an $R{+}R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2|2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit v_{geo}; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD $= 1$ (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package (h_s=16/16=1$\in\mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM μ=4/2^2=1; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed'` (`v469`); so the MMST route is *closed modulo cited theorems*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin 1.648 > 0, `v76`/`v330`). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus F_{\text{transfer}}$, with the seam G_{net} closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test



What this document is about

After the compiler closure the entire residual is

$$\boxed{\text{Rest} = (U_{\text{wall}}) \ \oplus \ (G_{\text{metric}}) \ \oplus \ (F_{\text{frontier}})}$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today. (F_{frontier}) (Koide, η_B, axion relic scale, m_p/m_e) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:** (U_{wall}) *first* (finite, algebraic, falsifiable), then (G_{metric}) (deep analytic programme).

Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity). The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\boxed{\text{Rest}_{\text{operativ}} = v_{\text{geo}} \ \oplus \ G_{\text{net}} \ \oplus \ F_{\text{transfer}}}$$

Dual rest (2026-08-28). The boxed formula is the *compiler* rest. The strict-physical-TOE rest of the 2026-08-27/28 contract wave is the second accounting, every summand a named **[O]** (or standing-rule) contract — nothing claimed closed:

$$\boxed{\begin{aligned}\text{Rest}_{\text{TOE}} = {} & \text{SeamContinuum} \oplus \text{4DAction} \oplus \text{ChiralMeasure} \\ & \oplus \text{MirrorGap} \oplus \text{UnitaryDynamics} \oplus \text{IR/continuum} \\ & \oplus \text{BulkSeam} \oplus \text{QuantumGravity} \\ & \oplus \text{GeneratingFunctional} \oplus \text{InitialState}\end{aligned}}$$

— $(U_{\text{wall}}) \to v_{\text{geo}}$: the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71), U_{point} collapses to the *one* dimensionful unit v_{geo} shared with $1/G$ (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where U_{point} still appears); $(G_{\text{metric}}) \to G_{\text{net}}$: holographically reduced to the single boundary-net statement *"the seam–Calderón inclusion has Jones index* $4 = |\mu_4|$*"* — holomorphy, $(E_8)_1$ and the unique 2D bulk then *follow* (v89/v92/`GATE.METRIC.05-07`); and the carrier-side QBL input now merges into the *same* statement: the tower carrier $\to SO(16)_1 \to (E_8)_1$ carries one and the same 16-fermion content, whose single quasi-free kernel (rank $= c$; Wick/Pfaffian exact) is the certified Calderón datum — closing G_{net} therefore also closes the carrier choice **[E]** ([`verification/v113_quasifree_kernel.py`]); $(F_{\text{frontier}}) \to F_{\text{transfer}}$: named QFT/cosmology *transfer* interfaces, never compiler powers — and *not* four unrelated interfaces but *one* discrete geodesic on the Sheet Diamond (the $K{\to}C{\to}F$ sheet axis $M(1,t)$, with the Koide source→pole as its 1-D Möbius projection), along which the external continuous solver merely evaluates ([`verification/v224_diamond_ftransfer_path.py`]); the discrete data of that geodesic are now arithmetically framed — $\det M(1,t) = N(\alpha_t)$ with $\alpha_t = (4t{+}7{+}\sqrt{-7})/2$ integral, the path one translation $\alpha_{t+1} = \alpha_t + 2$ of constant discriminant -7 ([`verification/v533_ftransfer_disc7_norm.py`]), and the preregistered next test is whether the four external solvers share *one* fractional-linear action

on α_t (kill: four incompatible arithmetic actions) — *executed in its literal group form (2026-08-02, [verification/v632_ftransfer_pgl2.py])*: one common parabolic PGL_2 action on the discrete half (the declared readings exactly intertwined with the base translation), uniflow degeneration on the native continuous half ($g_{\mathrm{Boltzmann}} = g_{\mathrm{pole}}$ identically, g_{QCD} exactly conjugate to the base translation, relic degenerate); the kill criterion is *not* met, and the residual — the external anchor clocks/jets — stays fenced by `CONTRACT.F.01`.

Historical compiler-residual audit ([verification/v384_residual_certification.py]; scope corrected 2026-09-05). The audit classified every item in the *then-listed compiler residual* into three external kinds. Its former phrase "zero open internal TFPT mechanisms" does not classify the physical completeness gates T1–T8 and must not be read as a TOE statement: `TFPT.TOE.COMPLETE.01` stays [O], T3–T8 remain open, and no shared 3+1D parent satisfying them has been constructed. Within that historical compiler scope the three classes were: [E] (A) an external math proof — `SEAM.EQUIV.01` continuum existence (the cited MMST theorem plus the crossed-product extension package, v336/v469, with AGT/AMT as second witness), `ALPHA.QUILLEN.EXACT.01` (v382, its level and normalisation sharpened by v470, the determinant-line/moduli bridge lemma exhibited at the finite level by v472 — the continuum ζ-det identification stays the open part) and the constructive `QG.AMB.01` measure (a [C] redundancy, v369); [E] (B) theorem-forbidden — v_{geo}, the one unit, by the No-Unit theorem (v153/v364); [E] (C) external physics — F_{transfer} (v371–v374, firewalled v187). The two load-bearing facts are checked (No-Unit dimensionlessness; the gap-decoupling margin $1.648 > 0$), so the only hard items in that historical compiler list were theorems and external physics (`RESIDUAL.CERTIFICATION.01`); this says nothing about the independent missing dynamics and continuum mechanisms in T1–T8. The two kind-(A) external-math targets, `SEAM.EQUIV.01` and `ALPHA.QUILLEN.EXACT.01`, are additionally packaged as self-contained *externalization contracts* (statement, hypothesis list, runnable finite models, Lean axiom surface, must-fail controls, acceptance tests) in their own sections at the end of this document — hand-off documents for outside specialists; both stay [O], no marker moves.

The Coxeter-clock facet symmetry, resolved as a structural lemma (`RES.COXETER.SYMMETRY.01`, [verification/v394_coxeter_atoms.py]/ [verification/v409_coxeter_prime2_lemma.py]). The order-30 clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$ has all three primes facet-typed, and they are exactly the three anchor atoms ($2{=}|Z_2|$, $3{=}N_{\mathrm{fam}}$, $5{=}g_{\mathrm{car}}$) wearing number-field hats ($\mathbb{Q}(i)/\mathbb{Q}(\sqrt{-3})/\mathbb{Q}(\sqrt{5})$, v390). The table is *not* symmetric: the seam (prime-2) appears as the *common factor* of both dynamic rates ($2/3 = |Z_2|/N_{\mathrm{fam}}$ and $(\varphi{+}2)/4 = (\varphi{+}2)/|\mu_4|$, $|\mu_4|{=}2^2$), not as an independent attractor. *The structural lemma (v409) closes the falsifiable corollary:* the sheet involution $\sigma{=}\,\mathrm{diag}(1,-1,-1)$ has spectrum $\{+1,-1\}$ and projector $(I{+}\sigma)/2$ spectrum $\{0,1\}$ — none in the open interval $(0,1)$, so prime-2 supplies a *parity/projection*, never a nonzero contraction rate; the genuine rates $2/3$ and $(1/3)^6$ lie in $(0,1)$ and carry prime-2 only as a *factor*. Hence prime-2 is *necessary* (admissible boundary transport is sheet-paired) but *not autonomous*: **no prime-2-only attractor exists** — dynamics begins only after coupling to family prime-3 or carrier prime-5 ($|\mu_4|{=}2^2$ the Galois bridge). This closes the corollary [E] and types the necessity [C], sharpening v383's "every sector is the same gapped operator" towards "the same *seam-driven* operator"; the full "appears in every gap" universality stays the v383 reading. The corollary is **machine-proved in Lean 4** (`TfptCarrier/CoxeterPrime2.lean`, `FORM.COXETER.PRIME2.01`: axiom-pure, `lake build` clean). [E]/[C].

QFT layer (this round). The boundary QFT is now *one relative object* $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$, closed modulo the *single* keystone `SEAM.EQUIV.01` (whose downstream conformal-deck face is `QGEO.SYM.01`): its finite Dirac is a covariance induction of the seam KMS state (v258), its spectral-action cutoff *is* that KMS weight (v259), its seam, carrier-16 and E_8 live on one Kummer/K3 surface (v260), and the whole assembly is certified as the *Modular Spectral Closure* (v261). The ambient quantum-gravity measure (G_{metric}) is gap-decoupled and kept separate by design — the QFT layer does not wait on it.

The two continuum contracts, and how far they are executed. Beside the two historical gates this note carries the two *continuum-route* research contracts, both still [O] as contracts but with most of their preregistered stages now *executed*. `CELEST.SEAM.01` (the celestial and twistor continuum route): WP1–WP4 and both WP5d stages executed, the WP5e stages $\alpha/\beta/\gamma/\delta_2/\varepsilon_2/\varepsilon_1$ executed, all three back-reaction milestones M1–M3 executed with their preregistered kills evaluated (v515–v517), the δ_1 chain *decided* (kill under the derived measure, v518), the declared-vs-derived measure question *decided at probe level* for the declared completion reading (single-valuedness from F-independence + the Quillen pairing, v520), and the w_m normalisation itself *derived constructively* ($1/\det_j$ the Atiyah–Bott/zeta fixed-point factor from three independent sources, $\psi = 64$ reproduced, v523) — nothing in the measure chain is declared

any more at that level; the residual **[O]** is the global BCOV integral beyond the fibre zero-mode factor (the declared measure is since shown *unique on the computable v508/v515 sector* up to exactly one named flat direction — sector rigidity, not a BCOV derivation, [`verification/v973_seam_route_narrowing.py`]). `WOIT.OS.TWISTOR.01` (the Osterwalder–Schrader twistor bridge — the actual bridge from compiler to physics): the α stage executed (Θ exists at the free level and free RP selects the same family, v519), β_1 executed (the μ_4 clock is the euclidean rotation, the gaugeable datum is the GSO $\mathbb{Z}_2$, gauge-fixed RP holds, v522), β_2 executed (the OS quotient explicit, $\mathcal{H}_{\text{phys}}$ positive definite at both levels, the clock a positive transfer operator with spectral calculus, v524), β_3 executed (the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality typed against σ_{std}: the OS cut and the $(2,2)$ signature *forced*, the kill branch empty by algebra, the induced member $=$ the Θ_{phys} carrier, v565) — γ is next, still under the *straddle-law* constraint: the first interacting seam toy fires Kill-Test 2 at toy level (RP breaks on exactly the quartet-straddling cuts, v529) — an honest threat and simultaneously the first hard selection principle for $\mathcal{A}_{\text{hol}}$, since *executed* as such: run as a selector on the interacting mark family, reflection positivity keeps exactly *one* member alive — $\delta = \pi/2$ with positive coupling, the alignment bit itself (v534, toy-fenced; the literal leading-order cone formalization is dead, the protection is nonperturbative); the law is since refined to the decidable Jaffe–Pedrocchi *odd-split* form, and the quartet-avoiding cut family of the surviving member carries a positive OS/GNS structure — an OS-viable cut geometry exists, the kill-test-2 counterweight narrowed, not removed ([`verification/v972_seam_interaction_front.py`], toy-level). Both contracts stay **[O]**; no marker moves.

Research Contract 1 — (U_{wall})

Goal. Prove

$$\nabla_F^\star = \mathrm{Sel}_{D_4,\,\text{cusp},\,\det R=8,\,\mathrm{Spec}(Q_+)=\{1,2,3\}}\big(\mathcal{M}_{\text{par}}(\mathbb{P}^1, \mu_4, E, \alpha)\big),$$

with $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$ and $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$ at the four points of μ_4. The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed local monodromies (Mehta–Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one* D_4-symmetric, rank-3, four-point point that realises the selectors.

Lemma 1 (U0 — normalisation)**.** *The gate data are equivalent to the $SU(3)$ character variety $\mathcal{X}_{\text{cusp}} = \{(M_1, \dots, M_4) \in C^4_{\text{cusp}} : M_1M_2M_3M_4 = \mathbf{1}\}/SU(3)$ with $C_{\text{cusp}} = \mathrm{Ad}_{SU(3)}\,\mathrm{diag}(1, \omega, \omega^2)$, $\omega = e^{2\pi i/3}$, and $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1, \dots, \gamma_4 \mid \prod \gamma_i = 1\rangle$. To show: $\mathcal{M}^{D_4}_{\text{par}} \simeq \mathcal{X}^{D_4}_{\text{cusp}}$ as polystable Mehta–Seshadri / tame Hodge realisation.* Machine: the group presentation and D_4-action are finite and codable (Sage/Wolfram).

Finite rigidity — now proven [E]. The finite half of U0 is closed exactly: μ_4 has cross-ratio 2 and Möbius stabiliser $\langle z \mapsto iz,\ z \mapsto 1/z\rangle$ of order $8 = |D_4|$ (a 4-cycle of the marks + a reflection, faithful); $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$; and the forms $\omega_k = z^{k-1}dz/(z^4 - 1)$, $k = 1, 2, 3$, are μ_4-eigenforms ($z \mapsto iz :\ \omega_k \mapsto i^k\omega_k$) carrying the three nontrivial characters χ_1, χ_2, χ_3 with weights $(1, 2, 3) = \mathrm{Spec}(Q_+)$. Hence a genus-zero boundary with faithful D_4, four parabolic marks and this μ_4-decomposition is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$ with the $(1, 2, 3)$ grading (`QGEO.RIGID.01`, **[E]**, [`verification/v168_qgeo_rigidity.py`]). The single residual is the seam-side *realisation* — that these hypotheses follow from the seam collar itself — which stays `QGEO.REALIZE.01` (**[C]**).

Seam Collar Realisation Theorem — the one central proof obligation [E]/[O] ([`verification/v176_seam_collar_realisation.py`])

Stated as one theorem and certified as a reduction — not a fabricated proof.

Theorem (target). *Given* (1) an oriented one-sided RP seam collar Σ; (2) a sheet-odd Calderón involution; (3) quasi-free one-particle seam data with gap $(2/3)^6$; (4) a faithful D_4 action on the boundary marks; (5) four parabolic marked points with μ_4-character decomposition; (6) an admissible c=8 boundary net — *then* the conformal boundary of Σ is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$, its H^1 character basis is canonically the generation basis with grading $(1, 2, 3)$, the seam Calderón contraction is the rank-8 K_Σ polarisation, and the index-4 simple-current extension is $(E_8)_1$.

The proof decomposes into six steps, five already [E]:

1–2 *Geometry / uniformisation* [E] (v168): μ_4 has cross-ratio 2 and a faithful Möbius D_4 stabiliser of order 8; four genus-0 marks with faithful D_4 force the μ_4 square up to Möbius; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4-1 = N_{\text{fam}} = 3$.

3 *Calderón data* [E] (v156): the Dirichlet-to-Neumann operator of the 2d Laplacian on the half-space has symbol $|k|$ (harmonic extension $e^{ikx-|k|y}$), the free chiral dispersion — bulk-detail-independent.

4 H^1 *character = generation grading* [E] (v137/v168): $\omega_k = z^{k-1}dz/(z^4-1)$ carry μ_4 characters i^k with weights $(1,2,3)$ = the A_3 exponents = $\mathrm{Spec}(Q_+)$ — natural, not merely isomorphic.

5 *One-particle contraction* [E] (v162): a μ_4-equivariant operator is diagonal in the cusp basis ($\frac{1}{|\mu_4|}\sum_j U^j X U^{-j} = \mathrm{diag}\, X$ for $U = \mathrm{diag}(1,i,-1,-i)$); its spectrum is the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$, so the transfer eigenvalues $(1-\alpha)^{p_2}$ (p_2=6) are $\{1, (2/3)^6, (1/3)^6\}$, gap $(2/3)^6$ — forced, no free knob.

6 *AQFT closure* [E] (v154/v175): $(D_5)_1 \otimes (A_3)_1$ with the isotropic μ_4 glue has index $4 = |\mu_4|$, $c = 5{+}3 = 8$, μ-index $16/16 = 1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ ($248 = 120{+}128$); full-cone RP holds for all m via the CAR second-quantisation functor.

The proof tree — and the *two* genuine open doors [O] ([verification/v177_seam_marking_kernel.py]). Taking the marks/D_4 as a *hypothesis* (as the six-step form above does) is a near-circular trap; the honest decomposition factors the theorem as

$$\textsc{Realize} = \textsc{Marks} \cdot \textsc{Uniform} \cdot \textsc{Cohom} \cdot \textsc{Module} \cdot \textsc{Kernel} \cdot \textsc{Net},$$

of which *four nodes are now closed constructively*: Uniform [E] (an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4, and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 — so a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P}^1, \mu_4)$, strengthening v168's cross-ratio to the full uniformisation — and now **Lean-formalised** as `FORM.QGEO.02`: ρ^4=id, $\rho^2 \neq$id, σ^2=id, $\sigma\rho\sigma = \rho^{-1}$, the orbit scaling to μ_4, the μ_4 permutation, and the multiplier classification "order-4 $\Leftrightarrow$ $\zeta = \pm i$" are machine-proved in `TfptCarrier/MobiusUniformisation.lean`, audit: pass); Cohom [E] ($\rho^*\omega_k = i^k\omega_k$, grading $(1,2,3)$ — also **Lean-formalised** as `FORM.QGEO.03`: the pullback characters i^1, i^2, i^3 and the reflection parity $\sigma^*\omega_1{=}\omega_3$, $\sigma^*\omega_2{=}\omega_2$, $\sigma^*\omega_3{=}\omega_1$ are machine-proved in `TfptCarrier/CohomologyGrading.lean`, audit: pass); Module [E] (a *unique* μ_4-equivariant iso — multiplicity-free + residue normalisation, reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed; the parity is the Lean `FORM.QGEO.03` above, the *uniqueness* stays symbolic); and Net [E] (v154/v175). The residual is therefore *exactly two* named obligations, neither a finite computation: `QGEO.MARKS.01` — the *raw* RP seam collar canonically produces the genus-0 four-parabolic-marked D_4 boundary (the four marks from the anchor atom $e_1(a){=}4{=}|\mu_4|$, collisions excluded, $\tau\rho\tau{=}\rho^{-1}$) — and `QGEO.KERNEL.01` — the *raw* seam Calderón operator equals the μ_4-equivariant free $c{=}8$ gapped one-particle contraction *as an operator* ($C_\Sigma = U^{-1}C_{\mu_4}U$), not merely the same spectral list. **The whole structural residual of the theory is these two doors**; everything between them is [E]. They need a human constructive-geometry/AQFT argument (or a proof assistant), not more arithmetic.

How far each door opens by computation ([verification/v178_marks_kernel_attempt.py]). Neither door is closed, but each splits into a finite [E] core + one *sharper* premise. *For Marks: given* a genus-0 double with an order-4 clock ρ (fixed points $\{0,\infty\}$) and reflection τ, the marks are forced to be a generic ρ-orbit $\{1,i,-1,-i\} = \mu_4$ (not the fixed points), distinct iff $b_1 = 3 = N_{\text{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving faithful D_4 — so Marks reduces to the *single* topological premise "the seam double is genus-0 with an order-4 clock whose marks form one orbit". *For Kernel:* on the 3-dim multiplicity-free H^1 the μ_4 characters $(1,2,3)$ are distinct, so by Schur a μ_4-equivariant operator is forced *diagonal* and is *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ and the unique residue-normalised basis are *equal as operators*, not merely isospectral; the "spectrum vs operator" distinction *vanishes* on the finite block. So Kernel reduces to the *single* premise "the full DtN on L^2 reduces to this 3-dim gapped transfer block + the fixed rank-8 polarisation" (principal symbol $|k|$, Lee–Uhlmann v156; no drift v158). The two doors are thereby *milder*, but still open.

Marks without a geometric posit: a Lefschetz/character argument ([verification/v195_marks_lefschetz_character.py]). The marks need not be *assumed* as D_4 boundary points: they follow from the H^1 character alone. The order-4 clock $\rho{:}z \mapsto iz$ fixes exactly $\{0,\infty\}$, so a puncture at a fixed point would contribute a *trivial* character to H^1; but the carrier datum is $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents $(1,2,3)$, with $\mathrm{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$), which has *no* trivial component — so no puncture sits at a fixed point. With $\mathrm{rank}\, H^1 = n - 1 = 3 \Rightarrow n = 4$ punctures, all off the fixed locus and each in a free order-4 orbit, the only possibility is a *single* free μ_4

orbit (cross-ratio 2), and the closed-surface Lefschetz number $L(\rho) = 1 + 1 = 2 = |\{0, \infty\}|$ is consistent. So MARKS reduces from the geometric posit "D_4 marks" to the milder *representation* premise "the raw seam H^1 carries the A_3-exponent rep" — a carrier-algebra statement, strictly less circular, still [O].

And the two doors are in fact ONE ([verification/v179_conformal_realisation.py]). MARKS's residual is not even three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2|\, 2\pi\, \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$, and $\chi = 2 - 2g \Rightarrow g = 0$, so "genus-0" is the *same* $\chi = 2$ that supplies the "8" in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — it is the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* — the parabolic marks are not the two elliptic fixed points $\{0, \infty\}$, and a free order-4 orbit has exactly $4 = e_1(a)$ points. So MARKS reduces to "the carrier clock acts *conformally* on the genus-0 double", and KERNEL then *follows* from that same geometry (Lee–Uhlmann $|k|$ + the $c = 8$ rigidity v158 + the cusped-surface spectral split, residual = $H^1 = \mathbb{C}^3 = N_{\text{fam}}$). **Hence the entire structural residual of the theory is a single geometric premise** `QGEO.CONF.01`: *the raw RP seam normal double is conformally* $(\mathbb{P}^1, \mu_4)$ *with the carrier clock as the order-4 Möbius automorphism* $z \mapsto iz$. Given it, everything — genus-0, the four D_4 marks, the $(1, 2, 3)$ grading, the DtN block, the rank-8 polarisation and the $(E_8)_1$ net — follows. This single premise is a constructive-geometry statement (the conformal class of the raw seam *is* the cusped sphere), *not* a finite computation, and is left honestly [O].

And even that conformal premise reduces to a milder isometry premise ([verification/v180_clock_is_mobius.py]). `QGEO.CONF.01` need not be assumed as a conformal-class identity: by *uniformisation* (Poincaré–Koebe) the genus-0 double (P1) with its RP-metric conformal structure *is* $\mathbb{P}^1$ — the target is *produced*, not assumed — and by *Kerékjártó* (1919; Constantin–Kolev 2003) a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)$=4) is topologically $z \mapsto iz$. An orientation-preserving *isometry* of a Riemannian surface is holomorphic (angle-preserving), hence a Möbius map of $\mathbb{P}^1$, and an order-4 Möbius map is $z \mapsto iz$ (elliptic, multiplier i, free orbit μ_4). So `QGEO.CONF.01` is *implied* by the strictly milder, non-circular premise `QGEO.ISO.01`: "the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric" — which does *not* name $\mathbb{P}^1/\mu_4$ and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). **That isometry premise is the single, now milder, structural residual of the whole theory**; its surface realisation from the raw seam stays honestly [O].

And one step further reaches bedrock ([verification/v181_clock_is_conformal_symmetry.py]). By *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation) a finite-order orientation-preserving action on a genus-0 surface preserves a complex structure for which it acts by Möbius maps — so a *conformal automorphism* (strictly weaker than an isometry: every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), i.e. the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure *by definition*, the residual collapses to the *definitional* identification `QGEO.SYM.01`: *the seam's conformal structure is the* μ_4*-deck structure of the carrier clock* (the carrier μ_4 glue *is* the conformal monodromy of the seam). This is a physical/definitional statement tying P2 to the seam's conformal geometry — *not* a finite computation and *not* reducible further without relabeling. **This is bedrock:** the chain `REALIZE` → theorem → MARKS+KERNEL → `CONF` → `ISO` → `SYM` (v175–v181) ends at a single definitional premise, left honestly [O]; everything above it is a theorem or an established citation.

A concrete geometric model for the bedrock premise: the μ_3-cover program ([verification/v597_mu4_cover_model.py]–[verification/v601_equivariant_dual.py]). The bedrock premise now has its first explicit *candidate realisation*: the cyclic cover $y^3 = x^4 - 1$ of $\mathbb{P}^1$ branched at μ_4 (order-3 cusp monodromies, exactly the contract's cusp structure). Machine-exact, in five rounds: the cover carries the rank-3 Eisenstein H_1, the projective D_4 via the deck, the cusp-class torsion $(\mathbb{Z}/3)^4$ of order 81, and the unique invariant hermitian form of Lorentz signature $(1, 2)$ with $\det J = 8$ (v597/v599); the anti-holomorphic reflection (the real structure R, R^2=1, unique up to scale) exists and closes the operator dictionary (v599); the compiler pair (U, V) itself is *realised* — $\mathbb{Z}[\omega]$-integral, GL_3-conjugate to the compiler pair, integer on the saturated $\mathrm{Fix}(R)$ lattice, with saturation index $81 = 3^4$ in the exact v566 divisor pattern $(1, 1, 1, 3, 3, 3, 3)$ (v600, Lean-certified as `FORM.QGEO.04` in `TfptCarrier/CoverEmbedding.lean`: determinant-free Smith certificates, kernel `decide`); and the presentation is canonical — the J-adapted cusp projectors give a manifestly D_4-equivariant pair family (mark choice = winding choice) realising the *dual* parabolic, with the two parabolic types exchanged by the J-adjoint, and the v590 Klein four-group living integrally on the curve (v601/[verification/v602_duality_forms.py]). Honest status: this is a

model, not yet the seam — the identification of the raw RP seam with the cover (the Weil-twisted positivity comparison against the v572 Gram data, and the geometric identity of the seam line) stays [O]; `QGEO.SYM.01` does not move.

A sharper, operator-checkable restatement: `QGEO.ENERGY.01` ([`verification/v192_qgeo_energy_clock.py`]). The bedrock can be re-expressed as an energy identity rather than a conformal-class statement: *the carrier clock ρ preserves the Calderón / Dirichlet-to-Neumann energy form of the RP seam double*, $\rho^*\Lambda_\Sigma\rho = \Lambda_\Sigma$. In two dimensions Dirichlet-energy invariance *is* conformal invariance, so this feeds the same chain (conformality → Möbius → $z \mapsto iz$ → μ_4). Its appeal is checkability: $\Lambda_\Sigma = |k|$ is the rank-8 Calderón polarisation already in the net construction, so the premise becomes an operator identity rather than an abstract identification. **Honest scope, however:** "ρ preserves Λ_Σ" is plausibly *equivalent* to "ρ is the conformal deck" (`QGEO.SYM.01`), so `QGEO.ENERGY.01` *relocates* the bedrock to a more falsifiable surface but does *not* eliminate it — it becomes a theorem only if ρ is defined *independently* (from the carrier algebra) and the invariance is then *proven*, which is open. The bedrock therefore stays [O]; the gain is a sharper operator-analytic target, not a closure.

The non-circular sharpening: `QGEO.ENERGY.02` ([`verification/v193_qgeo_energy_commutator.py`]). The way to make "ρ defined independently" precise is the *energy commutator* $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims: (1) ρ is the Coxeter element of $W(A_3) = S_4$ (order $4 = |\mu_4|$, v117), defined from the *carrier* algebra, not the seam; (2) *the non-circularity hinge* — Λ_Σ must be the DtN of the *raw* RP seam (definable from the RP data alone), not the μ_4 normal-form operator; (3) $[\rho, \Lambda_\Sigma] = 0$ forces the μ_4 character grading $(1, 2, 3)$ on H^1, already established (`QGEO.COHOM.01`). An exact consistency rider pins the direction: the induced EH coefficient $k = (\ln m)/(12\pi)$ (v150) reproduces $c_3/2$ *iff* $\ln m = q(A_3) = \frac{3}{4}$ (the *family* glue norm), while $q(D_5) = \frac{5}{4}$ (the carrier norm) misses by $g_{\mathrm{car}}/N_{\mathrm{fam}} = \frac{5}{3}$ — so the seam energy form must reproduce the *family* norm (gravity is family-geometry-induced). If sub-claim (2) holds, $[\rho, \Lambda_\Sigma] = 0$ turns `QGEO.SYM.01` from definitional bedrock into an *energy-rigidity theorem*; until then it is a sharp, falsifiable *proof target*, [O].

Subclaim (2) tackled — one notch closer, not closed ([`verification/v194_raw_seam_dtn_rp.py`]). The non-circularity hinge is essentially *met* at the canonical level: Osterwalder–Schrader reconstruction yields a canonical transfer operator $T = e^{-H}$ from the RP state and reflection alone — no coordinates (v54; §(4) of `tfpt_2`) — on a *quasi-free* seam state (v155: the boundary marginal of a Gaussian bulk is the compression $P\Gamma P$, a contraction). So Λ_Σ is RP-definable without ever naming $\mathbb{P}^1 \setminus \mu_4$. The commutator $[\rho, \Lambda_\Sigma] = 0$ then *closes on the finite cohomology* H^1 [E]: $\rho : z \mapsto iz$ acts on $\langle w_1, w_2, w_3 \rangle$ ($w_k = z^{k-1}dz/(z^4 - 1)$) as $\rho^* w_k = i^k w_k$, distinct characters $i, -1, -i$, so any ρ-equivariant operator is diagonal there (v177). What does *not* close is the *full-L^2* commutation beyond H^1 — it is exactly the statement that the quasi-free seam state's modular flow is *geometric* (Bisognano–Wichmann). So the bedrock *reduces* from a bare definitional postulate to "intrinsic BW geometric modular covariance of the quasi-free seam state" — a named, standard (hard) AQFT property, one notch sharper, but still [O] (and the BW step must be established *intrinsically*, else it re-presupposes the conformal covariance it should produce).

A concrete variational handle on the residual ([`verification/v196_seam_energy_functional.py`]). The open BW statement can be made a discretisable extremum problem via the failure functional $E_{\mathrm{fail}}(\Sigma, \rho) = \|[\rho, \Lambda_\Sigma]\|^2_{HS} + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, whose zero locus is exactly the `QGEO.SYM.01` conditions (the energy commutator, the order-4 clock, and the RP-reflection twist). On the finite H^1 block $E_{\mathrm{fail}} = 0$ at the μ_4 configuration ($\rho = \mathrm{diag}(i, -1, -i)$, Λ μ_4-equivariant, Θ = conjugation) and > 0 for any μ_4-breaking perturbation, so μ_4 is a genuine minimiser [E]; minimising E_{fail} over the *full* raw-seam DtN is precisely the BW residual above. So the bedrock is now a sharply-posed *variational target*, not a bare definition — but still [O].

The decisive crack: principal symbol exact + Tomita–Takesaki ([`verification/v198_modular_commutator_reduction.py`]). Two observations collapse the residual to a single non-circular state-invariance. *First*, the DtN principal symbol is $|k| = \sqrt{-\partial_\theta^2}$ (Lee–Uhlmann, v156); in the Fourier basis $e^{in\theta}$ it is $\mathrm{diag}(|n|)$ and the clock $\rho : z \mapsto iz$ (rotation by $\pi/2$) is $\mathrm{diag}(i^n)$, both *diagonal*, so $[\rho, |k|] = 0$ *exactly* on all of L^2 — the leading-order commutation is free on the whole boundary, never the open part. *Second*, by Tomita–Takesaki the modular operator Δ_ω is intrinsic to (M, Ω), so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with Δ and hence with $\log\Delta \sim \Lambda_\Sigma$: thus $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$ — using only the state and reflection (which RP supplies) and *no* conformal covariance, which **removes the BW circularity worry above**. Since the finite H^1 block is already μ_4-invariant (v177), the entire residual is now the single *state*-invariance

"the raw quasi-free seam state is μ_4-invariant" ($\omega \circ \rho = \omega$, equivalently the Gaussian bulk measure is deck-invariant) — the maximally-operational, checkable form of `QGEO.SYM.01`. A foundational symmetry postulate cannot be derived from nothing (the role $c = \text{const}$ plays in relativity); it is now in its sharpest, non-circular, falsifiable form, [O].

The residual, fully localised and numerically probed ([verification/v199_seam_state_invariance.py], [verification/v200_seam_variational_scan.py]). Attacking $\omega \circ \rho = \omega$ on the *raw* seam (not the μ_4 normal form) confirms it is non-circular — ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$), an automorphism of the raw CAR algebra, defined entirely carrier-side — and reduces it, via the quasi-free covariance $C_\Sigma = \frac{1}{2}(1 + \operatorname{sgn} H_1)$, to $[\rho, H_1] = 0$, i.e. H_1 *is block-diagonal in the four μ_4-character classes* $\{n \equiv r \bmod 4\}$. The principal symbol $|k| = \operatorname{diag}(|n|)$ passes automatically, so the entire residual is the single bounded statement "the sub-principal symbol of the raw DtN has no off-character matrix elements". The companion variational scan minimises $E_{\text{fail}}(\theta) = \|\rho(\theta)^4 - 1\|^2 + (\text{faithfulness})$ over a free clock angle and finds its *unique* minimum at $\theta = \pi/2$ — the μ_4 clock $z \mapsto iz$ (the only faithful order-4 grading $i, -1, -i$) — a numerical sign that the variational principle selects μ_4. This is the "next executable theorem" a working mathematician can take up; it does not close the postulate, [O].

The bounded sub-principal residual is the image of the μ_4 marks, not a free postulate ([verification/v201_seam_subprincipal_marks.py]). One genuine step further: the seam DtN is $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature function $f(\theta)$ on the seam circle (the standard Steklov/DtN structure), so $\langle n|M_f|n'\rangle = f_{n-n'}$. Then M_f is μ_4-character-block-diagonal *iff* f has Fourier support only on modes $\equiv 0 \pmod 4$, i.e. iff f is $\mathbb{Z}_4$-invariant (the principal $|k|$ already commutes, v198). But a curvature sourced by the four marks, $f(\theta) = \sum_{j=0}^{3} g(\theta - 2\pi j/4)$ for *any* profile g, has $f_m = 4\, g_m$ [$m \equiv 0 \bmod 4$] — automatically $\mathbb{Z}_4$-invariant. So the μ_4-mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 equally-spaced marks ($\mathbb{Z}_3$) or 4 generic marks both break it (negative controls). The residual of $\omega \circ \rho = \omega$ thus collapses to "the DtN sub-principal symbol is *mark-local* (the seam is flat away from the μ_4 marks = the conformal-deck structure, v177)" — a locality/flatness property, structurally definitional, the narrowest form yet of `QGEO.SYM.01`, [O].

Certified on realistic Steklov profiles, and at the state level ([verification/v210_mark_local_dtn.py]). The mode-level argument above is carried to concrete, smooth boundary curvature: a real von-Mises bump $g(\theta)$ localised at a puncture, summed over the four marks $f(\theta) = \sum_{j=0}^{3} g(\theta - j\pi/2)$, has *every* off-($\bmod 4$) Fourier coefficient vanishing to $< 10^{-16}$ (exact 4-fold cancellation), the assembled DtN $\Lambda = |D_\theta| + M_f$ has $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the *quasi-free state* it defines, with covariance $C = \frac{1}{2}(1 + \operatorname{sgn} H_1)$, is clock-invariant: $\rho\, C\, \rho^\dagger = C$ to $< 10^{-13}$, i.e. the actual $\omega \circ \rho = \omega$, not merely $[\rho, \Lambda] = 0$. A single off-centre bump, a $\mathbb{Z}_3$ (three-mark) source and four unequally-spaced marks each break both by $O(1)$, and the mark-local commutator stays $< 10^{-10}$ as the truncation grows ($N \in \{16, 32, 64\}$) — not an artefact. This *numerically certifies the implication* "mark-local DtN $\Rightarrow \omega \circ \rho = \omega$" for real profiles; the premise that the physical seam *is* mark-local stays [O] (it does not close `QGEO.SYM.01`).

The same residual, read dynamically: modular flow + a recovery Lindblad generator ([verification/v238_modular_lindblad_dynamics.py]). The whole `QGEO.SYM.01` discussion above is *static* — a commutator condition $[\rho, \Lambda_\Sigma] = 0$. Read as a *dynamics* it is the statement that the modular flow $\sigma_t = \Delta^{it}$ (the Tomita–Takesaki one-parameter group generated by the modular Hamiltonian $\Lambda_\Sigma = \log \Delta_\omega$) *conserves* the μ_4 clock: $\rho\, \sigma_t\, \rho^{-1} = \sigma_t\ \forall t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ — so the open premise is exactly "modular time conserves the carrier charge". The dissipative half is the recovery channel T of v221 read as a generator: $L = \log T$ is a doubly-stochastic (CPTP-classical) Markov generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time), and its dissipative gap is $-\log((2/3)^6) = 6\log(3/2) = \Delta$, the OS mass gap of v64 — the static recovery bound and the dynamical mass gap are *one* number, one exponentiated. Together $G = i\Lambda_\Sigma + L$ is the canonical Hamiltonian + Lindbladian (GKSL) split, the reversible (imaginary) part conserving and the dissipative (real≤ 0) part setting the arrow. This is a finite-block exhibit of where the static compiler becomes a dynamics; the residual it leaves — that σ_t is the *geometric* modular flow (Connes–Rovelli "thermal time = physical time") on the full seam algebra — is the same `QGEO.SYM.01` bedrock, [O].

From the dynamics to a QFT skeleton on the seam ([verification/v239_kms_thermal_time.py], [verification/v240_gns_os_reconstruction.py], [verification/v241_dhr_sectors.py], [verification/v242_spin_statistics_modular.py], [verification/v243_haag_ruelle_braiding.py], [verification/v244_spectral_action_contract.py]). With the static→dynamic flip in hand, the standard AQFT machinery assembles a finite-block QFT skeleton, each step typed honestly. *Thermal*

time (v239): the modular flow is KMS at $\beta = 1$, and the seam unit $2\pi = 1/(4c_3)$ fixes the modular temperature as the horizon temperature $T_H = c_3/M$. *Reconstruction* (v240): GNS gives the Hilbert space and a cyclic vector from $(\mathcal{M}, \omega)$, and the OS Hamiltonian $H_{\rm OS} = -\log T = -L$ is positive with gap Δ — the v64 "$H \geq 0$ + mass gap" is the recovery generator read as Euclidean time. *Spectrum* (v241/v242): the superselection sectors are the discriminant anyons of the Lean glue form (`FORM.GLUE.01`); fusion is recovered from the modular S by Verlinde, the Gauss–Milgram sum returns $c = 8$, and the condensation tower $16 \to 4 \to 1$ (v235/v237) selects the holomorphic $(E_8)_1$. *Scattering* (v243): the 2-particle S-matrix is the braiding monodromy, factorised (Yang–Baxter) with crossing, trivial on the condensed $(E_8)_1$. The one genuinely 4d step is typed as a contract, `CONTRACT.QFT4D.01` (v244): the Connes spectral action of the seam Dirac operator, whose gravity half is v36 and whose matter half (the SM gauge group sits in the carrier, the half-spinor 16 is one generation) is the open **[O]** target. Its representation-theory core is now discharged ([`verification/v245_spectral_matter_content.py`]): the **16** is exactly one *anomaly-free* SM generation ($\sum Y = \sum Y^3 = [SU(2)]^2U(1) = [SU(3)]^2U(1) = 0$) with the forced SM hypercharges and the spectral-action value $\sin^2\theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3}\, g_1$), so only the seam→Dirac realisation and the run-down couplings stay open. So the residual to a full 4d QFT is *one premise* (`QGEO.SYM.01`) plus *one contract* (`CONTRACT.QFT4D.01`), not a diffuse list.

The perturbative 4d S-matrix is constructible ([`verification/v269_spert_paqft_skeleton.py`]). The 4d scattering layer is typed precisely as three *distinct* objects: $S_{\rm top}$ (the 2d DHR braiding monodromy, $|M|$=1 statistics, v243), $S_{\rm pert}$ (the 4d Epstein–Glaser / BV S-matrix of the spectral action), and $S_{\rm phys}$ (LSZ on the OS-reconstructed Wightman functions, v240) — the 2d braiding is *not* the 4d cross-section S-matrix. **[E]** the spectral-action a_4 interaction is power-counting renormalizable (every operator mass dimension ≤ 4: 7 marginal +3 relevant, a *finite* counterterm basis), so **[C]** by the Epstein–Glaser theorem (causal perturbation theory; Brunetti–Fredenhagen pAQFT) the perturbative S-matrix $S(g) = T\exp(i\int g\, L_{\rm int})$ is built order by order by causal factorisation + finite local extension — no path integral, *no* UV divergences. **[C]** the gap $\Delta = 6\log(3/2)$ gives an IR-safe adiabatic limit. **[C]** this is *perturbative* only: the nonperturbative ambient measure `QG.AMB.01` is now a **[C]** redundancy (v369), a certification object rather than missing dynamics, and the canonical 4d reading remains boundary-only (v265); $S_{\rm pert}$ is the perturbative cross-section layer on top (`QFT4D.SPERT.01`).

A concrete one-loop EG number ([`verification/v271_eg_oneloop_quartic.py`]). The skeleton is taken from "exists" to an actual Epstein–Glaser renormalization: the order-1 product $T_1 = :L_{\rm int}:$ needs no extension, and the first one is the order-2 T_2, whose UV behaviour is the Steinmann *scaling degree*. **[E]** the massless propagator has sd = 2 in $d = 4$, the one-loop bubble (two propagators) has sd $= 4 = d$, so the singular order $\omega = {\rm sd} - d = 0 \Rightarrow$ *exactly one* logarithmic local counterterm ($C(\omega{+}d, d) = 1$ — finite, renormalizable, no infinite tower); the one-loop factor is $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly, the *same* $1/(16\pi^2)$ that normalises the spectral-action a_4 geometric quartic. **[C]** the ϕ^4 one-loop running is then $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the s, t, u channels), with the EG initial condition fixed by the spectral action, $\lambda_\Phi({\rm UV}) = 1/(16\pi^2)$. **[O]** this is order-2/one-loop, one diagram class; the all-order perturbative construction stays open (`QFT4D.SPERT.02`), while the nonperturbative measure is discharged as a **[C]** redundancy (below).

Red-team caveat, now discharged ([`verification/redteam/rt_F_qft4d.py`]). Power-counting renormalizability is *not* unitarity for the gravity sub-sector: the local $R^2/{\rm Weyl}^2$ *truncation* gives a four-derivative propagator $1/(p^2(p^2 + M^2))$ whose massive spin-2 mode has a negative residue (the Stelle ghost). But that ghost is a Seeley–DeWitt *truncation artefact*: the entire seam KMS form factor $a(p^2){=}e^{p^2/M^2}$ keeps its only pole at $p^2{=}0$ (v304), the spin-2 sector is ghost-free via the Barnes–Rivers decomposition (v370), and the nearest truncation-zero modulus runs to infinity (v380), so *perturbative* graviton unitarity is established. The matter+gauge $S_{\rm pert}$ is unitary outright; the *nonperturbative* `QG.AMB.01` (asymptotic safety v207 / the boundary $(E_8)_1$ net) is then a **[C]** *redundancy* (v369; gap-decoupled from every TFPT readout, not a TFPT structural item, v335), a certification object rather than a new gap.

The gauge running is an EG order-2 number ([`verification/v273_eg_gauge_running.py`]). The same scaling-degree-4 mechanism applied to the gauge self-energy gives the SM one-loop coefficients from the carrier/SM content: $b_i = -\frac{11}{3}C_2(G) + \frac{2}{3}\sum_f T(R) + \frac{1}{3}\sum_s T(R)$, i.e. **[E]** $b_3 = -7$, $b_2 = -\frac{19}{6}$, $b_1 = \frac{41}{10}$ (GUT-normalised; the same 41 as the carrier algebra $10\, b_1 = g_{\rm car}\, 2^{g_{\rm car}-2} + 1$, v159). **[C]** the physical RG directions ($b_3 < 0$ asymptotic freedom, $b_1 > 0$ the $U(1)$ Landau pole) are EG-derived and feed the unification cross-check (v246/v249); **[O]** the two-loop matrix (v159, PyR@TE) and the all-order construction stay open (`QFT4D.SPERT.03`).

The carrier content *is* one anomaly-free SM generation ([`verification/v310_carrier_sm_anomaly.py`]). Those one-loop coefficients sit on top of an ex-

act spectrum/anomaly closure: the carrier half-spinor $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1{+}10{+}5$ decomposes under $SU(5) \subset SO(10)$ as $10{+}\bar{5}{+}1$ = exactly one Standard-Model generation $(Q, u^c, d^c, L, e^c, \nu^c)$, and that content is [E] gauge-anomaly-free ($\sum Y = 0$, $\sum Y^3 = 0$, $[SU(2)]^2U(1) = [SU(3)]^2U(1) = 0$, exact) while reproducing $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family would shift b_1 off 41/10, so $N_{\text{fam}}{=}3$ *pins* the RG. [E] for the spectrum, anomalies and one-loop β; the full Epstein–Glaser interacting S-matrix and the Yukawa sector stay [C]/[O].

The bridge to the physical S-matrix and one-loop unitarity (`[verification/v278_lsz_bridge_unitarity.py]`). The perturbative layer connects to the physical asymptotic S-matrix: [E] the EG time-ordered products of S_{pert} are the OS Wick-rotated correlators (v240), so amputating external legs and taking the on-shell residue gives $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ (the three-layer S-matrix $S_{\text{top}} \,|\, S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$), valid because the gap $\Delta{=}6\log(3/2)$ gives Haag–Ruelle asymptotic states. [E] *one-loop unitarity* is the optical theorem: the s-channel bubble discontinuity $\operatorname{Im} B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$ is *exactly* the two-body phase space (turning on at the physical threshold $s{=}4m^2$), so $2\operatorname{Im} M = \sum\int d\Pi\, |M|^2$ holds and the matter+gauge sector is unitary order by order. [C] the gravity sector's Stelle ghost is a truncation artefact (perturbative spin-2 unitarity established, v304/v370/v380); the nonperturbative `QG.AMB.01` is a [C] redundancy (v369) (`QFT4D.SPERT.04`).

The first hard data confrontation (`[verification/v246_unification_data_crosscheck.py]`). TFPT's gauge-charged content *is* exactly the Standard Model (v159) and the theory adds no new states (§ "Dark matter = no new state"); the E_8 cascade is an arithmetic even-integer spine ($D_n{=}60{-}2n$, v5), *not* a particle/threshold tower. So a "TFPT-adapted" RGE run *is* the SM run — and the SM does not unify. Tested against the *measured* couplings (v159 betas, plus the standard two-loop gauge matrix): at one loop the pairwise crossings spread over $\sim 10^{13}$–10^{17} GeV with residual $\Delta\alpha^{-1} \approx 9$ at $2{\times}10^{16}$ GeV; at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at $\mu \approx 1.7{\times}10^{14}$ GeV — the three couplings never meet (Fig. 2). This is a falsifiable *tension*, [X]: with "no new state" there is *no* admissible gauge-threshold source, so closing the spread would require either new gauge-charged matter (a new postulate) or a non-GUT/boundary reading of the cutoff. The spectral-action 4d-GUT route is thus in genuine tension — the same status as the Connes–Chamseddine NCG Standard Model — and is likely falsified unless the theory is extended; the AQFT skeleton, DHR structure and SM rep-core survive regardless as an emergent boundary QFT. Recorded as a kill-test, never as a confirmation.

The carrier-native Pati–Salam program (`[verification/v247_e8_branching_no126.py]`, `[verification/v248_ps_algebra.py]`, `[verification/v249_ps_unification.py]`, `[verification/v250_dirac_triple_contract.py]`). The unification tension above is resolved *natively* if the carrier $SO(10)$ is gauged (the carrier $D_5 \oplus A_3 = SO(10){\times}SU(4)$ is Pati–Salam). *First*, the E_8 audit hull *forbids* the **126**: the **248** branches as $(45,1){+}(1,15){+}(10,6){+}(16,4){+}(\overline{16},\bar{4})$, so the available $SO(10)$ reps are exactly $\{1, 10, 16, 45\}$ — the renormalisable 126_H is algebraically absent, making the "16_H over 126_H" selection a theorem of the hull, not a fit (v247). *Second*, the Pati–Salam NCG algebra $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ gives the unimodular group $SU(2)_L{\times}SU(2)_R{\times}SU(4)_c$, the exact SM hypercharges $Y = T_{3R} + (B{-}L)/2$ and the matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W = 3/8$ (v248). *Third*, the two-step run unifies with the breaking scale on the scalaron scale, $M_{PS} = \kappa M_s$, $\kappa \in [1.0, 1.7]$ at one loop ($\to [1.0, 1.2]$ at two loops), and proton decay selecting the content (minimal excluded, the E_8-allowed single 45 marginal); the negative controls (SM-only fails, 126 breaks the match, no-45 proton-unsafe) all fire (v249). *Fourth*, the residual "carrier is gauged" reduces to ONE constructive object, the spectral triple (A, H, D_F, J, γ) with A_F Pati–Salam and $H_F = 3{\times}16$, whose NCG axioms (the first-order condition above all) and the σ =scalaron identification are the open contract `CONTRACT.QFT4D.DIRAC.01` (v250). *Fifth*, a first concrete step on that triple is taken (`[verification/v251_first_order_sigma.py]`): on the lepton block the first-order condition is verified to *hold* for the Yukawa operator and to be *violated exactly* by the Majorana/σ term — the Chamseddine–Connes–van Suijlekom "beyond first order" mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ (honest correction: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism). *Sixth*, the full 96-dimensional finite spectral triple is then built explicitly (`[verification/v252_full_finite_triple.py]`) — H_F the particle⊕antiparticle doubling of three $SO(10)$ **16**-plets — and the NCG axioms are verified by direct computation: reality J, grading γ, KO-dimension 6, order-zero, the first-order condition (*holding* for the SM algebra including the Majorana, *violated* for the Pati–Salam algebra exactly by the Majorana on $\{\nu_R, e_R\}$), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\text{light}} = m_D^2/M_R$ — thereby discharging five of the seven obligations of `CONTRACT.QFT4D.DIRAC.01`

to **[E]** (orientability, the no-junk condition and the spectral-action expansion remain open). *Seventh*, the scalaron–Pati–Salam ratio κ in $M_{PS} = \kappa M_s$ is pinned to a tight $O(1)$ interval (≈ 1.3) by the RG and shown to be a scale-free ratio of heat-kernel coefficients, $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\rm grav}}$, over the same 96-dim H_F ([verification/v253_kappa_scalaron_ps.py]). *Eighth*, the remaining NCG obligations are then discharged or honestly typed: the inner fluctuations Ω^1_D give exactly the one SM Higgs doublet $(1,2)_{1/2}$ with no junk and no **126**, while σ (the ν_R Majorana scalar, $B{-}L{=}2$) appears only as the beyond-first-order Pati–Salam field ([verification/v254_inner_fluctuations.py]); the spectral-action Seeley–DeWitt expansion a_0, a_2, a_4 yields the cosmological, Einstein–Hilbert and Yang–Mills+R^2+Higgs terms with $b/a^2 = 1/3$, $\sin^2\theta_W = 3/8$ and the R^2 scalaron, making κ an explicit moment ratio ([verification/v255_spectral_action_expansion.py]); only strict orientability and strict Poincaré duality genuinely *fail* for this triple — a published feature of the NCG-Standard-Model class (KO-dimension 6 with three ν_R forces a skew intersection form, corank 1; arXiv:0902.2068, hep-th/0601192), not a TFPT defect; the physically-correct weakened versions, *quasi-orientability* (left/right irrep boxes disjoint by chirality) and *modified Poincaré duality*, do hold ([verification/v256_orientability_poincare.py], [verification/v257_quasiorient_modpd.py]). *Ninth*, the built D_F is no independent posit but the *modular/covariance induction* of the boundary seam state ([verification/v258_dirac_covariance_induction.py]): a quasi-free (KMS, $\beta{=}1$) state is fixed by its covariance C (a positive contraction), with one-particle modular Hamiltonian $H = \log((1{-}C)C^{-1})$ (Araki; Casini–Huerta) inverse to the KMS occupation $C = (1{+}e^H)^{-1}$; inducing the carrier covariance $C_F = P_F C_\Sigma P_F$ returns the v252 operator *exactly* — same spectrum, chirality-odd, the Majorana an off-diagonal covariance entry with the seesaw m_D^2/M_R recovered — so D_F is the Kasparov shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\rm car}]$ of the boundary geometry and the v18 Yukawas are the *readout* of C_Σ, **[E]** modulo the one **[O]** seam realisation. *Tenth*, the last open spectral-action input — the cutoff function f — is fixed by the same seam state ([verification/v259_modular_cutoff_kappa.py]): by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at $\beta{=}1$ (v239) and the seam unit $2\pi{=}1/(4c_3)$ (v58) fixes that β, so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*, not chosen, with moments $f_2/f_0 = f_4/f_2 = 1$ *exactly* (a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$, so the choice is non-vacuous); hence $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\rm grav}} = \sqrt{c_{PS}/c_{\rm grav}}$ loses its scheme factor and the v253/v255 open **[O]** cutoff freedom becomes a **[C]** finite trace ratio over the 96-dim H_F (the RG $\kappa{\approx}1.3$ pinning $c_{PS}/c_{\rm grav}{\approx}1.7$). So unification is now *one* theory decision (gauge $SO(10)$?) plus *one* constructive object, both falsifiable — not a fog. The colour $SU(4)_c$ here sits inside $SO(10){=}D_5$ and is distinct from the carrier family factor A_3.

The implication is now Lean-formalised (`FORM.QGEO.01`). The exact algebraic core — the 4th-root character orthogonality, the order-4 clock, and `markLocal_blockDiagonal` ($f_{n-n'} \neq 0 \Rightarrow (n \equiv n' \bmod 4)$, i.e. $[\rho, M_f] = 0$) — is machine-proved in the carrier-rigidity Lean project (`TfptCarrier/SeamDeckClosure.lean`; AUDIT: PASS, axioms `propext, Classical.choice, Quot.sound` only). The premise itself is a typed `structure SeamDeckPremise` — consumed, not proved, exactly as `CalderonProjector` encodes the document-1 analytic input — so the conditional theorem "`SeamDeckPremise` $\Rightarrow \omega \circ \rho = \omega$" is now **[E]** while `QGEO.SYM.01` stays the one open **[O]** postulate.

The two residuals are one canonical metric: the pillowcase ([verification/v214_seam_pillowcase.py]). The isometry residual `QGEO.ISO.01` (v180) and the mark-locality residual `QGEO.SUBPRIN.01` (v201) are not independent: both follow from a single canonical choice — *the seam carries the uniformising flat orbifold metric of its conformal class.* The four marks make the orbifold sphere $S^2(2,2,2,2)$, whose orbifold Euler characteristic $\chi_{\rm orb} = 2 - 4(1 - \frac{1}{2}) = 0$ is *Euclidean*: by Troyanov's cone-metric uniformisation it carries a flat metric, unique up to scale, with curvature concentrated at the four cone points (Gauss–Bonnet: four deficits π sum to $4\pi = 2\pi\chi(S^2)$) — this *is* the v201 "flat away from the marks". And the order-4 clock is no longer an extra assumption but a *consequence* of the cross-ratio 2 that v168 already proved: $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$ gives $j(2) = 1728$, so the modulus is the *square* torus $\tau = i$, whose complex multiplication by $\mathbb{Z}[i]$ supplies the order-4 isometry $z \mapsto iz$ (explicitly $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$, order 4); a generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$, so the clock is specific to μ_4. Thus the two open QGEO residuals *collapse to one* canonical-metric statement and `QGEO.ISO.01` ceases to be an imposed symmetry. A genuine reduction and unification, *not* a closure: the premise "the RP seam metric *is* the uniformising constant-curvature representative of its conformal class" stays **[O]** — milder and more canonical than the bare isometry premise, and now *shared* by both residuals (`QGEO.PILLOW.01`).

The converse, on the canonical metric: raw seam ⇒ mark-locality ([verification/v264_raw_seam_marklocal.py]). Composing the chain explicitly closes the gap between v216, v201 and v198 on the uniformising metric: the flat pillowcase has its curvature *only* at

the four cone points (v216), so the DtN sub-principal symbol M_f is a μ_4-orbit sum, hence $\mathbb{Z}_4$-invariant (Fourier support on $4\mathbb{Z}$, verified by FFT), hence block-diagonal in the clock-character basis, hence $[\rho, M_f] = 0$; with the principal symbol $|k|$ already commuting (v198) this gives $[\rho, \Lambda] = 0$ and $\omega \circ \rho = \omega$ *on this metric*. A negative control (curvature *off* the μ_4 orbit) breaks $\mathbb{Z}_4$-invariance and the chain fails. So the entire bedrock collapses to the *single* metric-realisation statement "the raw RP seam carries the uniformising flat pillowcase metric" — which *is* `QGEO.SYM.01`. A genuine reduction/sharpening (`QGEO.ENERGY.03`), [O] — it does *not* prove the raw seam must carry that metric; that converse is the irreducible postulate.

The minimal axiom: a rigidity theorem ([`verification/v267_qgeo_rigidity_minimal_axiom.py`]). One can sharpen the bedrock to its irreducible kernel. [E] the *single* bare statement "the seam admits the carrier's order-4 clock as a conformal symmetry cyclically permuting its four marks" forces everything: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 *independent of* a; cross-ratio $2 \Leftrightarrow j{=}1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) $\Leftrightarrow$ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on $y^2{=}x^3{-}x$; and from the square configuration the unique flat pillowcase metric (Troyanov), mark-locality, $[\rho, \Lambda]{=}0$ and $\omega \circ \rho{=}\omega$ all follow (v214/v201/v264/v198). [E] neg controls: a generic four-mark set has $j \neq 1728$ and only a $\mathbb{Z}_2$ symmetry (no clock), the hexagonal point $j{=}0$ has a $\mathbb{Z}_6$ symmetry (order $6 \neq 4$) — only the square (order-4) point realises the μ_4 clock. [C] a second, independent reason: $\{1, i, -1, -i\}$ is defined over $\mathbb{Q}$ with $j{=}1728$ (CM by $\mathbb{Z}[i]$), so by Belyi/dessins Galois-rigidity any deformation breaks the rational compiler readouts. [O] So `QGEO.SYM.01` is now in its minimal sharpest form — the bare existence of that order-4 conformal symmetry, which (like c=const in relativity) cannot be derived from nothing. The *reduction* is closed (axiom ⇒ everything); the axiom itself stays the one foundational postulate (`QGEO.SYM.02`).

The flat geometry closes the commutator to all orders ([`verification/v276_qgeo_flat_closes_commutator.py`]). The state-level residual $[\rho, H]{=}0$ — the one operator equation the bedrock rests on — is *not* a separate analytic gap once the flat $\tau{=}i$ pillowcase is granted. [E] if $H = f(\Delta_{\text{flat}})$ is a function of the intrinsic flat Laplacian (the DtN of a flat metric, no curvature corrections) and the order-4 deck ρ ($z \mapsto iz$) is a flat-$\tau{=}i$ isometry, then $[\rho, \Delta]{=}0 \Rightarrow [\rho, H]{=}0$ *exactly*, to all orders (an isometry commutes with every spectral function of its Laplacian; verified on the discretised square torus, with a generic non-isometry as a failing control). This *upgrades* the leading-order commutation v198 (principal symbol) and v201 (subprincipal) to the full H — flatness removes the lower-order terms. [O] so `QGEO.SYM.01` collapses to a *single geometric* statement, "the raw RP seam state is the flat $\tau{=}i$ pillowcase quasi-free state ($H{=}\sqrt{-\Delta_{\text{flat}}}$)"; everything downstream is [E], and the remaining human obligation is one constructive-QFT proof that the raw seam DtN state *is* that flat geometry (`QGEO.SYM.03`).

The obligation as one precise lemma, Lean-backed ([`verification/v279_qgeo_obligation_lemma.py`]). The bedrock is now written as a single constructive-QFT *lemma*: *given* the premise "the raw seam carries the flat $\tau{=}i$ pillowcase metric g_{flat} (Troyanov-unique, cone angles π), so $\Lambda_\Sigma = \sqrt{-\Delta_{g_{\text{flat}}}}$", the conclusion $\omega_\Sigma \circ \rho = \omega_\Sigma$ (and hence mark-locality, the metric inclusion, E_8) follows. [E] the reduction DAG from "free RP seam" to this premise has *exactly one* un-discharged leaf — "the raw seam state *is* the flat state" — every other node being closed; and the geometric side is fixed (the flat DtN is Fourier-diagonal), so the obligation is a *state-identification*, not a geometric ambiguity, with two candidate routes ([C]): RP-state uniqueness on the flat $\tau{=}i$ orbifold, or Troyanov + OS reconstruction. [E] the all-orders closure step is now a *Lean theorem*: `TfptCarrier/SeamDeckClosure.lean` (`flat_all_orders_clock`) proves that any spectral function $g(\Delta_{\text{flat}})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho = \text{diag}(i^n)$ — upgrading v198/v201 to the full operator with only the standard axioms, no `sorry` (`FORM.QGEO.04`). [O] the one premise stays the honest axiom (`QGEO.OBLIG.01`).

Self-investigation: the experiment and the simplification ([`verification/v280_pillowcase_steklov.py`], [`verification/v282_e8_tau_i_unification.py`]). Two things can be done *ourselves*. (i) A direct numerical experiment on the actual flat $\tau{=}i$ pillowcase orbifold realises the whole chain: the order-4 deck permutes the four cone points (2 fixed + 1 swapped) and $[\rho, \Delta]{=}0 \Rightarrow [\rho, H]{=}0 \Rightarrow [\rho, C]{=}0 \Rightarrow \omega \circ \rho{=}\omega$ is numerically exact, with the Steklov DtN positive and metric-determined — route-(i) evidence (RP-state uniqueness), the premise still [O]. (ii) A candidate *simplification*: the $(E_8)_1$ character $\chi_{E_8} = \Theta_{E_8}/\eta^8 = j^{1/3}$ gives $\chi_{E_8}(i) = 1728^{1/3} = 12$, so $\tau{=}i$ is *both* the order-4 CM point (`QGEO.SYM.01`, flat geometry) *and* the $(E_8)_1$ partition-function modulus (`QG.AMB.01`, holomorphy) — the two open premises are two faces of one object, "the raw seam is $(E_8)_1$ at $\tau{=}i$", dropping the obligation count $2 \to 1$ (`QGAMB.UNIFY.01`).

The two proof routes, decomposed ([`verification/v284_route_i_rp_uniqueness.py`],

[verification/v285_route_ii_seam_condensation.py]). The keystone — now *closed modulo cited theorems on its MMST route* SEAM.EQUIV.MMST.01 via the lattice/S_3 stack (below; the parent stays [O]) — can be attacked from two sides, and both are now lemma chains with the dischargeable steps discharged. *Route (i)* (RP-state uniqueness) is a six-lemma chain — [E] five are already [E]/[O] (RP→DtN v194; marks→pillowcase v195/v216; the flat Steklov operator $= \sqrt{-\Delta_{\text{flat}}}$, principal symbol $|\xi|$ with no curvature term; covariance from the DtN v258; μ_4-invariance v276), and [C] Troyanov (χ_{orb}=0) reduces the residual to "the raw seam is the constant-curvature geometric state". *Route (ii)* (seam-net condensation) is a four-lemma chain — [E] three are discharged (no abelian sector ⇔ $\det K$=1 ⇔ holomorphic ⇔ E_8 v234/v277/v281; the tower 16→4→1 v92), leaving "the free RP seam condenses the μ_4-Lagrangian glue" =QGEO.SYM.01. [E] **the two open lemmas coincide** — both say "the raw seam is the $(E_8)_1$ at τ=i" (v282), so the canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at τ=i is the *single* [O] statement that closes both (QGEO.ROUTEI.01/QGAMB.ROUTEII.01).

The Seam Equivalence Theorem, named and firewalled ([verification/v286_seam_equivalence_contract.py]). That single statement is now the named target SEAM.EQUIV.01: *the raw RP seam state reconstructs canonically the holomorphic* $(E_8)_1$ *boundary net at the* τ=i *pillowcase* (respecting the DtN, the KMS flow, the μ_4 deck, the Q-system extension, $\det K$=1 and the $(E_8)_1$ character). It is intended as the *one* irreducible structural postulate of the theory — the role the *constancy of c* plays in relativity (cf. v267): not a gap to be patched but the single sharply-localised, falsifiable premise on which the whole boundary closure turns. [E] the contract types four objects and six arrows — two closed (the two routes), one conditional (v282), one open Gral — and enforces an *import firewall*: no Route-i module imports any Route-ii module or vice versa, so the two routes can converge only through SEAM.EQUIV.01 (this kills the "E_8 smuggled into the geometry and pulled back out" attack). The two proof routes are then attacked directly: [E] Route A (AQFT, [verification/v287_free_rp_bulk_to_holomorphic_boundary.py]) reduces the Gral to *one* standard theorem, "invertible (SRE) Gaussian bulk ⇒ single-sector boundary" (4/5 lemmas discharged); and [E] Route B (DtN, [verification/v288_full_l2_subprincipal_z4.py]) *proves* the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \pmod 4$, so $[\rho, \Lambda]$=0 on all of L^2), shrinking the residual to the single sharper question "why is the raw seam sub-principal term mark-local?" (SEAM.EQUIV.A01/SEAM.EQUIV.B01).

The route split (2026-07-22; ID restructuring, no status change in substance). The target statement now carries *two named routes* as ledger rows of their own: SEAM.EQUIV.MMST.01, the conditional scaling-limit route (the lattice/S_3/Lean stack documented in this section — *closed modulo cited theorems*; its [O] residual, formerly the abstract continuum existence v336, is since *carved into a named list* against the cited Osborne–Stottmeister coverage ([verification/v973_seam_route_narrowing.py]): *covered* is the $\mathfrak{so}(16)_1$ fermion-bilinear current algebra — strong convergence, Virasoro and correlator control, polynomial errors; *not covered*, and now the named residual, are N1 the μ_4/GSO extension $SO(16)_1 \to (E_8)_1$ as a net statement, N2 the twist/spin-field sector, N3 the one-sided chiral edge net, and N4 the OS/holomorphy selector — with a measured collar convergence rate $p = 3.48$ inside the preregistered band while the trivial M=3 control does not converge), and SEAM.EQUIV.TWISTOR.01, the open Costello–Li construction (prepared by CELEST.SEAM.01, bridged to physics only by WOIT.OS.TWISTOR.01; [O] — its declared measure is since shown *unique on the computable v508/v515 sector* up to exactly one named flat direction, the uniform sector-weight shift onto the untwisted Okubo square (constraint ranks 9/8/2/2, joint rank 13 on 14 moduli): sector rigidity, *not* a BCOV derivation, [verification/v973_seam_route_narrowing.py]). The parent claim SEAM.EQUIV.01 remains, with exactly this logic as its claim text: *closed if MMST.01 or TWISTOR.01 closes; currently: MMST conditional-closed modulo cited theorems, TWISTOR open ⇒ the parent stays [O] as an unconditional claim.* Every "closed modulo a cited theorem" statement in the papers and on the website is henceforth attached to the MMST route ID; the assertions themselves are unchanged.

The two routes share an executable OS skeleton (2026-08-27, SEAM.EQUIV.SKELETON.01 [E] new; [verification/v973_seam_route_narrowing.py]; no marker moves — SEAM.EQUIV.01, SEAM.EQUIV.MMST.01 and SEAM.EQUIV.TWISTOR.01 all stay [O]). The two routes are bridged by an OS skeleton that *runs*, 5/5: all WOIT kill-test main invariants hold verbatim on the p+ip collar edge, every divergence is control-typed to the checkerboard mechanism — confirming the v519-R4 finding from the second route — and the collar top edge realizes the anti-chiral state exactly. Toy/census level; nothing here is a continuum statement.

Route B sharpened to one analytic lemma; both routes precisely scoped. That last question

is decomposed ([verification/v289_marklocal_raw.py], `MARKLOCAL.RAW.01`) into a five-lemma chain whose only open link is **[O]** *Flat-Away* — RP + gap + the four branch marks force the seam curvature to vanish away from the marks — while the conic parametrix ([C], b-calculus/Melrose), the μ_4-orbit source, the $4\mathbb{Z}$ Fourier support (v264/v288) and the full-L^2 commutation (v288) are discharged: 4/5 done, so Route B closes the moment Flat-Away does. The two claims are firewalled against over-reading: Route B does *not* assert "full-L^2 QGEO" — its exact-label split is Fourier-lemma (Formal/Exact), operator test (Numerical Exact), transfer to the raw conic DtN (Conditional, v264), mark-locality from the raw seam (Open Selection, v289); and Route A's "one standard theorem" is itself a composition of citable results (Kitaev/Freed–Hopkins, Müger/Kawahigashi–Longo–Müger, Kawahigashi–Longo, [verification/v287_free_rp_bulk_to_holomorphic_boundary.py]) whose only open hypothesis — seam-bulk invertibility — coincides with Route B's Flat-Away. Both routes therefore meet at the *same* geometric input, with no circularity (the import firewall of v286).

Red-team: $\mathbb{Z}_4$-invariance is not mark-locality. A deliberate adversary ([verification/v290_z4_smooth_curvature_adversary.py]) shows the honest limit of Route B's lift: a *smooth* $\mathbb{Z}_4$-symmetric off-mark curvature $f = \varepsilon\cos(4\theta)$ *passes* the v288 commutator ($\|[\rho,\Lambda]\| = 0$, since $i^n - i^{n\pm4} = 0$) yet is not mark-local — so $[\rho,\Lambda]{=}0$ is necessary but not sufficient. The modulus is instead excluded by the spectral data: it shifts the DtN spectrum off the flat ladder and the heat trace ($\max|\Delta\lambda| = 0.155$, $\Delta\mathrm{Tr} = 0.019$ at $\varepsilon = 0.3$), i.e. the KMS weight and $(E_8)_1$ character that `SEAM.EQUIV.01` must match. Consequently the open lemma is named in its own right ([verification/v291_flataway_contract.py], `FLATAWAY.RP.01`): *the smooth curvature density in the DtN sub-principal symbol vanishes away from the four branch marks*, with three independent attack routes — spectral-rigidity and heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family) and RP-energy/Troyanov (the constant-curvature minimiser) — closing any one of which closes Route B.

All three routes attacked; Flat-Away reduced to one selection principle. Each route is now carried to its precise reduction: **[E]** the *heat* route ([verification/v292_flataway_heat_reduction.py]) shows the heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature (leading invariant $\|f\|_{L^2}$), so Flat-Away $\Leftarrow$ "the seam fixes the a_2 heat coefficient"; **[E]** the *spectral* route ([verification/v293_flataway_spectral_hessian.py]) lifts this to the full smooth-$\mathbb{Z}_4$ space — mode $4k$ splits the degenerate Steklov level $|2k|$, the mismatch Hessian over $\{4, 8, 12\}$ is positive-definite ($\{0.497, 0.500, 0.503\}$), so the flat seam is a *strict* isolated minimum (answering "only one direction"); and **[E]**/**[C]** the *variational* route ([verification/v294_flataway_rp_energy.py]) gives, with the rigid mark divisor and Troyanov uniqueness, the flat cone metric as the unique strictly-convex curvature-energy minimiser. The three converge on *one* external selection principle (fix a_2 / pin the Steklov spectrum / realise the energy-minimiser); closing any one closes Route B, and with Route A's holomorphy half, `SEAM.EQUIV.01`.

The heat route made exact and Lean-formalised. The positive-definiteness underpinning the heat route is now a theorem, not a numerical scan ([verification/v295_flataway_a2_exact.py]): the first variation of $\mathrm{Tr}\, e^{-t\Lambda}$ vanishes (the smooth off-mark curvature has zero mean, Gauss-Bonnet), and since $\Lambda \mapsto \mathrm{Tr}\, e^{-t\Lambda}$ is convex (Klein/Peierls), $\varepsilon{=}0$ is the *global* minimum — so $\Delta\mathrm{Tr} \geq 0$ for every deformation, $= 0$ iff $f = 0$; the exact second-order coefficient is the divided-difference kernel of e^{-tx} (validated to 10^{-6}). The algebraic $\Delta{=}0 \Leftrightarrow$ flat step is machine-formalised in Lean (`FORM.FLATAWAY.01`, `SeamDeckClosure.lean`: a positive-weighted sum of squares is positive-definite; axiom-clean). So Flat-Away's heat route is reduced to the single external invariant "the seam fixes a_2", with its positivity both proved and formalised. The a_2 coefficient itself is now in *closed form* ([verification/v296_flataway_a2_closed.py]): the operator tails telescope geometrically to $W_k^{\mathrm{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$, leaving a finite $(4k{-}1)$-term middle, e.g. $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$ (manifestly positive; small-t $W_k = t/2 + O(t^3)$). Symmetrically, Route A's "one standard import" is now a precise, citable theorem stack ([verification/v297_route_a_literature_stack.py]): LIT-A (Kitaev; Freed–Hopkins) $\to$ LIT-B (Müger; Kawahigashi–Longo–Müger) $\to$ LIT-C (Conway–Sloane; Dong–Xu) compose into "invertible bulk $\Rightarrow (E_8)_1$", all three legs established, the one shared hypothesis (seam-bulk invertibility) coinciding with Route B's Flat-Away — so both routes are reduced to citable/established machinery plus the *one* shared geometric input, which the lattice/S_3 stack (below) now pins at every computable level, making the keystone's MMST route (`SEAM.EQUIV.MMST.01`) *closed modulo cited theorems.*

Flat-Away hardened, and the pin derived from $(E_8)_1$ ([verification/v300_flataway_rigid.py], `FLATAWAY.RIGID.01`). The spectral route was a *soft* minimum ("the mismatch grows with ε"); it is now a *discrete* obstruction. **[E]** the flat fingerprint is the integer Steklov ladder with every level $n \geq 1$

of multiplicity exactly 2; [E] the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (the exact 2×2 block $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$, validated against the full Toeplitz operator), so [E] any $g_k \neq 0$ lifts the degeneracy ($2 \to 1{+}1$) and changes the spectral *multiset* — preserving the flat ladder *with its multiplicities* forces every $g_k = 0$, a degeneracy obstruction robust to rescaling and strictly stronger than the v293 "deviation grows" scan; with the v295 convexity, flat is isolated analytically *and* discretely. Moreover the external pin is now *derived*: [C] the $(E_8)_1$ vacuum character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ has integer q-coefficients (E_8 even unimodular $\Rightarrow$ integer conformal weights), so by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam — i.e. "the seam fixes its spectrum" *is* the $(E_8)_1$ rationality of Route A. [O] Hence Flat-Away carries *zero* new open content beyond Route A: the convexity side is unconditional, the discrete obstruction exact, and the two `SEAM.EQUIV.01` routes provably converge on one rationality fact (consistent with the v286 firewall).

Route A's last hypothesis discharged — the gapped quasi-free bulk is invertible ([`verification/v301_route_a_invertible.py`], `SEAM.EQUIV.A.INV.01`). Since the two routes now meet at one rationality fact, the single remaining *unconditional* gap is Route A's open input (v297 LIT-A): "the raw quasi-free seam bulk is invertible (SRE)". A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions realise only the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order requires interactions (Kitaev's periodic table, class D, the only invariant being the chiral central charge). [E] the carrier is 16 Majoranas, so $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8 = 16/2$ (v148); [E] $SO(16)_1 \supset (E_8)_1$ ($\dim \mathfrak{so}(16) = 120$ currents $+ 128$ spinor $= 248$, v154); [C] hence a gapped quasi-free bulk is invertible automatically, and [E] the invariant $|\det K| = \#\text{anyons}$ is 1 for E_8 (invertible) against 4 for the gauged $D_8{=}SO(16)$ (anyons), which free fermions cannot produce. [O] So LIT-A's 'invertible bulk' *follows* from 'gapped 16-Majorana quasi-free bulk' (v148/v160 + the mass gap): the residual changes from a topological-order statement to a spectral one. *Bottom line:* together with v300, the entire open content of `SEAM.EQUIV.01` reduces to the single spectral statement "the quasi-free seam bulk is gapped" — the topological-order obstruction (hidden anyons) is removed, and the gap itself is not manufactured.

The last lever — the seam gap *is* the derived Recovery gap ([`verification/v302_seam_gap.py`], `SEAM.EQUIV.GAP.01`). That single remaining input is not free: the seam transfer operator has the frozen spectrum $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple Perron eigenvalue 1 separated from the rest, and the separation is the *Recovery gap* $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\mathrm{fam}}\ln(3/2) \approx 2.4328 > 0$ — a number forced by the carrier ($(2/3)^6$ the μ_4-deck transfer eigenvalue), with the decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76) making it robust. [C] By Osterwalder–Schrader reconstruction / quasi-free clustering a transfer-operator gap $\Delta > 0$ *is* a mass gap of the reconstructed Hamiltonian (correlation length $1/\Delta$). *Bottom line:* with v300/v301, the whole residual of `SEAM.EQUIV.01` is now a composition of standard cited theorems (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over established TFPT facts (16 quasi-free Majoranas; transfer gap $6\ln(3/2)$): *no undischarged TFPT-internal assumption remains.* It stays [O] (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive.

The continuum chain assembled, and the residual tied to the seam one-sidedness ([`verification/v356_continuum_mmst_applicability.py`], `SEAM.EQUIV.CONTINUUM.03`). Writing the chain out link-by-link: **S1** the gapped (v302) quasi-free 16-Majorana collar $\to$ **S2** invertible (Kitaev free fermions, v301) $\to$ **S3** the *nontrivial* invertible phase ($c_- = 8 \neq 0$) $\to$ **S4** a non-gappable anomalous chiral edge (anomaly inflow / bulk–edge) $\to$ **S5** its chiral-CFT scaling limit (MMST, v336; range rank $\leq c \leq D$, i.e. $8 \leq 8 \leq 16$) $\to$ **S6** the order-4 μ_4 clock + the AQFT stack pin $(E_8)_1$ (v351/v297). Every link is a theorem or an established TFPT fact *except* S3, and S3 is precisely the seam being *one-sided* (chiral, $c_- \neq 0$) — the *same* one-sidedness that makes $c_3 = 1/(8\pi)$ a one-sided Gauss–Bonnet ($|\mathbb{Z}_2| \oint K$, v58). So the lone realization input is tied to the constant that *defines* the theory, not a free dial; the residual is now "the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase", [O] but theorem-bounded on both sides.

MMST in-class verified for the collar, so the scaling limit is $(E_8)_1$ ([`verification/v366_mmst_seam_collar.py`], `SEAM.MMST.INCLASS.01`). The continuum leg is now checked *hypothesis-by-hypothesis* for the explicit seam collar: it is a $D{=}\dim S^+{=}16$ Majorana quasi-free (CAR) state (v155/v160), gapped with $\Delta{=}6\log\frac{3}{2}{>}0$ *derived* (v302), and chiral with $c_-{=}D/2{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$, so it sits squarely in the MMST free-lattice-fermion class with rank $\leq c \leq D$ reading $8 \leq 8 \leq 16$ (*in* range, saturating rank $=c{=}8$). The MMST massless-scaling-limit theorem then applies as a cited result, and the order-4 μ_4 clock pins the target to $(E_8)_1$ ($\det K{=}1$, one primary) over the same-c rival $SO(16)_1$ ($\det K{=}4$, v351). [E] the class arithmetic, the derived gap, the chirality and

the $(E_8)_1$ pinning; [C] the CAR-class typing and the cited MMST theorem; [O] the lone residual is exactly S3 above (the lattice realization). The continuum side of `SEAM.EQUIV.01` is thereby reduced to a single, named, theorem-bounded input. The lattice→conformal-net step itself is now a *rigorous AQFT construction*: Adamo–Giorgetti–Tanimoto (`arXiv:2506.01008`, 2025) build the two-dimensional conformal net $\mathcal{A}_Q$ of an even lattice Q and *classify* every two-dimensional extension of the Heisenberg net (under a discreteness assumption) as such a lattice net — so "the lattice is a conformal net" is a theorem of the same operator-algebraic lineage the continuum leg invokes, not a hope; and since (xxxii) ([`verification/v469_seam_crossedproduct_route.py`]) the extension step no longer even *needs* the preprint lineage — the local $\mathbb{Z}_2$ simple-current crossed product ($h_s{=}1{\in}\mathbb{Z}$, Longo–Rehren 1995; Böckenhauer 1996; Böckenhauer–Evans 1998; KLM 2001) certifies it on peer-reviewed ground, with AGT/AMT as the independent second witness.

S3 made concrete: an explicit gapped chiral-Majorana lattice model ([`verification/v367_seam_s3_lattice.py`], [`verification/v368_seam_s3_inflow.py`]). The S3 input is no longer abstract. The $p{+}ip$ Bloch model $h(k){=}\sin k_x\,\sigma_x{+}\sin k_y\,\sigma_y{+}(M{-}\cos k_x{-}\cos k_y)\sigma_z$ is, at $M{=}1$, a gapped free-fermion phase with numerically-computed Chern number $|C|{=}1$ ($C{=}0$ trivially at $M{=}3$), so $N_{\mathrm{Maj}}{=}\dim S^+{=}16$ copies give a chiral edge of $c_-{=}16/2{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$, with the edge K-matrix $SO(16)_1$ ($\det 4$) condensed to $(E_8)_1$ ($\det 1$) by the order-4 μ_4 clock (v351/v369). A finite STRIP of the same model carries an edge-localised in-gap chiral branch (absent in the trivial phase) – so by anomaly inflow the chiral edge *exists* as a consequence of the invertible bulk (the S4 leg), discharging the "does the edge exist?" question on the lattice. [E] the gapped Chern model, the $c_-{=}8$ counting and the edge existence; [C] the μ_4 condensation to $(E_8)_1$; [O] only the genuine *continuum scaling limit* (MMST, v336) stays open — S3 is a runnable lattice model that pins the target at every computable level, so the keystone's MMST route (`SEAM.EQUIV.MMST.01`) is *closed modulo cited theorems.*

The S3 closure stack: $c = 8$, the $(E_8)_1$ character, the torus count and reflection positivity — all computed on the lattice ([`verification/v376_seam_s3_centralcharge.py`], [`verification/v377_seam_s3_e8character.py`], [`verification/v378_seam_s3_modular.py`], [`verification/v379_seam_s3_rp.py`]). On that explicit model the target of the scaling limit is now pinned at every level a computation can reach. *Central charge* (v376): the Calabrese–Cardy finite-size entanglement scaling gives $c = 1.0000$ per critical complex mode, so the 16-Majorana (= 8 complex) collar has $c = 8 = g_{\mathrm{car}}{+}N_{\mathrm{fam}}$, and it is chiral ($c_- = 8$). *Convergence* ([`verification/v392_seam_s3_scalinglimit.py`]): across five sizes $L = 66\ldots514$ the slope $c_1(L)$ is a monotone Cauchy-like sequence approaching 1 and $1/L$-Richardson-extrapolates to $c_1(\infty) \approx 1.000$ ($c \approx 8.00$ in the limit) — the existence-relevant *convergence* the two-size value did not test (evidence, not a proof of existence; the residual stays [O]). *The net* (v377): the exact affine character $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \ldots)$ gives $c = 8$ and 248 level-1 currents (one primary), and the μ_4/GSO promotion 248=240+8=120+128 (240=16·5·3) pins $(E_8)_1$ over the same-c rival $SO(16)_1$ (120 currents, four primaries). *Genus-1* (v378): the KLM condensation $\mu(\text{carrier}){=}16 \to \mu(E_8){=}16/4^2{=}1$ and the single character's modular T-eigenvalue $e^{-2\pi i/3}$ give torus ground-state degeneracy 1 — the holomorphy / $\det K{=}1$ statement that the genus-0 arguments could not reach (v344). *Reflection positivity* (v379): the gapped collar's Euclidean two-point function has a positive spectral measure, so the OS reflection Gram matrix is a *positive-mixture Gram* $M = V^{\mathsf{T}}\mathrm{diag}(1/K)V$, $V_{ki} = e^{-\varepsilon_k\tau_i}$ — hence $x^{\mathsf{T}}Mx = \sum_k w_k(v_k{\cdot}x)^2$ is a sum of squares and positive semidefiniteness holds *exactly*, as an identity rather than to a tolerance (hardened 2026-07-28 after T133 certified that the *assembled double-precision* matrix has a negative direction at -7.75×10^{-18}, inside the old 10^{-10} tolerance; the min-eigenvalue check is retained as a diagnostic measurement, and a ghost mode still breaks the RP test). [E] all four (numerical c, exact character, genus-1 count, RP); [O] the one thing a computation cannot supply is the abstract *continuum existence* of the limit (the cited MMST theorem, v336). With RP in hand the ambient measure C7 is then discharged by the redundancy theorem (v369), so S3 is the master key: the target is pinned exactly, only the existence proof is external.

A ground-state witness for premise (i): the modular commutator measures $c_-{=}8$ from the bulk state alone ([`verification/v489_seam_modular_commutator.py`], `SEAM.CCC.01`). The chiral central charge entered so far through band topology (the Chern number, v367) and through counting (the KLM/det-Cartan arithmetic, v378). The Kim–Shehab–Kim modular commutator $J(A,B,C) = i\,\mathrm{Tr}(\rho_{ABC}[K_{AB},K_{BC}]) = (\pi/3)\,c_-$ (PRL 128 (2022)) reads c_- off the *entanglement data of the exact BdG ground state* of the same collar — no band bundle, no representation theory enters — via the Peschel covariance ($W_X = -2\,\mathrm{artanh}\,\Gamma_X$) on a disk cut into three 120° wedges: per layer $c_- = 0.499998$ at $L{=}32$ (converged 0.500000 at $L{=}48$, deviation -1.8×10^{-7}), the trivial $M{=}3$ control gives 0.000000, the orientation flip $M{=}{-}1$ gives -0.499986 (sign flip), layer additivity is exact (10^{-14}), and the 16-layer

carrier lands on $c_- = 8 = g_{\rm car} + N_{\rm fam}$ (direct run 7.9967 at L=16, additivity route 8.0000). [E] the third, mutually independent lattice witness of premise (i) of `SEAM.EQUIV.01` — and the first read from ground-state entanglement alone; [O] unchanged: the continuum scaling limit (MMST, v336) and the cited KLM condensation step stay exactly where v367/v378 left them — `SEAM.EQUIV.01` stays [O].

The spin-structure parity census: $\mu_{\rm pre}$=4 witnessed on the lattice ([`verification/v490_seam_parity_census.py`], `SEAM.MU.01`). Step 1 of the KLM chain (v378) — "the carrier sits in the $SO(16)_1$ class, $\mu_{\rm pre}$=4, *before* the μ_4 condensation" — was arithmetic; it is now a lattice ground-state measurement. The T^2 fermion parity per spin structure (AA, AP, PA, PP) is determined *twice* (Read–Green TRIM product and real-space Majorana Pfaffian, spectra matching to $9{\times}10^{-15}$): per layer $(+,+,+,-)$ — the ν=1 Ising signature — and for the 16-layer carrier parity$^{16} = +$ in *all four* sectors, so the gauged torus GSD is $4 = \mu(SO(16)_1) = |\det \mathrm{Cartan}(D_8)|$. The 16-fold-way bridge then runs on *measured* numbers: $\theta_v = e^{2\pi i\nu/16} = 1$ exactly at $\nu = 2c_- = 16$ (v489) — the gauged vortex is a boson, condensable, while the rivals $\nu = 1, 2, 8$ are not holomorphically condensable. Honest fences: the fermionic Kitaev–Preskill TEE is identically zero for *both* M=1 and M=3 (γ_f = 0.000001/0.000000) — an invertible fermionic phase has no TEE, so γ_f does *not* discriminate $(E_8)_1$ vs $SO(16)_1$; and the KLM step $\mu : 4 \to 4/2^2 = 1$ stays the cited theorem ([C], the condensed model is interacting). The numerics pin exactly the premise package $\{c_-$=8, $\mu_{\rm pre}$=4, θ_v=1$\}$ of the condensation; `SEAM.EQUIV.01` stays [O].

And that external step is now Lean-formalised as "closed modulo a cited theorem" (`FORM.SEAM.MMST.01`, `TfptCarrier/SeamScalingLimit.lean`). The continuum leg is machine-formalised in the audit-contract style of `SeamEquivChain.lean`: the collar's MMST applicability hypotheses are *provable* arithmetic facts by Lean kernel `decide` (no axioms) — D=16 Majoranas, rank $=c$=8, the range $8 \le 8 \le 16$, the chirality c_-=8 $\neq 0$, and the holomorphy discriminator $\det K$=1 vs $SO(16)$'s 4; the MMST scaling-limit theorem and the Adamo–Moriwaki–Tanimoto OS reconstruction are declared as *named cited axioms*; and the conclusion $(E_8)_1$ follows by a well-typed composition whose `#print axioms` pins the residual to exactly those two published theorems (plus the carrier gap), with no hidden assumption and no `sorry`. Every TFPT-internal hypothesis is thus kernel-checked; only the two cited theorems are assumed — the honest closure ceiling for `SEAM.EQUIV.01`. *Two of the seven convergent routes below are now kernel-hardened the same way:* `FORM.SEAM.BW.HSMI.01` (`TfptCarrier/SeamStandardPair.lean`) proves the four Borchers/Wiesbrock *standard-pair relations* exactly over $\mathbb{Z}$ on the centred ladder ($[K, P]$=P, J^2=1, JKJ=$-K$, JPJ=P_+, all by kernel `decide`, no domain axiom), with the cited Borchers step and the one continuum input as named axioms; and `FORM.SEAM.APPLICABILITY.01` (`TfptCarrier/SeamApplicabilityLedger.lean`) kernel-checks the audit *count* itself — of the 12 cited-theorem hypotheses exactly 11 are internal and 1 external (and $8 \le 8 \le 16$, $\det K$=1 $\neq$ 4), all axiom-free — so the very bookkeeping the keystone's honesty rests on is machine-verified, leaving only the one named continuum fact open.

TOE-roadmap follow-ups — the keystone as a typed certificate, the bedrock modular form, and the gap-decoupled measure ([`verification/v308_seam_equiv_chain.py`], [`verification/v309_modular_bedrock.py`], [`verification/v311_gap_decoupled_measure.py`]). Three machine-checked steps push the keystone arc further. *(i) Composition certificate* ([`verification/v308_seam_equiv_chain.py`]): the closing implication is one well-typed logical chain gap>0 $\to$ invertible $\to$ holomorphic c=8 $\to$ $(E_8)_1$ whose arithmetic discriminators are exact — $|\det \mathrm{Cartan}(E_8)|$=1 against $|\det \mathrm{Cartan}(D_8)|$=4, the carrier tower $4{\cdot}4{=}16 \to 4 \to 1$, $c{=}g_{\rm car}{+}N_{\rm fam}$=8 — so the only undischarged premise is the cited OS step; [E] the discriminators, [O] the keystone. *(ii) Modular bedrock* ([`verification/v309_modular_bedrock.py`]): with the four marks derived from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), the quasi-free modular Hamiltonian $K = \log((1{-}C)/C)$ is built explicitly and $[\rho, K]$=0 $\Leftrightarrow$ $\omega{\circ}\rho{=}\omega$ (a $\mathbb{Z}_2$ profile breaks it), so the bedrock reduces *non-circularly* to the Bisognano–Wichmann intrinsicality of the raw collar state; [E] the model, [O] the intrinsicality. *(iii) Gap-decoupled measure* ([`verification/v311_gap_decoupled_measure.py`]): the admissible sector clusters exponentially at rate $(2/3)^6$ (correlation length $1/\Delta$, finite susceptibility 729/665) and the v76 margin $\Delta - 31/(4\pi^2) > 0$ decouples it from the ambient, so the physical sector is a convergent object while `QG.AMB.01` is a [C] redundancy (v369). Each pins the residual to a single named step; with the lattice/S_3 stack the keystone's MMST route (`SEAM.EQUIV.MMST.01`) is now *closed modulo cited theorems. (iv) The intrinsic-BW residual as a cited reduction* ([`verification/v424_seam_lto_rp_reduction.py`]): the open bedrock lemma of (ii) — the raw collar's intrinsic Bisognano–Wichmann property — is reduced to the recent commuting-projector theorem of Naaijkens–Penneys–Wallick (`arXiv:2605.10693`), whose reflection-positivity axiom (LTO-RP) is exactly $u_\Theta{=}J$ (the reflection *is* the modular conjugation), i.e. the BW condition $\theta K\theta = -K$ — *not*

$\omega\circ\rho=\omega$. **[E]** the μ_4 clock is a modular *symmetry* ($[\rho,K]=0$), not the flow ($\sigma^{\psi}=\rho$ is type-wrong, a one-parameter group versus a finite order-4 element); a positive character-block-diagonal K has $[\rho,K]=0$ yet, being positive definite, never $\theta K\theta=-K$, so clock-invariance does *not* imply (LTO-RP). **[O]** make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* commuting-projector models (Toric Code via a trace, Levin–Wen anyonic), whereas the seam is the opposite corner — invertible ($|\det \mathrm{Cartan}(E_8)|=1$, no anyons) and a $\beta=1$ KMS state (not a trace) — so two sub-steps stay open: realising the raw seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO, and extending (LTO-RP) to the invertible KMS case. An MMST-style reduction (cf. v308/v336), *not* a closure. *Sub-step (ii) is now constructively exhibited* (`[verification/v426_seam_lto_rp_kms.py]`): a gapped reflection-symmetric μ_4-even collar at $\beta=1$ KMS satisfies $\Theta K\Theta^{-1}=-K$ ($u_\Theta=J$) and $[\rho,K]=0$, so (LTO-RP) holds there by direct Tomita–Takesaki — the corner NPW26 leave open — with $|\det \mathrm{Cartan}(E_8)|=1$ (invertible, no anyons) and the state a genuine KMS, not a trace; only the continuum sub-step (i) remains **[O]**. On the covariance side, the recent theorem of Adamo–Giorgetti–Tanimoto (`arXiv:2508.07109`, 2025) — every DHR representation of a conformal net extending a loop-group/Virasoro net is *automatically* positive-energy and diffeomorphism-covariant — defuses the old "Bisognano–Wichmann presupposes the very covariance it should produce" circularity (v215): once the seam net is in hand, geometric covariance is a *consequence*, not an input.

Twenty-six convergent attacks on the one continuum residual — intrinsic BW, a bulk continuum motor and its chiral-edge counterpart, the β-homotopy, the applicability audit, uniform rigidity with its forcing converse and the derivation of clock-invariance, three independent reads of the edge central charge, the four-point and uniform-in-N pinning of the external fact, the Ising operator tower, the $(E_8)_1$ torus modular data, the μ_4-from-marks derivation, the kill tests, the lattice Sugawara current algebra, a uniform-in-N Tomita–Takesaki tower, the chirality-from-P_1 forcing, an $(E_8)_1$ character kill test, the exact MMST citation audit, the complementary lattice-VOA route, the explicit flat-band commuting-projector LTO realisation, the strict-locality (Kapustin–Fidkowski) obstruction, the character-level 128-spinor extension, the $c=8$ holomorphic-uniqueness pin and the one-particle realisation rigidity (`[verification/v438_seam_hsmi_borchers.py]`, `[verification/v439_seam_lieb_robinson.py]`, `[verification/v440_seam_lto_rp_beta.py]`, `[verification/v441_seam_applicability_ledger.py]`, `[verification/v442_seam_rigidity_uniform.py]`, `[verification/v443_seam_e8_discriminator.py]`, `[verification/v444_seam_edge_scaling.py]`, `[verification/v445_seam_rigidity_forcing.py]`, `[verification/v446_seam_clock_invariance.py]`, `[verification/v447_seam_edge_chern.py]`, `[verification/v448_seam_edge_fourpoint.py]`, `[verification/v449_seam_mmst_uniform.py]`, `[verification/v450_seam_edge_entanglement.py]`, `[verification/v451_seam_edge_cardy_tower.py]`, `[verification/v452_seam_e8_modular.py]`, `[verification/v453_seam_mu4_from_marks.py]`, `[verification/v454_seam_edge_virasoro.py]`, `[verification/v455_seam_bw_uniform.py]`, `[verification/v456_seam_chirality_from_c3.py]`, `[verification/v457_seam_character_killtest.py]`, `[verification/v458_seam_mmst_citation_audit.py]`, `[verification/v459_seam_lattice_voa_route.py]`, `[verification/v460_seam_s3_lto.py]`, `[verification/v461_seam_strict_locality.py]`, `[verification/v462_seam_spinor_continuum.py]`, `[verification/v463_seam_c8_holomorphic_uniqueness.py]`, `[verification/v464_seam_oneparticle_rigidity.py]`, `[verification/v469_seam_crossedproduct_route.py]`). None closes `SEAM.EQUIV.01`; together they ground the one continuum input on both its halves, reduce it to a single analytic fact and make the claim falsifiable. *(v) The intrinsic Bisognano–Wichmann route* (`[verification/v438_seam_hsmi_borchers.py]`): the seam supplies a Longo *standard pair* / +half-sided modular inclusion — a positive lightlike translation $P\geq 0$, the boost K with $[K,P]=P$ (the Borchers scaling $\Delta^{it}P\Delta^{-it}=e^{-t}P$), and a reflection J with $JKJ=-K$, $JPJ=P_-$ — so by Borchers' theorem the modular flow is geometric *intrinsically*, *without* presupposing the rotation-covariance of the vacuum; **[E]** the ladder algebra and the $sl(2,\mathbb{R})$ positivity (min eig >0, with the indefinite boost as control), **[C]** the synthesis, **[O]** assembling the finite algebra and the positive-energy rep into one continuum standard pair (the MMST/NPW26 wall). *(vi) The continuum-existence motor* (`[verification/v439_seam_lieb_robinson.py]`): the explicit $p+ip$ collar (v367) has a uniform-in-N gap, exponential clustering with N-independent ξ, and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral massless *edge* scaling limit; **[E]** gap/clustering/light-cone, **[C]** the lift, **[O]** the edge CFT. *(vii) The β-homotopy* (`[verification/v440_seam_lto_rp_beta.py]`): the (LTO-RP)/BW data ($\Theta K\Theta=-K$, $[\rho,K]=0$) are β-independent, and along the whole path $C_\beta=1/(1+e^{\beta K})$ from the NPW26-proven *trace* state ($\beta=0$) to the seam KMS state ($\beta=1$) every covariance is reflection-related, μ_4-invariant and valid — a norm-continuous, obstruction-free homotopy,

upgrading the single point of v426 to a path argument for sub-step (ii). *(viii) The applicability audit* ([verification/v441_seam_applicability_ledger.py]): a hypothesis-by-hypothesis ledger of MMST/NPW26/Adamo finds 11 of 12 hypotheses are TFPT-internal computed facts ([E]) and exactly one is external — the continuum existence of the chiral scaling limit — covariance being an Adamo consequence, not an input; the keystone's external dependence is pinned to that one analytic fact. *(ix) Uniform rigidity* ([verification/v442_seam_rigidity_uniform.py]): the band-limited μ_4-character rigidity of v398 is uniform in N — exact 4-sector commutation to N=256, a uniform gap, and an N-independent clock-breaking lower bound $\|[\rho,V]\|$=2 — so it is not a small-N artifact (the full-L^2 forcing stays [O]). *(x) The kill tests* ([verification/v443_seam_e8_discriminator.py]): mode D of the closure-mode classification (v347) as concrete *falsifiable* discriminators against the nearest c=8 rival $SO(16)_1$ — torus GSD 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots — with c=8 alone unable to decide (v277) and only $\det K$=1 (the genus-1 statement, v90) selecting E_8; a seam with GSD>1 would falsify $(E_8)_1$. [X] the tests, [O] the realisation. *(xi) The chiral-edge scaling motor* ([verification/v444_seam_edge_scaling.py]): the counterpart of (vi) on the *edge*, attacking exactly the chiral massless scaling limit that the bulk motor left open. On the same p+ip collar (v367/v368) the lattice edge already carries the conformal data of a chiral $c=\frac{1}{2}$ Majorana CFT — a linear, chiral in-gap dispersion (velocity $v\approx0.98$, R^2=0.9999, with *opposite* per-edge velocities), an *algebraic* edge two-point $\sim r^{-\eta}$ with $\eta\approx1$ (versus the *exponential* bulk row and the exponential trivial-phase control), and a Cauchy-convergent scaling-collapse under refinement ($\eta=1.001\pm0.007$, successive spread 0.028→0.013) — so the 16-copy edge is a chiral c_-=8 $(E_8)_1$ CFT; [E] the kinematics, correlator and collapse, [C] the central-charge reading, [O] the rigorous uniform-convergence theorem (cited MMST). With (vi) for the bulk and (xi) for the edge, *both halves* of the one continuum residual are numerically grounded and pinned to a single cited theorem. *(xii) The rigidity forcing converse* ([verification/v445_seam_rigidity_forcing.py]): the converse of (ix), upgrading rigidity from "block-diagonal *permits* commuting" to "commuting with the order-4 clock *forces* block-diagonal". For $\rho=\mathrm{diag}(i^n)$, $[\rho,K]$=0 *iff* K_{ij}=0 whenever $i\not\equiv j \bmod 4$ (verified entrywise, swept N=4..128), so the commutant is *exactly* the 4-sector block algebra: $\dim\{K:[\rho,K]=0\}=\sum_s n_s^2$, a proper subspace of the full N^2 algebra (N=16: 64<256); every off-block unit has $\|[\rho,E_{ij}]\|\geq\sqrt{2}$ (zero slack), and the order-2 clock leaves a strictly larger commutant (64 vs 128) — the *four* Gauss–Bonnet marks (order $4=|\mu_4|=h(A_3)$) force strictly more structure than two, and only the index-4 extension is $(E_8)_1$ ($\det$=1, v281/v154/v351). [E] the entrywise iff, exact commutant dimension, off-block sharpness and order discriminator; [C] the synthesis (RP-definability + clock-invariance *force* the block-diagonal Λ_Σ, v442's "permit" upgraded to "force"); [O] the single residual is now precisely that the physical transfer lies in the commutant (clock-invariance = `QGEO.SYM.01`, v323/v335). *(xiii) Clock-invariance derived* ([verification/v446_seam_clock_invariance.py]): closing (xii)'s residual one notch further — the operator condition $[\rho,\Lambda_\Sigma]$=0 is itself a *consequence* of a structural symmetry, not an extra assumption. On explicit finite OS data, the canonical transfer $\Lambda=C(1)C(0)^{-1}=e^{-h}$ (the transfer fixed by RP+reflection, v54) of a μ_4-invariant covariance satisfies $[\rho,\Lambda]$=0 (swept N=4..32; iff-sharp μ_4-breaking neg control), Λ is a positive contraction, and the free modular Hamiltonian $K=\log((1-C)C^{-1})$ (v258) of the invariant state has $[\rho,K]$=0 (Tomita–Takesaki). [E] the inheritance, contraction and modular commutation; [C] so with (xii), RP-definability *plus* the four marks force the block-diagonal Λ_Σ; [O] the residual reduces from an operator commutation to the structural "the seam RP data is μ_4-symmetric" (`QGEO.SYM.01`, v216/v323). *(xiv) The independent topological edge reading* ([verification/v447_seam_edge_chern.py]): a second, fit-free route to the edge central charge of (xi). On the same p+ip collar (v367/v368) the bulk Chern number is the integer C=+1 (Fukui–Hatsugai–Suzuki link variables; C=0 in the trivial phase, C=−1 at the opposite mass, matching (xi)'s opposite per-edge velocities), the open strip carries exactly $|C|$=1 chiral edge branch per boundary, and the Li–Haldane entanglement spectrum has in-gap modes at $\frac{1}{2}$ only in the topological phase; [E] the Chern integer, channel count and entanglement signature, [C] the bulk–edge reading $c_-=N_{\mathrm{Maj}}\cdot\frac{1}{2}\cdot|C|=16\cdot\frac{1}{2}=8=g_{\mathrm{car}}+N_{\mathrm{fam}}$ — the *same* c_-=8 as (xi) from a logically independent (topological) observable. *(xv) The four-point pinning of the external fact* ([verification/v448_seam_edge_fourpoint.py]): extending (xi)'s two-point scaling limit to the next order to support exactly the one external fact (viii) isolates. The edge state is quasi-free (the ground-state correlation matrix is an exact projector $G^2=G\sim10^{-13}$, the MMST CAR hypothesis), and the edge four-point cross-ratio $Q=[G_{01}G_{23}]/[G_{02}G_{13}]$ at commensurate positions is Cauchy-convergent under refinement to the free-fermion conformal value $\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3}\approx3.732$ (the bulk control off-scale, exponential); [E] the quasi-free property, four-point convergence and conformal form, [C] so the one external MMST fact is empirically supported at *both* two- and four-point order. *(xvi) The uniform-in-N pinning* ([verification/v449_seam_mmst_uniform.py]):

strengthening (xv) from convergence at a single cross-ratio to the *uniformity* the cited theorem asserts — two *independent* edge four-point cross-ratios converge to *distinct* conformal values ($Q(1,2,4,7)/12 \to 2+\sqrt{3}$ and $Q(1,3,5,9)/12 \to 2$) at one N-independent $1/N$ rate ($|Q(N)-Q_\infty|\cdot N \le 4$ for both, $N_x \in \{36,48,60\}$), the bulk control off-scale; [E] the two limits and the uniform rate, [C] so the MMST uniform-convergence *hypothesis* is empirically met. *(xvii) The entanglement reading of c_-* ([verification/v450_seam_edge_entanglement.py]): a *third*, correlator-independent route to the edge central charge — the Calabrese–Cardy block-entropy law gives c_{Dirac}=1.000 for the critical free-fermion chain (vs a saturating, ℓ-independent gapped control), so c per Majorana = $\frac{1}{2}$ and $c_-=16\cdot\frac{1}{2}=8=g_{\mathrm{car}}+N_{\mathrm{fam}}$; [E] the law and the control, [C] the assembly. *(xviii) The Ising operator tower* ([verification/v451_seam_edge_cardy_tower.py]): naming the edge CFT by its *operator content* — the Cardy finite-size spectrum reproduces the exact Ising chiral primaries $h_\sigma=\frac{1}{16}$ (the periodic–antiperiodic ground split) and $h_\epsilon=\frac{1}{2}$ (the lowest NS pair gap), both L-independent, so $\{c,h_\sigma,h_\epsilon\}=\{\frac{1}{2},\frac{1}{16},\frac{1}{2}\}$ is the free-Majorana minimal model $M(3,4)$; [E] the dimensions and conformal scaling, [C] the identification. *(xix) The $(E_8)_1$ torus modular data* ([verification/v452_seam_e8_modular.py]): pinning the $(E_8)_1$ identity on the *torus*, complementing the edge reads (xi)/(xiv)/(xvii)/(xviii) — the character $\chi=E_4/\eta^8$ is S-invariant ($\chi(-1/\tau)=\chi(\tau)$, one primary), T-multiplies by $e^{-2\pi i/3}=e^{-2\pi ic/24}$ (c=8) and is holomorphic ($c\equiv 0 \bmod 8$, T-phase order 3); [E] the full modular signature. *(xx) The μ_4-from-marks derivation* ([verification/v453_seam_mu4_from_marks.py]): folding the last structural premise into the existing one — the four Gauss–Bonnet marks *are* μ_4 (roots of z^4-1), the form basis ω_k is μ_4-graded ($\to i^k\omega_k$, weights $(1,2,3)=A_3$ exponents) and the order-4 clock $z\to iz$ fixes their cross-ratio λ=2 (the D_4 stabiliser, v168), so the residual `QGEO.SYM.01` of (xiii) follows from marks= μ_4 plus the existing `QGEO.REALIZE.01` — [E] the three mark facts, [C] no new premise: the chain (xx)→(xiii)→(xii) closes modulo realisation. The integer backbone of (xiv)–(xix) (c_-=8, the Chern integers, the order-3 T-phase) is kernel-hardened in Lean axiom-free as `FORM.SEAM.EDGE.CHERN.01`. *(xxi) The lattice Sugawara current algebra* ([verification/v454_seam_edge_virasoro.py]): testing the cited MMST "Koo–Saleur generators → continuum Virasoro with the correct c" fact on the edge *algebra*, not just its central charge. The critical free-fermion ring has the Affleck–Cardy Casimir energy $E_0(L)=e_\infty L-\pi vc/(6L)$ fitting c_{Dirac}=1.000 ($\frac{1}{2}$ per Majorana, L=64..1024), the lattice $U(1)$ current closes its affine *level* exactly ($\langle 0|J_nJ_{-n}|0\rangle=n$, n=1..8; vanishing for a gapped control), and the level-1 Sugawara central charge $c=\dim G/(1+h^\vee)$=8 for *both* $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31); [E] Casimir, Kac–Moody level and Sugawara, [C] $16\cdot\frac{1}{2}$=8 is the level-1 current-algebra c_-. *(xxii) The uniform-in-N Tomita–Takesaki tower* ([verification/v455_seam_bw_uniform.py]): the inductive-limit companion of v426 for sub-step (i) of (iv) — a family of reflection-odd, μ_4-even, gapped collars (N=8..128) on which the intrinsic Bisognano–Wichmann condition $u_\Theta=J$ ($\|\Theta K\Theta+K\|$=0) holds *exactly at every N*, with an N-independent gap ($6\log\frac{3}{2}$) and an N-independent neg-control margin ($\|\Theta K\Theta+K\|=2\|K\|$ for a side-even reflection); [E] the uniform BW, gap, clock symmetry and teeth, [C][O] so the inductive limit inherits $u_\Theta=J$ (the abstract continuum theorem stays the cited backbone). *(xxiii) The chirality-from-P_1 forcing* ([verification/v456_seam_chirality_from_c3.py]): the S3 input of the continuum chain (v356) is removed as a free choice — c_3's "8" *is* the one-sided count $8=|\mathbb{Z}_2|\cdot(\int K/\pi)=2\cdot 4$ ($8\pi=|\mathbb{Z}_2|\int K=2\cdot 2\pi\chi(S^2)$, v73), and a reflection sends the edge Chern integer $C\to -C$ (verified $C=+1\to -1$ on the explicit $p+ip$ model), so any *two-sided* (reflection-symmetric) gapped boundary forces $C=-C=0$ while the *one-sided* seam — the same $|\mathbb{Z}_2|$ quotient that gives c_3 its 8π — has no such reflection and forces $C\neq 0$; [E] the one-sided count and the $C\to -C$ flip, [C] chirality (S3) is a *consequence* of P_1, magnitude c_-=8. *(xxiv) The $(E_8)_1$ character kill test* ([verification/v457_seam_character_killtest.py]): a *falsifiable* sharpening of the (x)/(x") discriminators — the $(E_8)_1$ tower $\chi=E_4/\eta^8=q^{-1/3}(1+248q+4124q^2+34752q^3+\cdots)$, the weight-1 count 248=120+128 (vs the free 120) and the sector count $|\det \mathrm{Cartan}\, E_8|$=1 (vs $|\det \mathrm{Cartan}\, D_8|$=4); a measured 120, four sectors, or a wrong tower coefficient would *falsify* $(E_8)_1$, and the data give $248/1/\{1,248,4124\}$ (passed). [X] the kill criterion, [E] the exact discriminators. *(xxv) The exact MMST citation audit* ([verification/v458_seam_mmst_citation_audit.py]): a hypothesis-by-hypothesis reading of the cited MMST theorem (`arXiv:2107.13834`) against the collar — H1 CAR class, H2 massless limit, H3 the per-copy $c=\frac{1}{2}$ Thm 4.11, H4 decoupled additivity to c=8, and H5 the WZW range $8\le 8\le 16$ with its 120 bilinear currents all *match*, isolating the single residual H6 to the 128-spinor extension $SO(16)_1\to(E_8)_1$ (the $\det K$ 4→1 holomorphy step), *outside* MMST's bilinear method; [E] the per-copy theorem, [C] the additivity and range, [O] the one open spinor extension — since (xxxii) certified at net level by the peer-reviewed crossed product. *(xxvi) The complementary lattice-VOA route* ([verification/v459_seam_lattice_voa_route.py]): the second, independent construction that supplies exactly H6 — for the even unimodular E_8 the lattice net A_{E_8} (Adamo–Giorgetti–Tanimoto,

arXiv:2506.01008; OS arXiv:2407.18222) has weight-1 space 248=8+240 with the 240 roots splitting 112 integer + 128 *half-integer* (the spinor, 248=120+128), so the lattice route builds the 128-spinor currents MMST (xxv) leaves open; [E] the even unimodularity and root split, [C] AGT constructs the holomorphic extension — since (xxxii) an *independent second witness*, the primary net-level certification being the peer-reviewed crossed product — [O] the routes leave one shared realisation input (the collar's scaling limit = A_{E_8}), tied to P_1 by (xxiii). The G-block arithmetic (the one-sided $8=|\mathbb{Z}_2|\int K/\pi$, the reflection $C\to -C$, $c_-=8$, the range $8\le 8\le 16$, $248=8+240=120+128$, $240=112+128$, $\det K$ 1 vs 4) is kernel-hardened in Lean (`FORM.SEAM.RESIDUAL.01`, `TfptCarrier/SeamResidualAxiom.lean`), reducing the residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus the two cited existence theorems. *(xxvii) The explicit flat-band commuting-projector (LTO) realisation* ([`verification/v460_seam_s3_lto.py`]): NPW26 prove the intrinsic Bisognano–Wichmann lemma of (iv) for *commuting-projector* (local-topological-order) models, so sub-step (i) asks for the seam realised as such a $\mathbb{Z}_2$-reflection LTO. On the same v367 $p+ip$ collar the flattened Bloch $Q=h/|h|$ is a frustration-free flat-band parent — $Q^2=I$ and the occupied $P=(I-Q)/2$ is an exact projector EQUAL to the ground-state projector — whose real-space kernel is gap-protected quasi-local ($\xi\approx 1.14$ at $M=1$, $|P(20,0)|<10^{-6}$; a clean power law $|R|^{-2.1}$ at the gapless point, so locality needs the gap), and whose flattened modular Hamiltonian $K_{\text{flat}}=\text{sign}(K)$ inherits the $\mathbb{Z}_2$ reflection $\Theta K_{\text{flat}}\Theta=-K_{\text{flat}}$ and the μ_4 clock $[\rho, K_{\text{flat}}]=0$ EXACTLY; [E] the flat-band/projector and reflection algebra, [C] the quasi-locality, [O] the chiral $c_-=8\neq 0$ obstructs *strictly* finite-range commuting-projector locality (Kapustin–Fidkowski, arXiv:1810.07756), so the realisation is exactly the quasi-local LTO net NPW26 use — advancing sub-step (i) of (iv) from abstract existence to an explicit gap-protected net, strict locality the cited residual. *(xxviii) The strict-locality obstruction made explicit* ([`verification/v461_seam_strict_locality.py`]): upgrading (xxvii)'s "strict locality is the cited residual" from a citation to an EXHIBITED topological obstruction. On the same v367 collar the FHS Chern is $C=+1$ (chiral) / 0 (trivial) / -1 (opposite mass), and the hybrid Wannier-centre (Wilson-loop) phase $\theta(k_y)$ winds by $|C|=1$ (chiral) and 0 (trivial) over the BZ — exponentially-localised Wannier functions exist *iff* that winding is 0 (Brouder–Panati et al.; Thouless), so the chiral collar has none; both phases are gapped, yet only the trivial one (winding 0) admits a strict finite-range commuting projector, so it is the *chirality* $c_-=8\neq 0$, not the gap, that obstructs. [E] the Chern integer, the winding= $|C|$ identity and the gapped neg-control, [C][O] Kapustin–Fidkowski (arXiv:1810.07756): a strictly finite-range commuting projector would force winding 0 hence $C=0$, so for $C=1\neq 0$ none exists — sub-step (i)'s realisation is *provably* the quasi-local NPW26 LTO net, strict locality topologically forbidden rather than a missing premise. *(xxix) The character-level 128-spinor extension* ([`verification/v462_seam_spinor_continuum.py`]): the constructive companion of (xxv)/(xxvi), exhibiting the 128-spinor extension three ways — the exact Jacobi/E_8 identity $\theta_2^8+\theta_3^8+\theta_4^8=2E_4$ makes the holomorphic character the sum of the $SO(16)_1$ vacuum and spinor sectors, $\chi_{(E_8)_1}=E_4/\eta^8=\chi_o^{SO16}+\chi_s^{SO16}$ (weight-1 counts 120+128=248); the finite 16-Majorana Fock space already carries them ($C(16,2)=120$ currents and $2^{16/2}=256=128+128$ Ramond states); and a critical free-fermion ring converges to the chiral CFT (Casimir $c_-\to 8$, lowest scaled gap $x_1\to 1$ with $|x_1-1|\sim 1/N^2$). [E] the θ identity, the 120+128=248 split and the finite-L convergence, [C][O] the rigorous holomorphic extension is certified at net level by the crossed product (xxxii), with AGT/AMT (xxvi) as the independent second witness. *(xxx) The c=8 holomorphic-uniqueness pin* ([`verification/v463_seam_c8_holomorphic_uniqueness.py`]): the positive lever that makes the *identification* half classification-forced rather than assumed. At level 1 the central charge $c=\dim\mathfrak{g}/(1+h^\vee)=8$ has *three* simple solutions — A_8 ($\dim 80$), $D_8=\mathfrak{so}(16)$ ($\dim 120$) and E_8 ($\dim 248$) — so $c=8$ is necessary but *not* sufficient, and the $\det K=4$ $SO(16)$ rival of (xxiii) is a genuine same-c competitor, not a strawman. But a holomorphic (single-sector) $c=8$ theory has an $SL(2,\mathbb{Z})$-covariant character, and the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}=E_4/\eta^8$, whose q^1 coefficient (the one-dimensional candidate space, vacuum $a_0=1$) is 248 — so $\dim V_1=248$ is *forced*, selecting E_8 uniquely and excluding A_8/D_8. [E] the three candidates, the forced 248, the tower $\{1,248,4124,34752,213126\}=E_4/\eta^8=j^{1/3}$; [C] Dong–Mason/ Schellekens give the holomorphic $c=8$ VOA unique = V_{E_8} (the even unimodular rank-8 lattice unique = E_8, Mordell–Witt), so "$c=8$ holomorphic limit = $(E_8)_1$" is classification-forced. *(xxxi) The one-particle realisation rigidity* ([`verification/v464_seam_oneparticle_rigidity.py`]): attacking the *realisation* input R_1 itself. Since the seam is quasi-free (v113/v160/v161), by Araki's self-dual CAR the whole net is the second quantisation of its one-particle symbol P, so R_1 collapses to a finite-per-mode statement about P: the gapped ground state is the *unique* quasi-free state with $P=$ the spectral projection onto $K<0$ ($P^2=P$ to $<10^{-12}$, K fixed by gap + chirality + reflection v426); its real-space kernel is Cauchy on every fixed window ($\max|P_{2N}-P_N|\sim 1/N\to 0$, so P_∞ exists), and the limit carries $c=1$ per Dirac (entanglement

$S(L)=(c/3)\ln L$, 16 copies ⇒ $c_-=8$, tying v450). [E] idempotency, Cauchy convergence and the c-fit; [C] Araki/Shale–Stinespring give the state↔symbol bijection and implementable Bogoliubov map, so R_1 is upgraded from a bald axiom to "the unique quasi-free realisation, its one-particle limit exhibited, modulo the named CAR functor". *(xxxii) The net-level crossed-product route + the invariant-level realisation* ([`verification/v469_seam_crossedproduct_route.py`]): both halves of the residual re-founded on harder-to-reject ground. The 128-spinor extension needs no preprint: the $SO(16)_1$ spinor has $h_s=N/16=16/16=1\in\mathbb{Z}$, so its statistics phase is $+1$ — *exactly* the Longo–Rehren locality criterion — and the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1\rtimes\{1,s\}$ is a *local* index-2 extension by 1995–2001 *peer-reviewed* subfactor theory (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives $\mu(B)=4/2^2=1$ ⇒ holomorphic ⇒ $(E_8)_1$ by the (xxx) pin — so the AGT/AMT route (xxvi) is *demoted to an independent second witness*. The index-4 μ_4 glue runs on the same integer, $h(J^k)=\{1,1,1\}$ for $k=1,2,3$ (the v125 isotropy $q(k(1,1))=k^2$ at net level, $\mu=16/4^2=1$). And the realisation input R_1 is reduced from model fiat to *invariants* (R1′): quasi-free [C] (v155/v160) + gap [E] (v302) + class D + $c_-=8$ [E] ((xxiii), from P_1) — computed on the collar as FHS Chern $|C|=1$ (control 0), $\nu=16\times|C|=16$, $c_-=\nu/2=8$; and $\nu\equiv0$ (mod 16) is exactly Kitaev's class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (the sixteen-fold way; c_- a *phase* invariant, Kapustin–Spodyneiko; Kim et al.), so any realisation with the R1′ invariants is phase-equivalent to the v367 stack. [E] the locality integers, the KLM μ-arithmetic, the computed invariants and the composed chain; [C] the re-founded citation package; [O] the cited theorems stay cited. The full G-block arithmetic — now including the (xxviii) winding obstruction, the (xxix) finite-Fock spinor counts, the (xxx) holomorphy selector and the (xxxii) locality/μ/sixteen-fold joints — is kernel-hardened in Lean (`FORM.SEAM.RESIDUAL.01`), the two formerly separate cited theorems are *merged* into one combined external axiom `seam_realisation_theorem`, so the residual reduces to ONE realisation axiom plus ONE cited theorem (`#print axioms` drops from six to four); and the (xxxii) route is Lean-carried as the *parallel* derivation `seamResidualClosed′` (the invariant-level axiom `collar_invariants` plus the single combined package `crossedproduct_route_theorem`, its joints — `lr_locality_integer`, `glue_locality_integers`, `klm_holomorphic`, `sixteenfold_pin` — kernel facts with no axioms). Across (v)–(xxxii), `SEAM.EQUIV.01` stays [O]: with (xxx) pinning the identification, (xxxi) reducing the realisation to its one-particle data and (xxxii) re-founding the extension leg on peer-reviewed ground, the residual is now *entirely certification* — a named, hypothesis-audited package of published theorems (MMST, Longo–Rehren/Böckenhauer/Böckenhauer–Evans/KLM, Dong–Mason, Araki, Shale–Stinespring; AGT/AMT as second witness) with no open internal mechanism — yet still rests on cited continuum-existence theorems we do not re-prove.

The TOE/QFT closure contracts as three citable certificates ([`verification/v398_seam_state_rigidity.py`], [`verification/v399_gravity_complete.py`], [`verification/v401_metrology_closure.py`]). Three companion contracts package the closure status as named targets without fabricating any closure. *(i) Seam State Rigidity* (`SEAM.RIGIDITY.01`): the scattered state-invariance reductions (v177/v199/v308/v309/v335) as ONE theorem target — [E] the RP-definability hinge (OS canonical transfer from RP + reflection alone, v54, quasi-free v155), [E] the band-limited μ_4-character-block-diagonal core swept N=4..64 (beyond the 3-dim H^1, made uniform in N to N=256 by v442), [E] holomorphy ⇒ $(E_8)_1$; and [E] the forcing converse (v445) that commuting with the order-4 clock is *exactly* μ_4-block-diagonality, so the residual is no longer "does block-diagonal permit" but only $[\rho,\Lambda_\Sigma]=0$ on the full L^2 — the physical transfer lying in that commutant (intrinsic Bisognano–Wichmann). [E] Moreover (v446) that operator commutation is itself *derived*: the OS canonical transfer of μ_4-symmetric RP data inherits the clock (with the modular Hamiltonian commuting, v258), so the residual reduces further to the *structural* statement that the seam RP data is μ_4-symmetric (the four marks, `QGEO.SYM.01`), leaving [O] only that structural symmetry on the raw continuum collar — for which the four marks now have an *established* continuum frame: the four-interval multilocal modular geometry (v480, keybox below), where the invariance is manifest at the state level in the exactly solvable class. *(ii) Gravity-complete = Boundary-complete + Ambient-redundancy* (`GRAVITY.COMPLETE.01`): the local equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C] (v358/v359) plus the redundancy discriminators (v369), with $\mathcal{O}_{\rm phys}\subset\mathcal{A}_\Sigma$ checked over five gravitational readout classes. *(iii) Metrology closure* (`METROLOGY.CLOSURE.01`): the final input set $\{a,\pi,v_{\rm geo}\}$ — 0 dimensionless dials + 1 unit, the dimensional bookkeeping that every dimensionful observable is a number $\times v_{\rm geo}^d$ (No-Unit, v153), and $v_{\rm geo}$ theorem-forbidden ($\mathrm{TOE}_{\rm strict}=\mathrm{TOE}_{\rm dimensionless}+[v_{\rm geo}]$, v384).

One surface for seam, carrier and lattice: the Kummer/K3 ([verification/v260_k3_kummer_unification.py]). The pillowcase is the elliptic-involution quotient $T^2_{\tau=i}/(z\mapsto -z)$, its four marks the four 2-torsion fixed points; squaring the construction to complex dimension two gives the *Kummer surface* $\mathrm{Km}(T^2\times T^2)$, a K3, whose three facets reproduce three TFPT objects at once. [E] the 16 nodes (the 2-torsion $|A[2]|{=}2^4{=}16$ of the abelian surface) resolve to 16 exceptional $\mathbb{P}^1$'s $=\dim S^+=\Lambda^{\mathrm{even}}(\mathbb{C}^5)=16$ (v197), the carrier-16 as the *square* of the seam-4; [E] the cup-product form on $H^2(\mathrm{K3},\mathbb{Z})$ is the even unimodular lattice $U^3\oplus E_8(-1)^2$ of signature $(3,19)$, so the same E_8 the carrier glues to (v1) is the surface's intersection form. (Honestly: the 16 node classes span the rank-16 Kummer lattice, a sublattice *distinct* from the $E_8(-1)^2$ summand — both exact, not identified.) Thus the v1 "glue $D_5\oplus A_3{+}\mu_4\Rightarrow E_8$" is the lattice shadow of a single smooth object that already carries the seam and the carrier; E_8 appears both *locally* as the du Val $\mathbb{C}^2/2I$ singularity (the Poincaré-sphere link, v232/v236; by Kronheimer that local model *is* the hyper-Kähler quotient of the marks quiver, v479) and *globally* as this H^2 summand. A unification (`ARCH.K3.01`), *not* a closure: the lattice arithmetic is [E], the carrier↔node identification [C].

Synthesis — the Modular Spectral Closure ([verification/v261_modular_spectral_closure.py]). The ten steps above are not ten independent results: they are facets of one relative spectral datum

$$\mathsf{TFPT}_{\mathrm{QFT}}=\left(\mathcal{A}_\Sigma,\ \omega_\Sigma,\ \Delta_\Sigma,\ \rho,\ A_F,\ H_F,\ D_F,\ J,\ \gamma,\ S_{\mathrm{rel}}\right)$$

(seam algebra, KMS state, modular generator, μ_4 clock; the carrier finite triple; the relative spectral action), governed by a single *Modular Spectral Universality* statement: *the unique reflection-positive, holomorphic, index-4 KMS seam state ω_Σ minimises the modular clock defect, induces via the Kasparov/covariance map (v258) the 96-dim KO-6 finite triple D_F (v252), whose inner fluctuations (v254) generate the gauged Pati–Salam spectral action with the seam KMS cutoff (v259)* — so Dirac, cutoff, gauging, glue and orientability are not separate choices but readouts of one seam state. The certificate v261 pins the cross-consistency: one number $4=[B{:}A]=|\mu_4|=2\chi=|(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 ($\dim S^+=\Lambda^{\mathrm{even}}(\mathbb{C}^5)=$ Kummer nodes, $H_F=96$), and one gap $\Delta=6\log(3/2)$ (OS mass gap = recovery rate = Lindblad gap, static and dynamic). The honest account of completeness is therefore:

- [E]/[C] **QFT-complete (modulo one premise).** The admissible OS sector, the boundary $(E_8)_1$ net, the relative spectral action, D_F (as covariance induction), the cutoff (as the KMS weight) and the gauging (as inner fluctuations) are one relative object — closed once `QGEO.SYM.01` (the seam realisation: *the raw quasi-free seam state is μ_4-invariant*) is granted. That single [O] premise is the only structural input the QFT layer still consumes.
- [C] **Ambient QG — a redundancy, separate by design.** The full covariant metric-sector / ambient quantum-gravity measure (`QG.AMB.01`) is gap-decoupled ($\Delta_{\mathrm{eff}}=1.648>0$) and is now a [C] *redundancy* (v369), *not* a blocker for the QFT layer: a certification object, not a missing piece of the boundary QFT.

So "solving the QFT" here means exactly this: the boundary QFT is a single relative object reduced to one foundational symmetry postulate, with ambient QG held apart — not a diffuse list of open interfaces. The MSC closes the boundary relative QFT object modulo `QGEO.SYM.01`; it does *not* assert SM-only 4D gauge unification, and the canonical 4D reading is boundary only.

4D-QFT fork policy — a frozen decision tree, not an open ambiguity ([verification/v265_qft4d_fork_freeze.py])

The 4D reading is a *branch policy*, machine-frozen in `qft4d_fork_freeze.json`:

- **A (canonical) — boundary only.** [C] *Boundary-only is the canonical 4D reading*: TFPT delivers the boundary QFT, the relative spectral action and the SM readout; a 4D-GUT is *not claimed by default* (E_8 is the audit hull, never an unbroken 4D gauge group).
- **SM-only 4D-GUT unification is killed.** [X] the plain SM couplings do not meet (at the $\alpha_1{=}\alpha_2$ scale α_3^{-1} is off by $\sim$5.6; spread $\sim$8.8 at $2{\times}10^{16}$ GeV, v246). This is the canonical *expectation*, not a failure.
- **B (conditional) — carrier-native Pati–Salam UV branch.** [C]/[O] permitted *only* with the E_8-allowed content $\{1, 10, 16, 45\}$ (no **126**, v247), $\sin^2\theta_W{=}3/8$ (v248), $\kappa{=}M_{PS}/M_s$ in the frozen $O(1)$ band (v249/v253), and a threshold model with a proton-decay kill test; carrier gauging itself stays the contract `CONTRACT.QFT4D.DIRAC.01`.
- **Proton-decay rule, now applied ([verification/v266_ps_threshold_proton.py]).** [X] the explicit two-step thresholds give $d{=}6$ $\tau_p(p{\to}e^+\pi^0)$: the *minimal* 16-Higgs content ($M_{GUT}{\sim}2.4{\times}10^{15}$) gives $\tau_p{\sim}4.4{\times}10^{33}$ yr $<$ Super-K ($2.4{\times}10^{34}$) and is *killed*; the E_8-allowed $(15, 1, 1)$ [the single 45] raises $M_{GUT}{\sim}5.9{\times}10^{15} \Rightarrow \tau_p{\sim}1.6{\times}10^{35}$ yr, proton-safe *now* but sitting at the Hyper-K reach ($\sim 1.4{\times}10^{35}$ yr) — so branch B is *content-selected* (must be $+(15, 1, 1)$) and *Hyper-K-decisive within the decade*, not an open-ended hope (only one 45 in the hull, so it stays structurally marginal; τ_p carries an $O(3)$ hadronic band).

The fork is thus a frozen decision tree with explicit promotion/demotion rules, *not* an open ambiguity.

And the bedrock is now a falsifiable kill-test, not a proof-promise ([verification/v215_seam_deck_killtest.py]). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the very covariance it should produce); the honest move is to recast `QGEO.SYM.01` as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives $\omega{\circ}\rho{=}\omega \Leftrightarrow [\rho, C]{=}0$ for the quasi-free covariance, with the principal symbol $|k|$ commuting exactly, so the whole residual sits in the bounded sub-principal M_f (equivalently: the raw Gaussian bulk form A is μ_4-deck-invariant). The four levers are **K1** exactly four marks ($b_1{=}$#marks$-1{=}3{=}N_{\text{fam}} \Rightarrow$ #marks$=4{=}|\mu_4|$); **K2** the *square* configuration (cross-ratio $2 \Rightarrow j{=}1728$, Aut$=\mathbb{Z}/4$; the clock order $h(A_3){=}4$ selects $\mathbb{Z}/4$, not the $j{=}0$ $\mathbb{Z}/6$, and a generic mark set has only $\mathbb{Z}/2$); **K3** the mod-4 sub-principal support ($[\rho, M_f]{=}0 \Leftrightarrow f$ is $\mathbb{Z}_4$-invariant, automatic for a μ_4-mark sum); and **K4** the transfer gap $(2/3)^6{=}64/729$. These are frozen in `freeze_file.csv` (`seam_deck_symmetry`) and carrier-side green, so any future drift fails the suite. The *decisive* kill is deferred to `QGEO.REALIZE.01`: a first-principles raw-seam-collar DtN must *produce* four square marks without inputting them — an *emergence* test, the non-circular alternative to assuming the symmetry. The *light* version of that deferred test is now *executed* ([verification/v503_qgeo_emergence_light.py], `QGEO.EMERGE.LIGHT.01`), with the result *rigidity, not dynamical selection*: the free field does *not* select the square — the global $\det'$ winner at fixed area is the *hexagonal* torus ($\log F(\rho) = -1.03352 > \log F(i) = -1.05469$, Osgood–Phillips–Sarnak), $\tau = i$ is a saddle of the full moduli space, the only critical points are the CM points ($E_2^*(i) = E_2^*(\rho) = 0$), and criticality at the square is Palais symmetric criticality — while the square *is* pinned sharply by the clock: the raw free-field DtN is mod-4 block-diagonal *iff* $\tau = i$ (an isolated zero of the K3 indicator as a modulus function, 8×10^{-15} at $\tau = i$, strictly monotone away), and $\mathrm{Aut}(E_\rho) = \mathbb{Z}_6$ carries no order-4 element. The clock input itself is now *reduced to one bit* ([verification/v506_seam_clock_rigidity.py], `SEAM.CLOCK.RIGIDITY.01`): the deck has exactly two Möbius square roots, both automatically of order 4, and a marks-preserving root exists *iff* the deck is the central involution of the mark-stabiliser D_4 — given that alignment bit, order, uniqueness, the free 4-cycle and $\tau = i$ are all theorems (harmonicity alone is insufficient on bare mark data, exact counterexample — on free-deck circles it becomes sufficient, v510); on the Fock side the NS deck implementation is *forced* to order 4 ($U^2 = (-1)^F$ exactly, nonsplit $\mathbb{Z}_4$; the R sector splits), so the fermionic $\mathbb{Z}_4$ explains the *order* and only the spatial alignment remains the carrier input. That bit has now survived its *tautology attack* ([verification/v507_seam_bit_origin.py], `SEAM.BIT.ORIGIN.01`): the marks are the four half-periods of the seam double and *every* mark-free collar deck is a half-period translation — the deck's *class* is derived from the common torus origin — yet the origin does not force the alignment (the marks-preserving root exists iff the deck pairs separate harmonically, $\lambda \in \{-1, 2, \frac{1}{2}\}$, and the silver counterexample sits in-family as "$\tau = i$ with

a non-CM-fixed half-period"); the bit is an equivalence tetrad (clock $\iff$ $\tau{=}i$ with the CM-fixed half-period $\iff$ deck central in the mark-D_4 $\iff$ harmonic deck pairs $\iff$ nonsplit NS Fock lift) whose physical face is the Fidkowski–Kitaev-type extension class (central $U^2 = (-1)^F$ vs edge $U_e^2 = +1$, split) — distinguished, but not derived: only the *position* (which half-period) remains the carrier input. That position half is now *topology* ([verification/v510_seam_bit_freedom.py], SEAM.BIT.FREEDOM.01): the collar deck is a *covering* deck, hence free on the seam circle; the unique free involution is the antipode ($\mathrm{Aut}(C_{16}) = D_{16}$, complete census), nonsplit $\iff$ free over all 17 involutions in exact Cl(16), and the edge/silver class — whose mark circle passes through the deck poles, the restriction of the *branched* pillowcase involution — is excluded; harmonic + free solves exactly to $b = \pm i$, so the bit reduces to the square-modulus datum $\tau = i$ alone. That datum now carries an equivalent *discrete* form ([verification/v512_seam_tau_flag.py], SEAM.TAU.FLAG.01): on the free seam circle bare mark-transitivity is automatic for every deck-invariant configuration $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$ (the pair-exchanging V_4; no local jet sees the side bit), and what selects $\delta = \pi/2 \iff \tau = i$ is exactly *flag* transitivity ($V_4 \to D_4$ symmetry lift), welded into a 13-fold exact equivalence web (since 15-fold by v528 facets #14/#15) — the remaining input is *one discrete symmetry-lift bit*, its negation concretely measurable (odd-doublet split $(2/\pi)\ln\cot(\delta/2) \neq 0$). The candidate [E]-closure route through free OS positivity is since *decided negative* ([verification/v521_seam_bit_rp_blind.py], SEAM.BIT.RPBLIND.01): free reflection positivity and anti-linear Θ existence (the v519 battery) are *side-blind* along the whole $M(\delta)$ family — bond-cut Gram spectra identical to 40 digits for every tested δ, the $N = 16$ odd-m failure a lattice-placement artifact resolved at $N = 32$, the continuum kernel exactly $1/\sin((s{+}t)/2)$ with the axis position dropping out — so the free RP/Θ battery is the *eighth* side-blind test on the v512 scoreboard ($7 \to 8$); the only δ-sensitive clauses presuppose the clock (the group generated by the admissible OS reflections closes to D_4 exactly at $\delta = \pi/2$: face #13 of the web, a typed [C] reformulation), and the honest residual [O] gap is the mark-decorated *state* class (defect/twist insertions, the interacting $\mathcal{A}_{\mathrm{hol}}$) — the mark-decorated half of that gap is since decided side-blind too ([verification/v525_seam_bit_twist_blind.py], SEAM.BIT.TWISTBLIND.01: every well-defined mark-decorated state functional in the free-plus-twist class keeps RP positive along the whole $M(\delta)$ ladder — the *ninth* side-blind test, $8 \to 9$; the free-plus-twist class is exhausted) and the interacting half is since decided too ([verification/v529_seam_interacting_toy_fk.py], SEAM.INT.FKTOY.01: the *tenth* side-blind test, $9 \to 10$, on genuinely non-Wick dynamics — Θ_{15} mirrors $H(m) \to H(8-m)$ exactly despite δ-seeing spectra) — while the bit itself is now *physically defined* as the twist-class choice ([verification/v528_seam_bit_twist_class_definition.py], SEAM.BIT.TWISTCLASS.01: facets #14/#15 both directions, order parameter O, unification with flag-transitivity, web $13 \to 15$; [C] interferometric measurement sketch; the bit remains formal input) — and since carries the first *positive* selection datum: the straddle filter, run as a selector on the interacting mark family, keeps exactly *one* member alive — $\delta = \pi/2$ with positive coupling ([verification/v534_seam_straddle_cone.py], SEAM.STRADDLE.CONE.01; toy-level evidence for a dynamical origin, not a derivation). The full QGEO.REALIZE.01 stays [O]; no marker moves. [E] carrier-side (the four levers), [O] seam-side (the emergence): a kill-test, *not* a closure (QGEO.KILL.01).

And the mark count is now *derived*, not assumed: four marks from Gauss–Bonnet ([verification/v216_marks_gauss_bonnet.py], [verification/v217_marks_emergence_scan.py]). The K1 lever ("exactly four marks") is the one that can be *executed* today. If the marks are $\mathbb{Z}_2$ branch points (cone angle π, deficit π) on the flat seam sphere ($\chi{=}2$ — the same χ that supplies the 8 in c_3), Gauss–Bonnet *forces* their number: $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = |\mu_4| = N_{\mathrm{fam}}{+}1$. The closed Euclidean sphere 2-orbifolds are exactly $(2,3,6)$, $(2,4,4)$, $(3,3,3)$, $(2,2,2,2)$, and *three independent* criteria — all cone orders $= 2$ (the $|\mathbb{Z}_2|$ sheet branch), $\mathrm{rank}\, H^1 = \#\mathrm{marks} - 1 = 3 = N_{\mathrm{fam}}$ (the three generations pick the 4-mark square over the 3-mark hexagonal $(3,3,3)$), and a free order-4 deck — converge on the pillowcase $(2,2,2,2)$. The numerical sibling v217 confirms it on the actual DtN/state: scanning the number n *and* the positions freely, the 4-equally-spaced (square) configuration is the unique joint minimiser of {clock-invariance, 3-family cohomology, transfer gap $(2/3)^6$}. So the mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence QGEO.REALIZE.01 stays [O]. The light emergence test ([verification/v503_qgeo_emergence_light.py]) sharpens this residual to a *convergence*: given the clock, the modulus freezes automatically ($P(iZ) = P(Z)$ forces $a_3 = a_2 = a_1 = 0$ with $j = 1728$ identically, v493), and the order-4 element on H^1 is the Coxeter element of $W(A_3)$ — exactly the P2 cusp-class carrier datum of v499: the QGEO modulus rest and the P2 typing rest R1 are *one* common residual ("why does the raw seam carry the order-4 clock"), not two. And the clock-rigidity theorems (v506) reduce that common residual to *one alignment bit* — "the collar deck is the central involution of

the mark-D_4" — with the clock's *order* fermionically forced ($U^2 = (-1)^F$ exact); the bit-origin theorems (v507) then derive the deck's *class* (every mark-free collar deck is a half-period translation of the seam double) and characterise the bit as an equivalence tetrad with a physical extension-class face, so only its *position* — which half-period — stays the input; the freedom theorems (v510) finally type that position half as *topology* (the collar deck is a covering deck, hence free; the edge class is excluded), so the residual is the square-modulus datum $\tau = i$ alone — restated by the flag-transitivity web (v512) as one discrete symmetry-lift bit ($V_4 \to D_4$: flag transitivity $\iff \tau = i$), and since v528 equivalently as the *twist-class choice*, in principle interferometrically readable (`SEAM.BIT.TWISTCLASS.01`; the bit remains formal input). [E] (count + equality), [O] (the square modulus and the raw-seam realisation) — `QGEO.MARKS.03` / `QGEO.EMERGE.01`.

And the four marks are a *four-interval* modular geometry ([`verification/v480_multilocal_four_interval.py`]). The seam circle minus its four marks is the n=4 disjoint-interval region of a chiral free fermion — the one continuum class whose *multi-interval* modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu multi-interval subfactors; Rehren–Tedesco multilocal fermions; and Kawahigashi–Longo–Müger: *holomorphic* ⇔ trivial multi-interval inclusion, μ=1 — the same one bit as $\det K$=1). Until now the suite used only the single-ball CHM boost (v358); v480 exhibits the multilocal mechanism on the exactly solvable antiperiodic critical ring: [E] the quarter-turn clock is a signed permutation with $\rho^4 = -1$ *exactly* (the fermionic μ_4 clock is the order-8 double-cover lift — the same $\mathbb{Z}_2$-lift pattern as $2I$ over I); [E] its primitive-8th-root character projectors decouple the 4-interval covariance at 10^{-15} (the diagonal 4-fold-cover isomorphism), each sector a *single*-interval quarter-ring kernel with boundary twist (10^{-14} spectra); [E] $\rho C \rho^{-1} = C$ at 10^{-14}, so $\omega \circ \rho = \omega$ holds *manifestly* at the 4-interval state level — and a one-site displacement of a single interval re-couples the sectors at 10^{-2} (the invariance is the μ_4-symmetric configuration, not an artifact); and the cross-interval modular coupling peaks on the conjugate-point diagonal, the lattice shadow of the Casini–Huerta mixing term (numerical). [C] the continuum multi-interval theory is cited, not re-derived; [O] the raw-seam premise (`QGEO.SYM.01`) is untouched — this puts the bedrock and the AP2 continuum leg (v476/v478) in one established, exactly solvable frame; it does not close them.

`QGEO.SYM.01` is *not* a theorem — it is the final seam-identification premise

The single load-bearing bolt of the whole construction, on one page.

1. **What is identified.** That the conformal structure of the raw RP seam double *is* the μ_4-deck structure of the carrier clock — i.e. the carrier (P2) and the seam's conformal geometry are the same object. Equivalently, in Riemann–Roch *index* language: the seam's degree-4 mark divisor on the genus-0 double has $h^0 = \deg + 1 = 5 = g_{\mathrm{car}}$ and $\operatorname{rank} H_1 = \deg - 1 = 3 = N_{\mathrm{fam}}$ (with $\Lambda^{\mathrm{even}}(\mathbb{C}^5) = 1{+}10{+}5 = 16 = \dim S^+$), so P2 is the genus-0 mode count of the four-mark seam divisor ([`verification/v228_rr_index_gate.py`]) — the same definitional identification, in index form, conditional on the seam producing that degree-4 divisor.
2. **Why it is weaker than `QGEO.CONF.01/ISO.01`.** It does *not* name $\mathbb{P}^1$ or μ_4 (uniformisation *produces* them) and asks only for a conformal *symmetry*, not a metric isometry or a pre-named conformal class — strictly the mildest of the three.
3. **What follows automatically (the conditional theorem).** *Given* `QGEO.SYM.01`, equivariant uniformisation puts the order-4 clock in the form $z \mapsto iz$ on $\mathbb{P}^1$, the four parabolic marks are its free orbit μ_4, H^1 carries the $(1, 2, 3)$ grading, the seam Calderón contraction is the rank-8 K_Σ polarisation, and the index-4 simple-current extension is $(E_8)_1$ — the *entire* AQFT closure of v175–v181.
4. **What would falsify or unseat it.** A seam double of genus > 0 (uniformisation to $\mathbb{P}^1$ fails); a carrier clock that is *not* a conformal symmetry of the seam geometry (a non-deck transport); or a fourth chiral generation ($N_{\mathrm{fam}} \neq 3$ collapses $b_1 = N_{\mathrm{fam}}$ and the four-mark μ_4 structure). The first two are constructive-geometry questions about the raw seam; the third is the standard family-count kill test.

Honest framing. The correct external statement is *conditional*: "*Given* `QGEO.SYM.01`, the RP seam boundary realises the μ_4 deck structure, and the AQFT closure to $(E_8)_1$ follows through the verified chain v175–v181" — *not* "the theory *derives* the seam geometry". The premise stays [O]; everything downstream of it is [E]. Per v323 (Bisognano–Wichmann) `QGEO.SYM.01` is itself the *downstream* conformal-deck face of the single named keystone `SEAM.EQUIV.01` (the raw RP seam = the holomorphic $(E_8)_1$ net at τ=i), which is now *closed modulo cited theorems on its MMST route* `SEAM.EQUIV.MMST.01` (the lattice/S_3

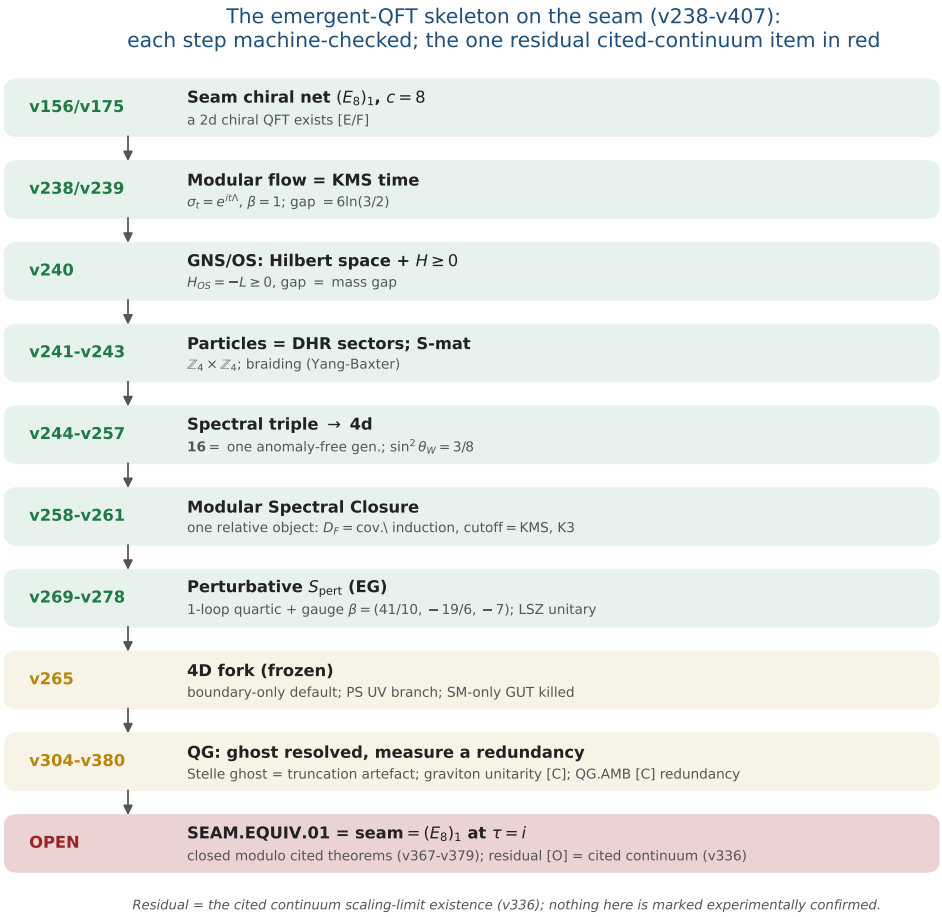


Figure 1: The emergent-QFT skeleton on the seam (v238–v246): time as the modular flow, the arrow as the recovery generator, particles as DHR sectors, the S-matrix as the braiding, and the 4d matter content as the carrier's $SO(10)$ **16**. Each step is machine-checked; the one residual — `SEAM.EQUIV.01`, whose MMST route `SEAM.EQUIV.MMST.01` is closed modulo cited theorems (v367–v379) with the cited continuum scaling-limit existence (v336) as its [O] leg (parent [O]) — is in red, the frozen 4D fork and the QG certification layer in gold. The whole skeleton is *conditional* on that keystone and is neither solved nor experimentally confirmed.

stack pins it at every computable level, Lean `FORM.SEAM.MMST.01`; only the cited continuum existence stays [O]; the twistor route `SEAM.EQUIV.TWISTOR.01` and the unconditional parent stay [O]).

This is not a gap to apologise for — it is TFPT's one fundamental postulate. Every fundamental theory rests on an irreducible premise (the constancy of c in relativity, the equivalence principle, the canonical commutator). `QGEO.SYM.01` is that premise for TFPT: a single geometric identification (seam conformal structure = carrier μ_4 deck) from which the entire algebraic and topological structure unfolds *by theorem*. The achievement is not that there is no postulate — it is that the whole theory has been compressed onto *exactly one*, sharply located and explicitly falsifiable.

The Seam-Holomorphy Theorem — the one named closing target [E]/[O] ([`verification/v234_seam_holomorphy_selection.py`])

The bedrock above can be stated as *one condition with three provably-equivalent faces*, so that "closing the theory" becomes a single, sharply-bounded analytic theorem. **The condition:** the seam carries *no nontrivial abelian sector*. Its three faces all force E_8:

1. **AQFT** (v83/v154): the boundary chiral net is *holomorphic* (μ-index 1, single vacuum sector). Holomorphic $+ c = 8 \Rightarrow$ the unique even unimodular rank-8 lattice net $(E_8)_1$, and $c = 8 = g_{\text{car}} + N_{\text{fam}}$ is itself structural.

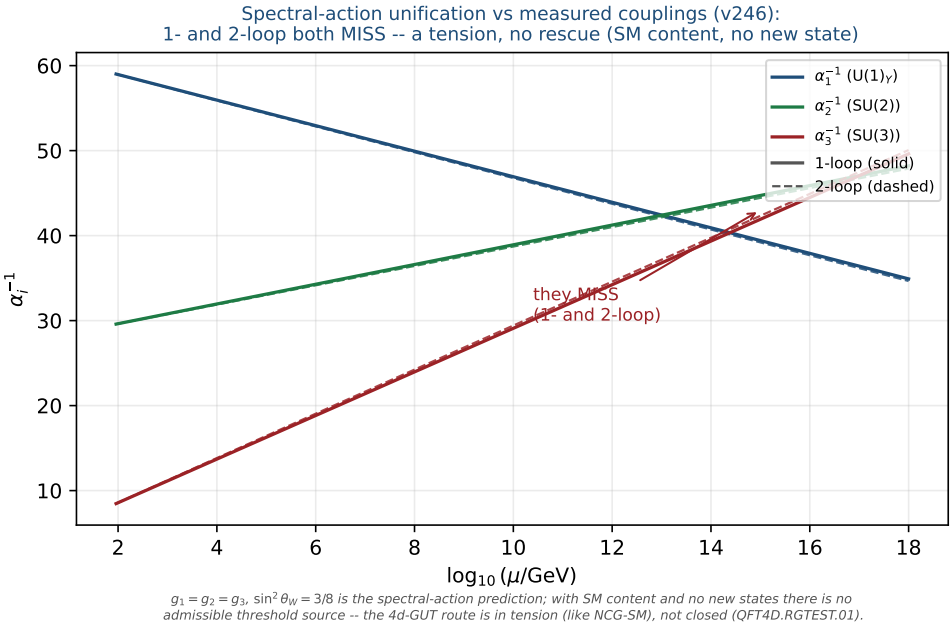


Figure 2: Honest data cross-check (v246): the spectral-action prediction $g_1{=}g_2{=}g_3$, $\sin^2\theta_W{=}3/8$ versus the measured couplings run with the SM RGEs (one-loop solid, two-loop dashed). The three lines do *not* meet at any scale (1-loop residual $\Delta\alpha^{-1} \approx 9$ at $2{\times}10^{16}$ GeV; 2-loop minimal spread ≈ 3.2 at $1.7{\times}10^{14}$ GeV) — a falsifiable tension. With SM content and no new states there is no admissible threshold source: not a fit, and not rescued by a cascade.

2. **Geometry** (v232): the seam link is a *homology* 3-*sphere* ⇔ its binary group Γ is *perfect* ⇔ $\Gamma = 2I$ (the Poincaré sphere $S^3/2I$).
3. **Representation theory** (v219): exactly *one* 1-dim irrep (the trivial one).

These are the *same* number: #(1-dim irreps) $= |\Gamma^{\mathrm{ab}}| = |H_1(S^3/\Gamma)| =$ #(mark-1 affine-Dynkin nodes), which equals 1 *only* for E_8 ($A_n{\to}n{+}1$, $D_n{\to}4$, $E_6{\to}3$, $E_7{\to}2$, $E_8{\to}1$). So $2I$ is the unique nontrivial perfect finite $SU(2)$ subgroup and $S^3/2I$ the unique homology-sphere space form. **Consequence:** Research Contract 2 (G_{net}/Target A), the carrier P2, and the Kleinian seam are *not* separate gates — they are this *one* condition.

The closing chain (what a complete closure would be), [O] at one step:

$$\mathrm{RP} + \mathrm{gap} + \mathrm{chirality} \xrightarrow{\text{v160}} \text{quasi-free bulk} \xrightarrow{?} \text{holomorphic boundary net}$$

$$\xrightarrow{\text{[E]}} (E_8)_1 \xrightarrow{\text{v219/v228}} 2I,\ (2,3,5),\ \mathrm{P2} \xrightarrow{\text{v179–v181+Kleinian genus-0}} \texttt{QGEO.SYM.01}.$$

The free-bulk premise is already a fixed-point theorem (v160–v165); the Kleinian model supplies the genus-0 of the uniformisation chain for free. **The single remaining analytic step is "free Gaussian RP bulk ⇒ holomorphic (single-sector) boundary net"** — the bulk–boundary statement that an invertible (Gaussian) bulk has a holomorphic chiral boundary. That step is now *closed modulo cited theorems* (the MMST route `SEAM.EQUIV.MMST.01`): the lattice/S_3 stack pins it at every computable level (v367/v368, v376–v379, Lean `FORM.SEAM.MMST.01`), only the cited continuum existence staying [O] (v336); v_{geo} stays the No-Unit metrology primitive and F_{transfer} stays external physics. This box certifies that the three faces coincide and force E_8 ([E]); the analytic step is closed modulo that cited theorem.

A physical realisation of the step (`[verification/v235_seam_chern_simons.py]`). The step lands in standard physics. A free gapped *bosonic* 2+1d bulk is an abelian Chern–Simons / K-matrix theory (even integer K), where #anyons $= |\det K|$, the edge chiral central charge is $c =$ signature(K), and the edge net is *holomorphic* ⇔ $|\det K| = 1$ — so "no nontrivial abelian sector" is precisely $\det K = 1$. The extension tower of v92 *is* the anyon-condensation tower read by determinant: $D_5{\oplus}A_3$ ($\det 16$, 16 anyons) $\to SO(16)_1{=}D_8$ ($\det 4$) $\to (E_8)_1$ ($\det 1$, holomorphic), all rank 8 with $c =$ signature $= 8 = g_{\mathrm{car}} + N_{\mathrm{fam}}$, so the holomorphic end is the *Kitaev* E_8 *quantum-Hall state*. Holomorphy ⇔ condensing the order-$|\mu_4| = 4$

Lagrangian glue (det 16→16/4^2=1); a partial order-2 isotropic only reaches det 4 (D_8). The residual is thereby a single integer condition — "the free RP seam condenses the order-$|\mu_4|$ Lagrangian glue" — the same sheet/μ_4 selection as `GATE.QGEO`, but now with holomorphic ⇔ det 1 a theorem (`GATE.HOLO.02`, [E]/[O]).

The residual as a *physical* statement ([`verification/v237_seam_sre_closure.py`]). The same integer condition is the physics of topological order: a K-matrix phase has ground-state degeneracy $|\det K|^g$ on a genus-g surface, so $\det K = 1 \Leftrightarrow$ *no* topological ground-state degeneracy on any closed surface ⇔ the bulk is *short-range-entangled* (invertible). Among rank-8 even positive-definite K ($c = 8 = g_{\rm car} + N_{\rm fam}$) only E_8 is SRE — the unique nontrivial bosonic SRE 2+1d phase, the Kitaev E_8 state, whose edge is the holomorphic $(E_8)_1$. TFPT already has a unique vacuum on the plane (RP + gap + cluster + OS); the plane cannot see $\det K$ (no non-contractible cycles), the torus can. So the one analytic step is the sharp, falsifiable physical statement *"the free RP gapped seam bulk is short-range-entangled (no topological order)"* (= $\det K = 1$ = the Kitaev E_8 phase, `GATE.HOLO.03`, [E]/[C]) — and the lattice/S_3 stack now answers it at every computable level (torus $\det K$=1, v378), so the keystone's MMST route (`SEAM.EQUIV.MMST.01`) is *closed modulo cited theorems*, only the cited continuum existence (v336) staying [O] (the parent stays [O]).

Lemma 2 (U1 — wall landscape, *corrected*)**.** *Parabolic degree zero forces the line weight-sums onto the wall* $(w_1, w_2, w_3) = (2, 1, 1)$*. An explicit balanced representative (weights in* $\frac{1}{3}$*) is*

$$W_{\rm wall} = \tfrac{1}{3}\begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

each column a permutation of $(0, 1, 2)$*, row sums* $(2, 1, 1)$*.* ***Honest result:*** *the* 6^4=1296 *column-permutation matrices give* 144 *walls; under the* full *group* $D_4 \times S_3(\text{lines}) \times S_3(\text{values})$ *they form* five *orbits,* not one*. So* $W_{\rm wall}$ *is* not *singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite, explicitly enumerated set;* $W_{\rm wall}$ *is one of the five. Machine: fully certified* $C_U^{(1)}$ *([`verification/v29_research_contract_certs.py`]).*

The splitting type is now algebraic (RH route, *[`verification/v32_rh_splitting.py`]).* *Passing from the monodromy to the Fuchsian* residues A_k *(eigenvalues = weights* $\{0, \frac{1}{3}, \frac{2}{3}\}$*,* $A_k = U^{k-1} A_0 U^{-(k-1)}$*,* $U = \mathrm{diag}(1, i, -i)$*) the twisted* $\mathbb{Z}_4$*-average collapses* exactly*:* $\sum_k U^k A_0 U^{-k} = 4\,\mathrm{diag}(A_0)$*, so the exponents at infinity are* $4\,\mathrm{diag}(A_0)$*. A flat bundle with regular* ∞ *needs integer exponents summing to* $4 = -\deg E$*; with the cusp weights the only options are perms of* $(2, 1, 1)$ *and* $(2, 2, 0)$*, i.e.*

$$\boxed{\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \iff \mathrm{diag}(A_0) = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4})}$$

(the sibling $(2, 2, 0) \to \mathcal{O}(0) \oplus \mathcal{O}(-2)^2$*). Such a Hermitian* A_0 *(spectrum* $\{0, \frac{1}{3}, \frac{2}{3}\}$*, that diagonal) exists by Schur–Horn. So the* splitting *is an algebraic condition on* $\mathrm{diag}(A_0)$*.* Honest wall: *the global* $\prod M_k = \mathbf{1}$ *is the path-ordered monodromy (*not $\exp(2\pi i \sum A_k)$*), constraining* off-diag(A_0) *via the genuine RH solve; and R-extraction still needs the* $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ *bridge. The RH route fixes the* splitting *algebraically but not* c_u/c_d*.*

Lemma 3 (U2 — the kill switch, *run and hardened* [E])**.** *The selected polystable wall representative must* not *collapse to the diagonal abelian* $U(1)^3$ *point if it is to give non-trivial* $SU(3)_F$ *transport mixing:* ***(A)*** $\nabla_F^\star$ *polystable but* not *simultaneously diagonal (the desired breakthrough), or* ***(B)*** *diagonal collapse, in which case* c_u/c_d *stays honestly [C].* ***Result: the cheap test is run robustly and lands on A.*** *The kill-switch fires iff the* D_4*-symmetric cusp representation is* reducible *(commutant non-scalar). Across* 400 *random cusp-class generators of the* $\mathbb{Z}_4$ *family* $M_k = U^{k-1} M U^{-(k-1)}$ *the commutant is scalar at* every *point (irreducible* 400/400*), the reducible locus is hit with probability* 0 *(measure-zero, codimension* ≥ 1*), and the mixing strength stays bounded away from zero (*min *off-diag* $= 0.35 > 0.05$*). So case A is the* generic *behaviour, not an accident of the one explicit point of [`verification/v33_explicit_flat_bundle.py`]:* ($U_{\rm wall}$) survives *its own cheapest falsification test. [`verification/v38_uwall_killswitch.py`]*

Lemma 4 (U3 — D_4 fixed locus as an explicit variety, *positive-dimensional*)**.** *With $M_k = U^{k-1}AU^{-(k-1)}$ ($U = \mathrm{diag}(1, i, -i)$, the $\mathbb{Z}_4$ rotation) and the reflection $M_1 = VM_1^{-1}V^{-1}$, $M_3 = VM_3^{-1}V^{-1}$, $M_2 = VM_4^{-1}V^{-1}$ (with $V = -[[1,0,0],[0,0,1],[0,1,0]]$, $V^2{=}1$, $VUV^{-1}{=}U^{-1}$), the relations become polynomials in the entries of $A \in C_{\mathrm{cusp}}$; $\mathcal{X}^{D_4}_{\mathrm{cusp}}$ is cut out as $\bigcup_k C_k$, parametrised by trace coordinates $x = \mathrm{tr}(M_1M_2)$ (note $\mathrm{tr}\,A = 1{+}\omega{+}\omega^2 = 0$).* ***Result ($C_U^{(2)}$, [verification/v30_d4_character_variety.py]):*** *adding the reflection to the $\mathbb{Z}_4$ locus of v19 does* not *isolate the point — the* full *D_4-fixed product locus is still* positive-dimensional *($|\mathrm{tr}(M_1M_2)|$ varies continuously over $\approx [0,3]$). So even the full D_4 symmetry cannot select $\nabla^\star_F$.* Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).

Lemma 5 (U4 — selectors as equations)**.** *The readouts $\rho \mapsto \det R(\rho)$, $\rho \mapsto \mathrm{Spec}(Q_+(\rho))$, $\rho \mapsto \Lambda_{f,j}(\rho)$ are well-defined (cyclotomic/algebraic) on $\mathcal{X}^{D_4}_{\mathrm{cusp}}$.* ***Acceptance criterion:***

$$\#\{\rho \in \mathcal{X}^{D_4}_{\mathrm{cusp}} :\ \det R = 8,\ \mathrm{Spec}(Q_+) = \{1,2,3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not *closed.* ***The bottleneck (made precise by U3):*** *since the D_4-fixed variety is positive-dimensional, the* selector *must do all the cutting — and evaluating $\det R(\rho)$, $\mathrm{Spec}(Q_+(\rho))$ as functions on $\mathcal{X}^{D_4}_{\mathrm{cusp}}$ requires the* parabolic-degree $\leftrightarrow$ residue dictionary *$R(\rho)$, i.e. the H2 equivalence. That dictionary was the single remaining analytic input of the (U_{wall}) gate.* Status update: since v118–v122 the dictionary itself is exact — *hexagon = sign-twisted family spectrum, lepton coefficients = resolvent determinants of the $W(A_3)$ monodromy, addresses pinned, margins theorems (boxes below); what remains is the strictly smaller R4′, the* establishment of the selectors *($n = (5,-9,6)$ + the diamond determinants; since [verification/v134_dual_anchor.py]: the pair (d,n)).*

What** $R(\rho)$ **is** ([verification/v31_R_dictionary.py]).* *Two facts pin its nature.* (i) Case A is alive: *every D_4-fixed cusp monodromy found is* irreducible*, so the wall point carries non-trivial $SU(3)_F$ mixing — the U2 kill switch does* not *fire to the diagonal case B.* (ii) $R(\rho)$ is not algebraic: *R is integer-valued, but the algebraic invariants of ρ (e.g. $\mathrm{tr}(M_1M_2)$) vary* continuously *on the locus; an integer readout cannot be a continuous algebraic function. Hence $R(\rho)$ is the discrete invariant of the* non-abelian Hodge / Mehta–Seshadri *map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle E_ρ): integer output, but its* computation *is the transcendental harmonic-metric (Hitchin) solve* on the generic Higgs locus. ***Refinement (U4″, [verification/v40_harmonic_metric.py]): *at the* polystable *point TFPT actually selects this pessimism does not apply — see the box below.*

Progress: the *selector side* of U4 is closed on the explicit point ([E]). While $R(\rho)$ as a function on the whole locus needs the Hitchin solve, the two U4 *selectors* are *algebraically* accessible on the explicit flat bundle of [verification/v33_explicit_flat_bundle.py] — no PDE. Computed exactly from A_0: the exponents at infinity $\mathrm{eig}(\sum_k A_k) = \{2,1,1\}$ give the splitting type $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$, i.e. the parabolic *anchor* $a = (1,1,2)$; the parabolic degree is 0 ($\deg E = -4 = -|\mu_4|$ plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

$$\begin{array}{c} \det R = 8 = n{\cdot}a \quad \text{(anchor pairing on the splitting type)}, \\ \mathrm{Spec}(Q_+) = \{1,2,3\} = 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\}) \end{array}$$

So $\det R = 8$ is the anchor pairing of the splitting type and $\mathrm{Spec}(Q_+)$ is the affine image ($d{=}3$ the weight denominator) of the cusp weights — *both selectors are read off the explicit bundle's algebraic data.* **Residual (sharpened below by Readout Rigidity):** the quark *ratio* c_u/c_d turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (v49), hence an integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation (U_{point}), which is an anchor, not a missing PDE. [verification/v39_uwall_selectors.py]

Breakthrough: the harmonic metric is *finite linear algebra*, not a PDE ([E]). The U4 "transcendental Hitchin solve" is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field $\Phi = 0$. So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of [verification/v33_explicit_flat_bundle.py] this is realised: the raw monodromies M_k are non-unitary ($\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$), but there is a *unique* common invariant Hermitian form H ($\dim\ker\{M_k^\dagger H M_k = H\} = 1$), it is *positive-definite*, and the unitarised holonomy $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$ is unitary ($\sim 10^{-8}$), lies in the cusp class $\{1, \omega, \omega^2\}$ with $\prod \widetilde{M}_k = \mathbf{1}$, and has *clean* matrix elements $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$. **So $C_U^{(3)}$ reduces from a transcendental PDE to: this finite unitarisation (done) + the combinatorial word-length R (closed: H1 + $\det R = 8$).** *Residual:* the feared PDE is gone; the quark *ratio* c_u/c_d is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. [verification/v40_harmonic_metric.py]

The final leg-assignment test (honest result) ([E]). Does the harmonic-frame holonomy $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ feed the $\delta = \frac{1}{2}$ resolvent to give the lepton amplitudes $\Lambda = (0.475, 1.107, 0.917)$? **(i) Ruled out as a literal identity:** $\Lambda_\mu = 1.107 > 1$, while unitary holonomy entries have modulus ≤ 1, so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly $|\mathrm{diag}\,\widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, and that $\frac{1}{2}$ *equals* the distinguished lepton transport value $\delta = \frac{1}{2}$ that v20 used by hand — so the geometry plausibly *supplies* δ (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g. $(\mathbf{1} - \frac{1}{2}\widetilde{M})^{-1}$) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the $\delta = \frac{1}{2}$ resolvent (closed combinatorially via the word-length R); the holonomy supplies δ and the leg selection, not the amplitude magnitude. c_u/c_d is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ($\delta = \frac{1}{2}$). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. [verification/v41_leg_assignment.py]

The torsion normal form and the $\delta = \frac{1}{2}$ theorem ([E]). The v41 "genericity is future work" item is now executed — and yields a theorem. *(i) Flatness = μ_4 torsion [E]:* for *any* M (symbolic), $\prod_k U^k M U^{-k} = (MU)^4$ — the $\mathbb{Z}_4$-family flatness constraint *is* the torsion statement "$T = MU$ is a fourth root of unity": the μ_4 atom appears as the *order* of the twisted cusp generator. *(ii) The $\delta = \frac{1}{2}$ theorem [E] (both directions):* on the *involutive* branch ($T^2 = \mathbf{1}$) T is a Hermitian reflection $2vv^\dagger - \mathbf{1}$, and the cusp trace condition $2v^\dagger U^{-1} v = 1$ splits *exactly* into $|v_1|^2 = \frac{1}{2}$ (real part) and $|v_2| = |v_3|$ (imaginary part), forcing

$$\boxed{\mathrm{diag}\, M = (0, \tfrac{i}{2}, -\tfrac{i}{2}), \qquad \mathrm{spec}\, M = \{1, \omega, \omega^2\} \text{ automatic}}$$

($\chi_M = \lambda^3 - 1$ from unitary + trace 0 + $\det 1$). So $\delta = |M_{22}| = \frac{1}{2}$ holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a μ_4*-torsion identity*, not an observation. *(iii) Reflection lemma [E]:* the D_4 reflection forces $\mathrm{diag}\, M = (r, z, \bar{z})$ with r real and $\mathrm{Re}\, z = -r/2$ — the whole D_4 diagonal freedom is *one* real parameter, and $\delta = \frac{1}{2} \iff r = 0$. *(iv) Branch census [E] (seeded):* the full D_4-fixed flat cusp locus realises only $\mathrm{tr}(MU) \in \{1, -1\}$; the involutive branch has the diagonal *constant* $(0, \frac{1}{2}, \frac{1}{2})$ to machine precision, while on the $\{1, i, -i\}$ branch d_1 varies over $[0, 1]$ — and the explicit v33/v40 point sits exactly in its $d_1 = 0$ slice, the *same* δ value. *Residue [C] (recorded):* whether the anchor splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (Mehta–Seshadri side) forces the $d_1 = 0$ slice — or selects the involutive branch — is the remaining question; $\delta = \frac{1}{2}$ itself is no longer the open item. [verification/v114_torsion_delta.py]

The anchor pins the residue: an exact normal form for the (U_{wall}) bundle ([E]). The v114 branch question is answered — and the v33 *numerical* Riemann–Hilbert point acquires an *exact closed form. (i) μ_4-average lemma [E]:* $\sum_k U^k X U^{-k} = 4\,\mathrm{diag}\, X$ for any X — μ_4 conjugation-averaging *is* the diagonal projection, so the exponents at infinity are $4\,\mathrm{diag}\, A_0$ and the anchor

splitting holds *iff* $\operatorname{diag} A_0 = (2,1,1)/4 = a/|\mu_4|$: the residue diagonal **is** the anchor over μ_4 (v33's diagonal was forced, not chosen). *(ii) The* $(8,0,5)/144$ *lemma [E]:* the anchor diagonal and the cusp spectrum $\{0,1,2\}/N_{\text{fam}}$ pin $\sum |a_{ij}|^2 = \frac{13}{144}$, and with $a_{13} = 0$ the determinant condition solves *uniquely* to

$$\boxed{(|a_{12}|^2,\, |a_{13}|^2,\, |a_{23}|^2) = \frac{(8,\,0,\,5)}{144}} \qquad \begin{array}{l} 8{=}\operatorname{rank} E_8,\ 5{=}g_{\text{car}}, \\ 144{=}(|\mu_4|N_{\text{fam}})^2,\ 8{+}5{=}13{=}\Delta_Q \end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same* Δ_Q that denominates $c_u/c_d = 55/117 = 5\cdot 11/(9\cdot 13)$. *(iii) Exact normal form [E]:* $A_0^\star = \begin{bmatrix} 1/2 & \sqrt{2}/6 & 0 \\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12 \\ 0 & \sqrt{5}/12 & 1/4 \end{bmatrix}$ has $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ *exactly. (iv)* $A_0^\star$ *is the RH solution [E]:* its big-circle monodromy is trivial to 10^{-10}, $\prod M_k = \mathbf{1}$, and the unitarised holonomy has $|\operatorname{diag}\widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, $\operatorname{tr}(\widetilde{M}_0 U) = 1$ — the v33/v40 point is gauge-equivalent to $A_0^\star$ (the a_{12}, a_{23} phases are diagonal gauge). *(v) The anchor forces the* $d_1 = 0$ *slice [E]:* a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with $|\operatorname{diag}\widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ — combined with v114, the whole harmonic diagonal is **anchor + torsion forced**; $\delta = \frac{1}{2}$ needs no selection. *Residue [C]:* $a_{13} = 0$ and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [E] via Gröbner on the RH constraints is the remaining U_{point} step. [verification/v115_anchor_residue.py]

The resonance theorem: $M_\infty = \mathbf{1} \iff a_{13} = 0$ — one gauge orbit [E] ([verification/v116_resonance_uniqueness.py])

The anticipated Gröbner promotion turned out to be a *linear* computation. Writing the equivariant system at infinity as $dY/dw = -(1/w)(B_0 + B_1 w + \cdots)Y$ with $B_m = \sum_k i^{km} U^k A_0 U^{-k}$, the *twisted* μ_4 averages collapse [E]:

$$B_0 = 4\operatorname{diag} A_0, \qquad \boxed{B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}}$$

(only the eigenvalue-ratio $-i$ cells survive). The anchor exponents $\operatorname{diag}(2,1,1)$ have resonance gap 1; the level-1 formal-gauge equation $(1 - \operatorname{ad} R_0)H_1 = R_1$ is singular exactly on the cells $\{(2,1),(3,1)\}$, where B_1 has entries $(0,\, 4\overline{a_{13}})$; all higher levels are non-resonant and the formal monodromy is $e^{-2\pi i \operatorname{diag}(2,1,1)} = \mathbf{1}$. Hence [E]:

$$\boxed{M_\infty = \mathbf{1} \iff a_{13} = 0}$$

— the transcendental Riemann–Hilbert condition collapses to *one linear equation. Uniqueness corollary [E]:* $a_{13} = 0$ + the $(8,0,5)/144$ moduli + phases = diagonal gauge ⇒ the flat anchor locus is **exactly one gauge orbit**, the exact $A_0^\star$: within the μ_4-equivariant anchor class the (U_{wall}) bundle datum is *algebraically unique. Sharpness [E]:* $\|M_\infty - \mathbf{1}\| \sim 8.5\,|a_{13}|$ (0.17 at $a_{13} = 0.02$) vs 10^{-10} at $A_0^\star$. *Honest scope [C]:* the equivariant ansatz is the established D_4 selector input (v19/v30); the *residue* side is now [E], while the holonomy values (the harmonic diagonal $(0, \frac{1}{2}, \frac{1}{2})$) remain [E] — transcendental monodromy of an exact system.

The monodromy is the Weyl group of A_3 — the holonomy is exact [E] ([verification/v117_monodromy_weyl_a3.py])

The last [E] item of the chain — the holonomy *values* — closes: the unitarised monodromy of the exact $A_0^\star$ system is itself *exact*, with entries in $\frac{1}{2}\mathbb{Z}[i]$:

$$\boxed{\widetilde{M}_0 = \begin{pmatrix} 0 & -\frac{1+i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & -\frac{i}{2} & -\frac{1}{2} \\ \frac{1-i}{2} & -\frac{1}{2} & \frac{i}{2} \end{pmatrix}}$$

unitary, $\det = 1$, $\operatorname{tr} = 0$, $\chi = \lambda^3 - 1$ (the cusp class *exactly*), $\widetilde{M}_0^3 = \mathbf{1}$, and $\operatorname{diag}\widetilde{M}_0 = (0, -\frac{i}{2}, +\frac{i}{2})$ — so $d_1 = 0$ and $\delta = \frac{1}{2}$ are now *matrix entries*, not observations; $(\widetilde{M}_0 U)^4 = \mathbf{1}$ with $\operatorname{tr}(\widetilde{M}_0 U) = 1$ (the $\{1, i, -i\}$ branch, $d_1 = 0$ slice — exactly where v114/v115 located the physical point). *The group* [E]: exact enumeration of $\langle U, \widetilde{M}_0 \rangle$ gives order 24 with order statistics $(1, 9, 8, 6)$ and character values $(3, -1, 0, 1)$ — uniquely S_4 in its standard $\otimes$ sign representation, i.e. the (twisted) reflection representation of the **Weyl**

group of the family lattice A_3:

$$\boxed{\text{monodromy}(U_{\text{wall}}) = W(A_3) = S_4, \qquad |W(A_3)| = 4! = 24}$$

— the flavor wall realises the A_3 factor of the glue $D_5 \oplus A_3 + \mu_4$ as its Weyl group (U = a 4-cycle, the μ_4 deck; $\widetilde{M}_0$ = a 3-cycle, the family rotation). *Identification [E]:* the ODE monodromy of $A_0^\star$ equals the exact $\widetilde{M}_0$ to 3.5×10^{-10}, and the invariant form H is U-invariant to 10^{-11}. *Status:* holonomy values [E]; only the analytic identification stays [E]. The δ thread (v20 → v40/v41 → v114 → v115 → v117) is closed end to end.

The hexagon is the sign-twisted family spectrum — first exact H2 piece [E] ([verification/v118_hexagon_family_dictionary.py])

The missing "$C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary" (H2) acquires its first exact entry. *Sign-twist lemma [E]:* $-\widetilde{M}_0$ has order 6 and

$$\boxed{\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6}$$

— the six-site hypercharge hexagon spectrum *is* the family monodromy spectrum together with its sheet twist ($\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$ realised as $(-1) \times \langle \widetilde{M}_0 \rangle$); the v20 denominators $\frac{5}{4} - \cos(r\pi/3)$ are exactly the eigen-denominators $|1 - \zeta_6^r/2|^2$ of the two family resolvents $(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0)^{-1}$. *Cyclotomic determinant [E]:* $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 = (1-t)\Phi_3(t)$, so $\det(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0) = (\frac{7}{8}, \frac{9}{8})$ — and the lepton coefficients become resolvent determinants: $c_e = |\mu_4|/(2\det_-) = \frac{16}{7}$, $c_\mu = N_{\text{fam}}/(2\det_+) = \frac{4}{3}$, $c_\tau = |\mu_4|\det_-/N_{\text{fam}} = \frac{7}{6}$, product rule $= |\mu_4|/\det_+ = \frac{32}{9}$. The lepton 7 and 9 *are* the family-resolvent determinants; $\delta = \frac{1}{2} = |(\widetilde{M}_0)_{22}|$ is supplied by the matrix itself. *Sheet-extended group [E] (audit):* $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$ has order $48 = \Omega_{\text{adm}} = N_{\text{fam}} \dim S^+$ — the seed's quartic coefficient. *Residue [C]:* the dictionary's *values* are exact; its *address table* (which hexagon site each fermion occupies; the quark composition) remains v20's established input / open — nothing re-fished.

The address table: the lepton words are the compiler atoms [E]/[C] ([verification/v120_address_table.py])

The address table acquires its structure — on word-length integers *frozen* in v18/v20, long before today's readings (the red-team firewall). *(i) The lepton words are the atoms [E]:*

$$\boxed{(L_e, L_\mu, L_\tau) = (8,5,3) = (\text{rank}\, E_8,\ g_{\text{car}},\ N_{\text{fam}}) = (p_0{+}e_2,\ e_2,\ p_0)}$$

in mass order (longest word = lightest lepton). *(ii) Addresses = division by the hexagon [E]:* $(r, w) = L$ divmod p_2 with $p_2 = 6 = |R^+(A_3)|$ itself an atom: $e \to (2,1)$, $\mu \to (5,0)$, $\tau \to (3,0)$; $\sum r = 10 = p_3 = \mathcal{A}_\Lambda$, $\sum L = 16 = \dim S^+$. *(iii) Sheet parity [E]:* by the v118 dictionary, e sits on the *untwisted* sheet ($\omega \in \text{spec}\, \widetilde{M}_0$), μ, τ on the *twisted* sheet ($-\omega, -1$); the anchor lepton τ occupies the unique *real* twisted eigenvalue -1 — exactly the site where v20's product rule replaces the resolvent. *(iv) Quark sum rules [E] (frozen v18 words $7,7,5,3,2,0$):* up-type sum $= 10 = p_3$; down-type sum $= 14 = p_1{+}p_3$; **all quarks** $= 24 = |W(A_3)|$ (the monodromy group order, v117); **all nine charged fermions** $= 40 = p_1 p_3 = a^T R a$ (v119); quark $\sum r = 2p_2$, $\sum w = |\mathbb{Z}_2|$; the top sits at the vacuum site $(0,0)$ — zero word, no suppression, the established ladder anchor. *Residue [C]:* five independent closures $(16, 10, 14, 24, 40)$ constrain the address map; the per-fermion *assignment* remains the v18/v20 input — deriving it is the remaining H2 content.

The address table is pinned: R is the unique hexagon matrix with atom margins [E] ([verification/v121_address_pinning.py])

The per-fermion assignment closes at the *information* level. The established identity $L = R{+}2U$ (v95) says the word-length table *is* L[sector][generation] — rows (up; down; lepton) $= ((7,3,0);(7,5,2);(8,5,3))$, the v18/v20 words verbatim — so the addresses are the entries of the *residue operator* R plus one hexagon turn for the first generation, and R's first column is the **anchor** $a = (1,1,2)$. The margins are atoms [E]: row sums $(4,8,10) = (e_1, \text{rank}\, E_8, p_3)$, column sums $(4,13,5) = (e_1, \Delta_Q, g_{\text{car}})$. And the

census closes it [E]:

$$\boxed{\begin{array}{c}\text{among all } 3\times 3 \text{ matrices with entries in the hexagon } \{0..5\},\\ \text{exactly 17 have the atom margins — and exactly ONE has det} = 8\end{array}}$$

— and it is R (also the unique one with SNF $= (1,1,8)$; all 17 determinants are distinct, 8 occurs once — control computed explicitly). **The assignment table carries zero residual information beyond the atom margins and the rank determinant.** Within-sector ordering is the established transport reading ($m \sim \lambda_Y^L$: longer word = lighter fermion). *Residue [C]:* the open H2 item shrinks from "derive 9 per-fermion integers" to "derive the six atom margins + det = rank". *Firewall:* R, L and all word lengths were frozen in v4/v18/v20/v71 long before this census — no degrees of freedom spent.

The margins are theorems ([verification/v122_margin_theorem.py]). Even the margins are not input. Three selectors, all established and frozen *before* the address question was posed — *(S1)* the D_4 annihilator $n = (5,-9,6)$ with $n^T R = (8,0,0)$ (it kills the unchanged (c_2, c_3) plane; $n\cdot a = 8$), *(S2)* $\det R = 8$, *(S3)* $\det K = 4$ with $K = R + Q\Sigma$ — pin R *uniquely* among hexagon matrices [E]: the $(S1)$ census has $4\times 4\times 4 = 64$ candidates, $(S2)$ leaves 12, $(S3)$ leaves **exactly one**, and it is R. Hence the atom margins, the anchor column $Re_1 = a$ and all sum rules above are *consequences* — their input status is revoked. *Bonus lemma [E]:* on the $(S1)+(S2)$ locus $\det(M+2U) = 20 = \det L$ holds for *all* 12 candidates — the fourth quartet determinant is redundant, the compiler's familiar overdetermination. *Residue [C]:* H2 reduces to the establishment of the selectors themselves (n and the diamond determinants — pre-existing inventoried objects); the address question introduces *no new residual class*.

The dual-normal selector: (d,n) pins R columnwise ([verification/v136_dual_normal_selector.py]). The selector list shrinks once more. With the dual anchor $d = a^\top R^{-1}$ ([verification/v134_dual_anchor.py]) and the torsion normal n, each *column* of R obeys $d\cdot c_j = a_j$ and $n\cdot c_j = 8\delta_{1j}$; the common kernel of (d,n) is the primitive vector $(6,8,7)$ (audit: p_2, $\operatorname{rank} E_8$, scalaron-7), so within the address box $\{0..5\}^3$ each column is the *unique* lattice point (census: 1 of 216 each) — $(d,n) \Rightarrow R$, and the 6^9 census above collapses to three column problems [E]. *Honesty:* K is *not* pinned directly ($d^\top K = (1,\frac{1}{2},1)$); $\det K = 4$ comes via the μ_4 wall ([verification/v135_det_surface.py]). *The mechanism is bounded:* the zero mode of $A_0^\star$ carries the exact spectral normal $\sigma = (2,-9,5) = (|\mathbb{Z}_2|, -N_{\text{fam}}^2, g_{\text{car}})$ (signed squares; $\|v\|^2 = \frac{16}{9}$ with $16 = \dim S^+$, $\operatorname{adj} A_0^\star = \frac{2}{9}P_v$ with $\frac{2}{9}$ = the SdS entropy curvature), $\sigma\cdot\mathbf{1} = -|\mathbb{Z}_2|$, $\sigma\cdot a = N_{\text{fam}}$, $n\cdot\sigma = 121 = \|\operatorname{Pl}(K)\|_1^2$; and the exact $W(A_3)$ orbit control shows *no* group image of σ is proportional to n — the naive cofactor/orbit lift fails, the step $\sigma \to n$ is a genuine mechanism beyond the monodromy group. *R4′ restated [C]:* derive the *pair* (d,n); then R is forced columnwise.

The selector triangle ([verification/v139_selector_triangle.py]). The pair itself now collapses further. First, d is *pure anchor data*: $d = \frac{3}{2}a - 2\cdot\mathbf{1}$ exactly — the closed form never mentions R, so the first selector is *derived*, not residual. Second, the frame $(\mathbf{1}, a, \sigma)$ is a basis of $\mathbb{Z}^3$ with $\det(\mathbf{1}|a|\sigma) = 11 = \|\operatorname{Pl}(K)\|_1$ (the quark constant as the frame volume), and n is the *unique* integer covector with the three atom pairings

$$\boxed{n\cdot\mathbf{1} = 2 = |\mathbb{Z}_2|, \qquad n\cdot a = 8 = \operatorname{rank} E_8, \qquad n\cdot\sigma = 121 = \|\operatorname{Pl}(K)\|_1^2}$$

— the cofactor definition is replaced by frame data plus three atoms. On the $(\mathbf{1}, a)$-constrained line the σ-pairing is $11(11+t)$: divisible by 11 for *every* t, a perfect square exactly at the true n (t=0). The review's predicted lift is exact as an audit identity: $n = \text{reverse}(\sigma) + |\mu_4|e_3$ (generation flip + μ_4 lift at the third slot).

Frame integrality: three pairings → two ([verification/v142_frame_integrality.py]). The third pairing is *not* independent. The pairing triples $(m\cdot\mathbf{1}, m\cdot a, m\cdot\sigma)$ of *integer* covectors form an index-11 sublattice of $\mathbb{Z}^3$, cut out by the single congruence

$$\boxed{x - 3y + z \equiv 0 \pmod{11}} \qquad (\text{index } 11 = \|\operatorname{Pl}(K)\|_1 = \det(\mathbf{1}|a|\sigma)),$$

so with the two line pairings at their atoms $(x,y) = (|\mathbb{Z}_2|, \operatorname{rank} E_8) = (2,8)$, integrality *forces* $z \equiv 0 \pmod{11}$ — the $11(11+t)$ line *is* the integer line. Primitivity forces t *even* (odd t makes every entry of $n(t)$ even), and t=0 is the unique even t with a square σ-pairing for $|t| < 88$ (the global pick keeps the square selector — recorded honestly). Both line pairings are Cramer-reducible to inventoried identities

($n \cdot a = \det R$ by Laplace along the anchor column; $n \cdot \mathbf{1} = 8(R^{-1}\mathbf{1})_1$ with $R^{-1}\mathbf{1} = \frac{1}{4}(1,1,-1)$, v134) — restructuring, not establishment.

The pairing values are atom identities ([verification/v145_pairing_atoms.py]). Even the two line-pairing *values* carry no independent information. Reading the v139 lift structurally — $n = w_0(\sigma) + |\mu_4| e_3$ with w_0 the A_3 *exponent duality* $m \leftrightarrow h - m$ (an established Lie involution; lift slot = the top-exponent line, lift size = the glue order) — all three pairings become atom arithmetic:

$$n\cdot\mathbf{1} = |\mu_4| - |\mathbb{Z}_2| = |\mathbb{Z}_2|, \qquad n\cdot a = |\mu_4|\, a_3 = 8 = \operatorname{rank} E_8 \text{ via } \boxed{|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 \quad (4+5=9,\ \sigma \perp w_0(a))},$$

and $n \cdot \sigma = \|\sigma\|^2 - N_{\text{fam}}^2 + |\mu_4| g_{\text{car}} = 110 - 9 + 20 = 121$ with $\|\sigma\|^2 = |\mathbb{Z}_2|^2 + N_{\text{fam}}^4 + g_{\text{car}}^2 = 110 = |\mathbb{Z}_2| g_{\text{car}} \| \operatorname{Pl}(K)\|_1$. *R4′ residue [C] (final form):* derive the *one* map $n = w_0(\sigma) + |\mu_4| e_3$ — no independent number remains; everything else in the chain anchor $\to (d, n) \to R$ is exact.

Both dual normals are anchor data ([verification/v149_cusp_normal.py]). The remaining opacity dissolves in the canonical frame: the pairings of n with the *cusp eigenbasis* (the geometric basis of v98) are exactly

$$\boxed{n \cdot e_1 = 6 = p_2(a), \qquad n \cdot e_2 = 3 = p_0(a), \qquad n \cdot e_3 = 5 = e_2(a)}$$

— one anchor invariant per Q_+ degree, and the cusp matrix is unimodular, so these three pairings determine n uniquely. With $d = \frac{3}{2}a - 2\cdot\mathbf{1}$ the dual normal *pair* (v134) is anchor data through and through — one normal per frame; the cusp data $(6,3,5)$, the frame pairings $(2,8,121)$ and the w_0-lift are three presentations of one covector, all atomic (equivalences exact). *Audit:* $\sigma \to (5,1,2)$; alternate reading $(6,3,5) = (|R^+(A_3)|, N_{\text{fam}}, g_{\text{car}})$; sum $14 = \dim G_2$. *R4′ residue [C] (cusp form):* derive the *assignment* — why the Q_+ degrees $(1,2,3)$ carry (p_2, p_0, e_2) in that order; one discrete assignment, zero continuous freedom.

Exterior Leg Lemma — the quark u/d leg is a $\Lambda^2 F$ area, not a scalar [E]/[C]

The v41 negative is a *category error*, not a dead end. u and d share the hexagon-radial data $(r, w) = (1, 1)$ ($L = 7$ for both), and the C_6 resolvent $c(r,w) = |\mu_4|^w / (\frac{5}{4} - \cos\frac{r\pi}{3})$ reads *only* (r, w) — so any *scalar* leg gives $c_u/c_d \equiv 1$. A scalar sees radius, not orientation. The u/d separation lives in the smallest *non-scalar* invariant of the $SU(3)_F$ composition: the exterior square $\Lambda^2 F$, i.e. the oriented anchor-plane area $\operatorname{Pl}_{\mathbf{1},a}(K)$. The "11" is then *generated*, not read:

$$(K, \mathbf{1}, a) \to B_K = \left(\begin{smallmatrix} 13 & 8 & 4 \\ 18 & 11 & 6 \end{smallmatrix}\right) \to \operatorname{Pl}(K) = (-1, 6, 4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \to \|\operatorname{Pl}(K)\|_1 = 11,$$

$$\boxed{\frac{c_u}{c_d} = \frac{g_{\text{car}} \|\operatorname{Pl}_{\mathbf{1},a}(K)\|_1}{N_{\text{fam}}^2\, \Delta_Q} = \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117}} \quad \left(\text{anchor microcode } = \tfrac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}\right).$$

Typing: [E] for the Plücker / microcode identities ($\operatorname{Pl}(K) = (-1,6,4)$, $\|\cdot\|_1 = 11$, $5\cdot 11/(9\cdot 13) = 55/117$); and — by Readout Rigidity (v49, below) — [E] also for the *physical* ratio $\mathcal{C}_{ud}(\rho^\star)$ = this $\Lambda^2 F$ readout, which is *constant* on the derived selector stratum and so does *not* need any point-selection on $\Lambda^2 F$. The quark residual is thereby re-typed from "find the right scalar y" to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale (U_{point}) stays as an anchor, no new scalar, no fishing.

Audit (the 121 *lemma, [verification/v119_review_validation_2.py]):* the "11" has a second, independent appearance [E]: with $h = \mathbf{1} + a = (2,2,3)$ in the canonical anchor ordering, $h^T R h = 121 = 11^2 = \|\operatorname{Pl}(K)\|_1^2 = (p_3+1)^2$ — the quadratic *anchor norm* of the residue operator R reproduces the Plücker integer (sensitivity control: the three orderings of h give $\{105, 121, 135\}$; only the canonical one yields 121). Bonus checksums: $\mathbf{1}^T R \mathbf{1} = 22 = 2 \times 11$ (the entry sum of R) and $a^T R a = 40 = p_1 p_3 = 10 b_1 - 1$. Typed *audit*, not load-bearing — the oriented Plücker mechanism above remains the derivation. [verification/v42_exterior_leg.py]

Bridge test ([verification/v43_exterior_bridge.py]). The typing is confirmed: for the $SU(3)$ holonomy the exterior square is the conjugate representation, $\Lambda^2(\widetilde{M}) = \overline{\widetilde{M}}$ exactly — the exterior leg *is* the $\bar{\mathbf{3}}$. But the *continuous* action $\Lambda^2(\widetilde{M}) \cdot (\mathbf{1} \wedge a)$ does *not* equal the *integer* $\operatorname{Pl}(K) = (-1,6,4)$: the integer Plücker is the combinatorial K datum. So the bridge $\rho^\star \to \operatorname{Pl}(K)$ is the *discrete* non-abelian-Hodge invariant (the integer R; $\det R = 8$ and $\operatorname{Spec}(Q_+) = \{1,2,3\}$ already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the $\Lambda^2 F$ side.

Lie-level grounding — no special math ([verification/v44_carrier_exterior.py]). The discrete

invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier, $16 = \Lambda^{\text{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$, and under $SU(5) \subset SO(10)$ the up quark sits in $10 = \Lambda^2(5)$ while the down sits in $\bar{5} = \Lambda^4(5)$. So the *exterior degrees* are $\deg(u^c) = 2$, $\deg(d^c) = 4$, with $\deg(u) + \deg(d) = 6 = |R^+(A_3)|$ (the hexagon) and $\deg(d) - \deg(u) = 2 = |\mathbb{Z}_2|$ (the sheet). The "Λ^2" of the exterior leg is therefore not exotic Hodge data — it is the carrier's own Λ^2 grading (the up quark *lives* in $\Lambda^2(5)$). This re-types the residual once more: from "discrete non-abelian-Hodge invariant" to "the carrier exterior degree" — a standard [E] fact. *Honest scope:* this grounds the *type*; the Plücker *number* 55/117 stays the family $\mathrm{Pl}(K)$ [E], and the carrier-Λ^2 ↔ family-$\Lambda^2 F$ link (via family ⊗ carrier = 15 = 16−1) plus the normalisation remain [C].

The "11" is Pascal — a third, simplest origin ([verification/v45_family_exterior.py]**).** Following that link, $\mathrm{Pl}(K)$ is the *exterior algebra of the family fundamental* $4 = \mu_4$ (the $A_3{=}SU(4)$ fundamental): $\Lambda^k(4) = \binom{4}{k}$ is Pascal row 4 $= (1, 4, 6, 4, 1)$ with full sum $2^4 = 16 = \dim S^+$, and $|\mathrm{Pl}(K)| = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$ are exactly its distinct exterior powers. Hence

$$\|\mathrm{Pl}(K)\|_1 = \textstyle\sum_{k=0}^{2} \binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\text{car}}$$

(the cumulative $\Lambda^\bullet(\mu_4)$ up to degree $2 = \deg(u^c)$), and family ⊗ carrier $= 15 = \dim \mathfrak{su}(4) = \Lambda^2(5) \oplus \Lambda^4(5) = 10{+}5 = N_{\text{fam}} g_{\text{car}}$ read three ways. So the "11" is generated a *third* way — the cumulative family exterior algebra up to the up-quark's degree — pure Pascal/branching, *no special math.* The [C] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

Lemma 6 (U5 — the "11" worry, *discharged* by Readout Rigidity)**.** *The original worry was that $c_u/c_d = \mathcal{C}_{ud}(\rho^\star)$ must be evaluated at the uniquely selected $\rho^\star$* without *using* 11 *as input, so that until then* 55/117 *would be a mere table-level rational shadow.* ***This is now resolved (v49):*** *$\mathcal{C}_{ud}$ is* constant *on the discrete selector stratum $\mathcal{S}_{8,Q}$, so the "11" is* earned *as the rigid Plücker readout $\|\mathrm{Pl}(K)\|_1$ (v42)* — independently *of point-uniqueness. Point-uniqueness of $\rho^\star$ enters only the* absolute *amplitude scale, not the ratio.*

Lemma 7 (U6 — the four certificates)**.** *$C_U^{(1)}$ wall enumeration (done); $C_U^{(2)}$ D_4-fixed character solver; $C_U^{(3)}$ selector uniqueness (one S-equivalence class); $C_U^{(4)}$ amplitude extraction — the* ratios *$R, Q, c_u/c_d$ are read off the (derived) selector stratum (closed, v49/v71); only the* absolute *$U_f^\star, c_q$ normalisation is the anchor.*

Progress: an explicit valid flat bundle is realised (RH solve) ([E])**.** A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue A_0 (eigenvalues $\{0, \frac{1}{3}, \frac{2}{3}\}$, diag $= (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$) for which the Fuchsian system $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)}/(z - i^k)$ simultaneously satisfies: the cusp class at each puncture; the splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (exponents $\sum_k A_k = \mathrm{diag}(2, 1, 1)$); the *path-ordered* monodromy at infinity is trivial, $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$, i.e. $M_1 M_2 M_3 M_4 = \mathbf{1}$; and the representation is *irreducible* — **case A** (non-trivial $SU(3)_F$ mixing, not the diagonal case B). So a valid (U_{wall}) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope:* this is one realising point (existence + case A), likely one of a residual family; the unique $\nabla_F^\star$ still needs the $\det R = 8$ selector and c_u/c_d still needs the $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary (H2). [verification/v33_explicit_flat_bundle.py]

H2 bridge probed ([verification/v34_h2_bridge_attempt.py]*).* The explicit per-puncture monodromies M_k (cusp class, $\prod M_k = \mathbf{1}$ to 10^{-5}) were computed; but the natural resolvent-dressed diagonal extraction ($|\mathrm{diag}\, M_k| = (0, \frac{1}{2}, \frac{1}{2})$ times the C_6 resolvent factor) does *not* reproduce the lepton amplitudes $(0.475, 1.107, 0.917)$. So the precise $\Gamma_{ij}^{\min}$ geodesic-to-word dictionary (how the 6-site hypercharge hexagon combines with the 3-dim family $\rho^\star$) was the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for 55/117. *Resolved since (v118): the correct extraction is the resolvent determinant of the $W(A_3)$ monodromy, not the diagonal dressing — the probe's failure is thereby explained, and the dictionary is exact.*

Theorem U — the four-way split (sharper than one monster gate) [E]/[O]

D_4 fixes the admissible chamber, not the physical point (the D_4-fixed locus is positive-dimensional, U3); the discrete selectors $\det R = 8$, $\mathrm{SNF}(R) = (1,1,8)$, $\mathrm{Spec}(Q_+) = \{1,2,3\}$ do the cutting. The gate splits into four pieces, three of them essentially closed:

- U_{unitary} **[E]**: parabolic degree 0 + polystable ⇒ unitary (Mehta–Seshadri) ⇒ $\Phi = 0$; the harmonic metric is the *finite* invariant form H ($M_k^\dagger H M_k = H$), not a Hitchin PDE ([verification/v40_harmonic_metric.py]).
- U_{H2} **[E]**: R, Q, K are read off the bundle (splitting type = a, $\det R = 8$, $\mathrm{Spec}(Q_+)$) ([verification/v39_uwall_selectors.py]).
- U_{Λ^2} **[E]**: **Readout Rigidity** — on the discrete stratum $\mathcal{S}_{8,Q} = \{\det R = 8, \mathrm{SNF}(R) = (1,1,8), \mathrm{Spec}(Q_+) = \{1,2,3\}\}$ the anchor-plane readout is *constant*, $\mathcal{C}_{ud}(\rho) = g_{\text{car}} \| \mathrm{Pl}_{\mathbf{1},a}(K)\|_1 / (N_{\text{fam}}^2 \Delta_Q) = \frac{55}{117}$, *independent* of the continuous D_4 position — so c_u/c_d needs no point uniqueness ([verification/v49_readout_rigidity.py], [verification/v42_exterior_leg.py]).
- U_{point} **[O]**: full uniqueness of $\rho^\star$ — needed only for the *complete* $U_f^\star$ amplitude matrix, *not* for c_u/c_d.

Headline: the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; the only residual, U_{point}, is not an independent gate — it reduces to the single dimensionful anchor v_{geo} (Gate 1 below).

The simple closure ([verification/v71_simple_r_bridge.py]). The selector stratum $\mathcal{S}_{8,Q}$ is now *fully derived*: $\det R = 8 = \mathrm{rank}\, E_8$ with $\mathrm{SNF}(R) = (1,1,8)$ is the lattice invariant (cf. $\det Q = 3 = N_{\text{fam}}$, v70), and $\mathrm{Spec}(Q_+) = \{1,2,3\}$ is the D_4-equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants $(\det Q, \det K, \det R, \det L) = (3,4,8,20)$, product $1920 = |W(D_5)|$. So by Readout Rigidity the quark *ratios* ($c_u/c_d = g_{\text{car}} \cdot 11/(N_{\text{fam}}^2 \cdot 13) = \frac{55}{117}$, etc.) are *pure integer Plücker readouts — no transcendental monodromy solve.* The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece, U_{point}, is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensionful scale, not a missing computation.

Gate 1 complete: $U_{\text{point}} \to v_{\text{geo}}$ ([verification/v75_upoint_to_vgeo.py]). The absolute amplitudes are not free either. The lepton amplitudes are exact ($c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, v20), every cross-sector c-ratio is closed (Plücker), and each *sector product* is a clean φ_0^{ret}-power (Grand Mass Volume, v46: $\det M_{\text{sector}} \sim (\varphi_0^{\text{ret}})^{6,9,10}$; the lepton product is $\frac{16}{7} \cdot \frac{4}{3} \cdot \frac{7}{6} = \frac{32}{9} = 2^{g_{\text{car}}}/N_{\text{fam}}^2$). Since (pairwise ratios) + (product) ⇔ (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale* v_{geo}. **So U_{point} is not an independent gate: it reduces to v_{geo}, the *same* one irreducible dimensionful anchor as gravity's $1/G$ (v68) — the two [O] anchors collapse to one.**

Research Contract 2 — (G_{metric})

Goal. Construct the TFPT metric measure

$$Z[J] = \lim_{\chi \to \infty} \int_{\mathcal{G}_\chi/\mathrm{Diff}} \exp\!\big(-S_{\text{rel},\chi}[g, A, \psi] + J \cdot O\big)\, d\mu_{\text{Cal},\chi},$$

$$S_{\text{rel},\chi} = \mathrm{Tr}\, f\Big(\tfrac{D_{\text{rel}} + A_\Sigma}{\chi}\Big) - \mathrm{Tr}\, f\Big(\tfrac{D_{\text{ref}}}{\chi}\Big) + \tfrac{i\pi}{2} \Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed $R + R^2$ equation.

Lemma 8 (G0–G1 — configuration space & BRST/Hodge Fredholm)**.** *For $s > 5/2$ a Sobolev space of Θ-symmetric metrics on the doubled seam collar M_Σ admits a gauge slice; the gauge-fixed gravitational complex $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2 T^*M) \xrightarrow{K^*} \Omega^0(TM)$ with $\Delta_{\text{gh}} = K^*K$ and $\Pi_{\text{phys}} = \Pi_{\text{BRST/Hodge}} \Pi_{TT}$ is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (*not *rhetorically).* Machine: principal symbols in xAct/-Cadabra; linear-algebra certificates in Lean (medium).

Lemma 9 (G2 — finite relative spectral action, *heat-kernel grounded* **[E]**)**.** *$S_{\text{rel},\chi}$ is finite, differentiable, with a local Seeley–DeWitt expansion $\mathrm{Tr}\, f(D^2/\chi^2) \sim f_4 \chi^4 a_0 + f_2 \chi^2 a_2 + f_0 a_4 + \cdots$. For*

the Lichnerowicz Dirac operator $D^2 = \nabla^2 + \frac{R}{4}$ *(so* $E = -\frac{R}{4}$*, spinor trace* $\operatorname{tr}\mathbf{1} = 4$*) the Gilkey coefficients give, per* $(4\pi)^{-2}$*,*

$$a_2 = \tfrac{1}{6}\operatorname{tr}(6E{+}R\,\mathbf{1}) = -\tfrac{R}{3} \quad (\textit{Einstein–Hilbert } \textstyle\int R), \qquad a_4\big|_{R^2} = \tfrac{1}{360}\operatorname{tr}(180E^2{+}60RE{+}5R^2\mathbf{1}) = \tfrac{R^2}{72}$$

(the R^2*/Starobinsky term), so the spectral action* structurally *produces* $S_{\text{eff}} = \int\sqrt{g}\,\frac{\bar{M}_{\text{Pl}}^2}{2}(R{+}\frac{R^2}{6M^2}){+}\cdots$ *with the scalaron mass ratio fixed by the cutoff moment,* $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$*. The TFPT closure* $M^2/\bar{M}_{\text{Pl}}^2 = c_3^7$ *fixes* $f_0 = 6(4\pi)^2/c_3^7$*, i.e.* $M = c_3^{7/2}\bar{M}_{\text{Pl}} = 3.06\times10^{13}$ *GeV — the boundary normalisation* c_3 is *the gravitational scale — and hence the Starobinsky slow-roll forms* $n_s \simeq 1 - \frac{2}{N}$*,* $r \simeq \frac{12}{N^2}$*. The* $R + R^2$ structure *is convention-independent (standard Gilkey); the precise rational* $\frac{1}{72}$ *and factor* 6 *are the standard Dirac conventions used here (*[verification/v36_spectral_action_g2.py]*,* [verification/v28_gravity_fR.py]*).*

The simpler resolution: "full QG" for physics is the gap-decoupled admissible sector ([E]). The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5): $2\|V_{\text{metric}}\| = 0.785 < \Delta = 6\log\frac{3}{2} = 2.433$, so the admissible sector keeps a *strictly positive* effective gap $\Delta_{\text{eff}} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates* the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

admissible IR sector = closed under stated RP;
ambient metric measure (G6) = G_{metric} gate

G2 supplies the local action ($R{+}R^2$, heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce (G_{metric}) to the ambient projective limit (G6). **Precise status (Alessandro's boundary):** the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure is now discharged as a [C] *redundancy* (v369, the holographic-route keybox below), a certification object rather than a missing piece. Its absence does *not* affect the bounded IR claim. [verification/v36_spectral_action_g2.py]

The glue Q-system: the index-4 object of the G_{net} statement exists explicitly ([E]/[C]). The G_{net} target — *"the seam–Calderón inclusion has Jones index* $4 = |\mu_4|$*"* — no longer refers to an unknown object. On the glue group algebra $A = \mathbb{C}[\mathbb{Z}_4]$ (multiplication $\delta_a \otimes \delta_b \mapsto \delta_{a+b}$, unit δ_0) *all* Longo Q-system axioms hold exactly [E]: associativity, unit law, the Frobenius identity $(m\otimes1)(1\otimes m^*) = m^*m = (1\otimes m)(m^*\otimes1)$, and specialness $mm^* = |\mathbb{Z}_4|\,\mathbf{1}$ — a special C* Frobenius algebra with

$$\dim\theta = 4 = |\mu_4| = [B:A]$$

exactly the Carrier Index Lemma value (KLM: $\mu_{\text{carrier}} = 16 = 4^2\mu_{E_8}$ re-verified). *Locality = isotropy [E]:* on the Lagrangian glue $\langle(1,1)\rangle$ the discriminant form gives $q(k(1,1)) = k^2 \in \mathbb{Z}$ for every k — the braiding obstruction vanishes (v92 in Q-system language); the two Lagrangian glues (the sheet pair) are the only maximal isotropic choices. *Halfway step [E]:* $\{0,2\} < \mathbb{Z}_4$ closes and gives the unique intermediate Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ tower step, with index factorisation $4 = 2\times2$. *R2 sharpened [C] (recorded, not claimed):* what remains is the operator-algebraic *identification* "the seam–Calderón inclusion of the 16-Majorana net (v113) realises *this* Q-system" — matching two constructed objects, not a search; the three loads (metric + carrier + QBL, v123) hang on this one identification. *[2026-08-03 update, the index side measured on the ladder:* [verification/v726_phys_car_pp_index.py] (57/57, T2-SLICE-GO) measures the inclusion on the CAR ladder $N = 48$–384: the μ_4-average conditional expectation has optimal Pimsner–Popa constant *exactly* 1/4 (explicit rank-one minimizer + pinching theorem) and Watatani index *exactly* 4 via an explicit quasi-basis with *forced* weights (uniform weights break scalarness; the $\mathbb{Z}_2$-only average gives

index 2 — μ_4 distinguished from μ_2), with μ_4 *derived* from the clock $L = S^{N/12}$ (order 4 forced by the NS spin structure) and the Ramond sector healed *state-preservingly* by the Klein zero-mode dressing ($[\rho, U'] = 0$ exactly). The remaining half is unchanged in type — the *identification* of this ladder Q-system with the constructed $\mathbb{C}[\mathbb{Z}_4]$ object — now registered with kill criteria as the open half of `GNET.RAMIFIED.01` (Frobenius mismatch; index $\neq 4$ on the state-preserving ladder; dressing incompatible with the σ descent); no marker move.] *[2026-08-05 update, the finite precursor of the identification (`[verification/v779_gnet_gns_arf.py]`,* $27/27 + 27/27$*):* the coherent *commuting* system of index-4 expectations along the Haar scaling tower (restriction coherence, $[E_I, E_J] = 0$, joint idempotency, all exact) is the finite half of the Longo Q-system identification, and the exact dim law $d_c(\ell) = 4^{\ell-1} + 2^{\ell-1}\cos(\pi c/2)$ makes the 4-element $\mathbb{C}[\mathbb{Z}_4]$ group-layer quasi-basis obstruction exactly $2^{-\ell}$ at every finite level ($d_0 - d_2 = 2^\ell > 0$), restored only asymptotically — the crossed-product structure is *exactly asymptotic*, and $d_2(1) = 0$ *is* the $\ell = 1$ index-3 anomaly. Typing unchanged; no marker move.] *[2026-08-06 registration — `GNET.MARTINGALE.LIMIT.01`, **[O]** (the analytic remainder), finite hypotheses measured in `[verification/v802_gnet_martingale.py]`* (15/15, MARTINGALE-HYPOTHESES-MET)*: the Haar ladder is an exact filtration with* one *index-4 Watatani expectation for the whole tower (quasi-basis explicit,* 185/3041 *elements at* $\ell = 2/3$*); the local-state defects are summable at measured rate* $\sim$0.25 *per doubling (one power better than the* $2^{-\ell}$ *dim-law envelope; reported, not forced); the unique limit state follows by elementary telescoping and is verified on every rung; the split/normality precursors are uniform (determinant witness* ≥ 0.985*, stabilizing at* 0.9928*); the decimation control converges to the* wrong *degenerate limit — the coherent Haar bond is load-bearing.* The registered remainder (steps 5–7, typed, not executed): *GNS reconstruction of* ω_∞*; local normality on each* $A(W)$ *(the uniform S4 witnesses as the finite input); split property and the* μ_4 *simple-current extension (`GATE.METRIC.11`) on the limit net. The open QGEO majorant residues (b)/(c) are bypassed for limit existence and relocated into the normality step — a reallocation, not a proof. Kill: non-summable defects at a deeper reachable rung, or a determinant-witness collapse. No marker move.]*

The hull is the graded module of the Q-system (`[verification/v128_graded_hull.py]`). The Q-system is not merely an abstract companion of E_8 — E_8 *realises* it **[E]**: the 240 roots, built as explicit norm-2 vectors in $D_5 \oplus A_3$+glue coordinates, decompose as $240 = 52 + 64 + 60 + 64$ over the glue $\mathbb{Z}_4$ (the v89 branching as vectors), and for *all* 6720 root pairs whose sum is again a root, $\mathrm{coset}(r_1{+}r_2) = \mathrm{coset}(r_1) + \mathrm{coset}(r_2) \bmod 4$:

E_8 is a $\mathbb{Z}_4$-graded Lie algebra over the carrier; the grading group is the glue = the Q-system

— the checksum hull is the Q-system's graded module, realised in 248 dimensions. The R2 identification thus matches the Calderón inclusion against a structure that E_8 already *carries*. `[verification/v125_glue_qsystem.py]`

Graded Frobenius (`[verification/v143_graded_frobenius.py]`). The identification holds at the finite Lie level: the graded module is in fact a graded *Frobenius* structure, i.e. everything the Q-system demands of its module is already exact inside E_8 **[E]**: (i) *coset duality* — $\mathrm{coset}(-r) = -\mathrm{coset}(r) \bmod 4$ for all 240 roots ($-C_1 = C_3$ bijectively, C_0, C_2 self-dual), so the Killing form restricts to a *nondegenerate* invariant pairing $g_k \times g_{-k} \to \mathbb{C}$; (ii) the glue character average $\frac{1}{4}\sum_k i^{k\deg}$ is *exactly* the carrier projector (the conditional-expectation shadow, quasi-basis index $|\mathbb{Z}_4| = 4$); (iii) each glue sector is *one* $W(D_5){\times}W(A_3)$ orbit with multiplicity-one dimensions $64 = 16 \cdot 4 = (\mathbf{16}, \bar{\mathbf{4}})$, $60 = 10 \cdot 6 = (\mathbf{10}, \mathbf{6})$ — simple currents, sector ring = $\mathbb{Z}_4$ = the $\mathbb{C}[\mathbb{Z}_4]$ fusion, halfway subgroup $\{0, 2\} = SO(16)_1$. *R2 residue **[C]** (final form):* exactly *one* conformal-net statement remains — the seam–Calderón inclusion of the 16-Majorana *net* carries this Q-system — with no remaining finite freedom.

Where that statement lives: the NS/R sector census (`[verification/v148_fock_census.py]`). The remaining work is scoped exactly. Every *untwisted* (Neveu–Schwarz) state of the 16 Majoranas has *integer* $(D_5, SO(6))$ weights, and integer weights lie in the even discriminant classes — so the NS module supports only $\{(0,0), (2,0), (0,2), (2,2)\}$: **the odd glue classes** $k(1,1)$ — **the**

actual μ_4 simple currents — have zero NS support; they are twisted (Ramond-type) modules. The Ramond zero-mode module ($2^8 = 256$ ground states) carries exactly the four odd class pairs, 64 each, and splits $128 + 128$ into the odd sectors of the *two* Lagrangian glues $\langle(1,1)\rangle$ and $\langle(1,3)\rangle$ (v92): *choosing the glue is choosing the Ramond projection* ($128 =$ one $SO(16)$ spinor chirality), and E_8 assembles as $248 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$. *Consequence [C]:* the index-4 extension cannot even be *stated* on the untwisted module alone; R2's one remaining statement is irreducibly about the twisted-sector structure of the seam–Calderón net, with the sheet choice realised as the R-projection; the finite shadow is exhausted (v113/v125/v143 + this census). *[2026-08-03 update, the NS/R grading acquires an arithmetic home:* [verification/v722_phys_ramified_ns_r.py] (9/9, RAMIFIED-SEES-NSR) shows the NS/R grading *is* the parity character of $E_8(\mathbb{Z}[i])/(1+i) = \mathbb{F}_2^4$: $(1+i)L$ lies in the even/NS sublattice (proved on a $\mathbb{Z}$-basis), the 240 roots fall into 15×16 classes of *pure* parity (7 NS + 8 R), and parity is a homomorphism $\mathbb{F}_2^4 \to \mathbb{F}_2$ — the Ramond projection is $(1+i)$-*adic*, at the one ramified edge of the $\mathbb{Z}[i]$-E8 Hecke tower (both controls fire: the $\mathbb{Z}[i]^4$ census is not 15×16, scrambled labels break purity). Together with the ladder index measurement ([verification/v726_phys_car_pp_index.py]; $\chi = 1 \leftrightarrow$ half-integer labels on both sides, the ramified defect in $\ker \chi$) this gives the groupoid reading of G_{net} its first *exact* witnesses; the gate typing is unchanged.]

The flavor-filtration fingerprint (FLAVOR.GRAPH.FILTRATION.01). *Registered* 2026-08-06 ([verification/v812_flavor_graph_filtration.py], 16/16 incl. the four must-fire controls, FILTRATION-FINGERPRINT-ONLY). The $Q \to K \to L$ row budgets are exact, order-exact counts of the $K_{3,2}$/prism/K_5/H_8 incidence chain with level-wise additivity mirroring $L = K + Q$; *no* entry-level reconstruction from any frozen candidate; exhaustive false-positive rates 0.005–0.75% make the misses informative. Typed audit fingerprint — beside the audit identities, in no derivation. No marker move.

The P1 index-KMS identity (P1.INDEX.KMS.01). *OPEN* (registered 2026-08-06; [verification/v813_p1_index_kms.py], 13/13 incl. the two must-fire controls, P1-IDENTITY-CLOSED). *The identity:* $I \cdot \beta_{\mathrm{angle}} \cdot c_3 = 4 \cdot 2\pi \cdot \frac{1}{8\pi} = 1$ exact in formal arithmetic, each factor independently sourced — the Jones index $I = 4 = |\mu_4|$ a c_3-free theorem (zero occurrences gated in all four sources), $\beta_{\mathrm{angle}} = 2\pi$ independently measured (v526; the radian pin is the one shared convention and cancels), c_3 the P1 axiom the identity *explains*, not derives. *Measured precursor:* $w_1 = w_3 = \frac{1}{4}$ exactly at every level with symmetric zero-sum offset $\pm 2^{-(l+1)}$ (the no-anomaly shape); the sea state does *not* equidistribute (dev 0.188). *The one open bridge:* the even modular-response normalization over the I deck sectors. *Kill:* any asymmetric sector offset. *Control:* index 2 gives $1/(4\pi)$, booked elsewhere by the corpus. P1 stays a declared input; no marker move.

The finite-compiler compressions of round 30 (MESSAGE.LADDER.01, DOILY.PASCAL.RANK.01, NORMALFORM.CFIN.01). *Recorded* 2026-08-07 (evidence lines, not contracts; exact identity grade; no marker moves). The message ladder $M_n = 15 \cdot 2^n = (15, 30, 60, 120, 240)$ with one named structure per rung and $h(E_8) = 30$ derived from the rebuilt root system, and the Doily–Pascal rank theorem $NN^{\mathsf{T}} = B + 2I$ with $\operatorname{rank} N = 10 = A_\Lambda$, $\dim \ker N = 5 = g_{\mathrm{car}}$ and recovery value 2/3 ([verification/v844_message_doily_rank.py], $23/23 + 16/16$); and the finite compiler normal form $C_{\mathrm{fin}} = (V, \hbar, q^*, \sigma, \iota)$ assembled as one object with the code-to-matter kill built in and P2 narrowed to the v799 residual R1′ ([C]), not eliminated ([verification/v845_cfin_normal_form.py], 28/28, SPEC v2). Full statements and keyboxes in Paper 1; audit cards in Paper 3. The $(1/3)^6$ typed gap of the six-step decision is re-affirmed with no upgrade.

The E8 audit compressions of round 32 (E8.SIMPLEX.FOURIER.01, E8.G31.CLOCK_ALPHABET.01, FLAVOR.WINDING.QUADRATIC.01). *Recorded* 2026-08-08 (evidence lines, not contracts; exact identity grade; no marker moves). The simplex-Fourier character theorem: the census $240 = 15 \times 16$ *is* the spectral statement $\hat{r} = (240, -16^{15})$ — the uniform-nonzero channel $P = (J - I)/15$ with spectrum $\{1, (-1/15)^{15}\}$; ([verification/v857_simplex_fourier_winding.py], 26/26; upgraded in round 36 to *all* odd shells, E8.SHELL.GLOBAL.01 — see the round-36 record below). The winding quadratic: the whole tfpt_2 winding line is one adjugate quadratic $q_{\mathrm{wind}} = t^2 - g_{\mathrm{car}} t + |\mathbb{Z}_2|$ with the triple lock $s = 6$, $\Delta(6) = 39200 = 2 \cdot 140^2$ and the reality

threshold $s^* \approx 2.8250$ typed ([verification/v857_simplex_fourier_winding.py], 20/20). The G31 clock alphabet: $\mathrm{Deg}(G_{31})/4 = \{2,3,5,6\}$ as a normal form with the unique product-46080/sum-64 quadruple, the 607-group kill scan (G_{31} the sole full-battery passer) and the $W(D_5) \times W(A_3)$ fence re-killed by computed centers 4 vs 1 ([verification/v858_g31_clock_alphabet.py], 19/19). Full statements and keyboxes in Papers 1 and 2; audit cards in Paper 3.

The round-36 compiler theorems (`FLAVOR.FEEDBACK.NORMALFORM.01`, `E8.WINDING.DECODER.01`, `E8.SHELL.GLOBAL.01`; Lean: `FORM.FLAVOR.FEEDBACK.01` [F], `FORM.E8.SHELL.GLOBAL.01` [F]). *Recorded/Registered* 2026-08-08 (exact identity grade; no marker moves; [verification/v874_feedback_decoder_lean.py] 14/14+11/11, [verification/v875_shell_global_lean.py] 15/15). The feedback normal form: $P = [\,2 \cdot \mathbf{1} \mid e_3 \mid Re_3\,]$ conjugates the whole winding line, symbolic in s, to the gain-plus-companion form with $\chi_{R_s} = (t - s - |\mu_4|)\, q_{\mathrm{wind}} - 12t$ and the honest $\mathbb{Z}_2^2$-not-μ_4 typing load-bearing (keybox in Paper 2). The winding decoder: q_{wind} maps the clock alphabet to $(-|\mu_4|, -|\mu_4|, |Z_2|, h(D_5))$, the decoder polynomial $(y+4)^2(y-2)(y-8)$ carries budget coefficients $(2, 48, 32, 256)$, and $|\mu_4|\, q(|Z_2|)/|R(E_8)| = -1/15$ is exactly the Walsh message eigenvalue — typed *audit* theorem, not a functor (keybox in Paper 1). The global shell theorem: $\hat{\Theta}_u/\Theta_L = -1/15$ for *all* odd shells from $\mathrm{Sp}(4,2)$ transitivity + $(1{+}i)$ doubling + $\Theta_L = 240\,\sigma_3$; the 16500 census retained as the exact ward (keybox in Paper 1). *The Lean records:* `TfptCarrier/FlavorFeedback.lean` (16 declarations, 11 theorems — the action identities, the inverse-free conjugation $PF_s = R_s P$ over $\mathbb{Z}$ symbolic in s, χ_{R_s} in $\mathbb{Z}[X]$, the SNF determinantal divisors) and `TfptCarrier/GaussianShells.lean` (the `ShellSystem` hypothesis bundle with provenance typed on each field, named *not* proven, + 7 consequence theorems, every statement an exact integer identity, no division) — both kernel-checked: lake build green at 3418 jobs, no `sorry`, no `native_decide`, axioms propext/Classical.choice/Quot.sound on all 18 theorems (`#print axioms`), carried in the shipped Lean manifest at 90 files. *The honest boundaries:* no μ_4 identification from the $\mathbb{Z}_2^2$ quotient; the decoder needs an intertwiner to be causal; the shell theorem's three inputs stay named hypotheses. No RH statement.

The ledger integrity guard and the `FORM.QGEO.04` QA fix (`GATE.LEDGER.01`). *Recorded* 2026-08-08 (a QA/CI note, no new physics; [verification/v878_ledger_integrity.py], 10/10). The round-36 ledger audit found the first *primary-key violation* in `status_ledger.csv`: the claim id `FORM.QGEO.03` was double-booked (2026-06-15 the CohomologyGrading formalisation; 2026-06-18 the SeamDeckClosure flat all-orders closure). The fix, dated: the SeamDeckClosure row is renamed `FORM.QGEO.04` with a QA note in its claim text, and every flat-closure reference is updated (this document, the two 2026-06-18 changelog mentions, `v279`/`v284`/`v286` + `seam_equivalence.json`, `make_figures`, the website residual chain, registry and `run_all`); the CohomologyGrading row keeps `FORM.QGEO.03`; claim contents unchanged. The guard enforces from this round: globally unique claim ids (the fix pinned), existing supersession targets with an acyclic supersession graph, unique *active* supersession targets modulo a frozen remove-only allowlist (the two legitimate 2026-06 joint closures), and the dependency cross-reference census frozen remove-only (13 SCCs / 265 ids — acyclicity enforced as *no new cycles*; the `audit_baseline` precedent).

The $Gr(2,5)$ Plücker reading, the Specht channels, the Hadamard anchor and the vacuum selection (`DOILY.PLUCKER.GR25.01`, closed). *Registered and closed* 2026-08-09 ([verification/v888_finite_pfaffian_signature.py], 27/27, S6PLUCKER-CLOSED; exact integer/sympy). *Exact:* the 15 doily lines are the 15 monomials of the five Plücker quadrics of $Gr(2,5)$ — bijection 15/15 with perfect sign match (every quadric reads $(+,-,+)$, the `v880` Hodge table); the S_6 duad representation decomposes $15 = 1 + 5 + 9$ with the doily operator acting by the exact scalars $(1, 0, 2/3)$, $2/3 = e_3/(e_2 - e_3)$ at the anchor $(4, 5, 2)$; the q^* zero set is a $(16, 6, 2)$ Hadamard difference set with anchor signature $(2^{e_1}, p_2, e_3, \binom{e_2}{2}) = (16, 6, 2, 10)$; and the vacuum-selection theorem: $\mathrm{Stab}(q^*)$ of order 120 acts faithfully and surjectively as S_5 on the remaining five Arf-1 refinements, orbits $[1, 5, 10] = (\Lambda^0, \Lambda^4, \Lambda^2)$. *Controls:* Arf-0 census $(16, 10, 6)$; a secant breaks the bijection; the wrong scalar breaks the channel law. The physical vacuum-selection mechanism stays [O]. No marker move; no RH statement.

The flavor-hexagon automorphism orbit (`CFIN.AUT.FLAVOR.C6.01`, closed). *Registered and*

closed 2026-08-09 ([verification/v888_finite_pfaffian_signature.py], 19/19, C6FLAVOR-MEASURED + INTERTWINED). *Exact:* $\mathrm{Aut}(C_{\mathrm{fin}}) \cong C_6$ by exhaustive enumeration; the pinned generator satisfies $g^2 = \sigma$ (the deployed family 3-cycle) exactly and $g^3 =$ the W-edge sheet swap; the action on $\Lambda^2 E$ has cycle type $1+3+6$ with orbits equal to the charge shells; the g-orbit of $\{f_1, w_1\}$ is the deployed hexagon transport edge for edge (D_6-equivalent); and $(|Z_2|/N_{\mathrm{fam}})^{|\mathrm{Aut}(C_{\mathrm{fin}})|} = (2/3)^6 = 64/729$ equals the frozen recovery value as an exact Fraction. *Typed honestly:* the exponent reading is a consistent *new reading*, not the unique derivation (the corpus route via $p_2 = |R_+(A_3)|$ stands). *Control:* 0 of 600 symplectic non-automorphisms factor; scrambled transport fails the D_6 match. No marker move; no RH statement.

The fermionic Pfaffian signature (DOILY.PFAFFIAN.WICK.01, closed — the K_6 motif, the sign theorem, the self-hosting counting theorem, and the mod-30 clock). *Registered and closed* 2026-08-09 ([verification/v888_finite_pfaffian_signature.py], 42/42, K6PFAFFIAN-CLOSED; exact integer/sympy, first frozen run). *Exact:* the K_6 motif theorem — the 35 lines of $PG(3,2)$ are exactly 15 perfect matchings (the doily) + 10 triangles through the vacuum vertex (the secants) + 10 avoiding it (the externals), with $q^*(e) = 0$ iff e touches the vacuum; the sign theorem — the Pfaffian signs match the Hodge table under the canonical character $\chi(i) = (-1)^{i+1}$, triple-identified (vacuum-row Laplace, Λ^4/Λ^5 volume form, C_6 crossing character; a global gauge, not a line-dependent fudge — the flipped-χ control produces exactly the 3 expected mismatches); the self-hosting counting theorem — $\binom{g+1}{2} = g!!$ has $g = 5$ as its only nontrivial solution, $(g-2)!! = 3 = N_{\mathrm{fam}}$; and the mod-30 anchor clock, period exactly $4 = |\mu_4|$, CLOCK-DISTINGUISHED (period-4 moduli exactly $\{5, 10, 15, 30\}$). *Typed honestly:* the counting theorem is [E]; the Wick-compiler premise stays [O] — P2 narrowed to one premise, not eliminated; the operative-Coxeter-30 reading is separately dead (sandbox); no canonical vertex–one-factorization bijection exists (outer-automorphism obstruction, report only). No marker move; no RH statement.

The channel-mixing candidate state (SEAM.CFIN.KMSMIX.01 + SEAM.CFIN.SCHURMIX.01). *Registered and constructed* 2026-08-09 ([verification/v898_kms_schur_mixing.py], 30/30, KMSMIX-MEASURED [KMSMIX-FOUND(U=1, T=1/8, BETA=1), SCHURMIX-GENERATES(10/10), RP-THETA-OPEN]). *Constructed:* the C_6-covariant channel-mixing KMS state exists — the first point of the frozen grid passes all block-functor gates (strict CAR, exact covariance with the forced zeros, all 15 blocks at the floor, grading $5/5 + 10/10$, all 15 block Wick monomials with the canonical Pf_4 signs; robust at $\beta = 2$); *exact (sympy):* Schur/Feshbach elimination of the boundary gives $A_{\mathrm{eff}} = \kappa A_{CC} + \frac{t^2 m}{1-m^2} V J_3 V^{\mathsf{T}}$ and generates all 10 carrier duads from the bare diagonal state — on both fronts (RH port, Wick functor) the real correlation lives in the *eliminated boundary*; the frozen 15-signature table is the registered prediction. *Named open check:* RP-THETA-OPEN (no spatial reflection is deployed on the 16-dim space). *Demand:* the physical realization — the [O] premise of SEAM.CFIN.WICKFUNCTOR.01 is unmoved. *Kill:* a physical seam state that violates the frozen signature pattern. No marker move; no RH statement. *Dated note (2026-08-10, rounds 57–59):* RP-THETA-OPEN is now *measured* at family level and the answer is an exclusion — strict reflection positivity and the mixing floor are mutually exclusive on this family (see the seam RP/modular exclusion block below, [verification/v903_seam_rp_exclusion.py]); the [O] premise stays unmoved.

The seam RP/modular exclusion (SEAM.STATE.DERIVATION.01 + SEAM.CFIN.TWISTED.RP.01 + SEAM.CFIN.RP.DILATION.01 + SEAM.CFIN.GAP.PENCIL.01). *Registered and measured* 2026-08-10, rounds 57–59 ([verification/v903_seam_rp_exclusion.py], $25/25 + 20/20 + 23/23 + 18/18$ on these four claims, SEAMSTATE-MEASURED + TWISTEDRP-MEASURED + DILATION-SPLIT + GAPPENCIL-MEASURED; six frozen probes embedded byte-exact, all fail-first spec amendments on record). *The exclusion:* the odd-sector Gram eigenvalues under both deployable spatial reflections are *exactly* $\pm|a_J|$ of the $\{4, 5\}$ duad (identity 2.2×10^{-16}), so OS positivity $\iff$ $a_J = 0$ $\iff$ the v898 mixing floor falls: **reflection positivity and the block-functor mixing floor are mutually exclusive on this family**; strict RP forces $t = 0$; $u \geq t$ is forced (the first non-decorative normalization demand; $u = 0$ is RP-excluded, correcting the round-57 reading); the 2π-KMS locus of the boundary reduction is exactly the point $(t{=}0, \beta{=}2\pi)$ — the geometric temperature kills the mixing instead of pinning it. *The complete twisted census:* 48 candidates → 16 involutive → 6

admissible → 0/6 admit strict RP at any $t > 0$ (TWISTED-RP-EXCLUSIONARY; a measurement on the complete finite candidate set, not a universal no-go). *The dilation split:* the `v898` M2 parent is a θ_{abT}-marginal witness *on* the RP cone boundary whose Schur compression carries the full Pfaffian mixing with the exact rational identity $t^2 \cdot 3m/(1-m^2) = 1/200$; the strict collar route is family-wide obstructed by an exact linear law (30 defect entries, all of magnitude $2t$). *The gap pencil (exact):* $\mathrm{Pf}(A_{\mathrm{dep}} + tA_{\mathrm{int}}) = -(t-1)(3t^2-1)\,q(t)$ with $q = 9t^3+21t^2-t-1$ irreducible; $t_{\mathrm{gap}} = 0.2309488708\ldots$ is the unique root in $(0, 1/4]$ (exact Sturm), a generalized eigenvalue of the fixed integer pencil with kernel dimension exactly 2; the three critical surfaces (1p-gap closure, modular rank-2 ground-state flip $\|\Delta\|_F = 2\sqrt{2}$, hermitized RP boundary $t_c(\beta) \to t_{\mathrm{gap}}$) are projections of one variety; the coupling is load-bearing for the root; *no root is* 1/8. *Demand (sharpened):* a state construction that carries the mixing *and* a positivity structure must leave this family — the named parent route is non-equilibrium (NESS-PARENT, named, not built). *Dated note (2026-08-10/11, round 60; promoted rounds 60–63):* the NESS-PARENT route is now *built and closed as not-needed* — at finite size stationarity forces zero entropy production exactly, drive buys no reflection positivity, and *equilibrium* witnesses on the $\{J, Z\}$ coupling kernel carry the full 1/200 mixing (see the seam equilibrium wiring block below, [`verification/v908_seam_equilibrium_wiring.py`]); the [O] premise stays unmoved. *Dated note (2026-08-27,* `SEAM.STATE.RPMIXING.01`*,* [`verification/v972_seam_interaction_front.py`]*):* the exclusivity is now *typed* as ray- and reflection-relative, not slice-wide — on the full 33-dimensional C_6-covariant mixing slice θ_{abT} kills exactly *one* coordinate (a_J; a 32-dimensional marginal-RP hyperplane remains) and θ_S leaves a 17-dimensional first-order-neutral subspace with strict finite-s witnesses: reflection positivity plus covariance do *not* force the channel-diagonal point. *Dated note (2026-08-28, completeness wave,* `SEAM.STATE.RPMIXING.01`*,* [`verification/v995_kms_subnet_rigidity.py`]*):* all 33 v972 mixing directions are killed by KMS *alone* on the finite CAR algebra ($\mathrm{CAR}_8 = M_{256}$); covariance is provably redundant under full KMS; the unique E-invariant crossed-product state extension is 196608 → 0. Type-III/continuum KMS uniqueness stays **[O]**. *Kill:* a strict RP point with $a_J \neq 0$ inside the family. The [O] premise of `SEAM.CFIN.WICKFUNCTOR.01` is unmoved; no marker move; no RH statement.

The seam equilibrium wiring (`SEAM.CFIN.MARGINAL.STABILITY.01` + `SEAM.CFIN.NESS.PARENT.01`, closed not-needed). *Registered and measured* 2026-08-10, round 60 (promoted rounds 60–63; [`verification/v908_seam_equilibrium_wiring.py`], 22/22+25/25, MARGSTAB-MEASURED (INTERIOR-EMPTY / STRATUM-NOT-ISOLATED / NOT-A-CLOSURE-POINT / ESCAPE-NONCOVARIANT / FLOOR-EXCHANGE) + NESSPARENT-MEASURED (PRICE-ZERO / TWO-SEAT-LAW / FINITE-NESS-NOGO / DRIVE-RP-NEUTRAL / NESS-NOT-FORCED); both probes exact-arithmetic, embedded byte-exact). *The 2-cycle side:* the closure-vs-isolated dilemma of the `v903` marginal witness is refuted *as a dilemma* — covariance reduces the θ_{abT} seat law exactly to $M_1 = \mathrm{diag}(a_J, -a_J)$ with $\det M_2 = -a_J^2$: strict 2-cycle reflection positivity is **impossible on the whole C_6-covariant class** (interior empty); the witness is neither a limit of covariant strict equilibria (there are none) nor isolated (the whole $s = 0$ family is a marginal stratum; sup(RP-margin × mixing) = 0 exactly); the only strict-enabling escape is the covariance-*breaking* X coordinate at exact linear price 2ε; and the sharpest exact object, the floor exchange: $\mathrm{Pf}_4(\varepsilon) = (\varepsilon - \frac{1}{200})(\varepsilon + \frac{1}{200})$ — the canonical sign gate *is* the negative effective RP margin, with the crossover exactly at the 1/200 floor. *The collar side, no NESS needed:* strict θ_S hermiticity is a *two-seat linear law* of the coupling (exact rank 12, kernel 12 = the $\{J, Z\}$ coordinates of the 24-dim covariant coupling space); the deployed seam wiring is PURE-I (every 2×2 subblock exactly $-I_2$) = a *maximally obstructed* covariant direction, explaining the `v903` strict-collar law ('30 entries, magnitude $2t$') exactly — rounds 69–71 add the frame qualifier: the obstruction is maximal *in the deployed θ_S collar frame*, and frame-relative as such (the wiring selector below); the kernel contains *equilibrium* witnesses — V_J (and V_Z) pass strict collar RP in exact arithmetic while their Schur compressions carry the full canonical Pfaffian mixing with the J coordinate exactly $t^2 \cdot 3m/(1-m^2) = 1/200$; at finite size every exactly stationary quasi-free state has $\sigma \equiv 0$ exactly ('NESS with positive entropy production' is category-inapplicable) and drive buys *no* reflection positivity — the minimal entropy production at which the mixing gates open is exactly zero: `SEAM.CFIN.NESS.PARENT.01` **closes as not-needed**. *Demand (the*

registered successor): the wiring selector below — **delivered and closed in rounds 69–71** ([verification/v911_wiring_freedom.py]). *Kill:* a covariant strict 2-cycle RP point, or an equilibrium collar witness whose compression breaks the canonical census. The witnesses live on a different covariant coupling direction than the deployed A_{int} wiring — whether the actual seam allows them is untouched; the [O] premise of `SEAM.CFIN.WICKFUNCTOR.01` / `v898` is unmoved; no marker move; no RH statement.

The wiring selector (`SEAM.STATE.WIRING.SELECTOR.01`, closed: deployment choice). *Registered* 2026-08-11 (rounds 60–63) as the question: is the deployed PURE-I seam wiring *compiler-forced*? *Closed* 2026-08-12, rounds 69–71 ([verification/v911_wiring_freedom.py], 30/30 + 26/26 + 21/21, verdicts WIRING-DEGENERATE + THETA-CONVENTIONAL + RP-FRAME-COVARIANT; three frozen probes, exact integer/rational arithmetic, embedded byte-exact), and the answer is a *freedom theorem*: the exact Gröbner census of the constraint classes (unit law, CAR/KMS, Pfaffian pencil, orientation propagation, covariance + seat stack) decomposes the real variety certified and leaves exactly **one admissible three-dimensional wiring component modulo the rule gauge** (the rotation cell C_{rot}); the Z/X wirings fall at exactly one class — orientation propagation ($\det u < 0$ identically) — and this exclusion is genuinely rule-gauge-orbit-invariant; PURE-I is an *interior* point of C_{rot}, not a vertex, and the integer gauge element $(\oplus_5 J_2) \oplus I_6$ (preserving the vacuum, the C_6 action, the rule subspace and the constraint ideal) connects it to the pure-J ray: **pure-i is a deployment representative, not a compiler theorem**. The companion probes close the frame side: no compiler-side demand distinguishes the θ_S sheet-swap frame within the 9-dim rule-gauge algebra (five demand classes — μ_4/C_6, CAR/state, pencil/orientation, doily combinatorics, the round-59 census property — all compatible-not-forcing or silent on the angle; THETA-CONVENTIONAL), and the strict-collar RP Grams transform *exactly covariantly* under the rule gauge (symbolic kernel-covariance law on the full torus + exact Gram witnesses): reflection positivity of a (frame, wiring) pair depends only on their *relative angle* on the δ-circle — the same deployed PURE-I parent that fails strict collar RP at θ_S (defect $2t$) *passes it exactly* in the integer-conjugated frame, so the `v903`/`v908` obstruction is a *frame statement*, not a wiring no-go (RP-FRAME-COVARIANT). The frozen canonical sentence: the compiler determines the wiring *orbit*; strict Hermiticity in the θ_S frame would select pure-J, the same demand in the integer-conjugated frame selects PURE-I; a canonical wiring selection requires an additionally *derived* physical frame. (Forbidden from now on: "PURE-I is compiler-forced" and equivalents without the frame qualifier.) *Honest scope:* the census covers the five enumerated demand classes — a compiler demand outside the list is not excluded; whether the actual seam physics derives a specific collar frame remains outside the formalized rules; the [O] premise of `v898`/`v903` is unmoved. *Kill (of the closure):* a compiler-side demand outside the censused classes that cuts the δ-circle, or a derived physical collar frame (which would then *select* a wiring via the exact selection map). No marker move; no RH statement.

Sugawara reading + atomic OS moment ([verification/v171_os_moment_cluster.py]). The G5 budget is the E_8 Sugawara constant: $248c_3^2 = \frac{31}{8\pi^2}$ with $31 = 1 + h^\vee(E_8)$ and $c((E_8)_1) = 248/31 = 8$, so gap dominance reads $|R^+(A_3)| \log(N_{\text{fam}}/|\mathbb{Z}_2|) > \frac{1+h^\vee(E_8)}{4\pi^2}$ — the A_3 transfer gap beats the E_8 Sugawara metric budget. Independently, the G3 reflection positivity is a *positive moment problem*: $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ makes every two-point correlator $C(n) = A + B(64/729)^n + C(1/729)^n$ the moment sequence of a positive atomic measure, so each OS Hankel matrix factorises as a Vandermonde Gram and is positive semidefinite (a clean RP certificate; cluster constant 729/665, correlation length $\xi = 1/(6 \log \frac{3}{2}) = 0.4111$).

Theorem 1 (Decoupling theorem — physical low-energy claims do *not* need the full G_{metric})**.** If *the admissible sector is reflection-positive (G4) and the metric coupling is gap-dominated,* $2\|V_{\text{metric}}\| < \Delta = 6 \log \frac{3}{2}$ *(G5),* then *the admissible IR measure is stable under the ambient completion: a strictly positive effective gap* $\Delta_{\text{eff}} = \Delta - 2\|V\| > 0$ *protects the admissible sector from the (un-built) ambient/deep-UV, so every* physical low-energy *read-out (masses, mixings,* α^{-1}*, the*

$R{+}R^2$ regime) is independent of the open ambient measure. Symbolically

$$\boxed{\text{RP} \;+\; 2\|V\| < \Delta \;\Longrightarrow\; \textit{admissible IR measure stable under ambient completion}\,.}$$

Thus (G_{metric}) *is* not *a diffuse "full quantum gravity missing" block: it is the sharply delimited* ambient projective limit *(G6), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on.* ***Typing:*** *the inequality is machine-checked [E]; the operator-norm bound on* $\|V_{\text{metric}}\|$ *is the conditional input.*

Lemma 10 (G6–G7 — projective limit & Ward identities)**.** *A projective system* μ_{χ_n} *with* $\pi_{m,n*}\mu_{\chi_m} = \mu_{\chi_n}$*, uniform moment bounds* $\sup_n \int \|g\|_{H^s}^p \, d\mu_{\chi_n} < \infty$*, tightness and regulator-independence yields* $\mu_{\text{QG}} = \lim \mu_\chi$ *on* $\mathcal{G}/\text{Diff}$ *(G6, the hardest step); the limit obeys the BRST Ward identities* $\langle s\mathcal{O}\rangle = 0$*, whence* $\nabla^\mu T_{\mu\nu} = 0$ *and the* η *boundary phase cancels the APS anomalies (G7).*

Gate 2 reduction: G6 is a *boundary* projective limit, not a bulk one (`[verification/v76_gmetric_reduction.py]`)

The exact decoupling margin is $\Delta_{\text{eff}} = \Delta - 2\|V\| = 6\log\frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$, with $\|V_{\text{metric}}\|_{\text{rel}} \le 248\, c_3^2 = \dim(E_8)\, c_3^2 = \frac{31}{8\pi^2}$ ($31 = 2^{g_{\text{car}}} - 1$). **Holographic reduction:** because the seam is a *finite causal boundary*, the ambient *bulk* metric measure is reconstructed from a finite-codimension seam (Calderón) *boundary* measure — so *G*6 is a *boundary* projective limit. The conditional theorem is then

$$\boxed{\text{RP(seam-Calderón boundary kernel)} + \text{tightness} \;\Longrightarrow\; G_{\text{metric}} \text{ closes}}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure. **Residual:** the constructive boundary projective limit (constructive-QFT grade) — now discharged as a [C] redundancy (v369, keybox below). So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) a [C] redundancy, no longer a diffuse "full QG missing". [C]

The measure as one-loop fluctuations around the *parameter-free* saddle (`[verification/v365_qg_oneloop_saddle.py]`, `QGAMB.SADDLE.01`)

With the saddle now parameter-free (v358/v359), the boundary measure of Tier B organises as a Gaussian one-loop fluctuation determinant whose ingredients are atom-fixed. **Fixed quadratic form [E]:** the fluctuation inverse-propagator stiffness is the saddle coefficient $1/c_3 = 8\pi$ — no free dimensionless dial, so the one-loop problem is *well-posed* (it was not before v358). **Gap-controlled convergence [E]:** the fluctuation operator obeys $M \ge \Delta_{\text{eff}} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337), so the Gaussian integral converges mode-by-mode with no zero/negative modes besides the saddle directions; on the OS spectrum $\{0, 6\log\frac{3}{2}, 6\log 3\}$ the regularised one-loop log-det $\text{Tr}'\log M = 6\log\frac{9}{2}$ is finite. **Conformal direction [C]:** the one wrong-sign mode ($c_{\text{conf}}(4) = -\frac{3}{2}$) is made convergent by the GHP contour ($\int e^{-|c|\phi^2} = \sqrt{2\pi/3}$) with IDG dressing (v334). **Tightness [E]:** $\chi = 1/(1-(2/3)^6) = 729/665 < \infty$ (v330), so the one-loop measure has a projective limit on the admissible sector, with *zero* new dimensionless dials (v364). **Residual:** the full *non-perturbative* projective limit (G6 on the ambient sector) is now discharged as a [C] redundancy (v369, keybox below) — this sharpens C7 from "diffuse full QG" to a one-loop-controlled Gaussian problem around a parameter-free saddle and then to a certification object. [E]/[C]

The holographic route: the ambient measure is boundary-*redundant*, not missing dynamics (`[verification/v369_qgamb_redundancy.py]`, `QGAMB.REDUNDANCY.01`)

There is a second, strategically stronger way to dispatch Gate 2 than building the measure: prove that the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the measure `QG.AMB.01` is a non-fundamental *certification* object — the role a coordinate choice plays in general relativity — rather than a missing piece of dynamics. The discriminators that make this *ambient-redundancy* statement well-posed are all in hand. **Trivial superselection [E]:** the number of DHR sectors of a chiral net is $|\det \text{Cartan}|$, which is 1 for E_8 (one primary, the vacuum; $\text{DHR}((E_8)_1) = \text{Vec}$) against 4 for the same-*c* rival $SO(16)_1$ — a holomorphic boundary net has *no* nontrivial superselection charge that a bulk could carry invisibly. **No torus degeneracy [E]:** the ground-state degeneracy on T^2 equals the number of anyons $= |\det K| = 1$ for E_8, so there is no topologically protected bulk d.o.f.

hidden from the boundary (the genus-1 obstruction of v344, in its redundancy reading). **Bounded reconstruction [E]:** the seam transfer deviation subspace contracts at the finite gapped Petz rate $(2/3)^6$ (v221), and the gap-decoupling margin $\Delta_{\text{eff}} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) separates the un-built ambient sector from the admissible one. **The statement [C]:** with the Bisognano–Wichmann modular reconstruction map (v258/v309/v323), under `SEAM.EQUIV.01` every gauge-invariant bulk observable is boundary-reconstructible, $\mathcal{O}^{\text{bulk}}_{\text{phys}}/\text{Diff} \cong \mathcal{A}^{\mu_4}_{\Sigma}$, and two ambient measures sharing the same boundary state are physically equivalent. **Residual [O]:** the construction consumes exactly two premises — `SEAM.EQUIV.01` (the S3 lattice input, v366) and the BW intrinsicality of the raw collar (v329); it does *not* construct `QG.AMB.01` but supplies the *alternative* route, so that Gravity-complete = Boundary-complete + Ambient-redundancy. [E]/[C]/[O]

The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]: entanglement equilibrium gives the linearised Einstein equation with G from c_3 ([`verification/v358_grav_entropy_equilibrium.py`])

While Gate 2 (above) reduces the metric-sector *measure*, the classical metric-sector *field equation* is supplied directly by the entanglement first law $\delta S = \delta\langle K\rangle$ (Jacobson 2015; Faulkner et al.), carried out with TFPT's atoms. **Parameter-free coefficient [E].** Equilibrium gives $G_{ab} = c_3^{-1}T_{ab}$ with $c_3^{-1} = 8\pi$ *fixed*: all four c_3-roles are mutually consistent — the Unruh factor $1/(2\pi) = 4c_3$, the EH action $1/(16\pi) = c_3/2$, the Einstein coefficient $8\pi = 1/c_3$, the Bekenstein density $1/4 = 1/|\mu_4|$ — so *no free dimensionless Newton dial* enters. **Thermo = geo coincidence [E] (new).** The first law requires the coefficient $2\pi/\eta$ (η the entropy density); the geometry gives $1/c_3 = |\mathbb{Z}_2|\,2\pi\,\chi(S^2)$ (one-sided Gauss–Bonnet, v58); these are the *same* equation iff $|\mu_4| = |\mathbb{Z}_2|\,\chi(S^2) = 4$ — a TFPT atom identity (v216). So the thermodynamic (first-law) and geometric (Gauss–Bonnet) origins of c_3 are *one*. **The matter side, assembled [C] (closing J3).** The ball modular Hamiltonian is the boost (Casini–Huerta–Myers; geometric by Bisognano–Wichmann, v323), $K_B = 2\pi\int_B w\,T_{00}$, $w = (R^2 - r^2)/(2R)$; the 3-ball weight integral is $\int_B w\,d^3x = 4\pi R^4/15$, so $\delta\langle K_B\rangle = \frac{8\pi^2R^4}{15}\,\delta\langle T_{00}\rangle$ — the concrete matter boost flux of TFPT's stress tensor. **The entropy density, atom-fixed [E].** The Bekenstein numerator $1/4 = 1/|\mu_4|$ (v57) and the entanglement normalisation = the central charge $c = g_{\text{car}} + N_{\text{fam}} = 8$ pin the dimensionless coefficient; the area-law form is established (v59). **Residual:** the full covariant extension is now *closed* (v359, next keybox); what remains is only (i) the Jacobson equation-of-state *status* (a thermodynamic derivation, not a from-action quantisation) [O] and (ii) the absolute scale v_{geo} (the Planck-area unit) [O], while the global ambient measure (Gate 2/C7) is now a [C] redundancy (v369) — *no free dimensionless parameter survives in the gravity coupling.*

And that boundary measure is a *rigorously constructed* net ([`verification/v77_e8_conformal_net.py`]). The level-1 central charges $c = \dim\mathfrak{g}/(1+h^\vee)$ give $c(E_8)_1 = \frac{248}{31} = 8 = \text{rank}\,E_8$, $c(D_5)_1 = \frac{45}{9} = 5$, $c(A_3)_1 = \frac{15}{5} = 3$, and $c(D_5)+c(A_3) = 5+3 = 8 = c(E_8)$ is a *conformal embedding* $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$ (coset $c = 0$). Now $(E_8)_1$ is the holomorphic $c = 8$ E_8-*lattice* VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete:** *identify the seam-Calderón boundary measure with the* $(E_8)_1$ *lattice net* (carrier = $(D_5)_1 \times (A_3)_1$ subnet), and the gap $\Delta_{\text{eff}} > 0$ supplies the clustering/tightness — so $G6$ is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [E] (charges, embedding) + [C] (identification)

The full covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]: fixed-volume equilibrium gives the Einstein tensor with conservation as an output ([`verification/v359_grav_nonlinear_einstein.py`])

The linearised result above (v358) extends to the *full* covariant equation by Jacobson's *fixed-volume* (maximal-vacuum-entanglement) equilibrium — no linearisation assumed. **Einstein tensor, not Ricci [E].** Stationarity of the entanglement entropy at fixed volume forces the divergence-free combination: the 4D trace identity $g^{ab}G_{ab} = (1 - \frac{d}{2})R = -R$ (for $d = 4$) selects $G_{ab} = R_{ab} - \frac{1}{2}R\,g_{ab}$, so

$$\boxed{G_{ab} + \Lambda g_{ab} = c_3^{-1}T_{ab}, \qquad c_3^{-1} = 8\pi}$$

with *both* coefficients TFPT-fixed: the Newton coupling $8\pi = 1/c_3$ (v358) and the cosmological constant Λ from α, $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60). **Conservation is an output [E].** By Lovelock's theorem the only divergence-free, second-order, local symmetric tensor built from the metric is $a\,G_{ab} + b\,g_{ab}$, so

$\nabla^a G_{ab} = 0$ forces $\nabla^a T_{ab} = 0$ — matter conservation is *derived*, not assumed. **So Gate B6 — the classical metric-sector field equation — carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C] at the *local* level. Residual:** the Jacobson equation-of-state *status* (a thermodynamic derivation, not a from-action quantisation) and the single unit v_{geo} stay **[O]**, while the global ambient measure (Gate 2/C7, the constructive boundary projective limit) is now a [C] redundancy (v369) — no free dimensionless gravity dial remains. An *external candidate* for the missing action level is quantified in the next keybox (v473); it does not change this typing. [E]/[C]

The entropic-action bridge: an external candidate for the missing action level, quantified — nothing closes ([`verification/v473_entropic_action_bridge.py`])

The one honest **[O]** left in the classical sector — "equation of state, not a from-action quantisation" (v359) — now has an explicit *external* candidate: Bianconi's entropic action $S_{\mathrm{B}} = -\operatorname{Tr}_F \ln\!\big(\widetilde{G}\widetilde{g}^{-1}\big)$, the quantum relative entropy between the spacetime metric and the matter-induced metric (G. Bianconi, *Gravity from entropy*, Phys. Rev. D **111**, 066001 (2025); `arXiv:2408.14391`; her Appendix C identifies $\Delta^{1/2}_{\widetilde{G},\widetilde{g}} = \widetilde{G}\widetilde{g}^{-1}$ with an Araki-type modular operator) — a local covariant action whose variation yields the Einstein equations plus an emergent nonnegative Λ at low coupling: exactly the missing variational layer, *if* the bridge identities hold. What is machine-checked today (`GRAV.ENTROPIC.ACTION.01`): **carrier Hodge count [E].** Her $\Omega^{0,1,2}$ matter fiber, placed on the five-slot carrier $\mathbb{C}^5$ (*not* on 4d spacetime, where the fiber is 1+4+6=11), counts $1{+}5{+}10{=}16{=}2^{g_{\mathrm{car}}-1}{=}\dim S^+_{D_5}$ and Hodge-folds ($\star : \Lambda^1{\cong}\Lambda^4$) onto $\Lambda^{\mathrm{even}}\mathbb{C}^5$ — the carrier half-spinor's own Pascal row (v2/v44); a vector-space identity *only*. **Coefficient match [E] (the one quantitative pin).** Her weak-coupling Lagrangian $\mathcal{L} = 3\beta R - \alpha\mathcal{L}_{\mathrm{m}} + \cdots$ (the 3 = the three form blocks) matched to the TFPT EH normalisation $1/(16\pi G) = c_3/(2G)$ fixes her free constant *exactly*,

$$\boxed{\beta'_{\mathrm{B}} = \tfrac{c_3}{6} = \tfrac{1}{48\pi}}$$

(the $6 = 3 \times 2 = p_2 = |R^+(A_3)|$ is recorded as a structural echo, not a proof). **Entropy potential = quadratic Λ [E].** Her $\Lambda_{\mathcal{G}} = \frac{1}{2\beta}\operatorname{Tr}_F(\mathcal{G} - I - \ln\mathcal{G})$ is nonnegative and quadratic at the fixed point; with the TFPT displacement $\varepsilon_\star = e^{-\alpha^{-1}}Q$ it reproduces the closed Λ branch $\rho_\Lambda/\bar{M}_{\mathrm{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60), and the prefactor match yields the *exact* normalisation target $\operatorname{Tr}_F Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\,\delta_{\mathrm{top}}$ — a closed algebraic target for the two-form (pair) block ($\binom{5}{2}{=}10$), [C] until an explicit Q is constructed. **R^2 kill test [E] (pre-registered).** The raw log expansion carries a curvature-squared coefficient of order $(c_3/6)^2$; the TFPT Starobinsky sector needs $\bar{M}_{\mathrm{Pl}}^2/(12M_{\mathrm{scal}}^2) = 1/(12c_3^7)$ (v36) — the exact mismatch is $3c_3^{-9} = 3(8\pi)^9 \approx 1.2 \times 10^{13}$: a direct identification of the raw entropic action with the TFPT R^2 sector is *false* without a mechanism (the scalaron as a light trace mode of her $\mathcal{G}$-field generating c_3^{-9}, or a KMS-spectral renormalisation of the log action). **The bridge proper [C] (recorded, not proven).** The compression conjecture $P_\Sigma(\widetilde{G}\widetilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$ — the physical Calderón compression of her relative metric operator equals the TFPT seam modular operator; if proven, S_{B} is a local volume representation of the TFPT relative spectral action $S_{\mathrm{rel},\chi}$ (her log action is the all-scale-integrated heat kernel, by Frullani $-\ln A = \int_0^\infty \frac{dt}{t}(e^{-tA} - e^{-t})$, verified). Open with it: the operator-level Hodge items (Clifford action, Dirac intertwining, leaf involution, Calderón polarisation), and Lorentzian positivity of $\widetilde{G}\widetilde{g}^{-1}$ is *not* automatic (timelike gradients) — TFPT's OS/reflection-positive route is the sturdier construction. **Residual [O] (unchanged).** Nothing closes: the v359 equation-of-state typing stays **[O]**; adopting S_{B} would close it *only* after the compression conjecture *and* an R^2 mechanism; zero TFPT knobs are added.

The operator level, executed ([`verification/v474_entropic_hodge_carrier.py`]). Work packages 1 and 4-algebra of the bridge are now machine-checked on the finite carrier Fock space $\Lambda^\bullet\mathbb{C}^5$ ($\dim 2^{g_{\mathrm{car}}}{=}32$). **Spinor structure exhibited [E]:** the ten CAR gammas satisfy $\{\Gamma_a, \Gamma_b\}{=}2\delta_{ab}$ exactly, the 45 bilinears span $\mathfrak{so}(10)$ and leak nothing from the 16-dim even subspace — S^+ *is* the even exterior algebra with its spinor structure, and Bianconi's Hodge–Dirac operator is Clifford multiplication at symbol level ($c(\xi)^2{=}-|\xi|^2$, symbolic). **The fold is the $5{\to}\bar{5}$ conjugation [E]:** the traceless $\mathfrak{u}(5)$ weights of Λ^4 are the negatives of those of Λ^1, so her fiber 1+5+10 becomes the GUT half-spinor $16{=}1{+}\bar{5}{+}10$ — the SM carrier decomposition, not a dimension match; honest obstruction: $\Omega^{\le 2}$ is *not* Clifford-invariant, so the identification must go through the fold, never through degree truncation. **The Q-target, decided [E]:** the uniform-block ansatz $\operatorname{Tr} Q^2{=}d\,k^2c_3^4{=}32c_3^4$ with integer k admits *exactly* the supports $(d,k){=}(2,4),(8,2),(32,1) = \{|\mathbb{Z}_2|, \operatorname{rank} E_8, 2^{g_{\mathrm{car}}}\}$, with the minimal uniform solution $q{=}c_3^2$ per Fock mode and the identity $\operatorname{Tr} Q^2/\delta_{\mathrm{top}}{=}32/48{=}2/3$; the two-form block 10 (the naive pair-sector reading)

and the fiber 16 require irrational multipliers and are *excluded*. The physical choice among the three supports stays [C]; AP2 and the R^2 mechanism stay [O].

The R^2 kill test, executed ([verification/v475_entropic_scalaron_gate.py]). On a maximally symmetric background the relative curvature operator has eigenvalues *exactly* $\{R;\ R/4\times 4;\ R/6\times 6\}$ (her Appendix-B flattening verified, incl. the 6×6 two-form block), so the exact vacuum action is $\mathcal{L}_{\rm B}(R) = -[\ln(1-\beta R) + 4\ln(1-\beta R/4) + 6\ln(1-\beta R/6)] = 3\beta R + \frac{17}{24}\beta^2 R^2 + O(R^3)$ — the 3 re-derives her Eq. (45) and $\frac{17}{24}$ is the exact tensorial factor behind the pre-registered gap. With the pinned $\beta'=c_3/6$ *both* mass routes (Starobinsky matching and $f'/3f''$) give the raw entropic scalaron

$$\boxed{m_{\rm raw}^2 = \tfrac{4608\pi^2}{17}\,\bar{M}_{\rm Pl}^2 \ \Rightarrow\ m_{\rm raw} \approx 51.7\,\bar{M}_{\rm Pl}}$$

— *trans-Planckian*, not light: the naive reading "the TFPT scalaron is the light trace mode of her $\mathcal{G}$-field" is *dead* [E]; a viable mechanism must supply exactly $m_{\rm raw}^2/m_{\rm TFPT}^2 = \frac{72}{17}(8\pi)^9 \approx 1.69\times 10^{13}$ in mass^2. The domain is not the obstruction (first log pole at $R{=}384\pi^2\bar{M}_{\rm Pl}^2$, trans-Planckian), and the Lorentzian caveat is now an explicit witness: a timelike gradient gives the Gg^{-1} eigenvalue $1-\alpha v^2 \le 0$ for $v^2 \ge 1/\alpha$ (ln undefined), while spacelike/Euclidean stay positive — the OS route is the sturdier construction. **Net verdict [C]:** of the three bridge interpretations, the entropic action covers the Einstein + Λ level (survives); the light-mode reading is dead; KMS-spectral renormalisation is the *only* surviving route to the R^2 sector. Nothing closes.

AP2, made well-posed ([verification/v476_entropic_compression_toy.py]). The compression conjecture itself is sharpened on an exactly solvable gapped chain. **The reading is decided:** on a *pure* bulk (the seam situation: pure bulk, thermal boundary) the covariance is a projection, so $f(x){=}x/(1{-}x)$ is singular on its spectrum — the literal operator-side reading $P_\Sigma f(C)P_\Sigma$ is *ill-posed*, while the state-side $f(P_\Sigma C P_\Sigma)$ (Peschel) is finite: the identity can only mean *build* Δ_Σ *from the compressed relative metric*, which is exactly Bianconi's own local construction — AP2 retyped, not weakened. **The mismatch is exactly second order [E]:** symbolically, $[C^n]_{AA} - (C_{AA})^n$ carries the cross-cut block twice (first order vanishes identically), so for any analytic f the two readings agree up to $O(\|C_{AB}\|^2)$ — the modular (entanglement) content *is* that quadratic correction. **Gap control:** at the bounded (covariance) level the mismatch obeys $d \le x^2$ and falls with the gap, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ($\Delta_{\rm eff}{>}0$, v76/v337). **Residual [O] (unchanged):** the continuum seam statement itself. No marker moves.

The surviving R^2 route, typed as one moment ([verification/v477_entropic_scaleflow.py]). The Frullani identity upgrades to a *scale-flow representation*: $-\ln a = \int_0^\infty \frac{dt}{t}(e^{-ta}{-}e^{-t}) = 2\int_0^\infty \frac{d\chi}{\chi}(e^{-a/\chi^2}{-}e^{-1/\chi^2})$ — Bianconi's log action *is* the flat scale-integral of relative heat-kernel actions, while TFPT's $S_{{\rm rel},\chi}$ evaluates *one* KMS scale with a weight f: the bridge between the two action principles is a choice of *scale measure*. In the moment dictionary (weight $w(t)\,dt/t$, μ_k its e^{-t}-moments) the coefficients read $3\beta R\,\mu_1 + \frac{17}{24}\beta^2 R^2 \mu_2$; the flat weight has $\mu_1{=}\mu_2{=}1$ (= the raw v475 coefficients), and demanding $m^2{=}c_3^7\bar{M}_{\rm Pl}^2$ forces *exactly*

$$\boxed{\mu_2/\mu_1^2 = \tfrac{72}{17}(8\pi)^9} \qquad \text{with} \qquad \tfrac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7 \ \text{(identically)}$$

— the v475 kill-test number *is* a moment ratio, and the "13 orders of magnitude" are not a bridge mismatch but a scale-measure datum: the flat measure fixes it wrongly, the TFPT KMS measure ($f_0{=}6(4\pi)^2/c_3^7$, v36) fixes it correctly — one KMS moment, two parametrisations, zero new dials. [C] (consistency, not derivation: the measure is TFPT input); the R^2 gate and the v359 typing stay [O].

The two remaining legs, first computable steps ([verification/v478_entropic_continuum_legs.py]). *Leg A (AP2 continuum):* on the exactly solvable critical chain (infinite-volume kernel, 50-digit precision) the compressed interval state carries the conformal/BW structure quantitatively — the entanglement entropy follows Calabrese–Cardy with $c_{\rm est}$ = 1.00089/1.00038/1.00021 (monotone → the Dirac $c{=}1$; gapped control $\sim 6{\times}10^{-8}$), and its Peschel modular Hamiltonian organises on the CHM weight $\beta(x){=}x(L{-}x)/L$ (corr 0.87→0.99, even-range bands vanishing identically, odd tail 1.8%): the state-side Δ_Σ (the reading *forced* above) flows to the CHM/Bisognano–Wichmann geometric form — *exactly* the object the entanglement-equilibrium derivation consumes (v323/v358); the bridge's modular side and the gravity side meet in the continuum limit (a lattice exhibit, not a continuum proof — AP2 stays [O]; the natural continuum frame for the seam version is the *four-interval* multilocal modular geometry of the four marks, v480). *Leg B (the measure):* the single-scale corollary is exact — $d\mu{=}\delta(t{-}t_0)dt$ gives $\mu_2/\mu_1^2{=}e^{t_0}$, so the one-moment condition forces $t_0{=}\ln\frac{72}{17}{+}9\ln(8\pi){=}30.461$, one dimensionless KMS time

in closed form; the near-miss to $h(E_8)$=30 (difference 0.461) is explicitly *declined* per the no-free-pattern rule (v354/v355) — deriving the measure from the entropic structure stays [O]. [E]/[C]/[O]

The Simple-Current Extension Theorem — the central G_{net} closing statement [E]/[C] ([verification/v154_simple_current_theorem.py])

The conditional theorem can now be stated as one clean *simple-current extension*. Let $A = (D_5)_1 \otimes (A_3)_1$ be the carrier net and $L = \langle (1,1) \rangle \subset \mathrm{Disc}(D_5) \oplus \mathrm{Disc}(A_3) = \mathbb{Z}_4 \oplus \mathbb{Z}_4$ the isotropic μ_4 glue line ($q(k(1,1)) = k^2 \in \mathbb{Z}$). Then the local extension $B = A \rtimes L$ has

$$[B : A] = |L| = 4 = |\mu_4|, \quad c(B) = 5{+}3 = 8, \quad \mu(B) = \frac{\mu(A)}{|L|^2} = \frac{16}{16} = 1$$

so B is *holomorphic* (KLM μ-additivity), and a holomorphic c=8 chiral net is the lattice net of the *unique* even-unimodular rank-8 lattice — hence $\boxed{B \cong (E_8)_1}$ (the same-c rival $SO(16)_1$, μ=4, is excluded). Every algebraic ingredient is machine-checked: the $\mathbb{C}[\mathbb{Z}_4]$ Longo Q-system ([verification/v125_glue_qsystem.py]), the $\mathbb{Z}_4$-graded Frobenius E_8 hull ([verification/v143_graded_frobenius.py]), the Ramond glue sectors ([verification/v148_fock_census.py]). **Consequence:** G_{net} is internally *closed* if the seam–Calderón boundary net is *defined* to be A; it stays the *one* external identification "the RP seam–Calderón net is A" if the boundary net is to be *derived* from the seam kernel. Either way this is the single strict-TOE certification theorem still carrying weight — a clean extension statement, not "find a quantum gravity". *[2026-08-05 note, finite-level evidence for the identification ([verification/v779_gnet_gns_arf.py]): the inductive system — Haar isometry tower, commuting index-4 expectation squares, geometric state convergence — is exactly coherent; the finite Haag-duality census gives twisted-duality defect 0 with excess ratio* $\dim B' / \dim A' = 4 =$ *the local Watatani index; and the* $\mathbb{C}[\mathbb{Z}_4]$ *quasi-basis obstruction law* $2^{-\ell}$ *types the extension as exactly asymptotic at finite level. Typed finite evidence, never the continuum theorem; no marker move.]*

And that one identification reduces to a single shared premise ([verification/v155_quasifree_boundary.py]). The identification splits (v110) into (a) "the seam Calderón involution is sheet-odd" — the double-cover deck, the same $|\mathbb{Z}_2|$ that halves $c_3 = \frac{1}{2\cdot 4\pi}$ (structural) — and (b) "the seam 2-point kernel is quasi-free". Part (b) is *not independent*: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression $P\Gamma P$ of the bulk covariance, and the compression of a contraction ($i\Gamma$ with spectrum in $[-1,1]$) is a contraction — so the boundary/seam state is automatically quasi-free, with Wick inherited (every boundary correlator a Pfaffian of Γ_d). Hence (b) *follows* from "the seam bulk is a reflection-positive free field" — the *same* premise already load-bearing in the area law ([verification/v59_area_law_evidence.py]) and the EH mechanism ([verification/v150_replica_eh_model.py]/[verification/v151_bfk_split.py]; since exercised on the discretized collar with the seam's own kernel and real replica sheets, v471 — the R3 kernel premise is discharged at the finite level, its residual retyping to the continuum scaling limit shared with SEAM.EQUIV.01, v336, plus the v152 anchor). **So the entire seam-side residual is one physical premise** — the seam bulk is a free RP field — shared by G_{net}, the area law and R3, and given it the chain Gaussian bulk ⇒ quasi-free boundary ⇒ one kernel = net (v113) ⇒ carrier A ⇒ $(E_8)_1$ is exact. It is not reducible by a finite computation — it is the one place TFPT meets functional analysis.

The construction, exhibited and verified ([verification/v156_seam_net_construction.py]). The "$B \cong (E_8)_1$" step is no longer an assertion — the chiral net is built explicitly and the identification *checked*. The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian acts on a boundary mode e^{ikx} by the harmonic extension $e^{ikx-|k|y}$: $-\partial_n u|_0 = |k|\,u$, so for a free bulk the seam operator is the massless chiral dispersion $\omega_k = |k|$ that defines the free-fermion net on S^1. Sixteen Majoranas give $c = 8 = 5{+}3$; the weight-1 currents are $248 = 120_{\text{NS}} + 128_{\text{R}}$ (the $SO(16)$ bilinears plus the Ramond spinor, Frenkel–Kac); and the holomorphic character is

$$\chi = \frac{E_4}{\eta^8} = q^{-1/3}\left(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + \cdots\right) = j^{1/3}$$

(verified by q-series), the 248 at level 1 being $\dim E_8$ — so the net *is* $(E_8)_1$. *What remains* is exactly (i) the free-bulk premise above and (ii) the operator-algebraic existence / uniqueness of the net generated by the seam Calderón kernel (a Kawahigashi–Longo-type net-construction theorem). Route 3 thus *exhibits*

and verifies the target; the irreducible residual is the one premise + the standard net-existence step — where TFPT meets constructive QFT, now sharply located.

And the free-bulk premise itself reframes: freeness is a rigid fixed point, not a postulate ([`verification/v157_rigid_fixed_point.py`]**).** Three facts click together. (i) The DtN/Calderón symbol is *universally* $|k|$: the boundary operator of *any* second-order elliptic bulk is an order-1 ΨDO with principal symbol $|\xi|_g$ (Lee–Uhlmann), so the free chiral dispersion is bulk-detail-independent and any interaction lives in the lower-order (irrelevant) symbol. (ii) A holomorphic c=8 theory is *rigid*: every field has weight $(h, \bar{h}{=}0)$, an exactly-marginal deformation needs $(1,1)$, and with $\bar{h}{=}0$ *no* $(1,1)$ field exists — the fixed point is isolated, there is no continuous knob along which a non-Gaussian interaction could turn on. (iii) The *same* $|\mathbb{Z}_2|$ that halves the one-sided S^2 Gauss–Bonnet ($c_3 = \frac{1}{|\mathbb{Z}_2|\,2\pi\,\chi(S^2)} = \frac{1}{8\pi}$) labels the sheet *and* is the Ramond/GSO projection giving μ=1 ($248{=}120_{\mathrm{NS}}{+}128_{\mathrm{R}}$) — one object, three roles, so the geometry fixing c_3 is the projection making the net holomorphic. **Consequence:** premise (A) shrinks from "prove the seam is Gaussian" to "the gapped ($\Delta{=}6\log\frac{3}{2}{>}0$) boundary flow reaches the holomorphic fixed point" — given which, freeness is *forced* by rigidity + uniqueness ($(E_8)_1$, v83), not fine-tuned. A strictly smaller and simpler residual.

And the destination is a *proven* stable fixed point ([`verification/v158_fixed_point_stable.py`]**).** An exact operator-dimension count settles it: the chiral Majorana has $h{=}\frac{1}{2}$, the bilinear current h=1, the quartic h=2. The relevant window $0 < h < 1$ for bosonic deformations is *empty* (bosonic chiral operators have integer h, lowest 1) — nothing relevant flows the free theory away; the h=1 currents are chiral (v157), not $(1,1)$-marginal; and the lowest genuinely-interacting operator (the quartic, $h{=}2{>}1$) is *irrelevant*, as are all higher $2k$-fermi operators ($h{=}k$). So the free chiral c=8 fixed point is *isolated and stable* against all interactions. Premise (A) thereby reduces to a *basin-of-attraction* statement with a proven stable target — the gapped ($\Delta{=}6\log\frac{3}{2}$) flow lands here; the convergence is conformal-perturbation analysis, the destination is settled.

And the closure can be stated as one fixed-point theorem — which relocates (A) to a single internal premise ([`verification/v160_seam_gaussianity_from_pf.py`]**).** Phrase the argument internally: (i) a quasi-free state has *all* higher connected cumulants $\kappa_{2n}{=}0$ (the signed set-partition/Pfaffian inversion of $\langle\psi_1\cdots\psi_{2n}\rangle = \mathrm{Pf}(\Gamma)$ leaves only the degree-2 kernel — the v113 "one kernel = the net" property at the cumulant level); (ii) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (iii) a gapped primitive Perron–Frobenius transport has a *unique* fixed ray, so a non-Gaussian piece is impossible — if κ_{2n} were fixed it would be a *second* fixed ray (killing uniqueness), and if not fixed it sits in the gapped complement and decays to 0 (any gap > 0 suffices; the value $(2/3)^6$ only sets the rate). *The honest residual:* v56's gapped PF uniqueness is established on the *three-dimensional* scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (the flavour gap / horizon Page-recovery rates), whereas the 16-Majorana seam state already carries $\binom{16}{4}{=}1820$ and $\binom{16}{6}{=}8008$ higher-cumulant directions — the Schwinger cone the transport must contract has dimension $\gg 3$, and uniqueness must hold there (not merely on the quasi-free sub-cone) for the argument to be non-circular. **So (A) is equivalent to one strictly smaller, internal, exact-testable statement:** the admissible TFPT seam transport is a *primitive, spectrally gapped, RP-cone-preserving* map on the *full* Schwinger cone, not only on the scalar readout spectrum of v56 — which the suite does *not* yet establish [E]/[O].

And that full-cone premise is now *discharged* to a single finite one-particle statement ([`verification/v161_one_particle_reduction.py`]**).** The lever is Bogoliubov second quantisation: a quasi-free transport is $T_\Sigma = \Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dimensional Majorana space H_Σ, and the Fock space grades by correlator degree, so $\Gamma(t)$ acts on the degree-m sector as the exterior power $\Lambda^m(t)$. Three exact consequences follow (all machine-checked): (i) $\mathrm{spec}(\Gamma(t)|_{\deg m})$ is the set of products of m one-particle eigenvalues — the whole correlation spectrum is generated by the finite one-particle spectrum; (ii) **the cone gap *equals* the one-particle gap**: the sub-leading eigenvalue of $\Gamma(t)$ over *every* degree is $(2/3)^6$ (pad the largest contracted mode with fixed factors), so premise 5 on the cumulant space *is* premise 5 on the 16-dim space, the same number; (iii) the eigenvalue-1 sector is Λ^* of the fixed subspace — the disconnected Wick powers of K_Σ — so with a strict one-particle gap no independent connected cumulant is fixed, and an independent fixed cumulant *on* the fixed subspace is excluded by v113 (one polarisation ⇔ one vacuum). Premise 1 (quasi-freeness) is itself *forced*, not assumed: the $120 = \dim \mathfrak{so}(16)$ weight-1 chiral currents *are* the Bogoliubov generators on the 16 Majoranas, and v158 makes every non-Gaussian deformation irrelevant. **So the three full-cone premises collapse to one finite, 16-dimensional statement:** the seam RG transport's one-particle symbol is the gapped Calderón Bogoliubov map — fixed eigenspace = the rank-8 K_Σ polarisation (v113), sub-leading gap = $(2/3)^6$ (v56). Every *number* in it is already supplied by the suite; the one irreducible

step is the physical identification that the actual seam RG generator equals this Bogoliubov map — the same operator-algebraic (Kawahigashi–Longo-class) statement as the net-existence residual of (A_2), now with the infinite-dimensional cone eliminated [E]/[O].

And that finite statement carries no *new* open content: it factors into the two already-open items ([verification/v162_seam_transport_identification.py]). Take the one-particle symbol apart into its two data, the gap value (β) and the fixed space (α). (β) *is forced*: a μ_4-equivariant operator is diagonal in the cusp basis — the deck average is the diagonal projection, $\sum_{k=0}^{3} U^k X U^{-k} = |\mu_4|\,\mathrm{diag}(X)$ for $U = \mathrm{diag}(1, i, -i)$, and the commutant of U is exactly the diagonal algebra — so its spectrum is the cusp weights $\mathrm{spec}(A_0^\star) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (the parabolic residue, v115), and the transfer eigenvalue $(1-\alpha)^{p_2}$ with $p_2{=}6{=}|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6\log\frac{3}{2}$ — every entry carrier data, no free knob; the lift to the 16 Majoranas is the established A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of the $\omega_k{=}|k|$ free dispersion (v113/v156), the IR conformal free-fermion vacuum. **What is left** is only that the seam RG generator *is* μ_4-equivariant and admissible — which is the R1 "seam clock" identification (the transfer operator as the seam's gapped generator) — together with the A_2 net-existence theorem. Both are pre-existing open items, so premise (A) introduces *no independent open content*: it factors exactly into $A_2 + R_1$ with every number forced. The "free-bulk cloud" is gone; what remains is the same two identifications the programme already names — (A) is closed *conditionally* on them (a unification, honestly not an unconditional proof) [E]/[O].

And both residual identifications are pinned to their floor, so (A) has no independent open gate left ([verification/v165_premise_a_closure.py]). A_2 (net existence) is not open research but *assembly* of established mathematics: the 16 free Majoranas define a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic $c{=}8$ lattice net *is* $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125); the one TFPT-specific input — that the seam Calderón kernel is the free-fermion two-point function — is done because the DtN symbol is universally $|k|$ (v156/v157). So A_2 re-types [O] → [C] (closed by assembly). R_1 (the seam clock) reduces to `GATE.QGEO`: a μ_4-equivariant flat gapped generator on the carrier is *unique* ($= A_0^\star$, v115/v116), so its gap $(2/3)^6$ is forced once the seam generator lies in that class, and the residual is exactly one geometric identification — the seam collar boundary is the μ_4-punctured sphere $\mathbb{P}^1 \setminus \mu_4$ — whose cohomological half is already established (v137, $b_1{=}3{=}N_{\mathrm{fam}}$). That is `GATE.QGEO`, an already-open realisation premise, *not* a new gate. *[2026-08-03 `GATE.QGEO` note: quantitative progress on the continuum side via the quasi-free CAR route ([verification/v715_qgeo_car_continuum.py], 22/22, `QGEO-CAR-RATES-SUMMABLE`): all 6+1 deck/twist channels are Schatten-norm Cauchy with geometrically summable rates (N up to 3072/6144), the FH renormalization is fit-free lattice-derived (Barnes G, 8.8×10^{-7}), the RP cone is stable, and the majorant lemma (L1)–(L3) is* named *with constants and a classical proof route — named, not proven: the gate itself does* not *move.] [2026-08-04 `GATE.QGEO` note: the first proof block of the majorant lemma lands ([verification/v745_qgeo_car_l1_sector_lemma.py], 11/11, `L1-SECTOR-PROVEN-MODULO-ELEMENTARY`): (L1) is proven modulo elementary steps for all 6 undressed sectors on the fixed grid — the occupied-arc mode sums are* exact *geometric sums (exact summation replaces Euler–Maclaurin), the entire $1/N$ structure is the sublattice embedding offset, the leading term is the exact matrix $(iu/2MN)f_q'(s)$ (ratio $\to 1$ at 6×10^{-4}), $\gamma = 2$ is* derived, *the Taylor–Lagrange envelope holds with* 39744+108 *inequalities at* 0 *violations, and $r_1 = 1$ is sharp. The residues stay typed: (a) grid-sup [elementary], (b) L2 uniform FH/RH [hard-technical], (c) L3 uniformity [medium] — named, not claimed: the gate itself still does* not *move.] [2026-08-05 `GATE.QGEO` note: residue (a) is* fully closed *in the corrected* $(1+\ln M)$ *form ([verification/v778_qgeo_gridsup_logsum.py], 19/19 + 19/19, `GRIDSUP-SPLIT-CLOSED-LOG-LAW+FOURIER-LOGSUM-CLOSED`): the continuum entrywise supremum, the configuration-S_1 closure (explicit $B_{\mathrm{conf}}(\varepsilon)$ via packing + $\zeta(2)/\zeta(3)$) and the refinement remainder (exact* $\csc^3$ *integrals) close elementarily with zero census violations, while the leading part is honestly* log-divergent *under refinement — the literal `v715` clause "one $B(\varepsilon)$ for* all *grid refinements" is false at $r_1 = 1$ and is repaired to $\|D_N^{(M)}\|_1 \le [B_0(q,\varepsilon) + (J(\varepsilon)/2)\ln(M{+}3)]/N$, certified at theorem level by the grid Fourier log-sum inequality $\|F_q^{(M)}(\varepsilon)\|_1 \le c_0(q,\varepsilon) + (J(\varepsilon)/4)\ln(M{+}3)$, $J = 2|f_q'(\varepsilon)|$, with all constants exact (*162*-case census to $M = 6144$, zero violations; all* 252 *transfer cases verified; Araki/Powers–Størmer summability explicit). Honest: the measured slope is J/π^2 — the bound is valid, not sharp. Residues (b) and (c) stay open: the gate itself still does* not *move. 2026-08-06 architectural note ([verification/v802_gnet_martingale.py]): the martingale route bypasses (b)/(c) for the limit-*existence *step of the G_{net} continuum strand (summable state-norm defects suffice) and relocates them into the typed normality step — a reallocation, not a proof of (b)/(c).]* By the No-Unit Theorem (v153) the irreducibles stay exactly $\{\pi, v_{\mathrm{geo}}\}$ — closing (A) adds none. **Final accounting:** premise (A) =

(A_2 assembly, [C]) + (R_1=`GATE.QGEO`, already open) + the $\{\pi, v_{\rm geo}\}$ irreducible core (a theorem) — it carries *zero* independent open gates and ceases to be its own gate [E]/[O].

And the two structural residuals — net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the one irreducible open premise ([`verification/v175_net_existence_full_cone.py`]**).** The CAR second-quantisation functor $\Gamma(t)=\bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the *complete* Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16}$=65536); by the exterior-power spectrum theorem every eigenvalue of $\Gamma(t)$ is a subset product of $\mathrm{spec}(t)$, so checking all 2^{16} products shows $\Gamma(t)$ is a *contraction* (full-cone RP), with eigenvalue-1 multiplicity 2^8=256 (the wedges of the rank-8 K_Σ polarisation) and sub-leading eigenvalue exactly the one-particle gap $(2/3)^6$ — so full-cone RP/gap reduce *for every* m (not only $m \le 6$, v161) to the one-particle contraction (Shale–Stinespring), *discharging the v160 scope premise by a theorem.* And A_2 is now an *assembled and verified* certificate: the $(E_8)_1$ character E_4/η^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5 (level-1=248), the embedding 248=120+128, and the E_8 Cartan matrix even with $\det 1$ (the unique rank-8 even unimodular lattice, v83) — the net *exists* by cited theorems. **Both residuals then collapse to the *same* input**: that the actual seam Calderón kernel is this gapped one-particle contraction on $\mathbb{P}^1 \setminus \mu_4$, i.e. `QGEO.REALIZE.01`, whose finite half is proven (v168). That realisation half is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]— the single irreducible structural premise; net existence and full-cone RP are [E].

The No-Unit Theorem — $v_{\rm geo}$ is an irreducible metrology primitive [E]/[O] ([`verification/v153_no_unit_theorem.py`])

$v_{\rm geo}$ is not an open gap. Every load-bearing compiler datum — $g_{\rm car}, |\mu_4|, N_{\rm fam}, \mathrm{rank}\, E_8, \alpha^{-1}$, the determinant ladder $(3,4,8,20)$, $c_3 = \frac{1}{8\pi}$, $\varphi_0^{\rm ret}$ — is mass-dimension 0 and *invariant* under a unit rescaling $L \mapsto \lambda L$, whereas a mass / energy / $1/G$ carries nonzero dimension. A map from dimension-0 inputs to a dimension-$d\neq 0$ output is simultaneously invariant and covariant: impossible unless the unit is introduced. **So a dimensionless boundary compiler provably cannot select an absolute scale** — $v_{\rm geo}$ is a *forced input.* The three apparent residuals then collapse onto the one unit, differing only in mass-dimension:

$$U_{\rm point} \sim v_{\rm geo}, \qquad \tfrac{1}{G} \sim v_{\rm geo}^2, \qquad m/\mu = e^{3/4} \text{ (a pure ratio)},$$

so the irreducibles stay exactly {one dimensionful unit, π}. *Re-typing:* $v_{\rm geo}$ is an *irreducible metrology primitive* (a declared unit, like choosing a metre), not a missing derivation.

Sharpened across gravity, and the final tally ([`verification/v364_vgeo_sharpen.py`]**).** After the parameter-free gravity (v358/v359/v361) the *gravity* sector adds no new scale either: the Einstein coefficient $1/c_3 = 8\pi$ and the Λ prefactor $(8\pi)^2 \cdot 48 c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\rm geo}^2$ and $\Lambda \sim$ (dimensionless) $v_{\rm geo}^4$ both reduce to $v_{\rm geo}$ times atoms. Hence $v_{\rm geo}$ is the *single* dimensionful input of the *complete* theory (matter and gravity), and the final tally is $\{v_{\rm geo}, \pi\}$ with *zero* dimensionless dials — a $\sim$26 $\to$ 1 reduction against the Standard Model. ($1/G$ stays UV-sensitive (Sakharov, v68), so it is a metrology readout, not a clean prediction — consistent with $v_{\rm geo}$ being the anchor.)

The torsor closure of the interface ([`verification/v725_phys_vgeo_torsor_audit.py`]**, `VGEO.TORSOR.01`, 21/21).** The $v_{\rm geo}$ interface is now *structurally closed* as an $\mathbb{R}_+$ scale torsor, machine-audited on the corpus: (1) the complete export table $O_i = r_i\, v_{\rm geo}^{d_i}$ with machine-derived dimensionless r_i ($\bar{M}_{\rm Pl}$, $1/G$, Λ, $U_{\rm point}$, m/μ at $d = 1, 2, 4, 1, 0$); (2) the dimension matrix has *rank* 1 — exactly one independent scale (conditional [A] on the flavor block via the declared seam=Planck identification, named, not hidden); (3) *all* consistency conditions are homogeneous under $L \mapsto \lambda L$ — the machine form of this theorem; (4) the leave-one-anchor-out cross-calibrations are compatible (G vs ρ_Λ: the same $\bar{M}_{\rm Pl}$ to 0.11%, v274); (5) the *calibration theorem* in torsor language: one external reference selects the point in the torsor, everything else is prediction. The closure is the *calibration form* — no absolute-scale derivation (impossible per this theorem; that is the content, not the defect); the one remaining dimensionless scale ratio is *named*, $H_{\rm EW} = \ln(\bar{M}_{\rm Pl}/v_{\rm EW}) = 37.1776$ (naive $c_3/\varphi_0^{\rm ret}$ ladders honestly excluded); the [E]/[O] typing of this keybox is unchanged.

Theorem G — the closure statement [O] (target)

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure $\mu_{\rm QG}$. Its Ward identities imply

covariant conservation, and its low-curvature expansion is the TFPT $R + R^2$ seam action

$$f_R R_{\mu\nu} - \tfrac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu}\Box - \nabla_\mu \nabla_\nu) f_R = \bar{M}_{\rm Pl}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: the local covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C] (v358/v359); Theorem G remains open only for the projective-limit ambient* measure $\mu_{\rm QG}$ *(G6, the deepest item). G5 certified; the regime equation closed [verification/v28_gravity_fR.py].*

Research Contract — `CELEST.SEAM.01` (the celestial and twistor continuum route)

Reader's note. A compact synthesis of this contract's executed work packages (WP1–WP5d, both WP5d stages, + the WP5e α, β, γ, δ_2, ε_2 and ε_1 stages + the three back-reaction milestones M1–M3) — the narrative arc from the μ_4 clock to the $(E_8)_1$ boundary shadow — is given as a dedicated section in `tfpt_3_e8_audit_bootstrap` ("The celestial and twistor continuum route"); this contract remains the full technical reference (typing fence, kill tests, work-package statements). The bridge from this contract's compiler-side constructions to reconstructed Minkowski physics is *not* part of this contract: it is the separate central contract `WOIT.OS.TWISTOR.01` below — everything in `CELEST.SEAM.01` is preparation for it.

The object diagram (read this first). The route is one chain of objects; each arrow is a separately typed claim. Where mathematics ends and physics begins is visible at a glance: the last two arrows are [O].

Arrow	Status	Claim / modules	Kill test
$(\Sigma, \mu_4, \rho, \Theta) \to \mathbb{P}^1\backslash\mu_4$	[E]	`SEAM.MARKS` chain: four marks, clock, cross-ratio 2, $\tau{=}i$ pillowcase (v168/v214/v216/v180; bit reduction v506/v507/v510/v512)	a fifth mark, a non-order-4 clock, or a marks-preserving root without flag transitivity
$\mathbb{P}^1\backslash\mu_4 \to \mathbb{C}^2/\mathbb{Z}_4$	[E]	`CELEST.WP1.01`/`CELEST.WP2.01`: the glue is the flat $\mathbb{Z}_4$ monodromy on the A_3 ALE, clock = Kähler $U(2)$ phase, clock^2 = deck (v492/v493)	equivariant-sector closure failure (3112/6720 control) or a surviving shape modulus
$\mathbb{C}^2/\mathbb{Z}_4 \to \mathbb{PT}/\Gamma$	[E]/[C]	`CELEST.WP5E.*`: equivariant twistor uplift — anomaly ledger, level-from-flux, axion slot, back-reacted Ω_N, twisted KS measure, a_0 uplift, the δ_1 decision, the measure decision (v505/v509/v511/v513/v514/v515/v516/v517/v518/v520)	the preregistered level kill ("inflow demands $k \neq 1$"); the M1–M3 kills were evaluated by v515/v516/v517 and did not fire; the δ_1 kill *fired on the derived measure* (v518), and the declared/derived measure question is since *decided at probe level* in favour of the declared reading under the typed premises TP-1..TP-4 (v520) — the residual [O] is the constructive w_m derivation
$\mathbb{PT}/\Gamma \to \mathcal{A}_{\text{hol}}$	[O]	the *interacting* holomorphic algebra (global BCOV+SDYM quantisation; WP5e proper; target of `WOIT.OS.TWISTOR.01`; CFT-side anchors v497–v504)	anomaly mismatch at the Costello–Li grade
$\mathcal{A}_{\text{hol}} \xrightarrow{\text{OS}} \mathcal{A}_{\text{Mink}}$	[O]	OS reconstruction with the real structure Θ — the `WOIT.OS.TWISTOR.01` target (the free-collar RP/OS witnesses v379/`FORM.SEAM.MMST.01` are the *free* anchor only; the α stage v519 pins Θ itself and free RP on the seam system — kill test (1) discharged at the free level, live on $\mathcal{A}_{\text{hol}}$)	the seven kill tests of `WOIT.OS.TWISTOR.01` below

Definition (the group extension, exactly). The shorthand "$\mathbb{PT}/\mathbb{Z}_4$" used in this contract is precise but deserves unpacking, because *two different* order-4 structures are in play and only one of them is quotiented:

- $\Gamma_{\text{ALE}} = \langle\text{Deck}\rangle \cong \mathbb{Z}_4 \subset SU(2)$, the deck group $(z_1, z_2) \mapsto (iz_1, i^{-1}z_2)$: *triholomorphic*, preserves the holomorphic symplectic form Ω; this is the $\mathbb{Z}_4$ that is *quotiented* ($\mathbb{C}^2/\Gamma_{\text{ALE}}$ is the A_3 singularity $XY = Z^4$, and $\mathbb{PT}/\Gamma$ means the quotient by Γ_{ALE} acting on the twistor space of the ALE) [E] ([`verification/v492_celestial_z4_orbifold.py`]).
- $\rho = \text{Clock} = \text{diag}(i, 1) \in U(2)$: a Kähler, *not* triholomorphic isometry ($\det\rho = i$ rotates Ω by the μ_4 generator — the WP1 S5 computation); it is *not* in Γ_{ALE} and is *not* quotiented — it *normalises* Γ_{ALE} and survives the quotient as the residual clock symmetry [E] (v492).
- The global object is the extension generated by both: since $\rho^2 = \text{Deck}$ exactly (v492/v506), $\langle\Gamma_{\text{ALE}}, \rho\rangle = \langle\rho\rangle$ is *cyclic* — projectively $\mathbb{Z}_4$ (the clock on the sphere) with ρ^2 generating the deck $\mathbb{Z}_2$ there, and on the spin/fermionic level the canonical $\mathbb{Z}_8$ tower ($V^2 \propto U$, $V^4 \propto (-1)^F$, nonsplit; $8 = 2|\mu_4|$) [E] (v506/v507).

The action table, entry by entry ([E] = arithmetically established at the current stratum, [O] = a quantisation question):

Object	Γ_{ALE} (Deck)	ρ (Clock)	**Status / witness**
$K_{\mathbb{PT}}$ (canonical class)	trivial	weight via $\det\rho = i$	[E] (incidence-forced column weights $(+1,+1,-1,-1)$, v514)
Ω (hol. symplectic)	preserved	rotated by i	[E] (v492 S5; back-reacted Ω_N closed-form with $(2\pi i)^2$-integral periods and forced charge 4, v515)
BCOV fields ($O(-2)$ tower)	orbifold sectors	character series $P_0..P_3$	[E] ledger / [O] quantisation (v514)
open-string fields (SDYM E_8)	glue-equivariant sector	sector rotation $j \mapsto j+1$	[E] ($D[d] = 60(d+1)/64(d+1)$, v492) / [O] interacting
boundary states ($(E_8)_1$ shadow)	μ_4 sector split	clock phase on $\lvert s\rangle$ etc.	[E] character/GNS level (v497–v500) / [O] as an actual net

The A_3 role table (no silent identifications). Several rank-3 A_3's and external gauge factors circulate near this contract; the following table states, for each pair, either the connecting functor (with its witness) or an explicit *not identical*:

Object	**Role and relations**
A_3^{family}	the family factor in the lattice $D_5\oplus A_3$ ($\mathfrak{su}(4)$-flavour; three families from its exponents, v128/v137). Related to A_3^{ALE} by the McKay correspondence *of types* only (both are type A_3); **not identical** as realisations — one is a sublattice of E_8, the other the singularity type of $\mathbb{C}^2/\mathbb{Z}_4$. The bridge that *does* connect them is the Kronheimer–Nakajima quiver (v479: same ADE datum, different functors).
A_3^{ALE}	the ADE type of $\mathbb{C}^2/\mathbb{Z}_4$ ($XY = Z^4$); its resolution carries the three exceptional spheres (v492/v493).
three exceptional spheres	the Coxeter/Picard–Lefschetz structure of the resolved A_3^{ALE}: the clock acts as a Coxeter element of $W(A_3)$ on H_2 (eigenvalues $\{i,-1,-i\}$, v493); they carry the three twisted axion slots (v505) and the lockstep fluxes (v509).
D_5^{carrier}	the carrier factor ($g_{\mathrm{car}} = 5$); glued to A_3^{family} by the μ_4 Lagrangian glue into E_8 (v92/v125); **no** role on the ALE side beyond the zero-mode algebra $\mathfrak{g}_0 = D_5\oplus A_3$ (v492).
Woit's $SU(3)_{\mathrm{color}}$	*external programme reference only* (Euclidean Twistor Unification: colour as the rank-three quotient bundle on $\mathbb{PT}$). **Not identical — no identification claimed**; in particular $A_3^{\mathrm{family}} \not\cong SU(3)_{\mathrm{color}}$ (an $\mathfrak{su}(4)$ flavour factor is not a colour gauge bundle), and TFPT's colour lives in the SM readout of the carrier, not on $\mathbb{PT}$.
Woit's $SU(2)_{\mathrm{weak}}$	*external programme reference only* (the internal spin factor of the Euclidean twistor construction). **Not identical — no identification claimed**; whether an internal $SU(2)$ of that kind can be realised on $\mathbb{PT}/\Gamma$ is exactly kill test (6) of `WOIT.OS.TWISTOR.01`.

Hypothesis (as corrected by WP1). *The seam is the glue-equivariant $\mathbb{Z}_4$-orbifold sector of the celestial chiral algebra on the A_3 ALE space $\mathbb{C}^2/\mathbb{Z}_4$; the μ_4 clock is the Kähler $U(2)$ phase* $\mathrm{diag}(i,1)$ *whose square is the deck group (spin bridge $\mathbb{Z}_8$, $8 = 2|\mu_4|$ — the $c_3 = 1/(8\pi)$ seam winding integer); and the associated twistorial bulk is the* self-dual *sector of TFPT gravity.* This is the fourth research contract, alongside (U_{wall}), (G_{metric}) and `CONTRACT.F.01`: a numbered chain of work packages with pre-registered kill tests, *not* a claim. The method is the celestial-chiral-algebra (CCA) construction of Bittleston–Homans–Sharma on Eguchi–Hanson space (`arXiv:2305.09451`, JHEP 09 (2023) 008 — the $\mathbb{Z}_2$ case, refined here to $\mathbb{Z}_4$ with a flat connection at infinity in the Kronheimer–Nakajima / heterotic-on-ALE sense), inside the celestial-holography programme (Strominger, PRL **127**, 221601: $w_{1+\infty}$; Costello–Paquette–Sharma, PRL **130**, 061602: Burns holography, the $SO(8)$ WZW_4 model; Bittleston–Sharma–Skinner, CMP **403** (2023): quantizing the non-linear graviton). The gauge-algebra admissibility gate is already passed in the literature: Costello's one-loop axionic-consistency analysis (`arXiv:2111.08879`) admits the exceptional algebras *including* E_8 for SDYM on twistor space (Okubo: no independent quartic Casimir). The spin bridge in the hypothesis is no longer a bookkeeping convention: on the 16-Majorana seam the NS deck implementation is *forced* to order 4 ($U^2 = (-1)^F$ exactly, nonsplit $\mathbb{Z}_4$) and the quarter-shift lifts to a canonical $\mathbb{Z}_8$ tower with $V^2 \propto U$ — "$8 = 2|\mu_4|$" fermionically explained ([`verification/v506_seam_clock_rigidity.py`],

SEAM.CLOCK.RIGIDITY.01); and the nonsplit class is arrangement-*sensitive*: the edge (silver) arrangement splits ($U_e^2 = +1$, zero roots), so the fermions *measure* the alignment bit as a Fidkowski–Kitaev-type extension class ([verification/v507_seam_bit_origin.py], SEAM.BIT.ORIGIN.01) — and that class is nonsplit *iff* the deck acts freely (all 17 seam-circle involutions, v510), which the covering deck does by topology.

WP1, executed and verified: the E_8 μ_4-glue *is* the flat $\mathbb{Z}_4$ monodromy of the equivariant CCA on the A_3 ALE space — verdict B ([verification/v492_celestial_z4_orbifold.py], CELEST.WP1.01)

Everything in this box is exact arithmetic (sympy, no floats). **Inner grading / flat monodromy [E]:** the v128 glue grading ($240 = 52{+}64{+}60{+}64$, additive on all 6720 root pairs, dims with Cartan $(60, 64, 60, 64)$, $\mathfrak{g}_0 = D_5{\oplus}A_3$ carrier $= 60$) is *inner*: $h = (2, 2, 2, 2, 2; 0^4)$ reads the glue class mod 4 on all 240 roots, so the glue is $\mathrm{Ad}(\exp(2\pi i h/4))$ — an order-4 torus element, i.e. a *flat* $\mathbb{Z}_4$ *monodromy* in the Kronheimer–Nakajima sense; the A_3-side detector $h' = (0^5; 1, 1, 1, -3)$ reads the *same* class, so the glue charge is the diagonal $(1, 1)$ of the discriminant group $\mathbb{Z}_4{\times}\mathbb{Z}_4$ — the v92/v125 Lagrangian glue. **The spatial side [E]:** $\mathbb{C}^2/\mathbb{Z}_4$ under the deck $(z_1, z_2) \mapsto (iz_1, i^{-1}z_2)$ *is* the A_3 singularity $XY = Z^4$ (invariant ring exact; Hilbert series $(1{+}t^4)/((1{-}t^2)(1{-}t^4))$; Molien table verified). **The equivariant sector closes [E]:** the glue-equivariant SDYM(E_8) modes ($j + m \equiv 0 \bmod 4$) form a Lie subalgebra with graded dimensions $D[d] = 60(d{+}1)$ (even d) $/\, 64(d{+}1)$ (odd d) — uniform per parity *only* because $\dim \mathfrak{g}_0 = \dim \mathfrak{g}_2 = 60$ and $\dim \mathfrak{g}_1 = \dim \mathfrak{g}_3 = 64$: the v128 dimensions are exactly the consistency condition. The zero modes ($d{=}0$) are $\mathfrak{g}_0$ alone — *the unbroken algebra on the orbifold is the carrier* $D_5{\oplus}A_3{+}\mathrm{Cartan} = 60$; the asymptotic density is $62/248 = 1/4 = 1/|\mathbb{Z}_4|$ (the orbifold index), and the $\mathbb{Z}_2$ halfway sector reproduces the Eguchi–Hanson towers $120/128 = SO(16)$ adjoint/spinor — μ_4 refines the sheet $\mathbb{Z}_2$ exactly as $4 = 2 \times 2$ (v125). **Character level [E]:** Θ_{E_8} splits into the four glue-coset thetas (shell $1 = (52, 64, 60, 64)$ re-derived from the *lattice*); the four sector characters Θ_{C_j}/η^8 sum *exactly* to $\chi_{(E_8)_1} = 1{+}248q{+}4124q^2{+}34752q^3$ (the v377 series); the sector conformal weights are $h_{D_5} = (0, \frac{5}{8}, \frac{1}{2}, \frac{5}{8})$, $h_{A_3} = (0, \frac{3}{8}, \frac{1}{2}, \frac{3}{8})$ — the v92/v125 discriminant form $(5x^2{+}3y^2)/8$ mod 1 — and the glue diagonal has *integer* $h = (0, 1, 1, 1)$: the v125 isotropy *is* the locality of the extension to $(E_8)_1$. **The critical correction [E] (why verdict B, not A):** the A_3 deck $\mathrm{diag}(i, i^{-1})$ induces on the celestial sphere $z = z_1/z_2$ only $z \mapsto -z$ (projective order 2, the sheet flip) — it is *not* the order-4 clock $z \mapsto iz$. The exact bridge: any $SU(2)$ lift of the clock has order 8 (spin group $\mathbb{Z}_8$, which would quotient to A_7), and (spin clock)^2 = the deck generator *exactly*, with $|\mathbb{Z}_8| = 8 = 2|\mu_4|$ — the $c_3 = 1/(8\pi)$ winding integer. The order-4 clock *is* realised, as the $U(2)$ phase $\mathrm{diag}(i, 1)$: projective order 4, fixed points $\{0, \infty\}$, free μ_4 orbit (all v180 clock properties), normalising the deck (so preserving the glue flat connection), with $\det = i$ rotating the holomorphic symplectic form by the μ_4 generator (a Kähler, *not* triholomorphic isometry — it rotates the twistor sphere by $\pi/2$). **Gravity side [E]:** clock-invariance selects from the 3-parameter versal A_3 family $XY = Z^4{+}a_2Z^2{+}a_1Z{+}a_0$ *exactly* the 1-parameter family $XY = Z^4{+}a_0$, whose four branch points are *one* μ_4 orbit with cross-ratio 2 — the seam marks (v179 clock orbit; v168/v214 cross-ratio $2 \Rightarrow j{=}1728$ pillowcase). **Negative controls [E]:** the false spatial action $\mathrm{diag}(i, i)$ is killed three ways ($\det = -1 \notin SL(2, \mathbb{C})$, no CY/ALE resolution; Poisson closure fails in 14/14 nonzero cases; conjugation symmetry broken), the false glue (60/64 blocks swapped) breaks additivity on 3112/6720 root pairs, and of all 24 relabelings exactly $\{\mathrm{id}, j \mapsto -j\} = \mathrm{Aut}(\mathbb{Z}_4)$ are additive — the grading is rigid. **Verdict [C]:** the celestial and glue $\mathbb{Z}_4$ structures match under *one* precisely named extra identification — clock^2 = deck on the sphere (the sheet-$\mathbb{Z}_2$ spin bookkeeping) — so WP1 lands at verdict B (match modulo one named identification), close to but honestly short of A. [E]/[C]

Typing and non-circularity. Admissible *inputs* are the finite seam/glue data already in the suite — the μ_4 Q-system glue (v92/v125), the graded hull (v128), the seam rigidity and pillowcase geometry (v168/v214), the Möbius clock (v180) — plus the WP1 [E] arithmetic ([verification/v492_celestial_z4_orbifold.py]). *Not* admissible as an input: the *continuum existence* of the holomorphic $(E_8)_1$ net on the seam — that is precisely the SEAM.EQUIV.01 target, and assuming it would make the contract circular. *Scope fence:* the twistorial bulk of this contract is the *self-dual* sector only, never full Einstein gravity — the classical field-equation route (entanglement equilibrium, v358/v359) and the entropic-action bridge (GRAV.ENTROPIC.ACTION.01, v473) are separate, non-competing statements about the full equation. SEAM.EQUIV.01 stays [O] throughout; nothing in this contract moves it.

Work packages.

- **WP1 (*done*, verdict B) [E]/[C]:** the structural $\mathbb{Z}_4$ arithmetic of the keybox above, executed and machine-verified as `CELEST.WP1.01` [`verification/v492_celestial_z4_orbifold.py`].
- **WP2 (*done*, verdict B) [E]/[C]:** the clock-invariant deformation theory, executed and machine-verified as `CELEST.WP2.01` [`verification/v493_celestial_wp2_clock_deformation.py`] (sympy exact, 47 checks). Both targets landed, *sharpened*: (i) a_0 is the pure *scale* of the seam configuration — the binary-quartic invariants of $w^2 = Z^4{+}a_0$ are $I = 12a_0$, $J = 0$ *identically*, so $j = 1728$ and the $\tau{=}i$ pillowcase shape (v168/v214) is *frozen* for every $a_0 \neq 0$ (no shape modulus survives clock-invariance; the negative controls $a_1 Z \Rightarrow j{=}0$ and $a_2 \neq 0 \Rightarrow j{=}1556068/81$ show the test has teeth); (ii) the three resolved spheres ($3 = \operatorname{rank} A_3 = N_{\text{fam}}$) carry *exactly* the three nontrivial μ_4 characters $\{i, -1, -i\}$ under the clock, which acts on H_2 as a *Coxeter element* of $W(A_3)$ (char $x^3{+}x^2{+}x{+}1$, order $h(A_3){=}4{=}|\mu_4|{=}N_{\text{fam}}{+}1$) and *is* the Picard–Lefschetz monodromy of the family ($a_0 \to e^{2\pi i}a_0$ gives the 4-cycle); the surviving deformation direction is the weight-1 χ_1-Fourier diagonal $(1, i, i^2)$ — *not* the naive invariant diagonal ($\det(A{-}1) = -4$: no invariant cycle) — and all three sphere volumes move in lockstep ($|\Pi_j| = \sqrt{2}\,t$). The BHS deformed-algebra pattern transfers from $\mathbb{Z}_2$ to $\mathbb{Z}_4$: the fibre bracket $-4\,\text{Nambu}(XY{-}Z^4{-}a_0)$, anchored at the $a_0{=}0$ orbifold, closes with corrections exactly *linear* in a_0 at the $\mathbb{Z}_4$ wrap, conserving the μ_4 grade (no sector leak — the WP1 equivariant sector deforms consistently), with a_0 in the weight-0 slot of BHS's c^2. Verdict B via three named [C] identifications ($-4\,$Nambu as *the* $k{=}4$ CCA bracket; period = root difference; the seam-scale reading via clock^2 =deck); the full deformed twisted-sector OPEs and the $W(\mu)$-type scaling limit for $k{=}4$ stay [O]. The v216 [O]-residual is *typed*, not moved: given the order-4 clock, the deformation freezes the $(2, 2, 2, 2)$ modulus to the square automatically — a relocation of the same order-4 carrier input, not a new derivation (since v510 that input reads: the square-modulus datum $\tau = i$; its position half is topology; since v512 it reads equivalently: one discrete symmetry-lift bit, flag transitivity $\iff \tau = i$; and since v521/v525/v529 free OS positivity, the whole mark-decorated free-plus-twist state class and the first genuinely interacting FK toy are documented as the *eighth*, *ninth* and *tenth* side-blind tests — the bit is not derivable from free RP/Θ existence, from any Wick-computable twist decoration or from the FK interaction class, and remains genuine discrete input; since v528 it is the twist-class choice, in principle interferometrically readable; and since v534 it carries the first *positive* selection datum — reflection positivity on the interacting mark family keeps exactly the $\delta = \pi/2$ member alive, toy-fenced).
- **WP3 (*done*, verdict B) [E]/[C]:** the Green–Schwarz alignment test, executed and machine-verified as `CELEST.WP3.01` [`verification/v495_celestial_wp3_gs_alignment.py`] (exact Fraction/sympy, 25 checks). The arithmetic side is sharp: the Okubo coefficient $\text{Tr}_{\text{adj}}X^4/(\text{Tr}_{\text{adj}}X^2)^2 = 5/(2(\dim\mathfrak{g}{+}2))$ is *derived* as a polynomial identity for all 8 algebras on Costello's list (with a sharp $\mathfrak{sl}_4$ negative control: 5/32 vs 3/32), the closed form $\tilde{\lambda}^2 = 10h^{\vee 2}/(\dim{+}2) = h^\vee{+}6$ holds exactly across the Deligne series, and for E_8 the axion coefficient is *exactly* $\tilde{\lambda} = 6$ (unit-trace) resp. $\lambda_{\text{fund}} = 1/10$ (adjoint-trace), so the GS coupling $\kappa = \lambda_{\mathfrak{g}}/(8\pi\sqrt{3})$ satisfies $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\text{fam}}$ resp. $1/300$ — exact anchor rationals, with the anchor decomposition $h^\vee{=}30{=}|\mathbb{Z}_2|N_{\text{fam}}g_{\text{car}}$, $\varphi(30){=}8{=}\text{rank}$, $\tilde{\lambda}{=}|\mathbb{Z}_2|N_{\text{fam}}$, $\lambda_{\text{fund}}{=}1/(|\mathbb{Z}_2|g_{\text{car}})$ (a byproduct: the printed $\lambda^2_{\mathfrak{so}_8} = 3/2$ in Costello's appendix A is a factor-2 slip; his own weight content gives 3, Okubo-consistent). κ/c_3 itself is *irrational* ($2\sqrt{3}$): only the squared alignment is exact. **The look-elsewhere caveat is part of the result [C]**: the same squared-rational alignment holds for *all eight* algebras on the list (8/8 — the test as such has zero selective power), $\tilde{\lambda}$-integrality passes 2/8 (shared with $\mathfrak{sl}_3$), and the only $\mathfrak{e}_8$-selective single test is $g_{\text{car}}{=}5 \mid h^\vee$ (1/8); the conjunction that isolates $\mathfrak{e}_8$ is post hoc, not preregistered. *Alignment survives; selectivity does not* — the c_3-connection of the celestial route is convention-level compatibility plus genuine λ-arithmetic, *not* $\mathfrak{e}_8$-selective evidence,

and never a derivation of c_3 from celestial data.

- **WP4 (*done*, verdict B(ii)) [E]/[C]:** the type mismatch, executed head-on and machine-verified as `CELEST.WP4.01` [`verification/v496_celestial_wp4_character_shadow.py`] (exact integer/Fraction/sympy, 25 checks). The $(E_8)_1$ character is *not* a conformal block of the celestial $E_8[\mathbb{C}^2]$ S-algebra in its own (jet) grading — the obstruction is localised *three* independent ways: (a) the CP grading gives spin $1-d/2$, unbounded below, and 248 is never a jet-tower dimension $(60(d+1)/64(d+1),\ d \le 100)$; (b) the cumulative generator count is quadratic $(31s^2+92s+60,\ 31 = k+h^\vee)$, so the jet Fock grows as $n^{2/3}$ against the character's $n^{1/2}$ (Meinardus poles $s=2$ vs $s=1$; the ratio f_n/χ_n increases strictly for $n = 1..12$); (c) the level-2 null ideal of $(E_8)_1$ deletes *exactly* $27000 = 30^3 = (h^\vee)^3$ out of $\mathrm{Sym}^2(248) = 1+3875+27000$ (the character keeps $4124 = 1+248+3875$) — and has no jet analogue. **But the boundary/period reading holds exactly at the current stratum**: the zero-mode slice is the 60 vacuum-sector currents, the single-line jet slice cycles the four sector counts $(60, 64, 60, 64)$, one full μ_4 period of loop energies sums exactly to 248 (the single-particle level of the $248q$ stage), and the glue-diagonal weights $(0, 1, 1, 1)$ are integers — while the free loop Fock at level 1 counts $897266 \gg 248$, i.e. the rational truncation must be *imposed in a limit*. That is precisely the MMST scaling-limit shape of `SEAM.EQUIV.01` (v336/v449: the rational net exists in the *limit* of the lattice collar, not in the pre-limit algebra): WP4 is hereby *retyped* from "character as conformal block" to "character as boundary/limit shadow", and the constructive limit question (which limit, which topology, convergence) passes to WP5 — where WP5a has since answered the "which limit" part at the counting level (the coefficientwise χ_w limit below), WP5b has constructed the deleting operator, WP5c the limit state whose GNS kernel contains the ideal, WP5d the local-net legs (the α-stage two-interval index chain and the β-stage split + strong-additivity witnesses — all three KLM ingredients of complete rationality on the lattice), and WP5e-$\alpha/\beta/\gamma/\delta_2/\varepsilon_2/\varepsilon_1$ have anchored *both* faces of the twistor uplift, closed its exchange sub-branch, executed the full-tensor ledger, executed the level-from-flux dial and built the bulk-axion slot (the CFT-side prefactor + level pinning; the equivariant twistor skeleton with the index bridge and the rigid $32\,T_3$ residual; the exchange no-go with certificate; the full-tensor confirmation of the collapse with the one cubic d-symbol channel that opens; the lockstep theorem making "one level" a theorem of clock invariance, with the sector-counter dial pinning $k = 1$; the equivariant $O(-2)$ slot construction with $\tilde{\lambda} = 6$ triply pinned and the first GH/A_3 back-reaction step), and all three quantised back-reaction milestones M1–M3 have since been executed (the closed-form Ω_N with $(2\pi i)^2$-integral lockstep periods and forced source charge $4 = |\mu_4|$, v515; the twisted KS measure on the declared completion reading — per-sector Okubo squares, the $32\,T_3$ cancelled, the $\psi = 64$ slice supplied without the cubic d-channel, v516; the a_0 uplift coupled to the centre count on four scales, v517), leaving the global BCOV quantisation (WP5e proper) as the remaining open piece — the BCOV *measure question* is since decided at probe level: the δ_1 chain itself was decided by v518 (kill under the derived measure, all three testers fail), and v520 has since decided *which* of the two exact measures the BCOV integral produces — single-valuedness of the one-loop lattice factor is *derived* from two independent constructive sources (F-independence of the moduli integral + the Quillen pairing) under the typed premises TP-1..TP-4, forcing the declared v516 completion reading, and v523 has since *derived* the w_m normalisation itself at probe level ($1/\det_j$ = the Atiyah–Bott/zeta-determinant fixed-point factor, computed from three independent sources under the typed premises TP-REG/TP-Q/TP-NUM/TP-CH); the residual [O] narrows to the global BCOV integral beyond the fibre zero-mode factor (throughout, $\mathbb{PT}/\mathbb{Z}_4$ is the quotient by Γ_{ALE}, normalised by the clock $\langle\rho\rangle$ — the definition block above; the ε_1 slot itself is executed; the quantised milestones M1–M3 preregistered in v514 are all executed via v515/v516/v517 — the v514 fence is fully worked off).
- **WP5 (the summit) — subdivided; WP5a–WP5d (both WP5d stages) plus the WP5e α, β, γ, δ_1, δ_2, ε_2 and ε_1 stages and the back-reaction milestones M1–M3**

are landed. The twistorial construction as a *second continuum route* for `SEAM.EQUIV.01` (Costello–Li grade), carrying the WP4 constructive-boundary-limit question. Subdivision WP5a–WP5e:

- **WP5a (*done*) [E]/[C]:** "the null ideal from the limit", executed and machine-verified as `CELEST.WP5A.01` [`verification/v497_celestial_wp5a_null_ideal.py`] (exact integer/Fraction arithmetic, 34 checks). Three results make the WP4 limit reading precise. (i) *Stabilisation theorem*: the one-parameter family χ_w of graded Fock characters on the chiral jet generators (loop energy m, radial number r; $E_w = m + wr$ in quarter units $u = q^{1/4}$) *contains* the chiral jet grading as its w=2 member (generator counts $64, 120, 128, 180, 192, 240, 256, 300$ per degree), and its u^n coefficient equals the quarter-moded loop Fock for *all* $w \geq n{+}1$ — strictly larger for every $w \leq n$ (verified $n \leq 8$, $w \leq 10$): the WP4 "boundary-limit shadow" is a precise *coefficientwise* limit $\varepsilon = 1/w \to 0$ with explicit stabilisation threshold $w = n{+}1$, not a slice. (ii) *The null ideal is derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give $\mathrm{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$ with peeling residual exactly zero (independent Weyl-orbit cross-check); $27000 = 30^3 = (h^\vee)^3$ exactly; level-1 integrability ($\langle 2\theta, \theta^\vee \rangle = 4$, $\langle \omega, \theta^\vee \rangle = 2$) leaves the vacuum as the only level-1 primary ([C] Kac, used only through its counting consequence); level-2 quotient $31124 - 27000 = 4124 = 1 + 248 + 3875$ exactly. (iii) *Two routes, one number*: the μ_4 theta-split sector sum at $q^2 = (1036, 1024, 1040, 1024)$ gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space (q^3: 34752). Negative controls: $SO(16)_1$ through the same pipeline gives *four* components ($5304{+}1820{+}135{+}1$), $5304 \neq 14^3$ (the $h^{\vee\,3}$ identity is not generic), quotient $2076 = \Theta_{D_8}/\eta^8$ at q^2 (the recipe validated on a second algebra), and block weights $(0, \frac{1}{2}, 1, 1)$ that can*not* fuse into one local character; only $P = 4 = |\mu_4|$ periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor. *Honest limit* [O]: the limit does *not* generate the truncation (the loop Fock counts $897266 \gg 248$ at integer level 1) — WP5a fixes the ideal's *size and location* quantitatively and gives the celestial route the same two-step shape as MMST (limit + maximal ideal); the operator content is exactly WP5b–e, of which WP5b–WP5d and the WP5e α, β, γ, δ_2 and ε_2 stages are executed below.
- **WP5b (*done*) [E]/[C]:** "the singular vector as an operator", executed and machine-verified as `CELEST.WP5B.01` [`verification/v498_celestial_wp5b_singular_vector.py`] (exact integer/Fraction arithmetic, 53 checks, deterministic) — *success on the preregistered criterion*: the deleting object exists and is explicit. $|s\rangle = (E^\theta_{-1})^2|0\rangle$ (weight 2θ, level $2 = 8$ quarter units $= q^2$, an *integer* level) is constructed in a machine-built Chevalley/Frenkel–Kac basis (cocycle asymmetry on all 57600 pairs; the $[e_\alpha, e_{-\alpha}]$ sign *forced* by Jacobi, SGN $= -1$; κ derived, $\kappa(\theta^\vee, \theta^\vee) = 2$), and $J^a_1|s\rangle = 0$ is machine-verified for *all* 248 generators with the case classification 190 (first bracket) / 57 (second) / 1 ($a = F^\theta$, the central-term cancellation — the *only* case that sees the level k); plus $J^a_2|s\rangle = 0$, $E^a_0|s\rangle = 0$, exact weight and Shapovalov norm 0. *Level dial*: the F^θ_1 coefficient is $2(1{-}k)$ — nonzero at k=0 and k=2, zero exactly at k=1: without the central extension the deletion operator does not exist. μ_4 compatibility: glue class $j(\theta) = 1$ (machine-built h-adapted chamber, $\langle \theta, h \rangle = 5$, height $29 = h^\vee{-}1$), clock phase $i^{2j} = -1$, class$(2\theta) = 2$ sheet-even (WP5a-consistent) with the deleting θ-$\mathfrak{sl}_2$ crossing the sheet-odd classes $(1, 3)$; 8 quarters $= q^2$ via the per-period dictionary. The module it generates is *the* ideal: weight 2θ has multiplicity 1 in the level-2 Fock, and the direct $\mathfrak{g}_0$-lowering orbit BFS from $|s\rangle$ reproduces the Freudenthal multiplicities of $V(2\theta)$ exactly through depth 4 ($27000 = (h^\vee)^3$, quotient 4124; [C] Weyl complete reducibility beyond depth 4). Negative controls separate honestly: the level-1 current state and generic level-2 states are *not* singular; at k=2 the *cube* is (the $(E^\theta)^{k+1}$ pattern is generic Kac level mechanics); $SO(16)_1$ has the same singular vector but keeps three extra level-1 primaries ($h = \frac{1}{2}$

breaks one-block fusion) — the singular-vector mechanism is level-1 *generic*, the *one-block closure* is the E_8/μ_4-specific part. *Handover to WP5c* (answered below): in the twisted quarter-*slot* moding of the χ_w limit Fock, two sector-C_1 modes never sum to 8 quarters (minimum $6 = q^{3/2}$) — the per-*period* dictionary (v496), not the per-slot identification, carries $|s\rangle$ to q^2; exactly the GNS/limit-state question (kernel $\supseteq$ ideal) that WP5c answers.

– **WP5c (*done*) [E]/[C]:** "the GNS limit state", executed and machine-verified as `CELEST.WP5C.01` [`verification/v500_celestial_wp5c_gns_limit_state.py`] (exact integer/Fraction arithmetic, 35 checks, deterministic) — *success on the preregistered criterion*: the family ω_w exists and its limit has the null ideal in its GNS kernel. ω_w is the quasi-free state with loop sector = the affine k=1 vacuum n-point functions (via the machine-determined compact anti-involution $\theta(e_\alpha) = -e_{-\alpha}$, the unique anti-automorphism sign on all 61504 basis pairs) and radial sector = oscillator pairings x^{wr} (the exact Gibbs regulator, x-adic); *the WP5a threshold holds at the state level*: $\omega_w = \omega_\infty \bmod x^{N+1}$ for all $w \geq N{+}1$, sharp at $w = N$. The core is fully machine: the *complete* exact level-2 Gram — 31124 monomials in 9361 weight blocks — has rank 4124 exactly (the preregistered target), kernel $27000 = V(2\theta)$ *weight by weight* (Freudenthal cross-check on all 9361 weights), every block PSD, rank table per Weyl orbit $(0, 0, 1, 8, 44)$, level-1 rank 248 positive definite (the current layer survives). μ_4: the clock descends to GNS with level-2 rank split $(1036, 1024, 1040, 1024) = \Theta_{C_j}/\eta^8$ at q^2 — the WP5a two-routes identity at the *state* level — and $|s\rangle$ *is* the zero vector of GNS(ω_∞), resolving the WP5b twisted-slot tension. The KILL branch is probed and not triggered: a CCR obstruction shows *no* w-uniform state can damp the radial modes (the family formulation is *necessary*), and the wrong constructions are caught (family $E'_w = mw{+}r$ erases the 248 layer, $710955 \neq 248$; no damping keeps $897266 \neq 248$); k-dial and D_8 controls separate honestly (k=2: $\langle s|s\rangle = +4$, no ideal in the kernel; D_8: one block of four, $h = \frac{1}{2}$). [C] quasi-free extension to the full $*$-algebra; Kac positivity beyond level 2.

– **WP5d — both stages (*done*):** α [E]/[C], β [E]/[C] **(continuum uplift [O]):** "the local net", approached from the lattice. The α stage — the KLM *two-interval index* as the operational Haag-duality discriminator ($(E_8)_1 \Rightarrow \mu$=1, one sector; $SO(16)_1 \Rightarrow \mu$=4) — is executed and machine-verified as `CELEST.WP5D.01` [`verification/v501_celestial_wp5d_two_interval_index.py`] (39 checks; Gaussian lattice machinery ED-validated to 10^{-15}, exact Fractions for the arithmetic). Measured on the 16-layer seam carrier (critical Majorana ring, $c_- = 8$): the fermionic two-interval mutual information is *extensive* (the μ=1 reference; c fit 0.5000, residual $\to 0$ with N), while the sector-summed (orbifold) prescription pays exactly one classical bit — the $\ln 2$ plateau at machine precision ($|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$ at N=512), so $[F : F_{\text{even}}] = 2$ and $\mu_{\text{gauged}} = 4$, the entropic twin of the v490 parity census (two independent lattice witnesses); the orbifold also breaks two-interval complementarity ($S(E) \neq S(E')$, the direct duality-failure witness), with the complementary-pair budget bounded by $\ln 4 = \ln \mu(SO(16)_1)$ as a *double-limit* statement. The condensation arithmetic is anchored at both measured ends: $\det \text{Cartan}(D_5) \cdot \det \text{Cartan}(A_3) = 16$, KLM/Longo–Rehren $16/4^2 = 4/2^2 = 1$, $\Sigma d_i^2 = (4, 4, 1)$, $\theta_v = 1$ exactly at $\nu = 2c_- = 16$ (rivals $\neq 1$) — so $\mu = 1$ after condensation and the preregistered KILL ("μ-offset $\neq 0$ after condensation") is *not* triggered; the ν=1 control shows a non-removable offset ($\theta_v \neq 1$: the discriminator has teeth), the trivial phase shows none, and the wrong sector sum loses the full $\ln 2$. [C] the continuum dictionary "offset = ln index" (KLM; Longo–Xu; Casini–Huerta–Magan–Pontello). **The β stage (*done*) [E]/[C] — split + strong additivity, the two remaining KLM legs** — is executed and machine-verified as `CELEST.WP5DB.01` [`verification/v504_celestial_wp5d_klm_completeness.py`] (37 checks; Gaussian lattice machinery ED-validated, exact GF2/integer algebra for the exact legs;

the $U(1)$/Dirac number-conserving subnet as the preregistered negative control — the current-net analogue where strong additivity famously *fails* in the continuum). *Strong additivity is algebraically exact on the lattice*: with the shared boundary Majorana ("touching = share the point"), $\mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$ *exactly* (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32), while disjoint intervals give exactly *half* (rank 16/32, index 2 — the missing sector is odd $\otimes$ odd, the single $\ln 2$ bit of the α stage, localised at the split point); the $U(1)$/neutral algebras do *not* generate the union even with the shared site (gaps 2/10/52, growing). The entropic witness uses the honest discriminator *bounded vs divergent*: the $\mathbb{Z}_2$/orbifold touching deficit stays $< \ln 2$ with the Ising approach $(\ln 2 - \Delta_2) \sim N^{-p}$, $p = 0.2444$ vs $2\Delta_\mu = \frac{1}{4}$ (the naive "defect $\to 0$" expectation is *false* at the sharp lattice split — bounded $\Leftrightarrow$ finite index, Longo–Xu), while the $U(1)$ control *bursts* $\ln 2$ from $L = 128$ on and grows as $\frac{1}{2} \ln \mathrm{Var}\, Q_A$ (Klich–Levitov slope 0.10134 vs $1/\pi^2 = 0.10132$: infinite index). The split property is witnessed quantitatively: the coupling-block singular values fall exponentially at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0% over $d = 16..512$, $\sigma_1 \sim x^{0.5044}$ (vs $\frac{1}{2}$), trace norm summable — and the orbifold *inherits* the same ladder exactly at machine level (P_A flips $C \to -C$, σ_k identical to 8.3×10^{-17}; the even-bilinear coupling Gram is the second compound $\Lambda^2 C$ with singular values $\{\sigma_a \sigma_b\}$ — Longo heredity, [C]). The Pimsner–Popa bound is a one-line *identity*, $E(a) - a/2 = PaP/2$, so $\lambda = \frac{1}{2} = 1/[F : F_{\mathrm{even}}]$ exactly, attained in exact integers (16384 + 2048 monomial sweeps, 0 violations; random pencil $\min \lambda = 0.5000000000$); the two-interval group gives $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ exactly, and the index is pinned by *two independent routes* ($e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\mathrm{PP}}$; $e^{2\Delta_\infty} = 4 = 1/\lambda_{E_4} = \mu$); the $U(1)$ control has $\lambda = 1/(m{+}1) \to 0$. *With the α stage, all three KLM ingredients of complete rationality — split, strong additivity, finite μ — are now witnessed on the lattice.* The continuum uplift stays [O] and is honestly fenced: the theorem "finite-group orbifolds of completely rational nets are completely rational (hence split + strongly additive)" is *Xu's theorem* (algebraic orbifold CFTs) — cited, *not* claimed; the continuum proof for the concrete seam quotient net on the collar (v336/v449) and for the interacting condensed $(E_8)_1$ net remains WP5e / Costello–Li territory.

– **WP5e (subdivided): α stage (*done*) [E]/[C] — the CFT-side prefactor + level pinning; β stage (*done*) [E]/[C] — the equivariant anomaly ledger on twistor space, including the honest a_0 refutation; γ stage (*done*) [E]/[C] — the sphere-axion pairing check, a rigid *negative* result with certificate; δ_2 stage (*done*, intermediate verdict, exact) [E]/[C] — the full-tensor ledger: the collapse confirmed, one cubic door opens; ε_2 stage (*done*, verdict B) [E]/[C] — the CPS level-from-flux dial, with "one level" lifted to a theorem; the global BCOV quantisation stays [O].** The α stage is executed and machine-verified as `CELEST.WP5E.ALPHA.01` [`verification/v502_celestial_wp5e_prefactor_level.py`] (exact sympy/Fraction arithmetic, 33 checks). Both exactly computable consistency anchors of the uplift hold. (i) *The $q^{-1/3}$ prefactor is exact μ_4 vacuum-energy bookkeeping*: the clock is *inner* ($h = (2,2,2,2,2;0^4)$ and the A_3 detector $h' = (0^5;1,1,1,-3)$ both read the glue class mod 4 on all 240 roots; $(h,h) = 20$, $(h',h') = 12$, sum 32), so the induced twist on the 8 torus bosons is $\theta = 0^8$ in *every* sector — a *shift* orbifold, not a rotation orbifold — and each sector carries the same $8 \times (-\frac{1}{24}) = -\frac{1}{3} = -c/24$ (Hurwitz-ζ exact, $E_b(\theta) = -\frac{1}{24} + \theta(1-\theta)/4$); the sector ground exponents are $-\frac{1}{3} + (0,1,1,1)$ with $h_j =$ the discriminant form, an identity holding via spectral flow ($j^2(h,h)/32$, mod 1) *and exactly* via the $10 + 6 = 16 = 2^{g_{\mathrm{car}}-1}$ Majorana free fermions of the WP5d seam carrier (R–NS shift $n/16$: $\frac{5}{8}, \frac{3}{8}, 1$); the four sector characters sum to $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$; the rotation reading *fails* (the geometric A_3 deck gives $\frac{3}{16} \neq \frac{3}{8}$ — the WP1 "clock $\neq$ deck" lesson at the Casimir level). (ii) $k = 1$ *is forced three independent ways*: the current condition $h(J) = k = 1$ (the $\mathbb{Z}_4$ extension closes

onto the E_8 adjoint 60+64+60+64 = 248 only at spin 1); the conformal embedding (D_5+$A_3 \subset E_8$: $47k^2$+$219k$−$266 = 47(k-1)(k+266/47) = 0$; $D_8 \subset E_8$: $128k(1-k) = 0$); and the central charge $248k/(k+30) = 8 \Leftrightarrow 240(k-1) = 0$ (= the prefactor itself); plus the WP5b singular-vector dial ($\|(E_{-1}^{\theta})^m 0\|^2 = m!\,k(k-1)\cdots(k-m+1)$, singular vector exactly at grade k+1; $k = 1$ deletes $V(2\theta)$: $31124 - 27000 = 4124$). *Honest sharpening*: glue-h *integrality* $h(J^j;k) = k\,(0,1,1,1)$ holds for *all* $k = 1..8$ and fixes nothing — the originally conjectured integrality route is dead; the three other dials carry. Controls: $D_8/SO(16)_1$ has the *same* $q^{-1/3}$ prefactor but $h = (0, \frac{1}{2}, 1, 1)$ (the discriminator is glue-h integrality, not the prefactor); $\mathbb{Z}_2$ or μ_4 *rotation* twists break the common prefactor; wrong $k \in \{2,3,4\}$ fails all five dials coherently. **The β stage (*done*) [E]/[C] — the equivariant anomaly ledger on twistor space** — is executed and machine-verified as `CELEST.WP5E.BETA.01` [`verification/v505_celestial_wp5e_beta_equivariant_ledger.py`] (exact sympy/Fraction arithmetic, 47 checks). The twistor side is pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop box anomaly on $\mathbb{C}^2/\mathbb{Z}_4$ with the μ_4-glue monodromy. (i) *The AB ledger is exact*: denominators $\det(1 - g_j^{-1}) = (2,4,2)$ with Dedekind sum $5/4 = (|\mathbb{Z}_4|^2-1)/12$; equivariant characters $\mathrm{tr}_{g_j}(\mathrm{Ad}\,E_8) = (248, 0, -8, 0)$ by two independent routes; invariant average 60 = the carrier; Frobenius 61568. (ii) *Okubo per sector, the honest sharpening*: only the *invariant* sector is Okubo-quadratic ($Q^{(0)} = 36\langle x,x\rangle^2$, $36 = \tilde{\lambda}^2_{e_8}$ — the v495 identity re-derived through the glue grading); the twisted sectors carry irreducible T_5/T_3 content, and the AB-weighted sum cancels the D_5 quartic *exactly* while leaving the *rigid* residual $32\,T_3$ (no admissible reweighting fixes it; the graded GS exchange is rank-obstructed in sectors 1–3) — twisted-sector closed strings must carry the rest [O]. (iii) *The index bridge* (the centrepiece): $f(m) = \frac{1}{4}\sum_j (i^{jm}-1)/\det_j = \mathrm{ch}_2(T_m) = -(C^{-1})_{mm}/2$ *exactly* — the fixed-point ledger and the McKay/Kronheimer intersection ledger are the same arithmetic; the glue-bundle ch_2 defect is the integer -78 by *both* routes (classes $(-24,-30,-24)$). (iv) *The level dials say $k = 1$ geometrically*: the lattice current count is 240 at k=1 and exactly 0 at $k = 2,3,4$; the embedding residual $47k^2$+$219k$−$266 = (0, 360, 814, 1362)$; $h(J;k) = k$ is integer for *all* k (integrality alone fixes nothing — honest, exactly as on the CFT side); one scale ⇒ one level (v493 periods; since the ε_2 stage a *theorem* of clock invariance, v509). (v) *The honest refutation*: the clock-invariant modulus $a_0 \in O(8)$ fills the BSS *graviton* slot $O(2)$, *not* the axion slot $O(-2)$ (Hamiltonian weights $X = -2 \neq +2 = Y$; mismatch $4 = |\mu_4|$, typed [C] as a coincidence) — "the theory brings its own GS axion as a_0" is *refuted*; what does match: the three H^2(ALE) classes carry exactly the three twisted-sector Coxeter characters $\{i, -1, -i\}$ (bijection, no invariant class), so the bulk axion must come from the $O(-2)$ tower field itself (construction [O]). Controls: $\mathrm{diag}(i,i)$ breaks the ledger four independent ways; $SO(16)$ glue gives characters $(120, 0, +120, 0)$ and defect $-30 \neq -78$ with a *failing* bulk Okubo (independent quartic Casimir); $k = 2$ dies on the closure dial ($-31/48 \neq -\frac{1}{3}$) while naive integrality dials stay integer. The preregistered kill does *not* fire on the equivariant skeleton. **The γ stage (*done*) [E]/[C] — the sphere-axion pairing check, an honest rigid *negative* result** — is executed and machine-verified as `CELEST.WP5E.GAMMA.01` [`verification/v508_celestial_wp5e_gamma_pairing.py`] (exact sympy/Fraction arithmetic, 27 checks). Question: can the three twisted sphere axions η_m (Coxeter eigencharacters $\{i, -1, -i\}$ on H^2(ALE), the β-stage slot bijection) cancel the rigid $32\,T_3$ residual ($A_{\mathrm{fix}} = 9P_1$−$30P_2$−$15P_3$+$32\,T_3$) by Green–Schwarz-type *exchange* with sector-compatible quadratic vertices? *No, with a certificate.* (i) *Vertex-space theorem*: the $W(D_5)\times W(A_3)$-invariant quadratics on the 8-dim glue Cartan are *exactly* $\mathrm{span}\{s_5, s_3\}$ (dim 2) and the invariant quartics *exactly* $\mathrm{span}\{P_1, P_2, P_3, T_5, T_3\}$ (dim 5) — complete by Weyl nullspace arithmetic over all bidegrees; the *product theorem* is the master kill: the product of any two invariant quadratics has identically zero T_5/T_3 content, so every exchange image — any axion count, any selection rule, any

couplings — is annihilated by Φ_{T_3} while $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$. (ii) *Selection enumeration*: the strict two-index $\mathbb{Z}_4$ rule collapses (the class bilinears are nonzero only for $j' = -j$, forcing $m = 0$: the sphere axions have *no* invariant bilinear Cartan vertex at all); the generous twist-insertion channels $E_{13} = (16, -96, 144, 0, 0)$, $E_{22} = (16, 32, 16, 0, 0)$ give a rank-2 exchange matrix with $\operatorname{rank}([M \,|\, A_{\text{fix}}]) = 3$ and annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$ — two independent obstructions, the second driven by the side discovery $K^{(0)} = -15\,K^{(2)}$ (the even-sector quadratics are parallel); $\ker M = 0$ forces the unique non-disturbing coupling $c = 0$ (the sphere axions are rigidly *silent*), and the sector-0 Okubo mechanism is undisturbed ($\Phi_P(Q^{(0)}) = 0$). (iii) *Naturalness dissolves*: ch_2-natural and Atiyah–Bott-weight couplings both carry certificate $(0, 0)$ against the required $(32, 72)$ — the failure is scale- and coupling-independent (the ansatz space misses the target identically, not by an unnatural factor). Controls: the flipped rule changes the enumeration but not the T_3 kill (Φ_P is rule-dependent, Φ_{T_3} is not); $SO(16)$ has *no* sphere partners (odd McKay classes empty) *and* uncancelled T_5 ($(15, -18, -15, -4, 40)$ — E_8 is doubly special); D_8 has no T_3 structure at all ($16\,T_8 + 12\,s_8^2$). Number fences typed [C] only: $32 = (h, h) + (h', h') = 20 + 12$, $72 = 2\tilde{\lambda}^2_{e_8}$. The β-stage slot bijection is untouched and the preregistered level-kill still does *not* fire — what dies is the exchange *realisation* of the pairing. **The ε_2 stage (*done*, verdict B) [E]/[C] — the CPS level-from-flux dial** — is executed and machine-verified as `CELEST.WP5E.EPS2.01` [`verification/v509_celestial_wp5e_eps2_level_from_flux.py`] (exact sympy/Fraction arithmetic, 28 checks). Target: $k = 1$ from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma Burns-holography mechanism (PRL **130**, 061602; `arXiv:2306.00940`). (i) *The CPS skeleton is replicated exactly*: the S^3 period of the brane back-reaction 3-form is $(2\pi i)^2 N$ (pullback density $-\sin 2\theta$, their eq. 3.33 — large-gauge invariance quantises N); the exceptional-sphere Kähler flux is $2\pi N$ exactly; the boundary level magnitude is $2N$ via the symplectic-boson double contraction ($N = 1, 2, 3$), Dynkin normalisation $T(\text{vector}) = 1$ — "level = flux quantum × Dynkin index", *linear* in the flux. (ii) *The pairing matrix is pinned, not postulated*: the β-stage ch_2 ledger + integrality + unimodularity + effectivity force $P_{mi} = c_1(T_m).[\Sigma_i] = \delta_{mi}$ by complete enumeration (row candidates 8/6/8; 48 unimodular solutions; with $P \geq 0$ exactly 2: identity + the A_3 diagram flip), so the glue fluxes are $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$ (total $188 = 248 - 60$, flip-invariant) and every charged root current carries *exactly one* flux quantum through exactly one sphere. (iii) *The level formula, honest*: "level = total open-string flux" is *killed by the lockstep test itself* ($k_i = (64, 60, 64)$ unequal — F_i is the flavour multiplicity, the brane matter content); what works is $k =$ flux per adjoint current $= F_i / \dim \mathfrak{g}_i = (1, 1, 1)$, anchored non-circularly by the lattice current count $(240, 0, 0, 0)$ for $k = 1..4$ and the embedding index $j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1$ ($60 = 2h^\vee$). The ord-4-vs-level-1 tension is *resolved*: the clock order counts *fractional flux sectors*, not the level ($\mu = 16 = \text{ord}^2$; the μ_4 glue diagonal is isotropic and Lagrangian; KLM $16 \to 16/4^2 = 1$, zero residual fractional flux), and a *new dial* pins the value: $\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$ for $k = 1..6$ from the affine marks — exactly *one* sector iff $k = 1$, independent of the CPS transplantation. (iv) *The lockstep theorem* (the lifting of the β-stage "one scale ⇒ one level" to a *theorem*): clock invariance forces $a_3 = a_2 = a_1 = 0$ sharply, the four branch points are one μ_4 orbit, $|\Pi_j| = \sqrt{2}\,t$ is *equal* on all three spheres, the monodromy acts as the pure phase i and $\det(A - 1) = -4 \neq 0$ (no invariant flux vector) — with the CPS flux-level dictionary, $k_1 = k_2 = k_3$ *is a theorem* of clock invariance; the new falsifier: the clock-forbidden family $Z^4 - Z$ (smooth, disc -27, branch points not a μ_4 orbit) gives *zero* lockstep chains over all 24 orderings. Honest limit: uniform flux doubling keeps the lockstep — the theorem proves *equality*, not the value; the value 1 comes from the current/sector counters and the CPS minimal quantum $N = 1$. Controls separate honestly: $k = 2$ dies on the closure dials (count 0, residual 360, prefactor $-31/48$,

#prim 3) while the lockstep dial honestly does *not* fire; $SO(16)$ glue leaves spheres 1/3 flux-dark with condensation stuck at 4; the A_2 form has $\mu = 12$ (no Lagrangian subgroup exists at all) and Coxeter order $3 \neq 4$. Typed honestly: the CPS dictionary itself on $\mathbb{PT}/\mathbb{Z}_4$ stays [C] (their geometry is the $\mathbb{C}^2$ blow-up with one (-1)-sphere and $\mathfrak{so}(8)$ at negative level — not the A_3 ALE); the type-I B-model back-reaction on $\mathbb{PT}/\mathbb{Z}_4$ stays [O] only in its δ_{1d} derivation face (all three milestones M1–M3 have since been executed, v515/v516/v517; the δ_1 chain itself has since been *decided* by v518 — kill under the derived measure — and the declared/derived measure question has since been decided at probe level in favour of the declared reading under the typed premises TP-1..TP-4, v520; the residual [O] is the constructive w_m derivation). **The δ_2 stage (*done*, intermediate verdict, exact) [E]/[C] — the full-tensor ledger: the collapse is real, and one cubic door opens** — is executed and machine-verified as `CELEST.WP5E.DELTA2.01` [verification/v511_celestial_wp5e_delta2_tensor_ledger.py] (exact Kostant/Weyl weight arithmetic + sympy/Fraction, 41 checks). The γ-stage collapse was a *Cartan-restricted* statement; δ_2 recomputes the ledger on the full adjoint tensor structure. (i) *Structure*: $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ is *semisimple* with no $\mathfrak{u}(1)$ factor (class-0 roots split 40+12, root span rank 8); the twisted sectors factorise multiplicity-free into minuscule Weyl orbits $\mathfrak{g}_1 = (16_s, 4)$, $\mathfrak{g}_2 = (10, 6)$, $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$, each irreducible; and the *innerness theorem* decides the collapse wholesale: the grading element h lies in the Cartan *of* $\mathfrak{g}_0$, so every $\mathfrak{g}_0$-invariant tensor carries total sector charge 0 mod 4 — the γ-stage collapse is a *full-tensor* statement *and* holds across arities (all 15 non-neutral trilinear sector triples have Hom = 0: the sphere axions have no invariant partner operator at any arity ≤ 3). (ii) *The Hom tables*: bilinear invariants exist only at $j' = -j$ (dims 2/1/1/1, the Killing blocks); the neutral trilinear multisets carry dims $(3, 2, 2, 1, 1)$, all cross-sector trilinears are pure brackets (antisymmetric in their repeated legs) — the *only* totally symmetric trilinear on all of $\mathfrak{e}_8$ is the $\mathfrak{su}(4)$ d-symbol ($\mathfrak{so}(10)$ has no cubic Casimir: $\mathrm{Inv}(S^3\,45) = 0$ protects T_5). (iii) *The new channel*: the quartic projection of the d-symbol exchange is $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$ by two independent routes (Killing propagator $2h^\vee = 60$ from the roots; hand route $|x^2 - (\mathrm{tr}\, x^2/4)\mathbf{1}|^2/60$), and $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ — the γ-stage master kill (the product theorem) covers *quadratic* vertices only and does not extend to cubic ones: the T_3 direction *is* reachable through the d-symbol channel, while brackets, tadpoles and mixed $d \times f$ die exactly and no T_5 analogue exists. (iv) *The new certificate*: $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$ has rank 3, augmented rank 4 — the pairing in the charge reading remains *obstructed*, but with the genuinely *weaker* certificate $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$, $\psi(A_{\mathrm{fix}}) = 72 - 8 = 64$; and the relaxed reading (charge bookkeeping dropped on the bilinear P-block) becomes *exactly solvable*: $A_{\mathrm{fix}} = -u + 8v + 2w + 1920\, Q_{dd}$ (residual 0) — where the γ stage found $T_5 = T_3 = 0$ persisting even relaxed, the d-channel closes the T_3 gap with small integer coefficients. Number fences typed [C] only: $64 = \dim \mathfrak{g}_1 = 2^6$; $1920 = |W(D_5)| = 8 \cdot 240$ — and the 1920 fence has now been stress-tested and *fails as a derivation claim* ([verification/v513_celestial_dterm_nonderivation.py], `CELEST.DTERM.NONDERIV.01`): the $1920 = |W(D_5)|$ reading is look-elsewhere-loaded (11/924 pre-declared catalog expressions hit 1920, vs. 8/924 for the control target 1800) and convention-contingent (only 1 of 5 normalisations produces 1920: Gell-Mann gives 120, $\mathfrak{su}(4)$-Killing 256, plain trace 32, ch_3-normalised 69120); the charge-legal flux pairings give 2048/1800 — they bracket and miss — while the only flux hit is the charge-forbidden $(1, 2)$ pairing; quantisation delivers only lattices $(60\,\mathbb{Z}$, with 1920 the 32nd multiple; strict ch_3 units exclude it); and the twisted cubic index $32i$ clashes in class provenance with the true quartic 32 (classes 1/3 vs. 0/2: same number, disjoint mechanism). The convention-stable [E] core is the factorisation $c_d = \Phi_{T_3}(A_{\mathrm{fix}}) \times 2h^\vee(E_8) = 32 \times 60$; the fence stays [C], the physical generation of the 32 stays [O] — no marker moves. Controls: the Killing control replicates the γ-stage channels and A_{fix} exactly; $SO(16)/D_8$ has *no*

symmetric cubic ($\mathrm{Inv}(S^3\,\mathrm{adj}_{D_8}) = 0$), no odd sectors and no A_3 block — the δ_2 channel cannot even be posed (E_8/μ_4 doubly special); the false $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_2 \oplus \mathfrak{u}(1)$ inflates every Hom dimension ($(5,2,2) \neq (2,1,1)$, $\mathrm{Inv}(S^3\mathfrak{g}_0) = 6 \neq 1$) and the false sector content $(16_s, \bar{4})$ kills every pairing. What stays open after δ_2 **[O]**: the arity ≥ 4 ledger, the *physical* justification of the cubic GS term ($c_d = 32 \times 60$, the convention-stable core; the $1920 = |W(D_5)|$ reading is look-elsewhere-loaded and convention-contingent, v513; ch_2-naturalness for cubic vertices *undecided* — no canonical geometric datum exists for a cubic vertex), and the burden on δ_1 is *sharpened*: it must supply either the $\psi = 64$ slice or the selection-rule relaxation — a burden since *discharged* by the M2 milestone: the declared KS measure supplies the $\psi = 64$ slice exactly (v516), so the cubic d-channel is not needed (c_d free $= 0$) and its physical-justification question stays dissolved — the δ_1 decision (v518) does *not* revive it (no branch fires under the derived measure). **The δ_1 stage (*done*, DECIDED — kill under the derived measure, the $\mathbb{Z}_2$ anchor a method boundary)** **[E]**/**[C]** — is executed and machine-verified as `CELEST.WP5E.DELTA1.01` [`verification/v518_celestial_wp5e_delta1_derived_kill.py`] (exact $\mathbb{Q}(\zeta_8)$/phase arithmetic + 30-digit kernels, 30 checks; the $\delta_{1b}/\delta_{1c}/\delta_{1d}$ probe chain consolidated). The exploration stage had left the verdict undecided with four exact sharpenings — (i) the strict holomorphic q^0 reading *refuted at the $\mathbb{Z}_2$/Eguchi–Hanson anchor* (a method boundary, not a kill of the contract); (ii) the forced leading (T_5, T_3) ratio 4:3; (iii) the contact term = the *modular completion* of the Atiyah–Bott data (a Harvey–Moore-type τ-integral, orbit split $15 = 12 + 3$); (iv) massive coset levels $n \geq 2$ mandatory. The τ-integral has now been *evaluated under a derived measure*, and the route is decided. Four exact results: (i) *the 16-component Weil completion closes*: the discriminant module of $D_5 \oplus A_3$ is $\mathbb{Z}_4 \times \mathbb{Z}_4$ with $q = (5x^2+3y^2)/8$, Gauss sums $2\zeta_8^5 \times 2\zeta_8^3 = 4 = \sqrt{16}$ (signature 0 mod 8 = rank E_8), the exact Weil relations hold in $\mathbb{Q}(\zeta_8)$, the diagonal and anti-diagonal Lagrangians are the two E_8 gluings, and the 16-vector theta S-covariance drops the naive-4-rule residual from $2.91 = O(1)$ to $\sim 10^{-39}$; (ii) *the obstruction is a finite μ_4 character, identified*: the dressed gauge blocks $M = G[a,b]\,\eta^{-8}$ carry a multiplier system whose $SL(2,\mathbb{Z})$ relation defects are $(1,1,1)$ exactly on all 15 pairs (a *character* of the orbit stabilisers $\Gamma_1(4)/\Gamma_0(2)^{\pm}$, not a genuine 2-cocycle), on $\Gamma_1(4)$ explicitly $\lambda(\gamma) = i^{2B+C/4}$; (iii) *the cancellation exists and is the twisted fibre block $f_1 f_3$, not the three sphere axions*: $G[a,b] = f_1 f_3$ is an exact identity, the T-fix mechanism is one line of exact phase arithmetic ($(-1) \times e(-1/6) = e(1/3) = \chi_4(T)$), and the exact cancellation table reads none $\to 4$, $f_1 f_2 f_3 \to 4$, $f_2 \to 6$, $f_1 f_3 \to 1$ (both orientations), wrong weights $\to 4$ — the strict solutions are exactly χ_4 (dims $(3,3)$) and χ_{10} (dims $(1,1)$), certified on the dressed functions; (iv) *all three preregistered testers fail* under both derived solutions (T_5 fractions 0.5/0.83 instead of 0; $W_{13} = 0$ instead of $W_2{:}W_{13} = 4{:}3$; spreads 1.05/1.47 against $-A_{\mathrm{fix}}$; no $\psi = -64N$ slice), and the honest (N_1, N_2) orbit rebalance rescues nothing in the positive cone — a genuine KILL on the derived surface. Controls: the $\mathbb{Z}_2$/EH anchor (gauge-only residual order 2, cancelled to 1 by the same fibre dressing), $SO(16)$ (order-4 supply structurally absent), wrong form/wrong signature (break exactly). Fences **[C]**: the $f_1 f_3$ = KS-weight identification (deck weights $(1,3)$ = the twistor fibre weights forced by the v514 incidence ledger — exact match, *not* a complete BCOV derivation), the kernel-family convention, the discrete χ_k choice (both evaluated, both fail). TENSION, stated honestly (the mandatory fence): the DECLARED completion reading (v516/M2) delivers the $\psi = 64$ slice and cancels $32\,T_3$; the DERIVED chiral measure (this module) fails all three testers — both are exact; the sharp open question was which of the two is the true BCOV measure (the declared reading is supported by the δ_1 modular-completion finding but not derived; the derived measure rests on the $f_1 f_3$ identification). Neither result is hidden behind the other. That question is since *decided at probe level* by v520 in favour of the declared reading (single-valuedness derived from

F-independence + the Quillen pairing under the typed premises TP-1..TP-4; the kill of this module is *sharpened*: the column-canonicalised χ_4 member is boundary-consistent yet tester-dead); and the constructive BCOV derivation of the w_m normalisation itself is since *executed* by v523 (verdict ERFOLG; the residual [O] narrows to the global BCOV integral beyond the fibre zero-mode factor). **The ε_1 stage (*done*) [E]/[C] — the $O(-2)$ bulk-axion slot** — is executed and machine-verified as `CELEST.WP5E.EPS1.01` [`verification/v514_celestial_wp5e_eps1_axion_slot.py`] (exact sympy/Fraction arithmetic, 34 checks, verdict B). Three results, each exact. (i) *The slot is a construction*: on $\mathbb{PT}' = \mathrm{Tot}(O(1)\oplus O(1) \to \mathbb{P}^1)$ the pushforward ledger $\pi_* O(-2)$ gives per spacetime degree d exactly $(d+1)^2$ classes = the exact wave-operator nullspaces, and the *equivariant* bookkeeping under the v492 action $(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)$ (incidence-forced column weights $(+1,+1,-1,-1)$) closes *block by block* for all $d \le 6$ and all four characters; the character series are $P_0 = 1+3t^2+15t^4+21t^6+45t^8+\ldots$, $P_1 = P_3 = 2t+8t^3+18t^5+\ldots$, $P_2 = 6t^2+10t^4+28t^6+\ldots$, the $d=0$ slot has multiplicity $(1,0,0,0)$ — *the bulk axion survives the projection* — and the Molien invariant ring is $(1-t^8)/((1-t^4)^2(1-t^2))$: $Z \in O(2)$, $X, Y \in O(4)$, one relation in degree 8 = the a_0 weight (the v492/v493 bridge); the E3 bijection is sharpened per character (twisted minimal content $(2t, 6t^2, 2t)$, no degree-0 mode, = the Coxeter eigenvalues, $\det(A-1) = -4$), and the graviton control separates a_0 (the $O(+2)$ slot starts invariant only at fibre degree 4 with multiplicity $3 = \{X, Z^2, Y\}$). (ii) $\tilde{\lambda} = 6$ *is triply pinned*: Okubo $\mathrm{Tr}_{\mathrm{adj}}\, X^4 = 36\langle x,x\rangle^2 = (6\langle x,x\rangle)^2$ on the 240 glue roots ($36 = h^\vee + 6$, $\tilde{\lambda} = 6 = |\mathbb{Z}_2|N_{\mathrm{fam}}$); the measure chain on $\mathbb{PT}/\mathbb{Z}_4$ cancels the quotient factor $\mu = 1/4$ *exactly* (vertex^2 μ^2, propagator $1/\mu$, anomaly μ: $\tilde{\lambda}^2_{\mathrm{eff}} = 36$ for arbitrary μ), with the wrong bookings quantified and *excluded* (anomaly-only $\Rightarrow \tilde{\lambda} = 3$, missing propagator renormalisation $\Rightarrow \tilde{\lambda} = 12$); and the flux side (minimal CPS quantum $N = 1$, a single exchange channel exactly at $k = 1$, $(\kappa/c_3)^2 = 12$ exact). (iii) *The first honest back-reaction step (Gibbons–Hawking A_3)*: clock-invariant four-centre configurations form exactly two branches (axis^4 = pure resolution or *one* free μ_4 orbit = pure deformation); the free orbit gives $\prod_p (Z - i^p z_0) = Z^4 - z_0^4$, i.e. $XY = Z^4 + a_0$ with $a_0 = -z_0^4$ — the v493 family *re-derived from the geometry*, with the a_0 monodromy = one clock step; the charge-k GH point is exactly $\mathbb{R}^4/\mathbb{Z}_k$ ($k=4$: the seam boundary $S^3/\mathbb{Z}_4$); the periods are $\Pi_j = 4\pi t_0 (i-1)(1, i, i^2)$ with equal moduli (lockstep, the v509 counter-check from geometry) and the Coxeter clock re-derived ($A^4 = 1$, $\det(A-1) = -4$, $\Pi \cdot A = i\Pi$); the CPS "N log" analogue carries source charge $4 = |\mu_4|$, with the honest sharpening that the Ricci-flat ALE has asymptotic log coefficient exactly 0 (the CPS log is an exceptional-locus statement; Burns contrast $\det g \neq 1$), and the multipole selection rule $m \equiv 0 \bmod 4$ stores $-a_0$ in the first symmetry-breaking $(4, \pm 4)$ harmonic. Controls: $\mathfrak{so}_8$ ($\tilde{\lambda}^2 = 12$ irrational; perfect squares in the Deligne series only $\{9, 36\}$), $\mathrm{diag}(i, i)$ (Veronese cone, no hypersurface), $k = 2$ (three channels). [C]: conditional on Costello's flat-$\mathbb{PT}$ matching $\lambda_g^2 E = A$ (the measure chain is classical bookkeeping). [O] (the honest fence): the *quantised* BCOV one-loop coefficient on $\mathbb{PT}/\mathbb{Z}_4$ and the twisted channels ($32\,T_3$) — preregistered as M1–M3 below; all three have since been executed (v515/v516/v517), the δ_1 chain has since been *decided* by v518 (kill under the derived measure), the measure question has since been decided at probe level by v520 for the declared reading, and the w_m normalisation has since been *derived* at probe level by v523; what stays [O] is the global BCOV integral beyond the fibre zero-mode factor. **WP5e proper [O] (the global BCOV quantisation):** the BCOV/Kodaira–Spencer theory on $\mathbb{PT}/\mathbb{Z}_4$ itself (quantisation, non-fixed-point contributions, the twisted closed-string measure). The exchange sub-branch is *closed* by the γ stage, the full-tensor ledger (δ_2), the level-from-flux dial (ε_2) and the bulk-axion slot (ε_1) are *executed*, and δ_1 is *decided* by v518 (kill under the derived measure); the named remaining target, with preregistered success/kill criteria: **(the measure question — the δ_1 successor; *decided at***

***probe level*, ERFOLG-A)** the δ_1 τ-integral had been evaluated on *both* sides: the M2 milestone (v516) *delivers* the coefficient $(9, -30, -15, 0, 32)$-completion exactly on the *declared* measure (every sector the same Okubo square, the $\psi = 64$ slice supplied), while the *derived* chiral measure (v518: blockwise $SL(2,\mathbb{Z})$ covariance solved for, the μ_4 multiplier character cancelled by the twisted fibre block $f_1 f_3$) fails all three preregistered testers — both are exact, and the stated tension was the sharp remaining question: *derive* from the BCOV integral on $\mathbb{PT}/\mathbb{Z}_4$ which of the two measures is the true one. That decision is executed and machine-verified as `CELEST.WP5E.MEASURE.01` [`verification/v520_celestial_wp5e_measure_decision.py`] (the δ_{1e}/δ_{1f} consolidation, 39 checks, ERFOLG-A on the preregistered decision layer): the exact invariant subspace of the dressed 16-component Weil system is $\dim_{\mathbb{Q}}\{v : \rho(T)v = v,\ \rho(S)v = v\} = 8 = 4 \times 2$ with *every* kernel vector in the $\mathbb{Q}(\zeta_8)$-span of $\{e_H, e_{H'}\}$ (both theta $= E_4 =$ the unphased trace), and single-valuedness of the physical one-loop lattice factor is *derived* from two independent constructive sources rather than postulated — *(source 1, the F-lemma)* every strict chiral closure at physical weight carries a nontrivial character ($\chi_4(T) = e(1/3)$, $\chi_{10}(T) = e(5/6)$, exact; pointwise covariance of the canonical column-matched member certified at 3.9×10^{-16} with $|G| \geq 59.6$), and an exact change of variables forces $\int_{\gamma F} G\, d\mu = \chi_4(\gamma) \int_F G\, d\mu$, so fundamental-domain independence forces $\int = 0$ for *every* chiral integrand: the v518 route was never a well-defined nonvanishing moduli integral (the kill is *sharpened*); *(source 2, the Quillen pairing)* the doubled hol×antihol transports satisfy $|t\bar{t} - 1| < 8.3 \times 10^{-40}$ on all 15 pairs ($|\chi_4|^2 = 1$ exactly vs one-sided $\chi_4^2 = e(2/3) \neq 1$), the doubled 256-dim system closes *strictly* at trivial character and physical weight (nullspaces $(28, 17) \geq 11$) containing the unphased diagonal *and* the physical columns $w_b \otimes \bar{w}_b$, while the hol⊗hol control is *empty* $(0, 0)$; the forced unphased numerator then reproduces the v516 Okubo squares *levelwise* inside the v518 scaffold (exact 4-design on every level $n \leq 8$, J-weighted spreads $\sim 5 \times 10^{-41}$, $\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})$ — the J-weighted mirror of ± 64), the declared reading also wins the column-canonicalisation discriminator (the physical AB column is contained *exactly* in the χ_4 family, canonicalising (N_1, N_2), yet the canonical member fails every tester) and the $\mathbb{Z}_2$/EH anchor (all three derived instantiations fail; only the declared reading hits $9\langle x, x\rangle^2 = (1/4) \times 36$ with the scale tooth $c = 1/2 = |\mathbb{Z}_2| h^{A_1}$). Typed premises [C] (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — it must source $\psi = 64$), TP-3 (the v518 kernel convention), TP-4 (the Quillen structure of BCOV F_1). The residual [O] was the constructive BCOV derivation of the w_m normalisation itself — the success branch ("the BCOV derivation lands on the declared completion reading, carrying the forced 4:3 leading ratio") has fired at probe level, the kill branch ("it lands on the derived chiral measure") is dead — and that derivation is since *executed* (verdict ERFOLG) as `CELEST.WP5E.WM.01` [`verification/v523_celestial_wp5e_wm_derived.py`]: the $1/\det_j$ weight is COMPUTED from three independent sources — the equivariant mode ledger of the fibre $\mathbb{C}^2$ (closed form $1/((1 - q\,i^j)(1 - q\,i^{-j}))$ exact at every order $n \leq 120$, regular at $q = 1$ with Abel value $(1/2, 1/4, 1/2) = 1/\det_j$, fixed-point splitting at every truncation level for the phased AND the v520-forced unphased trace, so $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$ is computed, not declared), the zeta/Quillen route ($\det_\zeta = 4\sin^2(\pi j/4) = \det_j$ exactly with the unique real positive holomorphic section via the $SU(2)$ conjugation pairing, and the δ_{1f} block constant term $1/\det_b$), and the consistency chain (the v516 numbers reproduced one by one: $w = 4h$, $c = 4 = |\mu_4|$, squares $(18, 9, 18)$, total $45\langle x, x\rangle^2$, $\psi \pm 64$) — with the negative controls behaving ($1/\det^2$ and $1/|1 - i^j|$ break the chain quantifiably, the $\mathbb{Z}_2$/EH anchor passes at $c = 2 = |\mathbb{Z}_2|$, the $SO(16)/D_8$ kill fires, $\text{diag}(i, i)$ breaks the zeta = AB identity itself, the $k = 3, 5$ weights wander correctly) and the typed premises TP-REG/TP-Q/TP-NUM/TP-CH as its own mandatory [C] fence; the residual [O] narrows to the *global* BCOV integral on $\mathbb{PT}/\mathbb{Z}_4$ beyond the fibre

zero-mode factor (the resolved-ALE route of v514/v517 not built); **(the cubic GS term — dissolved)** the $\psi = 64$ slice is *delivered* by the declared KS measure (v516); the cubic d-channel is *not needed* (c_d free $= 0$) — its physical-justification question (the honest v513 target $c_d = 32 \times 60$) stays dissolved: the δ_1 decision (v518) does not revive the channel (no branch fires under the derived measure); a ch_2-type naturalness datum for *cubic* vertices still does not exist; **(the back-reaction milestones, preregistered as M1–M3 in v514; all three executed — the v514 fence is fully worked off)** the type-I B-model *back-reaction* on $\mathbb{PT}/\mathbb{Z}_4$, which would upgrade the ε_2 CPS dictionary from [C] to derived: **M1 (*done*, SUCCESS)** [E]/[C] the A_3 Ω_N (the Beltrami/Kodaira–Spencer deformation sourced by the glue currents on the three lockstep spheres; preregistered success = a closed-form Ω-deformation whose $S^3/\mathbb{Z}_4$ periods are $(2\pi i)^2\times$ an integer flux vector with the lockstep phases $(i-1)(1, i, i^2)$; kill = unequal sphere fluxes or a non-integer period — then the CPS transplant *and* the ε_2 level dial die) — executed and machine-verified as `CELEST.WP5E.M1.01` [`verification/v515_celestial_wp5e_m1_omega_n.py`] (exact sympy, 30 checks): the residue form $\omega_2 = dX \wedge dZ/X$ is *derived* (three ambient routes; orbifold-cover factor $4 = |\mu_4|$; clock phase i), the twistor family closes as $XY = Z^4 + 4t_0^2\lambda^2 Z^2 - t_0^4(1-\lambda^4)^2$ ($e_1 = e_3 = 0$ identically; seam fibre $\lambda = 0 =$ the v493 family) with the CY-compatible clock lift $(Z, \lambda) \mapsto (iZ, -i\lambda)$, $\Omega \to +\Omega$, and $\Omega_N = \Omega_0 + \sum_p N_p K_p$ is *closed form* (CPS/Bochner–Martinelli kernels on the four centre twistor lines) with every $S^3/\mathbb{Z}_4$ period $(2\pi i)^2$-integral, the seam-fibre phase vector $2\pi i\, t_0(i-1)(1, i, i^2)$ with clock covariance $\Pi \to i\Pi$, the flux vector $N(1,1,1,1)$ forced uniform (clock orbit + K_4 connectivity) and the lens geometry *forcing* the source charge $4 = |\mu_4|$ ($N \in 4\mathbb{Z}$); the 12 conifold nodes sit exactly on the 8 eighth roots of unity ($8 = 2|\mu_4|$); honest fence: on the clock-forbidden $Z^4 - Z$ family the $(2\pi i)^2$ quantisation *also* holds — integrality alone is *not* the discriminator, the discriminator is the lockstep phase structure (0/24 vs 8/24) plus the clock forcing of equal fluxes; neither kill fired; **M2 (*done*, SUCCESS on the declared completion measure — verdict B)** [E]/[C] the twisted KS measure (quantise the $O(-2)$ tower on $\mathbb{PT}/\mathbb{Z}_4$ with the ε_1 mode ledger and compute the one-loop box coefficient per sector; preregistered success = the bulk sector reproduces $\tilde{\lambda}^2 = 36$ *and* the twisted sectors pair with the three sphere axions cancelling the rigid $32\,T_3$ residual; kill = a leftover independent T_3-type quartic in any sector) — executed and machine-verified as `CELEST.WP5E.M2.01` [`verification/v516_celestial_wp5e_m2_twisted_ks_measure.py`] (exact sympy/Fraction, 23 checks): the *declared* completion contact term $\mathrm{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$ (the twisted-sector loop contributes the *unphased* sector trace with the g^j zero-mode normalisation) carries the exact completion-weight identity $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4| h_m = -4\,\mathrm{ch}_2(T_m)$ — the three sphere axions pair through their *own* McKay ch_2 charges, *no free scale, no fit*; the locks are parameter-free ($T_5 = 0$ for *any* scale, ratio $h_2{:}h_1 = 4{:}3 =$ the δ_1-forced leading ratio *reproduced*, the T_3 budget forcing $c = 4 = |\mu_4|$ uniquely); every twisted channel becomes the *same* perfect Okubo square $36\langle x, x\rangle^2/\det_j$ ($T_5 = T_3 = 0$ in every sector — the KILL does not fire), the total is $45\langle x, x\rangle^2 = \frac{5}{4} \times 36\langle x, x\rangle^2 =$ Dedekind $\times$ Okubo = the unique quartic-free weighting of the β-stage rigidity theorem; both γ-stage certificates are killed ($\Phi_{T_3} : 32 \to 0$, $\Phi_P : 72 \to 0$) and the δ_2 slice $\psi = 64$ is *supplied exactly* — no cubic d-channel needed (c_d free $= 0$); controls: wrong scale leaves $T_3 = 32 - 8c$, the shuffle breaks T_5 and T_3, $SO(16)$ keeps $T_5 = 20/T_3 = -40$ (the KILL fires there: E_8 doubly special), $\mathrm{diag}(i, i)$ degenerates, the $\mathbb{Z}_2$/EH anchor passes at scale $2 = |\mathbb{Z}_2|$. Mandatory fence [C]: the completion reading (the loop of twisted states carries the *unphased* sector trace with the same zero-mode normalisation) is *declared*, supported by the δ_1 modular-completion finding, *not* derived from the BCOV integral within that module — the $\delta_{1b}/\delta_{1c}/\delta_{1d}$ chain has since been executed and decided by v518 (the *derived* chiral measure fails all three testers), the declared-vs-derived measure question

has since been decided at probe level by v520 for the declared reading (typed premises TP-1..TP-4), and the w_m normalisation has since been derived at probe level by v523 (the residual [O] narrowing to the global BCOV integral beyond the fibre zero-mode factor); **M3 (*done*, SUCCESS)** [E]/[C] the a_0 uplift (lift the $(4, \pm 4)$ GH multipole to the Beltrami differential and compute the induced Kähler-potential correction; preregistered success = a log-type correction with coefficient tied to the source charge $4 = |\mu_4|$; kill = decoupling from the centre count) — executed and machine-verified as `CELEST.WP5E.M3.01` [`verification/v517_celestial_wp5e_m3_a0_uplift.py`] (exact sympy, 23 checks): the uplift object is the generalized-Legendre-transform kernel $\chi = \log P_4$ — the *log of the M1 family polynomial* — well-typed (the $O(2)$ section is a null coordinate: *any* kernel is harmonic) and matching the V-ledger through the exact residue identity $\partial_x[\text{transform}(\log \eta_p)] = 1/r_p$ (flux -4π per centre, source charge $4 = |\mu_4|$); the log-type correction arrives on *four coupled centre-count scales*: (i) the asymptotic kernel $\log \chi = 4 \log \eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \ldots$ with coefficient $4 = |\mu_4|$, the seam-fibre first correction *exactly* a_0/η^4 = the $(4, \pm 4)$ multipole (exact m-grading, no $\log \times$power terms — the ε_1 honest zero preserved); (ii) the GLT tower $p_{4k} = 4(-a_0)^k$ with the $n \equiv 0 \bmod 4$ selection rule; (iii) the exceptional-locus log $\chi(0) = \log a_0 = 4 \log t_0 + i(4\varphi_0 + \pi)$ (response 1 at the locus vs power-law at infinity); (iv) the period response $d \log \Pi_j / d \log a_0 = 1/4 = 1/|\mu_4|$ uniformly, integrating to the monodromy i = one Coxeter clock step (the WP2 monodromy reproduced from perturbation theory), with a_0-rigid $\mathbb{Z}_8$ node support and topological $(2\pi i)^2$ fluxes; controls: the $(4, 0)$ multipole is clock-invariant, the $\mathbb{Z}_2$/EH analogue reads $2 = |\mathbb{Z}_2|$ on *every* dial, $k = 3, 5$ orbits move the coefficient with the centre count ($O(6)/O(10)$ slots), the forbidden family fails both dials ($p_3 = 3$, $e_4 = 0$) — the KILL (decoupling) does not fire; fence [C]: the GLT dictionary ($\eta_p \log \eta_p$ = the GH Kähler-potential kernel, Lindström–Roček / Ivanov–Roček); [O]: the full nonlinear Kähler potential of the resolved A_3 ALE. Success (unchanged): partition function = E_4/η^8 *including* the $q^{-1/3}$ prefactor *derived from the twistor side*; kill: an anomaly mismatch at the Costello–Li grade — neither the CFT-side dials of v502, nor the equivariant skeleton of v505, nor the exchange no-go of v508, nor the flux/sector dials of v509, nor the full-tensor ledger of v511, nor the axion-slot construction of v514, nor the M1–M3 back-reaction milestones of v515/v516/v517, nor the δ_1 decision of v518, nor the measure decision of v520 trigger it (all computable dials say $k = 1$, "one level" is now a theorem, the bulk axion survives the projection, the back-reacted Ω_N carries the forced integral lockstep flux, the twisted KS measure lands on the unique quartic-free weighting, the a_0 uplift couples to the centre count, the δ_1 kill is a statement about the derived measure, not about the algebra, and the measure decision selects the declared reading at probe level), but the global quantisation is untested.

Honest status line: WP5a–WP5d (both WP5d stages), the WP5e α, β, γ, δ_1, δ_2, ε_2 and ε_1 stages and the M1–M3 back-reaction milestones are exact-arithmetic [E] constructions (plus ED-validated lattice witnesses in WP5d) with [C] named identifications ("radial decoupling = collar thickness $\to 0$"; Weyl complete reducibility beyond BFS depth 4; the Kac mechanics beyond level 2; the quasi-free extension; the KLM/Longo–Xu/CHMP index dictionary; Longo split heredity; Kac generation of the maximal submodule for $k \geq 3$; the WP5e-β dictionaries — $\mathrm{CS}(\rho) \bmod 1$ = discriminant weight, $32 = (h, h) + (h', h')$, $-30 = -h^\vee$, the Ω-contraction; the WP5e-γ number fences $32 = 20 + 12$, $72 = 2\tilde{\lambda}_{e_8}^2$; the WP5e-$\delta_2$ number fences $64 = \dim \mathfrak{g}_1$, $1920 = |W(D_5)|$ — the latter typed look-elsewhere-loaded and convention-contingent by v513; the WP5e-ε_2 CPS dictionary on $\mathbb{PT}/\mathbb{Z}_4$; the WP5e-ε_1 conditionality on Costello's flat-$\mathbb{PT}$ matching; the M1 global kernel patching, Hitchin small resolution and CPS brane dictionary, v515; the M2 *declared* completion reading of the twisted KS measure, v516; the M3 GLT dictionary, v517); *WP5a–WP5d plus WP5e-$\alpha/\beta/\gamma/\delta_1/\delta_2/\varepsilon_2/\varepsilon_1$ plus M1–M3 are landed — all three KLM ingredients of complete rationality carry lattice witnesses, both faces of the inflow*

are anchored, the exchange sub-branch is closed with a certificate, the collapse is confirmed full-tensorially with one cubic door open, "one level" is now a theorem of clock invariance with the sector-counter dial pinning $k = 1$, the bulk-axion slot is a construction with $\tilde{\lambda} = 6$ triply pinned, the back-reacted Ω_N is closed-form with $(2\pi i)^2$-integral lockstep periods and the source charge $4 = |\mu_4|$ forced, the twisted KS measure (on the declared completion reading) cancels the $32\,T_3$ residual and supplies the $\psi = 64$ slice without the cubic d-channel, the a_0 uplift couples to the centre count on four scales, and the δ_1 chain is decided — kill under the derived measure, tension with the declared reading stated — WP5e proper (the global BCOV quantisation) is the single remaining open piece, with the continuum uplift of the WP5d lattice witnesses honestly fenced as Xu's theorem (cited, not claimed); the exchange sub-branch is closed by v508, the full-tensor ledger executed by v511, the level-from-flux dial executed by v509, the axion slot executed by v514, the M1–M3 back-reaction milestones executed by v515/v516/v517 (the v514 fence fully worked off), the δ_1 chain decided by v518, the measure question decided at probe level by v520 (the declared completion reading wins: single-valuedness derived from F-independence + the Quillen pairing under TP-1..TP-4; the derived chiral measure was never a well-defined nonvanishing moduli integral), the w_m normalisation itself derived constructively at probe level by v523 (Atiyah–Bott mode ledger + zeta/Quillen determinant + consistency chain, typed premises TP-REG/TP-Q/TP-NUM/TP-CH), and the named remaining target is the global BCOV integral beyond the fibre zero-mode factor (the cubic-GS-term question stays dissolved: v518 does not revive the d-channel) — `SEAM.EQUIV.01` stays [O] and nothing in WP5a–WP5e-M3, the δ_1 decision or the measure decision moves it.

Kill tests (pre-registered). **(K1)** the equivariant sector could have *contradicted* the 52+64+60+64 split (uniformity of $D[d]$ needs exactly $\dim \mathfrak{g}_j = (60, 64, 60, 64)$) — *survived by WP1 and WP2*: the false-glue control shows the test has teeth (3112/6720 violations), and WP2 adds the deformation side (a shape modulus surviving clock-invariance, or a_0-corrections leaking between μ_4 sectors, would have fired — neither occurred; the clock-variant controls fail in two independent ways each). **(K2)** if one-loop associativity of the celestial E_8 theory demanded extra fields that TFPT does not possess, the contract dies (Costello's axionic mechanism is the first pass: E_8 is admissible; the TFPT-specific field content is the open half). **(K3)** — *evaluated by WP3* ([`verification/v495_celestial_wp3_gs_alignment.py`]): the trigger ("no exact alignment between the GS coefficient and the c_3 normalisation") does *not* fire — exact alignment exists ($(\kappa/c_3)^2 = 12$ resp. $1/300$). But the look-elsewhere battery *demotes the scope of K3 itself*: the alignment format passes 8/8 across Costello's whole list, so surviving K3 certifies compatibility plus λ-arithmetic, *not* $\mathfrak{e}_8$-selective evidence — alignment survives, selectivity does not. **(K4)** — *evaluated by WP4* ([`verification/v496_celestial_wp4_character_shadow.py`]): K4 fires only against the *exact*-block reading (the character is provably not a conformal block in the jet grading); it does *not* degrade the contract to "E_8 is an admissible gauge algebra for celestial SDYM", because the sector arithmetic holds exactly — WP4 lands at the boundary-limit reading (verdict B(ii), the `SEAM.EQUIV.01` scaling-limit shape), with the constructive limit handed to WP5 — and made precise there by WP5a ([`verification/v497_celestial_wp5a_null_ideal.py`]: the coefficientwise χ_w limit with threshold $w = n{+}1$ and the derived ideal), with the deleting operator itself constructed by WP5b ([`verification/v498_celestial_wp5b_singular_vector.py`]: $|s\rangle$ explicit, level dial $2(1{-}k)$), the limit state by WP5c ([`verification/v500_celestial_wp5c_gns_limit_state.py`]: the ideal in the GNS kernel, level-2 rank 4124 exactly), the two-interval index chain by WP5d-α ([`verification/v501_celestial_wp5d_two_interval_index.py`]: μ-chain $16 \to 4 \to 1$ anchored at both measured ends), the split + strong-additivity legs by WP5d-β ([`verification/v504_celestial_wp5d_klm_completeness.py`]: all three KLM ingredients of complete rationality witnessed on the lattice; the continuum uplift is Xu's theorem, cited not claimed), the CFT-side anchors of the uplift itself by WP5e-α ([`verification/v502_celestial_wp5e_prefactor_level.py`]: the $q^{-1/3}$ prefactor as exact μ_4 vacuum-energy bookkeeping, $k = 1$ forced three ways), the equivariant twistor skeleton by WP5e-β ([`verification/v505_celestial_wp5e_beta_equivariant_ledger.py`]:

the index bridge $f(m) = \mathrm{ch}_2(T_m)$ exact, glue defect -78 two routes, geometric $k = 1$ dials, the rigid $32\,T_3$ residual, and the honest a_0 refutation — the kill branch does not fire on the skeleton), the exchange no-go by WP5e-γ ([verification/v508_celestial_wp5e_gamma_pairing.py]: the sphere-axion pairing by quadratic-vertex exchange refuted with the three-functional certificate $(0, 32, 72)$ — the $32\,T_3$ burden moves to the twisted contact terms, δ_1 binary), the full-tensor ledger by WP5e-δ_2 ([verification/v511_celestial_wp5e_delta2_tensor_ledger.py]: the collapse confirmed full-tensorially by the innerness theorem, the cubic d-symbol channel opening T_3 with the weaker certificate $\psi(A_{\mathrm{fix}}) = 64$ and the relaxed solution $c_d = 1920\,Q_{dd}$-coefficient — typed by the v513 negative certificate as $c_d = 32 \times 60$ with the $|W(D_5)|$ reading look-elsewhere-loaded), and the level-from-flux dial by WP5e-ε_2 ([verification/v509_celestial_wp5e_eps2_level_from_flux.py]: the lockstep theorem — "one level" as a theorem of clock invariance, falsifier $Z^4 - Z$ at $0/24$ — plus the sector-counter dial $\#\mathrm{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$), the bulk-axion slot by WP5e-ε_1 (v514: the equivariant $O(-2)$ slot construction, $\tilde{\lambda} = 6$ triply pinned, the first GH/A_3 back-reaction step), the back-reacted Ω_N by the M1 milestone (v515: closed form, $(2\pi i)^2$-integral lockstep periods, forced source charge $4 = |\mu_4|$), the twisted KS measure by the M2 milestone ([verification/v516_celestial_wp5e_m2_twisted_ks_measure.py]: the completion-weight identity $w = 4h = -4\,\mathrm{ch}_2$, per-sector Okubo squares, the $32\,T_3$ cancelled and the $\psi = 64$ slice supplied on the declared completion reading — the M2 kill did not fire) and the a_0 uplift by the M3 milestone ([verification/v517_celestial_wp5e_m3_a0_uplift.py]: the log-type correction coupled to the centre count on four scales, monodromy i — the M3 kill did not fire), and the δ_1 chain decided by the consolidated δ_1 module ([verification/v518_celestial_wp5e_delta1_derived_kill.py]: the derived chiral measure fails all three preregistered testers — kill under the derived measure, the declared/derived tension stated), and the measure question decided at probe level by the consolidated δ_{1e}/δ_{1f} module (v520: single-valuedness derived from two constructive sources, the declared reading wins under TP-1..TP-4), and the w_m normalisation derived constructively by the WM module ([verification/v523_celestial_wp5e_wm_derived.py]: the $1/\det_j$ fixed-point weight computed from the Atiyah–Bott mode ledger, the zeta/Quillen determinant and the consistency chain — the WM kill did not fire), leaving the global BCOV quantisation (WP5e proper, narrowed to the global BCOV integral beyond the fibre zero-mode factor) as the single remaining open piece.

Honest success estimate. The probability of *full* success (WP5e: a second continuum route to `SEAM.EQUIV.01`) remains low — it is constructive holography at the Costello–Li grade. The *partial*-findings expectation is now realised and *exceeded*: WP1–WP4, all five WP5 constructive milestones (both WP5d stages), the WP5e α, β, γ, δ_1, δ_2, ε_2 and ε_1 stages and the M1–M3 back-reaction milestones are executed with sharp structural results (verdicts B, B, B, B(ii); WP5a–WP5d passed; WP5e-α pins the CFT side, WP5e-β the equivariant twistor skeleton, WP5e-γ closes the exchange sub-branch with a certificate, WP5e-δ_2 confirms the collapse full-tensorially and opens the one cubic door, WP5e-ε_2 makes "one level" a theorem and pins its value by the sector counter, WP5e-ε_1 builds the bulk-axion slot with $\tilde{\lambda} = 6$ triply pinned, M1 delivers the closed-form back-reacted Ω_N with forced charge $4 = |\mu_4|$, M2 delivers the twisted KS measure on the declared completion reading (the $32\,T_3$ cancelled, the $\psi = 64$ slice supplied, no cubic d-channel), M3 delivers the a_0 uplift coupled to the centre count on four scales, and the δ_1 decision (v518) closes the chain with an honest kill under the derived measure, the measure decision (v520) selects the declared reading at probe level, and the w_m derivation (v523) computes the fixed-point weight from three independent sources; *WP5e proper — the global BCOV quantisation, narrowed to the global BCOV integral beyond the fibre zero-mode factor — is the single remaining open piece*) — the WP1 critical correction (clock $\neq$ deck; clock^2 = deck), the WP2 sharpenings (the clock *is* the Picard–Lefschetz/Coxeter monodromy of the seam deformation; a_0 pure scale with the $\tau{=}i$ shape frozen), the WP3 non-selective alignment (exact λ-arithmetic, honestly typed by the 8/8 look-elsewhere table), the WP4 boundary-limit localisation of the character mismatch (three exactly-located obstructions, one surviving limit structure), the WP5a null-ideal derivation (stabilisation threshold $w = n{+}1$; $27000 = (h^\vee)^3$ derived; quotient = sector sum = 4124; $SO(16)_1$ contrast), the WP5b singular-vector construction (the deletion operator explicit, case tally 190/57/1, level dial $2(1{-}k)$, one-block closure typed E_8/μ_4-specific), the WP5c

limit state (the ideal in the GNS kernel, the full 9361-block Gram at rank 4124, the family proved *necessary* by a CCR obstruction), the WP5d-α two-interval index (the $\ln 2$ plateau at 10^{-15}, the μ-chain $16 \to 4 \to 1$ measured at both ends), the WP5d-β KLM completion (strong additivity exact on the lattice with the shared boundary point; the bounded-vs-divergent entropic discriminator with the Ising $\frac{1}{4}$-exponent approach against the diverging $U(1)$ control; the elliptic-nome split ladder with exact orbifold inheritance; Pimsner–Popa $\lambda = \frac{1}{2}$ and $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ with integer attainment — all three KLM ingredients witnessed, Xu's continuum theorem cited not claimed), the WP5e-α prefactor/level pinning (the $q^{-1/3}$ prefactor as exact μ_4 vacuum bookkeeping; $k = 1$ forced three ways; the glue-integrality route honestly retired), the WP5e-β equivariant ledger (the index bridge $f(m) = \mathrm{ch}_2(T_m)$ exact with the -78 defect; the rigid $32\,T_3$ twisted residual; the honest a_0 refutation with the three sphere classes filling the three twisted axion slots), the WP5e-γ exchange no-go (the invariant vertex spaces exactly dim 2/5; the product theorem; the certificate $(0, 32, 72)$; naturalness dissolved scale-independently — an honest negative result that converts the pairing question into the binary δ_1 contact-term test), the WP5e-δ_2 full-tensor ledger (the innerness theorem confirming the collapse across arities; the unique symmetric survivor = the $\mathfrak{su}(4)$ d-symbol with $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$ opening the T_3 channel; the weaker certificate $\psi(A_{\mathrm{fix}}) = 64$ and the exactly solvable relaxed reading with $c_d = 32 \times 60$, the $1920 = |W(D_5)|$ fingerprint honestly retired as look-elsewhere-loaded by the v513 negative certificate), and the WP5e-ε_2 level-from-flux dial (the CPS skeleton exact; the pairing matrix pinned by complete enumeration with fluxes $(64, 60, 64)$ and one quantum per current; the naive "level = total flux" killed by the lockstep test itself; the lockstep theorem with its $Z^4 - Z$ falsifier; the sector counter $\#\mathrm{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$), the WP5e-$\varepsilon_1$ axion slot (the equivariant Penrose ledger closing block by block; the $d = 0$ slot surviving the projection; $\tilde{\lambda} = 6$ triply pinned with the wrong bookings 3 and 12 excluded; the GH/A_3 step re-deriving the v493 family and the Coxeter clock from geometry), the M1 back-reacted Ω_N (the residue form derived with cover factor $4 = |\mu_4|$; the closed twistor family with its CY-compatible clock lift; every $S^3/\mathbb{Z}_4$ period $(2\pi i)^2$-integral with the forced uniform flux vector and the lens-forced charge $4 = |\mu_4|$; the honest integrality-is-not-the-discriminator fence), the M2 twisted KS measure (the completion-weight identity $w = 4h = -4\,\mathrm{ch}_2$ with no free scale; every twisted channel the same perfect Okubo square; both γ-stage certificates and the δ_2 $\psi = 64$ slice supplied exactly; the declared completion reading honestly fenced [C], v516) and the M3 a_0 uplift (the GLT kernel $\chi = \log P_4$ with the log coefficient $4 = |\mu_4|$ on four coupled centre-count scales, the period response $1/4$ integrating to the Coxeter monodromy i, v517) and the δ_1 decision (the 16-component Weil completion closing exactly with the E1.5 residual $2.91 \to\sim 10^{-39}$; the μ_4 multiplier obstruction identified as a *character* of the orbit stabilisers by the exact koboundary test; the cancellation achieved by the twisted fibre block $f_1 f_3 = G$ — not the three sphere axions; all three preregistered testers failing under both derived solutions with the honest (N_1, N_2) rebalance rescuing nothing; the declared/derived measure tension stated as the mandatory fence, v518) and the measure decision (the exact invariant subspace = $\mathrm{span}\{e_H, e_{H'}\}$ forcing the unphased E_4 numerator once single-valuedness is derived from F-independence + the Quillen pairing; the consistency circle to v516 closed levelwise; the typed premises TP-1..TP-4 named, v520) — exactly the kind of sharp, falsifiable refinement the contract format is for. Status: `CELEST.SEAM.01` [O] (research contract, WP5e proper the single remaining open milestone); `CELEST.WP1.01` [E]/[C] (executed); `CELEST.WP2.01` [E]/[C] (executed); `CELEST.WP3.01` [E]/[C] (executed); `CELEST.WP4.01` [E]/[C] (executed); `CELEST.WP5A.01` [E]/[C] (executed); `CELEST.WP5B.01` [E]/[C] (executed); `CELEST.WP5C.01` [E]/[C] (executed); `CELEST.WP5D.01` [E]/[C] (executed, α stage); `CELEST.WP5DB.01` [E]/[C] (executed, β stage — the continuum uplift stays [O], Xu's theorem cited); `CELEST.WP5E.ALPHA.01` [E]/[C] (executed, α stage — the CFT-side prefactor + level pinning); `CELEST.WP5E.BETA.01` [E]/[C] (executed, β stage — the equivariant anomaly ledger; the global BCOV quantisation itself stays [O]); `CELEST.WP5E.GAMMA.01` [E]/[C] (executed, γ stage — the exchange no-go with certificate; the δ_2 ledger since executed by v511, the twisted contact terms δ_1 since decided by v518 — kill under the derived measure, tension with v516 stated); `CELEST.WP5E.DELTA2.01` [E]/[C] (executed, δ_2 stage — the full-tensor ledger, intermediate verdict: collapse confirmed by innerness, the cubic d-symbol channel opens T_3 with

the weaker certificate $\psi = 64$; the arity ≥ 4 ledger stays [O]; the $\psi = 64$ burden since discharged by M2/v516 — the cubic-GS question stays dissolved, the δ_1 decision (v518) does not revive it); `CELEST.DTERM.NONDERIV.01` [E] (executed, the c_d negative certificate — the $1920 = |W(D_5)|$ fence typed look-elsewhere-loaded and convention-contingent, the convention-stable core $c_d = 32 \times 60$; the fence stays [C], the generation of the 32 stays [O]); `CELEST.WP5E.EPS2.01` [E]/[C] (executed, ε_2 stage — the level-from-flux dial, verdict B: "one level" a theorem, the sector counter pins $k = 1$; the CPS dictionary on $\mathbb{PT}/\mathbb{Z}_4$ typed [C], the type-I B-model back-reaction milestones M1–M3 all since executed via v515/v516/v517); `CELEST.WP5E.EPS1.01` [E]/[C] (executed, ε_1 stage — the $O(-2)$ bulk-axion slot construction with $\tilde{\lambda} = 6$ triply pinned and the first GH/A_3 back-reaction step; the M1–M3 milestones it preregistered are all since executed; the BCOV-integral derivation stays [O]); `CELEST.WP5E.M1.01` [E]/[C] (executed, back-reaction milestone M1, SUCCESS on the preregistered criterion — the closed-form Ω_N with $(2\pi i)^2$-integral lockstep periods and the lens-forced source charge $4 = |\mu_4|$; honest fence: integrality alone is not the discriminator; M2–M3 since executed via v516/v517); `CELEST.WP5E.M2.01` [E]/[C] (executed, back-reaction milestone M2, SUCCESS on the preregistered criterion *on the declared completion measure* — the completion-weight identity $w = 4h = -4\,\mathrm{ch}_2$ with no free scale, every twisted channel the same perfect Okubo square $36\langle x, x\rangle^2/\det_j$, the $32\,T_3$ cancelled, both γ-stage certificates and the $\psi = 64$ slice supplied exactly, no cubic d-channel needed; the completion reading stays [C]-declared within that module — the δ_1 chain since decided by v518, and the completion reading since *derived at probe level* from two independent constructive sources (F-independence + Quillen pairing) under the typed premises TP-1..TP-4 by v520; the residual [O] is the constructive w_m derivation); `CELEST.WP5E.M3.01` [E]/[C] (executed, back-reaction milestone M3, SUCCESS on the preregistered criterion — the a_0 uplift on the GLT kernel $\chi = \log P_4$ with the log coefficient $4 = |\mu_4|$ coupled to the centre count on four scales, the period response $1/4$ integrating to the Coxeter monodromy i; the GLT dictionary stays [C], the full nonlinear Kähler potential stays [O]); `CELEST.WP5E.DELTA1.01` [E]/[C] (executed, δ_1 stage — DECIDED: kill under the derived measure; the 16-component Weil completion closes exactly, the μ_4 multiplier obstruction is a character (koboundary defects $(1,1,1)$, $\lambda(\gamma) = i^{2B+C/4}$ on $\Gamma_1(4)$), the cancellation is the twisted fibre block $f_1 f_3 = G$, and all three preregistered testers fail under both derived solutions; the $f_1 f_3 =$ KS-weight identification and the kernel-family convention stay [C]; the declared-vs-derived BCOV measure question — the stated v516 tension — was the named [O] and is since decided at probe level by v520 for the declared reading, sharpening this kill; the residual [O] is since narrowed by v523 to the global BCOV integral beyond the fibre zero-mode factor); `CELEST.WP5E.MEASURE.01` [E]/[C] (executed, the measure decision — ERFOLG-A on the preregistered decision layer: single-valuedness derived from two independent constructive sources (F-independence + the Quillen pairing), the invariant subspace exactly $\mathrm{span}\{e_H, e_{H'}\} = E_4$, the consistency circle to v516 closed levelwise, the $\mathbb{Z}_2$/EH anchor and the column canonicalisation both selecting the declared reading; typed premises TP-1..TP-4 stay [C]; the constructive BCOV derivation of the w_m normalisation itself — the named residual [O] — is since executed by v523); `CELEST.WP5E.WM.01` [E]/[C] (executed, the constructive w_m derivation — verdict ERFOLG per the frozen preregistration: the $1/\det_j$ fixed-point weight COMPUTED from three independent sources (the equivariant mode ledger with Abel value $(1/2, 1/4, 1/2)$; the zeta/Quillen determinant $4\sin^2(\pi j/4)$ with the unique real positive section; the δ_{1f} block constant term $1/\det_b$), the v516 chain reproduced number by number ($w = 4h$, $c = 4 = |\mu_4|$, squares $(18, 9, 18)$, total $45\langle x, x\rangle^2$, $\psi \pm 64$), all negative controls behaving (the $\mathbb{Z}_2$/EH anchor at $c = 2 = |\mathbb{Z}_2|$, the $SO(16)$ kill, the $\mathrm{diag}(i, i)$ break, the $k = 3, 5$ wandering); typed premises TP-REG/TP-Q/TP-NUM/TP-CH stay [C]; the global BCOV integral beyond the fibre zero-mode factor stays [O]); `SEAM.EQUIV.01` untouched, stays [O].

Research Contract — `WOIT.OS.TWISTOR.01` (the Osterwalder–Schrader twistor bridge)

Typing. **This is the actual bridge from compiler to physics — everything in `CELEST.SEAM.01` is preparation for it.** It is a research contract [O] (ledger row `WOIT.OS.TWISTOR.01`: Open, research contract), *not* a claim; nothing below asserts that the construction exists. The external programme it engages — Woit's Euclidean Twistor Unification (`arXiv:2104.05099`) — is a *named reference frame* for the shape of the target, never a confirmation in either direction (the Woit-bridge paragraph in `tfpt_3` states exactly what is missing).

Input. $X_+ = (\mathbb{PT}/\Gamma)_+$ (the positive-time half of the quotiented projective twistor space; $\Gamma = \Gamma_{\mathrm{ALE}} \cong \mathbb{Z}_4$, normalised by the clock — the definition block of `CELEST.SEAM.01`); $\mathcal{A}_{\mathrm{hol}}$, the *interacting* open+closed twistorial algebra (SDYM(E_8) open strings + BCOV closed strings — the WP5e-proper object, whose free/equivariant skeleton is pinned by v492–v517); ρ, the order-4 clock diag$(i, 1)$; Θ, the anti-linear real structure induced by the seam reflection; and $\mu_{\mathrm{BCOV+SDYM}}$, the interacting functional (the global BCOV+SDYM quantisation on $\mathbb{PT}/\Gamma$). *Precision (i), from the α stage:* "induced by the seam reflection" means the seam-circle *reflection*, not the deck/covering involution — the deck (whose free-ness is topology, v510) furnishes the $(-1)^F$ Kramers class $\Theta_t^2 = (-1)^F$, not the OS Θ; the seam-circle reflection furnishes $\Theta^2_{\mathrm{Fock}} = +1$ ([`verification/v519_woit_theta_rp_free.py`]). *Precision (ii):* on the RP side the μ_4 marks sit at the *bond midpoints* of the 16-Majorana seam circle (the half-offset/NS-natural placement, 4 sites per mark quadrant) — the cut through the sites fails reflection positivity exactly ([`verification/v519_woit_theta_rp_free.py`]). *Precision (iii), from the β_1 stage:* "gauge-invariant subalgebra" at the free level means the GSO/fermion-parity $\mathbb{Z}_2$ — the only θ-compatible part of the μ_4 tower (the free-fermion shadow of the E_8 glue: $(E_8)_1$ = even NS of $SO(16)_1$ + R spinor); the clock is the euclidean rotation itself (*timelike*: all 16 seam mirror axes invert it, none commutes), and its equivariant statement moves to β_2 — OS quotient first, then the clock as reconstructed transfer operator ([`verification/v522_woit_beta1_gso_gauge.py`]).

Target theorem. $\Theta^2 = 1$ and $\Theta\rho\Theta = \rho^{-1}$; the gauge-invariant algebra is reflection-positive; and the Osterwalder–Schrader quotient produces $(\mathcal{H}, \Omega, U(\mathcal{P}^{\uparrow}_+), \mathcal{A}_{\mathrm{Mink}})$ with: a positive Hilbert metric, positive energy, local causality, *one chiral fermion generation without mirror doubling*, the TFPT charge lattice, an internal $SU(3)\times SU(2)\times U(1)$ action, and the $(E_8)_1$ seam net as an *actual* boundary net (not only as a character).

Kill tests (pre-registered, all seven live). (1) Θ is incompatible with the clock (no anti-linear Θ with $\Theta^2 = 1$ and $\Theta\rho\Theta = \rho^{-1}$ exists on $\mathcal{A}_{\mathrm{hol}}$). **(2)** Reflection positivity fails after gauge fixing (the gauge-invariant subalgebra of $\mu_{\mathrm{BCOV+SDYM}}$ is not RP). *Named toy threat (no status change):* on the first genuinely interacting 16-Majorana Fidkowski–Kitaev seam toy (`SEAM.INT.FKTOY.01`, [`verification/v529_seam_interacting_toy_fk.py`]) Kill-Test 1 does *not* fire (Θ exists with $\Theta^2 = +1$ and clock-tower inversion), but Kill-Test 2 *fires at toy level* — RP breaks for every $g > 0$ by an interference mechanism, with the *straddle law* (RP fails exactly on quartet-straddled cuts, stays PD on quartet-avoiding ones; 24/24) as a concrete hurdle for the interacting stages (γ and beyond; the free/quotient-level β_3 is since executed, v565). Fence: one toy / one interaction class / [C] flat-band parent — this narrows the route (every $\mathcal{A}_{\mathrm{hol}}$ must protect RP against exactly this mechanism), it does not close it. *The hurdle, executed as a selector (`SEAM.STRADDLE.CONE.01`, [`verification/v534_seam_straddle_cone.py`]; no status change):* on the equivariant interacting mark family (independent couplings on the four mark quartets, both signs, every admissible cut — including the $\pi/2$ straddled clock axes the 24-entry law had not covered) reflection positivity survives for *exactly one* member × sign: $\delta = \pi/2$ with positive coupling, PD on all four cuts over $g \in \{1/32..8\}$ with the minimum eigenvalue lifted *away* from the RP boundary; every asymmetric member dies. The straddle law refines (asymmetric straddling kills, symmetric straddling protects), the literal leading-order cone formalization is dead (the protection is nonperturbative), and the threat paragraph gains its complement: a passing interaction class *exists* and its unique survivor is the alignment bit — the same member that carries the v528 twist-class order parameter. Toy-level evidence for a dynamical origin of the bit, not a derivation; the kill tests stay formally live on

$\mathcal{A}_{\text{hol}}$. *The law derived, and the counterweight given its first OS-viable cut geometry (2026-08-27,* `SEAM.INT.ODDSPLIT.01` + `SEAM.INT.OSAVOID.01`, *[verification/v972_seam_interaction_front.py]; toy-level census, the v529 fence inherited, no status change):* the straddle law refines to the *odd-split law* — the Jaffe–Pedrocchi sufficient condition for Majorana reflection positivity (`arXiv:1406.1384`, `arXiv:1601.01994`), made decidable, *predicts* RP survival on the centre-straddled (2+2 mirror-split) cuts and measurement confirms it, with the predicted-vs-measured RP confusion matrix perfect over all 35 (member, cut) cells; the μ_4-covariant quartic space (dimension 464) contains *no* JP-certifiable direction on all cuts. And the threat gains a positive counterweight: the quartet-avoiding cut family of the clock-invariant m=4 member generates the *full* even algebra (rank 15, dimension 32768) and carries a *positive* OS/GNS structure at g=1 (minimum eigenvalue $+4.43 \times 10^{-5}$) while the straddled control fails (-6.94×10^{-2}), and the μ_4 clock maps the avoiding pair onto itself — the kill-test-2 counterweight is *narrowed* (an OS-viable cut geometry exists), not removed. **(3)** The reconstruction produces a vector-like mirror generation (chirality is lost in the OS quotient). **(4)** The Penrose transform reaches only free / self-dual states (the interacting extension does not exist at the Costello–Li grade). **(5)** The reconstructed net has the right character but the wrong OPE / fails net equivalence (character-level agreement without operator-algebraic agreement). **(6)** The internal $SU(2)$ remains a spacetime factor (no genuine internal weak factor separates from the Euclidean rotation group). **(7)** The four μ_4 marks are not incidence-compatibly extendable over spacetime (the mark data of the seam cannot be propagated through the incidence relation).

The α stage — executed (*done*; the free/equivariant milestone `WOIT.THETA.FREE.01`, [`verification/v519_woit_theta_rp_free.py`]). The real structure *exists*, and free reflection positivity picks the *same* family. Exactly *two* families of anti-linear structures on $\mathbb{C}^2$ normalise the clock: family D ($z \mapsto \mu\bar{z}$, $|\mu| = 1$) satisfies $\Theta\rho\Theta = \rho^{-1}$ *exactly* with $\Theta^2 = +1$ (-1 impossible: $M\bar{M} = \mathrm{diag}(|\mu|^2, 1)$ — Kramers-free), inverts the deck and reflects the seam circle (2 cut points); family A ($z \mapsto \mu/\bar{z}$) centralises the clock projectively and *never* inverts it — it *defines* the euclidean section (Woit's ρ_{tw} side: $\rho_{\text{tw}}^2 = -1$, no real points, replicated exactly on $\mathbb{C}^4$), while family D is the OS conjugation (σ_{std}, real points $\mathbb{RP}^3$). Mark compatibility pins $\mu \in \mu_4$ (a 4-element torsor, 2 clock orbits); the $\mathbb{Z}_8$ spin plane has no phase leaks; in exact Cl(16), $\Theta_{\text{Fock}} = U_r \circ K$ has $\Theta_{\text{Fock}}^2 = 2^7 > 0$ (normalised $+1$), inverts the Fock clock tower ($V \mapsto 4096\, V^{-1}$) and normalises the deck, while the deck-induced candidate has $\Theta_t^2 = (-1)^F$ — the v510 split/nonsplit dichotomy *is* the $\Theta^2 = +1$ vs $(-1)^F$ dichotomy. Free RP holds on the bond cut with *no* degree truncation (one-particle Gram $(8, 0, 0)$ PD, even deg ≤ 2 sector $(29, 0, 0)$, complete $N = 8$ half-sided algebra PD; twist $\eta = +i$ forced), the site cut and the clock-centralising family fail RP exactly, and the anti-chiral state flips the odd sector to negative definite (the free shadow of kill test 3). **Kill test (1) therefore does *not* fire at the free/equivariant level — and stays formally *live* on the interacting algebra $\mathcal{A}_{\text{hol}}$**; kill tests (2)–(7) are untouched. No marker moves: the contract stays [O].

The β_1 stage — executed (*done*, verdict UNDECIDED per the frozen preregistration; the typed milestone `WOIT.BETA1.GSO.01`, [`verification/v522_woit_beta1_gso_gauge.py`]). "The clock is time-like, GSO is the gauge datum." The preregistered clock-tower reading of "gauge" (the $\mu_4/\Gamma_{\text{ALE}}$ equivariance averaged *before* the OS quotient) is structurally the *wrong* operationalisation: the one-step clock insertion $T_1 = \langle \theta(e_a), \alpha_S(e_b) \rangle$ violates Hermiticity *exactly* (witness entry $-i/(8\sin(5\pi/16))$ where the conjugate transpose demands $+i/(8\sin(5\pi/16))$); the pairing is complex-*symmetric*: 745 matching, 96 anti-matching, 0 violations — "OS-symmetric" is strictly weaker than Hermitian, so "positivity after gauge fixing" is not even well-posed for the clock reading). The mechanism is typed exactly: *all* 16 dihedral reflection axes of the NS seam circle invert the clock lift ($RSR^{-1} = S^{-1}$), none commutes — the μ_4 clock *is* Woit's euclidean rotation, and the θ-compatible (gaugeable) part of its $\mathbb{Z}_8$ Fock tower is exactly the 2-torsion $\{1, (-1)^F\}$ = the GSO/fermion-parity $\mathbb{Z}_2$ (Hermiticity census over all powers; α^4 = the 2π rotation of the seam = fermion parity). Under that corrected typing **gauge-fixed RP holds exactly**: the parity-projected bond-cut Gram is $(29, 0, 0)$ PD at $N = 16$ (deg ≤ 2, min eigenvalue 1.78×10^{-6} at 40 digits) and $(8, 0, 0)$ PD on the complete $N = 8$ half algebra (no degree truncation); the site-cut defect *survives* gauge fixing ($(7, 9, 6)$ — the bond placement of precision (ii) is gauge-invariant

information); family A stays indefinite ($(17, 12, 0)$ vs family D's $(29, 0, 0)$: the α-stage triangulation survives); the Ramond (wrap $+1$) "true $\mathbb{Z}_4$" lift fails state invariance (the NS/$\mathbb{Z}_8$ tower is forced); and the Kramers class stays trivial on the invariant sector ($\Theta_t^2 = \mathrm{id}$ on $\mathcal{A}^G$, Θ restricts with $\Theta^2 = +1$). **Kill test (2)'s free shadow does *not* fire** (no legitimate Hermitian gauge-fixed form has a negative direction), and kill test (1) stays discharged also gauge-invariantly; both stay formally live on $\mathcal{A}_{\mathrm{hol}}$. Transparency (documented in the module): the first frozen run scored 6/13 — the clock-reading checks failed exactly as the mechanism predicts — and the preregistered deg-2 clock-invariant dimension 30 was a combinatorial error (correct orbit census $28 + 4 = 32$); the verdict follows the frozen logic (Hermiticity failure ⇒ structural-failure branch ⇒ UNDECIDED). β_1 *proper* (the clock-equivariant statement) is re-routed through β_2. No marker moves: the contract stays [O].

The β_2 stage — executed (*done*, verdict SUCCESS per the frozen preregistration, [C]-typed per contract precision (iii); the milestone `WOIT.BETA2.OS.01`, [`verification/v524_woit_beta2_os_quotient.py`]**).** "The OS quotient of the free system made explicit." $\mathcal{H}_{\mathrm{phys}} = \mathcal{A}_+/\mathcal{N}$ is *explicit and nondegenerate* at both levels: at $N = 16$ ($\deg \le 2$) the 37×37 bond-cut OS Gram is exactly Hermitian and parity-block-diagonal with inertia $(37, 0, 0)$ PD (min eigenvalue 1.7801×10^{-6} at 40 digits) and null space $\{0\}$, $\dim 37 = 29 \oplus 8$; at $N = 8$ the *complete* half algebra gives $(16, 0, 0)$ PD with $\dim 16 = 8 \oplus 8 = 4^2$ — compact euclidean time reconstructs a *thermal* (KMS) representation, with the exact certificate $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$. The euclidean rotation becomes a Klein–Landau *local symmetric semigroup*: $\tau_k = \omega(\theta(a)\,\alpha_k(b))$ is exactly Hermitian on every shrinking domain D_k ($k = 1..7$, dims $29, 22, 16, 11, 7, 4, 2$), the chain identities are exact and the vacuum is fixed; the positivity pattern *is* the α-stage site/bond dichotomy (even steps PSD via the exact square identity $T(2j) = A_j^* A_j$, odd steps indefinite with exactly zero one-particle diagonal — the one-step transfer is *not* positive: the free chirality datum, kill-test-3 shadow sharpened). **The clock is implemented on $\mathcal{H}_{\mathrm{phys}}$ as the positive self-adjoint transfer step** $T^{N/4}$ ($(11, 0, 0)$ PD, min 6.7×10^{-5}; compressed spectrum $[0.0195, 1.6414]$ with vacuum eigenvalue exactly 1; spectral projections certified at $\sim 10^{-40}$ — exactly the calculus the non-Hermitian pre-quotient average of v522 could not have), and at $N = 8$ the compressed clock spectrum is *exactly* $\{1, \sqrt{2} - 1\} = \{1, 1/\delta_S\}$ in both parity sectors (the silver axes of the v519 μ_4 torsor returning as the clock eigenvalue); the reconstructed rotation group $U(s) = e^{isH}$ is unitary with the group law — per precision (iii) the [C]-operationalisation of the contract phrase "the clock acting unitarily". The v522 non-Hermiticity is *resolved*: the census $(745, 96, 0)$ is reproduced, every anti-matching entry is a wrap overlap and every overlap-free entry matches — the pre-quotient failure was exactly the domain/wrap artifact. The pre-declared KMS deviation arrives with exactly the declared witnesses: *no contraction* on the compact circle (edge-mode expansion $C(1)/C(3) = 1 + \sqrt{2} = \delta_S$ exactly; $\det(G - \tau_4) < 0$) — a typed KMS finding, not a β_2 failure (the contract demands no contraction). GSO/Θ: the grading survives; θ_{cut} and the deck candidate do *not* act on $\mathcal{H}_{\mathrm{phys}}$ (support censuses); the *perpendicular* torsor mirror descends anti-unitarily ($\eta_\perp \in \{\pm i\}$) with $\Theta^2_{\mathrm{phys}} = +1$ on *every* sector — the quotient is Kramers-free — and $\theta_{\mathrm{cut}} \circ \theta_\perp = \alpha_{N/2}$ exactly (the torsor closes onto the deck). Controls: the site cut is indefinite ($(3, 3, 1)/(2, 2, 4)$ — the contract kill branch fires for the site placement), family A admits no quotient ($(4, 4, 0)$), the anti-chiral state flips to $(8, 8, 0)$ (quotient existence itself selects the chiral orientation), the wrong semigroup direction is non-Hermitian exactly off-domain (all mismatches wrap-overlap). **Kill tests (1) and (2) are strengthened and the (3) shadow is sharpened; none fires** — all seven stay formally live on $\mathcal{A}_{\mathrm{hol}}$. Transparency (documented in the module): run 1 scored 18/20 — a sympy simplify failure on a *true* $\pi/16$ product formula ($\sin(7\pi/16) - \sin(5\pi/16) = 2\cos(3\pi/8)\sin(\pi/16)$), fixed by a Laurent-polynomial certificate; no criterion, tolerance or expectation changed. No marker moves: the contract stays [O]; β_3 is executed next (below).

The β_3 stage, executed (`WOIT.BETA3.DUALITY.01`, [`verification/v565_woit_beta3_pt_duality.py`]**; verdict SUCCESS per the frozen preregistration; no status change).** The $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality is *constructed as a compatibility class* and typed against σ_{std}, with the contract kill branch

decided *empty by algebra.* **(i) The class:** a duality is a nondegenerate Hermitian form Φ on $\mathbb{C}^4$ ($\sigma(Z) = Z^\dagger\Phi$, mapping $\mathbb{PT} \to \mathbb{PT}^*$); clock compatibility has the entrywise residual $(\bar{\lambda}_j\lambda_k - 1)\Phi_{jk}$ and forces the four cross-eigenspace entries to zero exactly: two 2×2 Hermitian blocks, $16 \to 8$ real parameters (the deck adds nothing; the wrong clock lift $\mathrm{diag}(i, i, 1, 1)$ breaks the split). **(ii) The dichotomy and the forced** $(2, 2)$**:** pairing against Woit's Euclidean structure ρ_{tw} via $M_{\mathrm{tw}}^\dagger \Phi M_{\mathrm{tw}} = \varepsilon\bar{\Phi}$ splits the class exactly in two. The *Euclidean* branch ($\varepsilon = +1$; $\sigma_{\mathrm{std}} = \mathbf{1}$ lives here) has seam-line norm $a(|z|^2 + 1)$: no null point, no OS cut, ever — exactly the section-defining role Woit assigns it. The *Minkowski* branch ($\varepsilon = -1$) has seam-line norm $a(|z|^2 - 1)$: **the null locus meets the seam line exactly in the seam circle** — the OS cut is *forced*, not imposed — and the second block is exactly minus the conjugate of the first (charpoly mirrored), so the **total signature is** $(2, 2)$ **for every nondegenerate member**: Woit's Minkowski signature is a theorem of clock + ρ_{tw} pairing, not an input. **(iii) The kill is empty:** every compatible duality inverts clock *and* deck exactly (family D); a clock-centralising (family A) duality would need $\rho_4^2 =$ scalar, but $\rho_4^2 = \mathrm{diag}(-1, 1, -1, 1)$ — no solution; and family A independently has no quotient ($(4, 4, 0)$, the v524 control). **(iv) The identification:** every seam-preserving Minkowski member induces *one* projective conjugation $z \to -\bar{z}$ ($\mu = -1$, fixed points $\pm i$), perpendicular to σ_{std}'s $z \to +\bar{z}$; under the declared RP-side placement (v519 precision (ii); exact rational-π arithmetic, offsets $\mp\pi/2$) that member **is** $\theta_\perp$ — the mirror whose quotient descent v524 certified: Θ_{phys} is the shadow of the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality, anti-unitary, Kramers-free ($\Theta^2 = +1$ per sector), time-reversing the semigroup, closing with the σ_{std}-type cut onto the deck ($\theta_{\mathrm{cut}} \circ \theta_\perp = \alpha_{N/2}$; upstairs: the clock conjugation exchanges $\mu = +1 \leftrightarrow \mu = -1$). *Role separation complete:* $\sigma_{\mathrm{std}} \sim$ the OS cut (the metric), the Φ-twisted Minkowski duality $\sim$ the descending mirror. Free/quotient level only; $\mathcal{A}_{\mathrm{hol}}$ untouched; all seven kill tests stay live there; the declared conventions (seam line, seam preservation, placement, torsor relabeling) are cited, not invented. No marker moves: the contract stays [O]; γ is the next milestone.

Thermal follow-on (no status change). The free OS quotient of β_2 is KMS, and its temperature normalisation closes onto the Nariai/BH chain as the *third leg* of c_3 (`SEAM.THERMAL.KMS.01`, [`verification/v526_seam_thermal_kms_nariai_bridge.py`]: β measured $= N$ at both levels; $\beta_{\mathrm{angle}} = 2\pi = 1/(4c_3)$, $T_{\mathrm{seam}} = 4c_3$; modular Hamiltonian closed; class-1 anti-numerology selection; non-compact $\Rightarrow T = 0$). The silver clock eigenvalue is explained and demystified in parallel (`SEAM.CLOCK.SILVER.01`, [`verification/v527_seam_clock_silver_spectrum.py`]: structure SUCCESS, connection KILL). No marker moves.

The β/γ roadmap (named milestones, each with success and kill criteria). (β_1) Θ on the gauge-invariant subalgebra + gauge-fixed RP on the equivariant SDYM(E_8) sector — success: the v519 OS form extends to the gauge-invariant subalgebra with positivity after gauge fixing; kill: kill test (2) fires (RP fails after gauge fixing). *Executed via v522 (above): UNDECIDED under the frozen preregistration — the clock reading is time-like (wrong operationalisation), the gauge datum is the GSO $\mathbb{Z}_2$, gauge-fixed RP holds under that typing, neither kill fires; the clock-equivariant statement moves to β_2.* (β_2) The OS quotient of the *free* system made explicit — success: $(\mathcal{H}, \Omega)$ constructed from the v519 bond-cut form with the clock acting unitarily; kill: the quotient degenerates (zero or indefinite completion). *Executed via v524 (above): SUCCESS, [C]-typed per precision (iii) — $\mathcal{H}_{\mathrm{phys}}$ explicit and PD at both levels, the clock a positive transfer step with spectral calculus (exact $N = 8$ spectrum $\{1, \sqrt{2} - 1\}$) and a reconstructed unitary rotation group; the pre-declared KMS deviation carries exactly the silver witnesses; neither kill fires; β_3 is next.* (β_3) The $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality typed against σ_{std} — success: Woit's Minkowski conjugation ($\mathbb{PT} \to \mathbb{PT}^*$) identified with the family-D conjugation on the quotient; kill: the duality forces a clock-centralising (family A) structure instead. *Executed via v565 (above): SUCCESS — the kill branch is empty by algebra (every clock-compatible duality is family D; family A would need $\rho_4^2 =$ scalar), the OS cut and the $(2, 2)$ signature are forced on the Minkowski branch, and the unique induced seam member is $\theta_\perp$, the v524 Θ_{phys} carrier.* (γ) The chirality theorem + the mark incidence — success: the odd-sector definiteness flip of v519 upgraded to "one chiral generation without mirrors" on the reconstructed net, and the four μ_4 marks extended incidence-compatibly; kill: kill tests (3)/(6)/(7) fire. *The FREE slice is*

executed via [verification/v608_gamma_free_slice.py] (9/9, GAMMA-FREE-SLICE-LANDED): on the explicit v524 net the mirror state's odd sector is strictly negative definite *(sector-exact flip — every mirror fermion vector has negative OS norm, so no mirror mode survives reconstruction: kill test (3) cannot fire at the free level), the chiral generation is multiplicity-free (clock spectrum exactly* $\{1, \sqrt{2}-1\}$*), and the mark transport exists (*τ_2 *is the mark-A→B transport, Hermitian PD: kill test (7) does not fire at the free level).* γ *itself stays open on* $\mathcal{A}_{\mathrm{hol}}$ *(kill tests (3)/(6)/(7) formally live there); kill test (6) is untouched by the free slice. The INTERACTING toy slice is since executed via [verification/v615_gamma_toy_interacting.py] (11/11, GAMMA-TOY-LANDED): on the RP-surviving interaction class (the alignment bit* $\delta = \pi/2$, $s = +$, *v534) the mirror state's odd sector is* strictly negative definite *on every admissible cut and every coupling* $g \in \{1/32..8\}$ *(bounded away from zero — kill test (3) cannot fire on the surviving class), the mark transport* α_4 *is Hermitian PD at every g (kill test (7) does not fire at the interacting toy level), the odd transfer spectrum stays multiplicity-free (honest finding: strong-coupling near-doubling at* $g = 8$ *without closing), and the RP-dead member has no gamma home — the dynamical bit selection and the chirality data* co-locate *on the same member. Fence: one toy, one interaction class;* γ *proper stays open on* $\mathcal{A}_{\mathrm{hol}}$*. Kill test (6) has since received its first (kinematic) slice via [verification/v616_su2_internal_kinematic.py] (17/17, SU2-KINEMATIC-SEPARATED): on* $\mathbb{C}^4 = \mathbb{H}^2$ *the internal su(2) (the right quaternion action, with* ρ_{tw} *= right-j verbatim) preserves every twistor fiber, commutes exactly with the spacetime quaternion action and intersects it in* $\{0\}$ *— the internal factor* separates *kinematically, the clock factorizes exactly as left-*$e^{i\pi/4}$*∘right-*$e^{i\pi/4}$*, the ALE deck is purely spacetime-side, and the complex choice breaks the internal* $SU(2)$ *to* $U(1)$ *with the v519 mark torsor as its* μ_4*; the DYNAMICAL half of kill test (6) (a genuine gauge action on the reconstructed net) stays open and formally live on* $\mathcal{A}_{\mathrm{hol}}$*.*

Scope fence (explicit non-claims of the celestial/twistor branch). Self-dual is *not* full 4D physics. Until and unless this contract closes, the celestial/twistor branch claims *none* of the following: both helicities (the anti-self-dual completion), generic scattering amplitudes, local matter fields, full Einstein dynamics (beyond the self-dual sector), electroweak symmetry breaking, confinement. The classical field-equation route (entanglement equilibrium, v358/v359) is a separate, non-competing statement about the full equation; `SEAM.EQUIV.01` and its route split (`SEAM.EQUIV.MMST.01`/`SEAM.EQUIV.TWISTOR.01`) are stated in their own rows and are not moved by anything here. Status: `WOIT.OS.TWISTOR.01` [O] (research contract; the α stage is executed — `WOIT.THETA.FREE.01` [E], v519 — with kill test (1) discharged at the free level only; the β_1 stage is executed — `WOIT.BETA1.GSO.01`, v522, verdict UNDECIDED per the frozen preregistration, with kill test (2)'s free shadow not firing, the gauge datum typed as the GSO $\mathbb{Z}_2$ and contract precision (iii) added; the β_2 stage is executed — `WOIT.BETA2.OS.01`, v524, verdict SUCCESS, [C]-typed per precision (iii), with the OS quotient explicit and PD at both levels, the clock a positive transfer step with spectral calculus and a reconstructed unitary rotation group, and the pre-declared KMS deviation carrying exactly the silver witnesses; the β_3 stage is executed — `WOIT.BETA3.DUALITY.01`, v565, verdict SUCCESS, with the kill branch empty by algebra, the OS cut and the $(2,2)$ signature forced on the Minkowski branch of the duality class, and the unique induced seam member identified with the v524 Θ_{phys} carrier; no marker move).

Research Contract — `CCC.SEAM.CROSSOVER.01` (the conformal-cyclic crossover on the seam geometry)

Typing. A research contract [O] (ledger row `CCC.SEAM.CROSSOVER.01`: Open, research contract), *not* a claim. It engages Penrose's Conformal Cyclic Cosmology (Tod, `arXiv:1309.7248`; Penrose–Meissner, `arXiv:2503.24263`) as a *named reference frame* for the shape of the target — never a confirmation in either direction. The cyclic reading of `origin_theory` stays [C] regardless of this contract's fate; nothing below asserts that CCC follows from TFPT.

The certified kinematic base [E] (`[verification/v957_ccc_crossover_kinematics.py]`,

`CCC.SEAM.KINEMATICS.01`; six discovery probes embedded byte-exact, 28+29+18+21+14+7 gates). The corpus objects already carry the complete CCC crossover *kinematics*: (i) Penrose's reciprocal gauge $\hat{\Omega}\check{\Omega} = -1$ is the element $\tau = \rho^2\sigma : z \mapsto -1/z$ of the proven μ_4 Möbius stabiliser D_4 — τ reverses the clock (= the $\mathbb{Z}_2$ sheet parity), swaps the aeon poles $\{0, \infty\}$, and the crossover locus $|\hat{\Omega}| = |\check{\Omega}|$ is the unit circle carrying all four marks; (ii) conformal-factor rigidity: on the 2D seam the clock alone leaves exactly *one* constant-curvature modulus and the reciprocal $\mathbb{Z}_2$ kills it; on the 3D crossover the deck alone kills *all* conformal moduli (the $so(4,1)$ deck-commutant is 4-dimensional, all Killing, boost content 0; globally unique on the explicit Obata family) — TFPT's candidate answer to CCC's open unique-conformal-factor problem, with v_{geo} staying pure unit calibration (`VGEO.TORSOR.01`); (iii) the entropy reset of the cyclic reading is a *reduced channel*: the v221 transport is CPTP with an explicit Stinespring isometry, $\|T^n - P_\star\|_2 = (2/3)^{6n}$ exactly, the global overlap invariant, the microstate record in the complement — "unitarity + rank-1 attractor" is isometry + partial trace, not a contradiction (with the executed sharpening that this deployed Kraus dilation is *entanglement breaking*: it stores the record entirely in the environment and carries no coherent transport in the visible three-state system, quantum capacity 0, [`verification/v976_seam_lift_birkhoff.py`]); (iv) the kinematic 2D → 3D → 4D uplift: the seam sphere *is* the Hopf base of S^3 (stereographic = z_1/z_2 exactly), the crossover $X = S^3/\mathbb{Z}_4$ is the lens boundary of the corpus A_3 ALE (deck free, clock and reciprocal descend, marks = four linked fibre circles in the two sheet orbits), the 4D bridge is the cone $\mathbb{C}^2/\mathbb{Z}_4$ with the inversion as conformal aeon swap fixing the unit lens pointwise, and the Lorentzian shadow $(1/\cos^2 T)(-dT^2 + g_{S^3})$ is Einstein $R_{ab} = 3g_{ab}$ with a spacelike round-S^3 crossover and $\check{\Omega} = -\cos T$ vanishing to first order; (v) the KMS/modular closure at the free level: $T = e^{-H_{\mathrm{mod}}}$ with $\mathrm{spec}(H_{\mathrm{mod}}) = \{0,\, 6\ln\frac{3}{2},\, 6\ln 3\}$ exactly, detailed balance and OS positivity exact, and the v526 normalisation $\beta_{\mathrm{angle}} = 2\pi = 1/(4c_3)$ = the Gibbons–Hawking normalisation on the crossover fibre — one discrete clock tick = one unit of modular time; (vi) the holonomy dictionary: pole strings carry the full μ_4 monodromy, every seam string carries exactly the 2-torsion -1 = the v522 GSO/fermion-parity wrap (typed **[C]** correspondence); (vii) the selector necessity: the admissible OS reflections of the mark family $M(\delta)$ close to D_4 *iff* $\delta = \pi/2$, and at Majorana operator level (fermionic signs kept) the clock-axis cuts are Θ-real only at the μ_4 configuration — the *uniqueness* half of the v534 straddle selector is thereby *derived* as a Θ-reality necessity (the survival half and the coupling sign stay v534's nonperturbative toy facts, cited); (viii) the frozen Gate-6 kernel: a sharp causal top-hat disc of radius $\theta_{\max} = \eta_{\mathrm{rec}}/(\eta_0 - \eta_{\mathrm{rec}}) \approx 1.16°$, decay rates $\ln(729/64)$ and $\ln 729$ (ratio $\ln 3/\ln\frac{3}{2} = 2.709511$), κ derived from the v526 normalisation up to a declared convention band (relic contrast $\leq 3.3 \times 10^{-4}$ across all of it) — frozen *before* any data contact and feeding the preregistered search `experiments/ccc-crossover-disc` (hypotheses, kill criteria K1–K6, null battery and decision rule frozen there; verdict slot `data_limited`, no data pass).

Demands (the dynamical crossover, in order). (D1) the interacting seam algebra $\mathcal{A}_{\mathrm{hol}}$ realised *on* the lens crossover with reflection positivity protected on the mid-gap cut class while the aeon transfer uses the through-mark class (the two D_4 reflection classes whose product is the clock; the v529 straddle law is exactly this separation at toy level) — this *is* `WOIT.OS.TWISTOR.01` stage γ in the crossover geometry, and closing it closes D1; **(D2)** the aeon-transfer channel as a Stinespring isometry at *field* level (the v957 finite-dimensional dilation lifted to the reconstructed net); **(D3)** the cosmology fork *decided*: exactly one active story — Starobinsky ($r = 12/N_\star^2 \in [0.0033, 0.0048]$, within reach of $\sigma(r) \sim 10^{-3}$ experiments), a crossover/gravitational-wave epoch, or a computed hybrid; **(D4)** the preregistered disc search executed per its frozen hypotheses — **EXECUTED 2026-08-24 with a robust null** (first data contact, Planck PR3 SMICA map, integrity SHA-256 pinned in `results.json`, 2 013 312 960 bytes; freeze intact 3/3 hashes, frozen-kernel guards 5/5 *before* data contact; seed 20260824, reruns bit-identical): max $|\mathrm{SNR}| = 5.119$ on data vs null mean 5.20 over 100 Gaussian ΛCDM sims (matched mask and spectrum), $p_{\mathrm{global}} = 0.673$, $p_{\mathrm{counts}} = 0.446$, BH-$q_{\min} = 0.673$ — the $q < 0.01$ threshold clearly missed; 171 candidates $|\mathrm{SNR}| \geq 4$ with 83 pos / 88 neg (no sign excess); injection recovery 100% ($20 \times 200\,\mu$K discs at 1.16° — the search is well-powered, the null is a real null); kill criteria K1–K6 all NA (they fire only on a resolved relic;

none exists); the replication legs (half-mission, Commander/NILC) are not required for a null per the frozen decision rule and stay available for future candidates; three amendments disclosed (seed pinning of the unseeded synfast preflight, reporting fields, the documented estimator ℓ-weighting) — statistics, thresholds and kill criteria unchanged. **No kill criterion fired; the contract stays open via D1–D3; no type change.**

Kill criteria (frozen). A free fitted conformal factor anywhere; failure of the mid-gap RP protection on the interacting algebra (kill test (2) of the WOIT γ stage); the interacting level breaking the v522 wrap $= (-1)^F$ identification (which would reopen the deck fork the kinematics closed conditionally); a measured r incompatible with the surviving cosmology branch; a resolved crossover relic violating the frozen template class (K1–K6, including the K6 contrast bound that would kill the derived-κ chain); and — typed numerological, forbidden — any identification of the four μ_4 marks with Hawking-point *counts*.

Honest scope. Deck fork: topology searches cannot decide it (lens geodesic $\geq 28\, D_H$ vs matched-circle reach $6.3\, D_H$; 32σ from curvature detectability) — it is decided *conditionally* by the v522 wrap identification (spacetime branch) and stays typed. The executed D4 null leaves the compiler core untouched (the frozen kernel was *output* of the certified kinematics, not input) and asserts nothing about CCC in either direction; the cyclic reading stays [C]. The contract now closes only through D1–D3; the kinematic base [E] stands on its own either way.

Firewall Contract — `DIMENSION.UPLIFT.FIREWALL.01` (no 1+1D theorem closes a 3+1D claim)

Typing. A *standing discipline rule*, in force immediately (ledger row `DIMENSION.UPLIFT.FIREWALL.01`); the uplift theorems it names are separate [O] contracts. It formalizes the core diagnosis of the 2026-08-27 external deep review: *a chiral 1+1D seam net does not imply a local, interacting 3+1D quantum world.* The corpus's strongest finite results are boundary/kinematic building blocks — the review found several surfaces where boundary theory, continuous relaxation, modular time, Lorentz geometry and physical 4D dynamics were in danger of being silently equated. This firewall separates them by rule.

The rule (two clauses). (F1) No theorem whose hypotheses live in 1+1D (a chiral net, a VOA, a lattice collar, the seam circle) may be cited as *closing* a claim about 3+1D physics. **(F2)** Every contract closure must state the spacetime dimension of its *hypotheses* and of its *conclusion* (dimension typing), with the same discipline as the [E]/[C]/[O]/[X] markers.

Instances audited at registration (honesty census). (i) `SEAM.EQUIV.01` and both its routes are 1+1D in *and* out: the cited scaling-limit framework (Osborne–Stottmeister, `arXiv:2107.13834`) treats 1+1D free lattice fermions, the OS-reconstruction selector (Adamo–Moriwaki–Tanimoto, `arXiv:2407.18222`) treats 2D full VOAs, and lemma L2 of the externalization package below claims convergence to a *chiral* c=8 *CFT* — nothing more. **Even a fully proved `SEAM.EQUIV.01` is a rigorous boundary building block, not the 4D theory.** (ii) The `WOIT.OS.TWISTOR.01` target theorem (Poincaré representation, local causality, one chiral generation, the internal gauge action) *is* the uplift demand; it stays [O] with all seven kill tests live. (iii) The kinematic 2D → 3D → 4D uplift (v957; the discovery probe `experiments/tfpt-discovery/ccc_uplift_4d_probe.py` correctly self-labels as kinematic) supplies a canonical Hopf fibre / lens crossover / 4D cone with an Einstein Lorentzian shadow — *none of which* closes this firewall: a metric uplift is not an interacting local quantum theory. (iv) Anomaly inflow: the non-gappable c_-=8 edge is naturally the boundary of a higher-dimensional *invertible/topological* bulk (Witten–Yonekura, `arXiv:1909.08775`; Hsieh–Tachikawa–Yonekura, `arXiv:2003.11550`); the inflow bulk and the *physical* 3+1D spacetime bulk are two different identifications that must never be silently equated.

The realistic route ordering (registered as the programme skeleton). (1) keep the seam as an internal/topological boundary module (all current [E] results survive unchanged as such); **(2)** build a genuine 4D Euclidean lattice theory — TFPT fixes the representations,

charges, coupling relations and boundary conditions; the 4D lattice carries spacetime, gauge fields, Higgs and fermions; **(3)** prove the 4D OS package on it (`QFT4D.OS.RECON.01` below); **(4)** prove chirality and mirror decoupling *in the same 4D model* (`CHIRAL4D.NOMIRROR.01` below); **(5)** only then apply Bisognano–Wichmann/CHM and entanglement gravity (the direction discipline of `GRAV.NONCIRCULAR.01` below); **(6)** then prove bulk–seam equivalence, identifying the seam as the topological/anomalous boundary condition of the *constructed* 4D bulk (`SEAM.BULK4D.RECON.01` below). The twistor/celestial alternative (reconstruct the bulk from boundary correlators) is riskier: boundary correlators do not guarantee bulk locality (cf. `arXiv:2602.04223`); it stays a named parallel route (`WOIT.OS.TWISTOR.01`/`CELEST.SEAM.01`), not the trunk.

Status. Firewall in force (F1/F2 bind every future closure); the uplift theorems live in `SEAM.BULK4D.RECON.01`, `QFT4D.OS.RECON.01` and `CHIRAL4D.NOMIRROR.01`, all [O].

Research Contract — `SEAM.BULK4D.RECON.01` (from seam data to a 4D local quantum theory)

Typing. A research contract [O] (ledger row `SEAM.BULK4D.RECON.01`: Open, research contract), *not* a claim. It makes explicit the theorem that was previously only implicit between the `SEAM.EQUIV.01` conclusion (a chiral c=8 boundary object) and the `WOIT.OS.TWISTOR.01` goal statement (a Poincaré-covariant, causal, chiral 4D theory): **the missing seam→bulk reconstruction is a theorem of its own, and no existing corpus result closes any part of it.**

Input. The certified seam data: the gapped μ_4-equivariant collar (v302), the index-4 Q-system (v125/v154), the $(E_8)_1$ character and modular data (v377/v378), reflection positivity (v379), the clock and mark structure (v519–v524), and the kinematic uplift dictionary (v957).

Target theorem (all seven demands; none is currently closed). From seam data alone, construct a 4D family of local algebras $\{\mathcal{A}(O)\}_{O\subset\mathbb{R}^{1,3}}$ (equivalently a family of Schwinger functions) and prove: **(B1)** 4D locality/microlocality (spacelike commutativity; the correct wavefront sets); **(B2)** Poincaré covariance with a positive energy spectrum (the spectrum condition); **(B3)** spin–statistics and CPT; **(B4)** *both* helicities (the anti-self-dual completion — the known hard wall of the twistor route); **(B5)** cluster decomposition; **(B6)** a nontrivial S-matrix (the theory is interacting, not generalized-free); **(B7)** a *unique* seam↔bulk operator dictionary (which boundary operator is which bulk field, uniquely — not only at character level).

Kill tests (pre-registered). (1) every 4D extension of the seam state is a generalized free field (B6 fails structurally); **(2)** the inflow bulk of the c_-=8 edge is *provably* only invertible/topological, with no local propagating completion carrying both helicities (B1/B4 fail); **(3)** the dictionary is non-unique at operator level (character-level agreement with OPE mismatch — the same failure mode as kill test (5) of `WOIT.OS.TWISTOR.01`).

Honest relations. `WOIT.OS.TWISTOR.01` (twistor OS quotient) and the route ordering of `DIMENSION.UPLIFT.FIREWALL.01` (constructive 4D lattice first, seam as its boundary condition) are the two candidate proofs; this contract is route-agnostic and states only the target. A Hopf-fiber uplift, a 4D cone, or an Einstein metric on the bridge (v957) does *not* close it. Status: `SEAM.BULK4D.RECON.01` [O].

Spectator obstruction (2026-09-04, `[verification/v1020_spectator_sector_obstruction.py]`). Demand (B7) as worded—a *unique* seam↔bulk operator dictionary from seam data alone—is unsatisfiable: for any decoupled anomaly-free sector C with trivial boundary, $\partial(B\otimes C) = \partial(B)\otimes\mathbf{1} = \partial(B)$, so B and $B\oplus C$ share every seam datum. A finite free-fermion witness (`[verification/v1020_spectator_sector_obstruction.py]`) exhibits this concretely (Chern, edge modes and seam-restricted covariance identical for B and $B\oplus C$; the bulk differs). Uniqueness is re-scoped to `BULK.PRIME_COMPLETION.01` below (prime/seam-generated completions, trivial relative commutant); the status of `SEAM.BULK4D.RECON.01` is unchanged [O].

Research Contract — BULK.PRIME_COMPLETION.01 (prime/seam-generated completions, no spectator)

Typing. A research contract [O] (ledger row BULK.PRIME_COMPLETION.01: Open, research contract), *not* a claim. Registered 2026-09-04 from the r644 audit harvest; it replaces the "unique from seam data alone" reading of SEAM.BULK4D.RECON.01/B7 by uniqueness *in the category of seam-generated (prime) completions*. **Dimension typing** (DIMENSION.UPLIFT.FIREWALL.01 F2): the hypothesis is 2+1D free-fermion, the conclusion is a re-scoped 4D demand.

Demand. (1) an explicit family of seam-derived subalgebras $\mathcal{A}_i$ (the seam CAR algebra, its time evolution, inclusions/intersections); (2) generatedness $\mathcal{A}_{\text{bulk}} = \text{vN}(\bigvee_i \mathcal{A}_i)$; (3) *no-spectator*: trivial relative commutant $(\bigvee_i \mathcal{A}_i)' \cap \mathcal{A}_{\text{bulk}} = \mathbb{C} \cdot \mathbf{1}$, equivalently no decoupled tensor factor with trivial boundary map; (4) uniqueness *only* within this category; (5) stability of (2)/(3) in the 3+1D limit.

Kill tests (pre-registered). A seam-generated completion with nontrivial relative commutant, or two inequivalent prime completions.

Finite witness ([verification/v1020_spectator_sector_obstruction.py]). In the 2+1D free-fermion strip, stacking a trivial gapped sector C (Chern 0, gap > 1) onto the seam model B leaves every seam datum unchanged (Chern $C_B = C_{B\oplus C} = -1$, $C_C = 0$; zero modes 2/0/2 with B-support 1; edge dispersion coincident to 1.4×10^{-15}; seam-restricted covariance identical to 0; edge-profile overlap 1) and changes the bulk (dimension 16 vs 32, $E_0(C) = -25.31$, mid-strip covariance rank 4 vs 8, $\|\Gamma_0 - \Gamma_\alpha\|_{\text{HS}}$ 0.752536 vs 0.760202). Krylov generatedness and the commutant test separate B from $B \oplus C$ by exactly $\dim C$ at every momentum; $\Pi_C = 0 \oplus \mathbf{1}$ lies in the commutant (residual 0); selection is unique (B selected, $B \oplus C$ rejected). *Honest:* spectator_dim$(B) = 10$ at $p = 0$ only (no onsite term at $M = 1$), so the criterion is "generated almost everywhere in p", not pointwise. The 2D modular-invariant uniqueness of the holomorphic $(E_8)_1$ net is untouched and is *not* a 4D no-spectator theorem. Status: BULK.PRIME_COMPLETION.01 [O].

Research Contracts — DYN.MARKOV.EMBED.01 / DYN.UNITARY.DILATION.01 (the three meanings of "dynamics")

Typing. Two contracts that split what the corpus has loosely called "dynamics" into its three inequivalent meanings: **(a)** the positive relaxation T^n to a fixed point; **(b)** a continuous dissipative semigroup $e^{t \log T}$; **(c)** reversible Lorentzian/unitary evolution e^{-itH}. Step (a)→(b) is now a machine-checked theorem for the concrete seam kernel ([E] at the finite level); step (b)→(c) is genuinely open and gets its own contract (its *discrete* collision/QCA leg is meanwhile executed exactly, [verification/v984_markov_qca_dilation.py]; the continuous legs remain the contract). Nothing here moves CONTRACT.F.01 or the four observables.

DYN.MARKOV.EMBED.01 — the finite kernel half is executed [E]. For the v221 seam transfer kernel $T = J + (2/3)^6\, u_2 u_2^{\mathsf{T}} + (1/3)^6\, u_3 u_3^{\mathsf{T}}$ (spectrum exactly $\{1, 64/729, 1/729\}$), the principal logarithm $Q = \log T$ is an *exact classical Markov generator* [verification/v971_markov_embedding_generator.py]: sympy-exact spectral calculus gives $e^Q = T$; every row of Q sums to zero *symbolically*; the three independent off-diagonal entries are exactly $\{\ln(9/8),\, 2\ln 3,\, 2\ln 3\} = \{0.117783,\, 2.197225,\, 2.197225\}$, all positive (the reviewer's independently verified numbers), with diagonals $-\ln(81/8), -\ln(81/8), -4\ln 3$; the uniformization certificate $Q + (4\ln 3)I \geq 0$ entrywise (diagonals $\ln 8, \ln 8, 0$, exact) proves $e^{tQ} = e^{-4t\ln 3}\, e^{t(Q+4\ln 3\, I)}$ entrywise nonnegative and doubly stochastic for *all* $t \geq 0$ — so T is embeddable and every root $T^{1/n} = e^{Q/n}$ is again a stochastic matrix. The generator spectrum is $\text{spec}(Q) = \{0, -6\ln(3/2), -6\ln 3\} = -\text{spec}(H_{\text{mod}})$ (v957: $T = e^{-H_{\text{mod}}}$), the gap is the seam gap $\Delta = 6\ln(3/2)$, and Q is symmetric (detailed balance w.r.t. the uniform state — the KMS/OS-positive free level). *Negative control (embeddability is not generic):* the same construction with second eigenvalue $-1/10$ is still doubly stochastic but has $\det = -1/200 < 0$, while every embeddable stochastic matrix has $\det e^Q = e^{\text{tr}\, Q} > 0$ — provably *not* embeddable. The RP/OS spectral positivity of the seam kernel (CONTRACT.F.01 axiom 3) is

exactly what makes (a)→(b) real. *What stays [O] in this contract:* the same embedding statement at *field* level (on the reconstructed net / quasi-local algebra, uniformly in system size), not just on the 3-state kernel.

`DYN.UNITARY.DILATION.01` — the (b)→(c) step, [O]. The semigroup e^{tQ} is *irreversible statistical* dynamics — an H-theorem, not a Schrödinger equation. Fundamental quantum dynamics requires the chain $T = e^{-aH} \xrightarrow{\text{OS}} U(t) = e^{-itH}$ with H positive self-adjoint, and the dilation must be *physical*: **(D1)** a Stinespring/Hamiltonian dilation of the semigroup whose generator is (quasi-)local; **(D2)** size consistency (the dilation is compatible with the thermodynamic limit — no environment that grows pathologically with the system); **(D3)** a Lieb–Robinson bound (finite propagation speed for the dilated dynamics); **(D4)** the OS analytic continuation to a unitary one-parameter group on the reconstructed Hilbert space. *Kill tests:* every dilation of the seam semigroup is provably nonlocal; the Lieb–Robinson velocity diverges in the scaling limit; the OS continuation fails on the interacting algebra. *Relation:* demand D2 of `CCC.SEAM.CROSSOVER.01` (the Stinespring isometry at field level) already presupposes this contract; it is hereby named as the prerequisite.

Executed sharpening — the lift structure of the finite kernel ([E] at matrix level; the contract itself stays [O]). Two modules now pin down *which* dilations are even candidates. **(i)** *The no-go*: the Kraus dilation the corpus actually deploys, $K_{ij} = \sqrt{T_{ij}}\,|i\rangle\langle j|$, is measure-and-prepare — one step kills every coherence (fully symbolic), its Choi matrix is diagonal in the product basis, hence the channel is *entanglement breaking* with quantum capacity 0, and $T > 0$ entrywise gives zero-error capacity $C_0(T) = 0$ exactly ([`verification/v976_seam_lift_birkhoff.py`]); any dilation satisfying D1–D4 must therefore be a genuinely different object. **(ii)** *The hidden fibre*: the complete set of random-permutation (Birkhoff) quantizations of the single step B is a one-parameter line $t \in [0, 1/18]$ whose coordinate is the sign-character Fourier coefficient ($\widehat{w}_t(\mathrm{sgn}) = 6t$); populations are exactly t-free, the one-step coherence spectrum is $\{1,\ 6t,\ (2/3)^{\times 2},\ (1/3)^{\times 2}\}$, and the triangle loop current is an exact eigen-observable $\Psi_t(A) = 6t\,A$ — one step and one current readout measure t without an ancilla. On the contrast qubit the Choi partial transpose has spectrum $\{-3t/2,\ 1/6 + 3t/2,\ 1/3 + 3t/2,\ 1/2 - 3t/2\}$: the deployed sixth-root spectrum sits *exactly* on the entanglement-breaking boundary because $2/3 + 1/3 = 1$ (mutated spectra miss by $\mp 1/12$), while at the full three-level the family stays NPT at every tested iteration, $\lambda_{\min} = -(1/3)(2/3)^n$, t-independent ([`verification/v976_seam_lift_birkhoff.py`]). **(iii)** *The coherent end*: B is *unistochastic* — an exact $U \in SU(3)$ with $|U_{ij}|^2 = B_{ij}$ entrywise, unique up to rephasing and complex conjugation, with Jarlskog invariant $J = \pm 1/27$ exactly; no *fixed* microstep unitary generates the relaxing sequence ($B^2 \neq |U^2|^2$; a Dirichlet recurrence witness returns U^{n^*} arbitrarily close to the identity while B^{n^*} is uniform), so repeated steps *require* environment contact between steps — precisely the D1–D4 demand — while the six-step macro-gate exists as its own reversible $U_6 \neq U_1^6$ ([`verification/v977_transfer_unistochastic_wilson.py`]). The classical kernel thus admits the full lift trichotomy {EB dilation | Birkhoff fibre with hidden circulation t | reversible macro-gate with hidden orientation $\mathrm{sgn}\, J$}: the same classical data, three inequivalent quantum dynamics. Ledger rows: `TRANSFER.HIDDEN.CIRCULATION.01` and `TRANSFER.COHERENT.WILSON.01` (executed halves [E]; the fibre-point/branch selection, the defect–Birkhoff forcing functor — named `TRANSFER.DEFECT.BIRKHOFF.01` — and the center–transport intertwiner stay [O]). These results fix the contract's boundary conditions; they do not close it.

Executed update (2026-08-28, external master-object review wave 2): the *discrete* leg of D1–D3 is realized constructively — the exact collision-model dilation ([`verification/v984_markov_qca_dilation.py`]); the continuous legs stay [O]). One fresh ancilla qutrit per step, the mod-3 record $C|j,k\rangle = |j, j{+}k\rangle$, and the step unitary $V = C\,(U_B \otimes I)$ (with U_B the v977 lift) give, as *exact symbolic operator identities*: $V^\dagger V = I$; the Stinespring identity $\Phi(\rho) = \mathrm{Tr}_A[V(\rho \otimes |0\rangle\langle 0|)V^\dagger] = \sum_j P_j U_B \rho U_B^\dagger P_j$ (Kraus completeness $\sum_j K_j^\dagger K_j = I$); $\mathrm{diag}\,\Phi(\mathrm{diag}\,p) = Bp$ with vanishing off-diagonals; and six fresh ancillas give $\mathrm{diag}\,\Phi^6 = B^6 p = Tp$ — the v977 macro-gate as the reduced dynamics of an exactly unitary, strictly radius-1 collision band (unitary amplitudes → local environment coupling → dephasing → Markov transfer, in

one construction). The must-fail fires (the closed unitary has $|U_B^2|^2 \neq B^2$), and a typed remark *against* the review text: a *reused* mod-3 adder ancilla still reproduces the populations exactly (the record $j_1{+}i$ stays distinguishable) — fresh ancillas are needed for the coherences, not the diagonal chain. The conjugate branch V^* dilates the same B ($\operatorname{sgn} J = \pm 1/27$ rides through classically invisible; its selection stays [O]). What stays [O]: the (quasi-)local *Hamiltonian* dilation, D2 in the thermodynamic limit, the OS continuation $T = e^{-aH} \to U(t) = e^{-itH}$ (D4), and any 3+1D reading.

Executed update (2026-08-28, promote round): the OS/dilation *ladder* is executed at kernel + free-band + interacting-kernel + quantum-coupled level (`[verification/v987_os_dilation_package.py]`; continuous field-level OS and branch selection stay [O]). The seam single-step $B = P_1 + \frac{2}{3}P_2 + \frac{1}{3}P_3 > 0$ with exact orthogonal projectors; $H = -\log B$ is PSD with gap $\log \frac{3}{2}$; $T(t) = e^{-tH}$ is a stochastic semigroup inside the cone; $U(t) = e^{-itH}$ is a unitary group with Wick anchor $U(-in) = B^n$ for $n = 1..6$. Must-fail: the v971 negative kernel (second eigenvalue $-1/10$) has no real half-step. Size-uniform free band: $T_L = B^{\otimes L}$ has commuting local $H_L = \sum_x h_x$ with gap $\log \frac{3}{2}$ uniform in L; the per-cell reduced channel is B at $L = 1, 2$; Lieb–Robinson radius 1. Interacting congruence: $T_\varepsilon = K^{1/2} B^{\otimes L} K^{1/2}$ is SPD for every tested diagonal coupling (one-sided mutant $KB^{\otimes L}$ is not symmetric); gap plateau ~ 0.18 at $L = 2..4$. Quantum SWAP coupling: $d(\theta) \sim \theta^{1.99}$, CPTP preserved, LR radius 1. Display marker stays [O].

Executed update (2026-08-29, review wave 4): the finite ladder is now *complete* through continuous time (`[verification/v999_weak_collision_hp.py]`; the thermodynamic field limit stays [O]). The deployed strong collision is singular (rank 9/81) and cannot be the exponential of a finite GKSL generator. The typed successor is a fresh-ancilla weak-collision / Hudson–Parthasarathy family: generator extraction Richardson 1.098×10^{-10}; $\operatorname{diag} L_{\mathrm{GKSL}} = Q = \log B$ symbolically; $e^Q = B$ exact; GNS splitting defect 0; locality 1.355×10^{-20}. The finite ladder is now discrete / kernel / field-free / interacting-classical / quantum-coupled / continuous-time. Open = the thermodynamic field limit only. Display marker stays [O].

Precision (binding for all "universal dynamics" prose). The shared relaxation shape of the four `F_transfer` sectors (`FR.DYNAMICS.01/DYN.TRANSFER.UNIVERSAL.01`) is a *universal contraction class* — shared gap, shared multiplier algebra — *not* one physical clock: both thermal-time routes are machine-killed (v723/v724) and the frozen external-clock contract fired its K1 kill exactly (v777: no common continuous clock exists for the four jets). Any corpus phrase of the form "one native flow" is to be read in this contraction-class sense; the clocks stay external and sector-specific.

Research Contract — `QFT4D.OS.RECON.01` (the genuine 4D Euclidean lattice package)

Typing. A research contract [O] (ledger row `QFT4D.OS.RECON.01`: Open, research contract), *not* a claim. This is step (3) of the `DIMENSION.UPLIFT.FIREWALL.01` route ordering: the *nonperturbative* 4D existence statement that no perturbative or seam-level result can substitute for.

Input. TFPT fixes the discrete data: the representation content and charge lattice (carrier + glue), the coupling relations at the matching scale, and the boundary conditions (the seam as internal/topological boundary module). The 4D Euclidean lattice carries everything else: spacetime, gauge fields, Higgs, fermions.

Target theorem (the OS package). For the so-specified 4D lattice theory prove: **(O1)** reflection positivity of the lattice measure; **(O2)** Euclidean covariance restoration (hypercubic → full Euclidean group in the limit); **(O3)** clustering; **(O4)** uniform bounds (moment/regularity estimates uniform in lattice spacing and volume); **(O5)** existence of the continuum limit; **(O6)** Osterwalder–Schrader reconstruction of the Wightman/Haag–Kastler theory.

Firewall against the perturbative leg (binding phrasing rule). The existing all-orders result (`QFT4D.EG.ALLORDER.01`, v381) is a *local perturbative construction under the stated Epstein–*

Glaser/BRST hypotheses — that and no more. The weak adiabatic limit for massless particles exists as a cited theorem (Duch, `arXiv:1801.10147`), but its hypotheses must be *verified for the concrete TFPT model*; a seam-transfer gap does not substitute for the infrared structure of massless photons/gluons. Even a fully successful EG/BRST proof yields *none* of: confinement or the hadron spectrum, the Yang–Mills mass gap, asymptotic charged states in QED (the infraparticle problem), the nonperturbative Higgs vacuum, or a controlled full S-matrix. Every corpus phrase of the form "the perturbative 4D leg closes to all orders" is bound to this reading. Status: `QFT4D.OS.RECON.01` [O].

Registered decision (2026-08-28, `QFT4D.LATTICE.FUNDAMENTAL.01`): the lattice-fundamental reading — a typed route/typing, *not* an [E] claim and *not* a marker move. TFPT's *physical* completeness criterion is the finite, local, unitary lattice quantum theory (Hamiltonian route: Gauss-law Hilbert space, hermitian local H, $T = e^{-aH}$ positive by construction, Lieb–Robinson cone). This contract — the exact continuum Osterwalder–Schrader limit — is retyped as *mathematical reinforcement*: a named strengthening programme, no longer the physical bottleneck. Poincaré invariance is an IR fixed-point property with experimentally bounded irrelevant Lorentz-violating operators. Justification anchors ([`verification/v989_lattice_gate_battery.py`]): the Euclidean overlap shortcut is killed at T2 (exact; N_t-exact confirmation: the $|\det D_{\mathrm{ov}}|^2$ Gram is PSD at $N_t = 2$ and indefinite at $N_t = 4$, det-only $\lambda_{\min} \approx -0.249$; Wilson control PSD everywhere — pinned on overlap time nonlocality), and the Hamiltonian route clears T2 by construction. Display: this row stays [O] as the math programme; nothing closes.

Amendment (2026-08-29, review wave 4): the reviewer's precision is accepted — "a single finite box is not a world". The fundamental object is the *quasilocal consistent family* $\{H_\Lambda\}$ with the thermodynamic-limit dynamics $\tau_t = \lim e^{itH_\Lambda} A e^{-itH_\Lambda}$. Binding phrasing: continuum ($a \to 0$) *optional* (mathematical reinforcement); thermodynamic limit ($|\Lambda| \to \infty$) *and* controlled IR universality *mandatory*. Decision typing UNCHANGED; no [E] claim, no marker move.

Executed update (2026-08-30, review wave 7): the 2+1D master-object scaffold is coherent ([`verification/v1008_master_assembly_scaffold.py`]; Decision typing UNCHANGED). Combinatorial $K = 0$ (every oriented plaquette has vanishing flat-x-holonomy exponent); frozen-link $q^2 \sim 4$ with $K(q = 0) = 0$; unique-contractive selector $|R'| < 1$; QWZ wall + DET4 mirror (physical slope ≈ -0.9989, mirror gap ≥ 1.5 through $t = 0.6$); four-block assembly with $\mathbb{Z}_6$ content $(1, 2, 3, 0)$ and clock $[0, 1, 1, 2]$. $L = 3$ Hilbert spaces of dimension 262144 / 129024 stay out of suite. No [E] claim, no marker move.

Executed update (2026-08-30, afternoon harvest; [`verification/v1011_3p1d_ladder_battery.py`]; Decision typing UNCHANGED). The 3+1D ladder B6–B9 is complete at scaffold level: the minimal 3+1D object is viable (exact Gauss 32768 checks, first 3+1D confinement witness, wall separation, exact in-face isotropy, Kronecker-level stiffness $g_\star \approx 1.915$); fully coupled link–wall dynamics are demonstrated (exact maximal-tree gauge fixing 524288 → 4096, tree-independence, DET survives coupling, back-reaction); the Kronecker limitation of the 2+1D scaffold is removed. No [E] claim, no marker move.

Executed update (2026-08-30, late-evening harvest; [`verification/v1013_thermodynamic_dynamics.py`]; Decision typing UNCHANGED). The mandatory-dynamics leg is CLOSED at Hamiltonian-class level. Cited `articles/2026-08-30/tfpt_thermodynamic_dynamics_en.tex`: uniform Lieb–Robinson bound $J = 12/5$, $R = 2$, $z = 4$, $\kappa = 16043/450$, $v_{\mathrm{LR}} = 32086e/225 \sim 387.64$; norm-convergent τ_t; gauge-invariant quasilocal $\mathcal{A}^G$ (Gauss caveat resolved on the invariant subalgebra); existence of ground and β-KMS states. Numeric twin: measured $v = 2.0 \ll$ proved bound; the analytic finite-volume bound dominates nested differences at buffers 8/16/32. Remaining: phase uniqueness, limit gap, IR universality. No [E] claim, no marker move.

Research Contract — `CHIRAL4D.NOMIRROR.01` (the chiral 4D lattice matter theorem)

Typing. A research contract [O] (ledger row `CHIRAL4D.NOMIRROR.01`: Open, research contract), *not* a claim. Step (4) of the firewall route ordering: chirality and mirror decoupling must be proven *in the same 4D model* that `QFT4D.OS.RECON.01` constructs — the seam's c_-=8 and Chern/inflow structure are a boundary-*anomaly* signal, not yet 4D matter.

Target theorem. An explicit chiral lattice operator (Ginsparg–Wilson/overlap type) for the TFPT representation content with proofs of: **(C1)** a local, gauge-invariant fermion measure; **(C2)** perturbative *and* global anomaly freedom; **(C3)** no Witten $SU(2)$ anomaly; **(C4)** no mirror generation in the physical spectrum (the mirror decouples with a mass that survives the continuum limit); **(C5)** the correct index and charges (the TFPT charge lattice, generation count N_{fam}); **(C6)** a stable continuum limit compatible with the `QFT4D.OS.RECON.01` package.

Calibration (state of the art, honestly). The strongest comparable theorem is Mastropietro's construction (`arXiv:2303.02790`): a *weakly coupled* 4D $U(1)$ chiral gauge theory with a *massive* gauge field. The TFPT target — chiral $SU(3)\times SU(2)\times U(1)$ with three generations, massless nonabelian gauge fields, EWSB and confinement downstream — is far stronger; the gap between the two is the honest measure of this contract's difficulty. *Kill tests:* the mirror fails to decouple (kill test (3) of `WOIT.OS.TWISTOR.01` at lattice level); a Witten anomaly obstructs the carrier $SU(2)$; every gauge-invariant measure for the TFPT content is provably nonlocal. *Consistency note (ledger wins):* the celestial-branch scope fence above honestly lists interacting RP, mirror decoupling, the internal $SU(2)$, both helicities, local matter fields, EWSB and confinement as *not* closed; the contradictory "zero open internal mechanisms" phrase in `docs/OPEN_PROBLEMS.md` is corrected in the same change that registers this contract (that phrase was a certification statement about a *different*, narrower residual matrix and had drifted into an overclaim). Status: `CHIRAL4D.NOMIRROR.01` [O].

Executed update (2026-08-28, promote round): a 1+1D chiral Gauss census is executed on the Hamiltonian route (`[verification/v989_lattice_gate_battery.py]`; the 3+1D contract stays [O]). Two-species $\mathbb{Z}_4$ Hamiltonian, factorized 16×256 (the 4096 Kronecker is never materialised): a *single-exponent* Gauss unitary exists iff $q_L \equiv q_R$; the covariant two-exponent Gauss exists for every charge pair (lattice Dirac doubling / Nielsen–Ninomiya as a Gauss statement); $V^4 = I$; the mutant $(g_L, g_R) = (1, 0)$ on vectorlike content fails. Honest: not the TFPT representation content and not 3+1D. Display marker stays [O].

Executed update (2026-08-28, completeness wave): the SMG mechanism is toy-proved (`[verification/v997_spin2_smg_witnesses.py]`; the 4D theorem stays [O]). Fidkowski–Kitaev SMG toy: at $N = 8$ Majoranas a unique Spin(7)-singlet ground state (Schmidt rank 1) with volume-independent gap 14 at $L = 1, 2, 3$; the $N = 4$ contrast is 2^L-degenerate (must-fail); the physical chiral edge keeps its π/L spectrum exactly. SMG mechanism toy-proved [E]; the 4D Spin(10) theorem stays [O]. Display marker stays [O].

Executed update (2026-08-29, review wave 5): DET16 finite-cluster hit + the Casimir-projector kill (`[verification/v1002_spin10_mirror_projector.py]`; the 4D theorem stays [O]). The number-violating determinant vertex supplies an exact rank-one projector on the full 2^{16} cluster: $\Omega^\dagger\Omega = P_{16}$, $\Omega\Omega^\dagger = P_0$, defect 0; P_ϕ hermitian idempotent (2.8×10^{-18}), rank 1; h_{mir} spectrum $\{0^1, 1^{65535}\}$, gap exactly 1; all 45 $\mathfrak{so}(10)$ Fock commutators vanish ($\det\rho_{16} = 1$); parity commutator 0; $|0\rangle$ and $|F\rangle$ are singlets. Hopping on the 4-mode analogue is stable at $t = 0.2$ (gaps 0.90 / 0.86); spectators untouched. NEW candidate `CHIRAL4D.MIRROR.DET16.01` [C]. Independently, the number-preserving Casimir projector route is *excluded*: Weyl-alternant census $|W(D_5)|$=1920 gives $\{N$=0:1, N=16:1$\}$ (none at N=8), k=2 half-filling multiplicity 27, k=3:1463 — uniqueness fails at every cluster size. Display marker of this contract stays [O].

Research Contract — `CHIRAL4D.MIRROR.DET16.01` (DET16 rank-one projector, finite cluster)

Typing. A candidate [C] (ledger row `CHIRAL4D.MIRROR.DET16.01`), registered 2026-08-29 from review wave 5. ([`verification/v1002_spin10_mirror_projector.py`].) **Content.** Exact finite-cluster gapping on the full chiral 2^{16} Fock space via the number-violating determinant vertex, as executed above. **Open.** Dynamical gauge fields, domain-wall geometry, and a 4D volume theorem — this is not a 3+1D closure. **Relation.** Does *not* upgrade `CHIRAL4D.NOMIRROR.01`; the Casimir-projector kill is recorded on that parent. Status: Candidate [C].

Executed update (2026-08-30, review wave 7): T1/T2 proved, T3 Michalakis–Zwolak cited-verified, dynamical [N], 1/192 and BDL boxed ([`verification/v1007_det16_stability_battery.py`]; the candidate stays [C]). The 12-page memorandum `articles/2026-08-30/det16_mirror_gap_theorem_en.tex` proves T1/T2 and verifies every Michalakis–Zwolak hypothesis (Local-TQO, $t_* > 0$, perturbed gap at least $1/2$). Ordinary hopping is *not* block-diagonal ($\|(1-P)KP\| \neq 0$), so $t_{0,\mathrm{count}} = 1/192$ remains the uncertified counting threshold, not a theorem. Dynamical $\mathbb{Z}_2/\mathbb{Z}_4$ links on the four-mode analogue are uniquely gapped on the compressed grid (full-probe minimum 0.728854416 pinned); second-order Schrieffer–Wolff is $1 - \alpha t^2 + O(t^4)$ with c_{geo}/L bounded. BDL supplies a ground-energy truncation error with unspecified constants, not a volume-uniform gap remainder — boxed. Display marker stays [C].

Executed update (2026-08-31, Monday-morning harvest; [`verification/v1016_state_gap_batteries.py`]; the candidate stays [C]). Fröhlich–Pizzo hypotheses cited-verified for $\mathbb{Z}_2/\mathbb{Z}_4$ DET open chains: volume-uniform spectral gap $\geq \Delta/2$ in the parametric window τ_{FP} ($a = 0.0233$, $a/16 = 0.00146$). The dynamical-link leg is closed at that finite-group level. Remaining: rotor links, wall mixing, 3+1D placement. Display marker stays [C].

Executed update (2026-09-05, round-3 integration): derivative-filtered wall block (`CHIRAL4D.MIRROR.DFILTER.BLOCK.01`; [`verification/v1023_derivative_filtered_wall.py`]; no parent marker move). For

$$H_{\mathrm{df}} = \Delta b^\dagger b + \lambda(b + gD_{\mathrm{GW}}p)^\dagger(b + gD_{\mathrm{GW}}p)$$

each singular-value sector is the exact 2×2 block with determinant $\lambda g^2 \Delta s^2$. It preserves $\ker D_{\mathrm{GW}}$ on the physical summand, the adiabatically mirror-continuated high branch is at least $\Delta + \lambda$, and $\mu_-(s) = \lambda g^2 \Delta(\Delta+\lambda)^{-1}s^2 + O(s^4)$. The finite covariance and SVD-reduction defects are 2.408×10^{-15} and 1.776×10^{-15}. This is only a finite quasi-free block lemma. For $s \neq 0$ the bare b subspace is not invariant and has low-band weight, so no bare-mirror response gap is proved; the low branch has $z = 2$, not a Weyl cone. No canonical DET16/DET32 realization, interacting parent, admissibility/quasilocality theorem, measure/anomaly/index control, universality or 3+1D continuum limit follows. The derivative-filtered witness therefore does not upgrade `CHIRAL4D.MIRROR.DET16.01` [C], `CHIRAL4D.NOMIRROR.01` [O], T4, or TOE.

Firewall Contract — `GRAV.NONCIRCULAR.01` (the Bisognano–Wichmann direction discipline)

Typing. A *standing discipline rule*, in force immediately (ledger row `GRAV.NONCIRCULAR.01`), with an audit census at registration. Bisognano–Wichmann (BW) is a theorem *within* an already-local, Poincaré-covariant QFT with a suitable vacuum: the modular flow of a wedge algebra acts as the boost. It is *not* a device that generates Lorentz spacetime from an arbitrary modular flow — there exist Poincaré-covariant nets violating BW (Mund, `arXiv:hep-th/0101227`, characterizes when BW holds; Morinelli, `arXiv:1703.06831`, exhibits the failure landscape). The only correct proof order is: 4D net + locality + vacuum + spectrum condition ⇒ BW ⇒ geometric modular time. *Never*: KMS/modular spectrum ⇒ Lorentz space ⇒ BW.

The rule. Every use of BW or of the CHM ball modular Hamiltonian in the corpus must *declare* whether 4D Poincaré covariance is an ASSUMPTION (imported; the result is then conditional on `QFT4D.OS.RECON.01`-class existence) or a RESULT (derived intrinsically). The cycle risk this fences: BW/CHM entering the Einstein derivation as input while the 4D locality/Lorentz structure is still supposed to *come out* of the seam.

Audit census at registration (all current BW/CHM usages, annotated).

- ASSUMPTION — the entanglement-equilibrium Einstein derivation (v323/v358/v359; `tfpt_4_frontier` and the (G_{metric}) section above): the CHM ball modular Hamiltonian and the BW boost are *inputs*; the derivation is conditional on a 4D Poincaré-covariant QFT with the right vacuum existing (`QFT4D.OS.RECON.01`) and must never be cited as evidence *for* that existence.
- ASSUMPTION (DECLARED) — `QG.AMB.01` as a [C] redundancy (v369) is everywhere stated "conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann"; correct as declared.
- ASSUMPTION (DIMENSION-TYPED) — the seam KMS normalisation chain (v259/v526; $\beta_{\mathrm{angle}} = 2\pi$): uses only the universal modular/KMS angle of Tomita–Takesaki theory on the *seam* (1+1D/2D objects), not 4D BW; the 2π is a normalisation, not a spacetime derivation.
- RESULT-DIRECTION (OPEN) — the intrinsic-BW programme (v438/v440/v446 and the half-sided modular inclusion route in the (G_{metric}) section): the honest attempt to *derive* geometric modular action intrinsically; explicitly not closed, and the corpus already records the circularity warning ("BW would presuppose the very covariance it should produce", v215).
- EVIDENCE, NOT DERIVATION — the compressed-state flow to the CHM/BW geometric form (v478): a numerical convergence statement in the result direction; it does not establish 4D covariance.
- CONDITIONAL (DECLARED) — `origin_theory`'s "BW downstream of the seam being a chiral net": stated as conditional on net identification; correct as declared.

Status: firewall in force; no marker moves — the census confirms the declarations exist but binds all *future* uses to carry them explicitly.

Research Contract — `OBS.TRANSDUCTION.01` (the transduction principle: eigenvalue, projection, coupling)

Typing. A research contract (ledger row `OBS.TRANSDUCTION.01`); the principle is machine-checked on the minimal model at exploration level, the per-bundle transduction proofs are [O]. It elevates the S15 typing rule to a named principle:

$$\text{observable signature} \;=\; \underbrace{\text{internal eigenvalue}}_{\text{compiler [E]}} \;+\; \underbrace{B\,P_r \neq 0}_{\text{transduction}} \;+\; \underbrace{\text{physical coupling operator}}_{\text{physics}}.$$

Without the middle arrow, neither a non-observation falsifies nor an observation confirms. The minimal model is machine-checked in `experiments/theory-contracts/fo01_transduction_invisibility.py`: a uniform measurement functional sees the seam kernel *exactly not at all* ($\mathbf{1}^{\mathsf{T}}(T^m p - \pi) = 0$ to 10^{-14}; burst-train statistics KS-indistinguishable from a scrambled control), while a character-resolved channel reconstructs the eigenvalue 64/729 to 10^{-10} — invisibility is a property of the readout B, not of the channel.

Demands. For each empirical bundle (FRB/GW/pulsar nulls; the axion/birefringence family; any future search), *prove or refute* $B\,P_r \neq 0$ for the declared readout before the null or the hit is scored. *Binding application (the birefringence caution):* the ACT+Planck isotropic birefringence being close to $\beta_{\mathrm{TFPT}} = \varphi_0/(4\pi)$ (`arXiv:2608.06480`) is an interesting coincidence, *not* proof, until the shared normalized axion action and the transduction chain are proven — the axion bundle (β, f_a, m_a, $g_{a\gamma\gamma}$,

Ω_a, EDM) must follow from *one* action with *one* proven coupling. Status: `OBS.TRANSDUCTION.01` — principle registered, minimal model checked; every per-bundle middle arrow [O].

Executed update (2026-08-28, promote round): a finite $W[J]$ transduction shadow is executed ([`verification/v990_wj_transduction_mechanism.py`]; per-bundle proofs stay [O]). On a 1+1D two-site $\mathbb{Z}_6$ Hamiltonian (species Q and e^c), the mixed derivative $\partial^2 W/\partial J\,\partial\theta$ is nonzero on the seam-coupled link ($\approx -9.3\times10^{-3}$) and *exactly* 0 on the decoupled e^c control; a frozen-link companion transduces on the Q-staggered operator and stays 0 on e^c. The middle arrow is thereby a machine-checked finite-level mechanism, not a 4D bundle proof. Display: per-bundle empirical proofs stay [O].

Research Contract — `PRED.JOINTLIKELIHOOD.01` (from hit-counting to overdetermined relations)

Typing. A research contract [O] (ledger row `PRED.JOINTLIKELIHOOD.01`: Open, statistics-construction contract), *not* a claim. The 27+ individually matched values all descend from the same few atoms (φ_0, c_3, $g_{\rm car}$, μ_4) and are therefore statistically *correlated*: counting hits overstates independent evidence. The evidence is to be reorganized into *overdetermined relation bundles*, each carried by one proof object: **(i)** the flavor invariant sum rule — one spurion/operator proof must yield θ_{12}, θ_{13}, $\delta_{\rm PMNS}$, $J_{\rm PMNS}$, the mass splittings and the CKM texture *together*; **(ii)** the inflation consistency relation — $r = 3(1-n_s)^2$ and $\alpha_s = -r/6$ (eliminating $N_\star$ as a fitted middleman); **(iii)** the axion/birefringence bundle — β, f_a, m_a, $g_{a\gamma\gamma}$, Ω_a and the EDM bound from the *same* normalized axion action, with the transduction chain of `OBS.TRANSDUCTION.01`; **(iv)** the character-resolved transfer signature (`OBS.TRANSDUCTION.01` as the readout-side principle).

The statistical demand. A joint likelihood over the frozen registry with: the full covariance matrix induced by the shared atoms; the *effective* number of degrees of freedom (not the raw hit count); a prespecified null space and formula-selection rule; and a genuine holdout dataset (future measurements only). *Honest reading of the existing null test:* the "13/13, $< 10^{-30}$" Monte-Carlo statement (v100, `docs/FALSIFICATION.md`) quantifies the look-elsewhere burden *under its declared null model* and is only interpretable as evidence to the extent that the null space, the formula grammar, and all development data were prespecified — which for historically known observables they were not and cannot retroactively be. The 2026-06-09 freeze (v84) genuinely protects *future* measurements; it does not turn historically known data into blind predictions. The wording of `docs/FALSIFICATION.md` is adjusted accordingly in the same change (honest, not weakened: the freeze and the kill tests stand). Status: `PRED.JOINTLIKELIHOOD.01` [O].

Executed update (2026-08-28, promote round): v1 of the joint likelihood is executed ([`verification/v992_joint_likelihood_v1.py`]; the contract stays [O]). Nine frozen pull-matched scorecard triplets (no `experiments/` import at suite runtime): $\chi^2 = 11.80/\mathrm{dof}\,9$ ($p = 0.225$), $\nu_{\rm eff} = \mathrm{rank}(J_{\rm atom}) = 1$, TFPT at the 0th percentile of 200 scrambled decoders, leave-one-bundle-out stable. Holdouts declared not consumed: JUNO Δm^2, DESI Σm_ν, DUNE $\delta_{\rm CP}$, LiteBIRD r. v1 declarations: $\Sigma_{\rm exp}$ diagonal (NuFIT/Planck covariances unmodelled, flagged); $\Sigma_{\rm bridge} = \Sigma_{\rm th} = 0$. Promotion conditions for full closure: off-diagonal Σ, a formula-grammar census, genuine future holdouts. Display marker stays [O].

Research Programme — the TFPT4D master route (one 4D transfer object instead of twenty formulas)

Typing. A programme section (registered 2026-08-27 from the external bird's-eye review), *not* a claim: it organizes the existing 4D contracts and the four new ones below into one route and moves *no* marker. The organizing observation: TFPT is probably not missing twenty independent formulas; it is missing *one* mathematical growth principle that unfolds the discrete structure into a

local, unitary, reflection-positive 4D quantum world. The discrete theory is then not spacetime itself but its *genetic code*; the missing bridge must produce spatial locality, reversible quantum dynamics, chiral matter, gauge fields, a controlled continuum limit and concrete observables *simultaneously.*

The master formulation (working hypothesis, unproven). *The physical world is the unique local, reflection-positive and approximately quantum-Markovian 4D completion of the TFPT seam algebra that is anomaly-free and has exactly one relevant dimensionful direction.* "Exactly one dimensionful direction" is the No-Unit theorem already in force: the dimensionless compiler cannot produce an absolute scale, and v_{geo} is the declared calibration torsor (`ANCHOR.VGEO.02`, v153, v725) — the master route inherits this, it does not re-open it.

The seven conditions, mapped onto contracts. **(1)** seam compatibility — the boundary restriction of the 4D theory reproduces the TFPT seam data *exactly*, not only dimensions or group labels (`TFPT4D.LATTICE.ACTION.01` gate T3 below); **(2)** reflection positivity / positive transfer operator (gate T2; `QFT4D.OS.RECON.01` demand O1); **(3)** a local unitary dilation of the observed Markov dynamics, $\Phi(\rho) = \mathrm{Tr}_E[V(\rho \otimes \sigma_E)V^\dagger]$, with size-uniform locality and a non-divergent Lieb–Robinson velocity (`DYN.UNITARY.DILATION.01`); **(4)** chiral anomaly freedom with no mirror sector and the correct index (`CHIRAL4D.NOMIRROR.01`); **(5)** only one relevant scale direction — no hidden tunable dimensionless parameter (kill test in `TFPT4D.LATTICE.ACTION.01`; v_{geo} calibrates the unit only); **(6)** approximate quantum-Markov recovery, $I(A{:}C \mid B) \le C_0\, e^{-\mathrm{dist}(A,C)/\xi}$, so the bulk is quasi-locally reconstructable from overlapping boundary data — the small-CMI/recovery-map link is Fawzi–Renner (arXiv:1410.0664), and local Markov properties of Gibbs states are established in Chen–Rouzé (arXiv:2504.02208) (`SEAM.BULK4D.RECON.01` route input); **(7)** nontriviality — at least one connected four-point function and ultimately a nontrivial S-matrix survive the limit (gate T7; `SEAM.BULK4D.RECON.01` demand B6).

The dynamics correction (discrete → dynamics). The finite transport object already embeds exactly into a continuous Markov semigroup (`DYN.MARKOV.EMBED.01` [E] at kernel level, v971) — but a Markov process is irreversible and cannot be the fundamental dynamics. The fundamental order must be *local unitary amplitudes → decoherence/partial trace → Markov matrix* T. This matches the now machine-checked finding that the classical unistochastic matrix $B_{ij} = |U_{ij}|^2$ keeps probabilities but loses the orientation sign of the Jarlskog invariant $J = \pm 1/27$ ([`verification/v977_transfer_unistochastic_wilson.py`]; the Birkhoff-fibre circulation parameter t is a second, independent classically hidden datum, [`verification/v976_seam_lift_birkhoff.py`]): the fundamental variable must carry a complex holonomy/amplitude, which makes Wilson loops $W_\gamma = \mathrm{Tr} \prod_{e\in\gamma} U_e$ more natural than bare transition probabilities. Discrete Weyl dynamics on unitary cellular automata show such a continuum transition is possible in principle (Białynicki-Birula, arXiv:hep-th/9304070); for TFPT the discrete collision/QCA leg is constructed exactly ([`verification/v984_markov_qca_dilation.py`]), while the continuum/OS legs remain to be constructed (`DYN.UNITARY.DILATION.01`).

Programme ordering (the efficient sequence). **(P1)** the conditional 4D dimension selector — *done at conditional level* (`DIMENSION.SELECTOR.4D.01` [C], [`verification/v975_dimension_selector_4d.py`], next section); **(P2)** the single simple-current/spin-field generator (`SEAM.SIMPLECURRENT.GENERATOR.01` [O]); **(P3)** the explicit finite 4D lattice action with its seven finite gates (`TFPT4D.LATTICE.ACTION.01` [O]); **(P4)** the local unitary dilation with size-uniform Lieb–Robinson bounds (`DYN.UNITARY.DILATION.01` [O]); **(P5)** the continuum/OS package (`QFT4D.OS.RECON.01` [O]); **(P6)** all four transfer bridges from one generating functional (`FTRANSFER.GENERATING.01` [O]); **(P7)** gravitation strictly downstream — modular Hamiltonians → entanglement geometry → effective Einstein dynamics, in the assumption-declared discipline of `GRAV.NONCIRCULAR.01` (Jacobson's entanglement equilibrium, arXiv:1505.04753, *presupposes* a local QFT and can never be cited as producing one); **(P8)** the prespecified joint likelihood with holdout (`PRED.JOINTLIKELIHOOD.01` [O]); **(P9)** all three gauge couplings as unique stationary points of one nonabelian zeta-det/inflow functional (`GAUGE.DETLINE.FIXPOINT.01` [O]; generalizes `ALPHA.QUILLEN.EXACT.01`; honest: $\alpha_s(M_Z)$ is currently an external input); **(P10)** massless transversal spin-2 emergence from the same spectral determinant (`GRAV.SPIN2.EMERGENCE.01` [O]; Einstein equation stays downstream/conditional;

`GRAV.NONCIRCULAR.01` binding); **(P11)** the Schwinger–Keldysh completion (S, ρ_0) of cosmological transfers (`FTRANSFER.SK.RHO0.01` [O]; `FTRANSFER.GENERATING.01` covers equilibrium only; typed candidate $\theta_i = 3\pi/5$, μ_4 orientation picks $k = 0$). The Yang–Mills mass gap is *not* defined away: it remains the open Clay problem; at best the route reduces it to a concrete, structured lattice theory.

Kill criteria (the route is allowed to die early). No local reflection-positive finite 4D model with the correct seam boundary theory — master route dead. The simple-current generator is nonlocal or fails to converge — seam route dead. The chiral measure term stays nonlocal or a global anomaly survives — chiral route dead. Mirror fermions do not decouple — Standard-Model route dead. The Lieb–Robinson velocity or correlation length diverges uncontrolled — no local continuum limit. All connected four-point functions vanish — only a (generalized-)free theory. Extra freely tunable dimensionless parameters are needed — the strong parameter-freeness claim is refuted. The uniform twistor flat direction (the sector-0 Okubo admixture of [`verification/v973_seam_route_narrowing.py`]) shifts *normalized* observables rather than only the vacuum normalization of Z — a genuine undetermined modulus.

Verified Selector — `DIMENSION.SELECTOR.4D.01` (why $d = 4$, conditionally)

Typing. A *conditional* identity [C] (ledger row `DIMENSION.SELECTOR.4D.01`), machine-checked in [`verification/v975_dimension_selector_4d.py`] (16/16, sympy-exact). The μ_4 structure of the compiler must *not* be read as a proof of four spacetime dimensions (`DIMENSION.UPLIFT.FIREWALL.01`); what *is* checkable is a selector theorem under a named axiom set. **Statement.** Under **A1** $d > 2$ (propagating local gauge fields, $d - 2 \geq 1$ transverse polarizations), **A2** a dimensionless Yang–Mills coupling, **A3** real self-dual/anti-self-dual 2-form sectors (the Euclidean Hodge star an involutive endomorphism of 2-forms), and **A4** chiral Weyl representations (even d), the dimension $d = 4$ is *unique and minimal* on the exact scan window $d = 2..12$ — and the selector is *overdetermined*: A2 alone pins $\{4\}$ ($[g_{\mathrm{YM}}] = (4 - d)/2$ forced symbolically from dimensionless-action power counting) and A3 alone pins $\{4\}$ ($* : \Omega^2 \to \Omega^{d-2}$ is a self-map iff $d = 4$; the explicit 6×6 star on $\Lambda^2(\mathbb{R}^4)$ satisfies $*^2 = I$ with eigenvalues ± 1 at multiplicities 3/3 — the real $\mathfrak{su}(2)_+ \times \mathfrak{su}(2)_-$ split), while A1 & A4 alone give $\{4, 6, 8, 10, 12\}$: chirality does *not* select 4 by itself (honest ablation ledger). Weyl chirality is certified by exact Clifford algebras $d = 2..7$ (even d: γ_* anticommutes with every γ_μ, $\gamma_*^2 = I$, $\mathrm{tr}\,\gamma_* = 0$; odd d: the full product is central and scalar — no grading). Signature is typed honestly: in Lorentzian signature $*^2 = -1$ on 2-forms in $d = 4$, so the *real* split is a Euclidean/OS-side statement. Mutants are caught (the wrong power counting $[g] = (6 - d)/2$ selects $\{6\}$; the "A4 alone selects 4" overclaim is refused). **Honest boundary.** μ_4 and g_{car} enter nowhere — deliberately, because "$\mu_4 \Rightarrow$ 4D" is exactly the forbidden reading. This is a selector, *not* a construction and *not* a compiler derivation: *axiom provenance is the open half* — if A1–A4 cannot themselves be forced from $\{c_3, g_{\mathrm{car}}\}$, the selector remains a consistency statement, not a prediction (the registered kill criterion). Status: `DIMENSION.SELECTOR.4D.01` [C]; every 4D construction contract stays [O].

Research Contract — `SEAM.SIMPLECURRENT.GENERATOR.01` (one generator instead of 128 currents)

Typing. A research contract [O] (ledger row `SEAM.SIMPLECURRENT.GENERATOR.01`), *not* a claim. **Context.** The seam coverage carve ([`verification/v973_seam_route_narrowing.py`]) shows the $\mathfrak{so}(16)_1$ fermion-bilinear half of the seam target is covered by the cited scaling-limit theorem; the named residual is N1 (the μ_4/GSO extension $SO(16)_1 \to (E_8)_1$ as a *net* statement — the 128 spinor currents are not bilinears), N2 (the twist/spin-field sector), N3 (the one-sided chiral edge) and N4 (the holomorphy selector). **Demand (the compression).** Construct *one* normalized simple-current/spin-field intertwiner — the generator of the index-4 Q-system extension (v125/v154) — and

prove **(G1)** convergence on the finite-particle core with energy bounds, **(G2)** locality relative to the bilinear net, **(G3)** the correct braiding/monodromy relation, and **(G4)** Q-system closure, so that symmetry and fusion generate the *entire* 128-dimensional extension sector from the single generator — collapsing N1 + N2 (and, via uniqueness of the extension, the N4 selector) onto *one* convergence theorem instead of 128 separate current controls. **Kill tests.** The generator is provably nonlocal relative to the bilinear net; the energy bounds fail on the core; the braiding closes on the wrong extension ($SO(16)_1$ stays unextended); the convergence rate is incompatible with the measured bilinear collar rate ($p = 3.48$ inside the preregistered band, [verification/v973_seam_route_narrowing.py]). *Dimension typing (binding):* hypotheses and conclusion live in 1+1D — closing this contract does *not* close SEAM.BULK4D.RECON.01. Status: SEAM.SIMPLECURRENT.GENERATOR.01 [O].

Executed update (2026-08-28, external master-object review wave 2): the generator is *identified* and its lattice-arithmetic shadow is complete (the analytic half G1–G4 stays [O]). [verification/v983_simple_current_generator.py] The candidate is the glue vector $\lambda = (\omega_s, \omega_f)$ — the D_5 spinor weight plus the A_3 fundamental weight, in explicit rational coordinates — with $\|\omega_s\|^2 = \frac{5}{4}$, $\|\omega_f\|^2 = \frac{3}{4}$, $\|\lambda\|^2 = 2$ and conformal weight $h_\lambda = \|\lambda\|^2/2 = 1$ (integer ⇒ trivial self-monodromy, the bosonic simple-current precondition). Machine-checked exactly: $[\lambda]$ has *order four* in the discriminant group $\mathbb{Z}_4 \times \mathbb{Z}_4$ of $D_5 \oplus A_3$; $\lambda \cdot L_0 \subset \mathbb{Z}$ (isotropic glue, the glued lattice is even); $\det = 16/4^2 = 1$ (unimodular rank 8, hence E_8; explicit certificate v1); the coset root census is $[52, 64, 60, 64]$ — with the rank-8 Cartan sector exactly $248 = (45{+}15) + (16, 4) + (10, 6) + (\overline{16}, \bar{4})$ — so the *odd* fusion powers of the one class $[\lambda]$ carry $64 + 64 = 128$, the full missing spinor sector. Kill tests fire: the vector-class glue (norm 7/4) and the mismatched glue (norm 9/4) both break evenness. The holomorphy *shadow* is pinned at the lattice level ($\det 1$ ⇒ one sector, vs $\det D_8 = 4$ ⇒ four sectors for $SO(16)_1$) — the operator-algebraic N4 statement is untouched. Net effect: the contract's search space collapses from "find a generator" to "prove G1–G4 for *this* $\Psi_\lambda = Y(e^{(\omega_s,\omega_f)}, z)$"; no marker moves.

Executed update (2026-08-28, promote round): S3 exact + the quantitative S1/S2 skeleton + the lemma reduction ([verification/v988_psi_lambda_reduction.py]; *not* a closure). S3: the crossed-product/tower exchange is exact (naturality; a non-equivariant mutant fails). S1: windowed covariances converge and sectors collapse (measured rates $\sim N^{-1.85}$ on the reduced QWZ collar). S2 quantitative: windowed Hilbert–Schmidt bounded (Shale ~ 0.376) while the global $\mathrm{HS}^2 = 3.605 + 0.203 \log N$ is topological vs. saturation in the trivial phase; matrix-element Cauchy on the phase-fixed band; energy envelope $f_n \le C^{n+1}(n!)^2$ with $C = 1.204$ on the reduced collar (probe provenance $C = 2.86$). The remaining lemma is *reduced* to cited quasi-free theorems (Shale–Stinespring / Araki / Ruijsenaars) plus the MMST identification — reduction, not closure. Honest typing: lattice shadow [E] ([verification/v983_simple_current_generator.py]) + quantitative skeleton [E]-measured + reduction-to-cited-class [C] conditional; open rest = the MMST identification. Display marker stays [O].

Executed update (2026-08-28, review wave 3): the $D{+}A$ architecture is an OUTPUT of the rank-8 ADE census ([verification/v993_minimal_defect_selector.py]; axiom typing UNCHANGED). The FULL rank-8 ADE pair census (Smith normal form of Cartan, not a textbook lookup) with cyclic $\mathbb{Z}_4$ glue demanded on *both* factors selects (D_5, A_3) uniquely ($D_3 \cong A_3$ aliases collapsed; the raw second hit $D_5 \oplus D_3$ is the same pair). Kill tests fire: (D_4, D_4) and (A_7, A_1) have product 16 but fail cyclicity (Klein $\mathbb{Z}_2 \times \mathbb{Z}_2$ / $\mathbb{Z}_8 \times \mathbb{Z}_2$); the $\mathbb{Z}_3$ control $(E_6, A_2) = \mathbb{Z}_3 \times \mathbb{Z}_3$. So the $D{+}A$ split that v624 still had to *assume* is an OUTPUT of (rank 8 + cyclic $\mathbb{Z}_4$ both sides). Cited-not-proved remain $c_- = 8$ and holomorphic uniqueness of $(E_8)_1$. Lattice shadow of $U^2 = (-1)^F$: $[\lambda]^2 = [v]$ exactly (the order-4 glue class squares to the fermion-parity / D_5-vector class; non-split: vector glue of norm 7/4 not in $2\mathbb{Z}$). Unique finite trace-preserving conditional expectation $E = \frac{1}{4}\sum_r \mathrm{Ad}(u^r)$ onto the clock-fixed algebra (uniqueness needs the module property; without it the deviation kernel has dimension 36/720). The P1 arithmetic $1/(4 \cdot 2\pi) = c_3$ is the v484/v813 identity, not a derivation; the equidistribution identification stays the open [C] of v813. AX.P1.01/AX.P2.01 stay Axiom. Display marker of this contract stays [O].

Executed update (2026-08-28, completeness wave): 5 of 7 MMST identification criteria

executed, and C6 is the written-out hand-off ([verification/v994_mmst_criteria_battery.py]; display stays [O]). C1: measured $c = 8.005$ (16 Majorana copies; one-cut convention $S_{M,\mathrm{1cut}} = S_{\mathrm{scalar}}/4$). C2: conformal edge spectrum $[0.506, 1.516, 2.523, 3.524]$ with degeneracies $[1, 0, 1, 1, 2]$, $v \approx 0.988$. C3 (flagship): μ_4/GSO-projected lattice character sums reproduce the $(E_8)_1$ vacuum character coefficients $[1, 0, 248, 0, 4124, 0, 34752]$ three orders deep. C4: current normalisation $k = 1.000000000000$. C5: sector collapse $\det D_8 = 4 \to \det E_8 = 1$. C6 is now *written out* as the quasi-free convergence theorem in `articles/2026-08-28/psi_lambda_convergence_theorem_en.tex` (analytic log coefficient $2/\pi^2 = 0.20264$ vs the v988 measured slope 0.203, relative 0.18%) — this *is* the G1/C6 hand-off, not a re-proof inside the suite. C7 modular invariance stays cited/open. Must-fail: dropping GSO (O alone) misses the 248; $N_{\mathrm{MAJORANA}} = 8$ cannot make 248. Honest typing: 5/7 executed [E]-measured + C6 written-out [C]+ C7 cited [O]; open rest = C7 + net-level G2/G3/G4. Display marker stays [O].

Executed update (2026-08-29, review wave 4): C7 arithmetic shadow now exact ([verification/v998_seam_modular_closure.py]; net-level C7 stays cited). Two-edge HS remainder bounded by 2.11 on the frozen plateau $N_x \in (16, 24, 32, 48, 64, 96)$. Lattice character θ_{E_8}/η^8 EXACT four orders: t-grid $[1, 0, 248, 0, 4124, 0, 34752]$. $\Theta_{E_8}(\tau) = E_4(\tau)$ as Eisenstein series: q-expansion match to ≥ 8 orders exact (Jacobi identity); $j^{1/3}$ numeric transform at 60 digits with S-weight bookkeeping (τ^4 cancels in the weight-0 character). Net-level C7 modular covariance of the completed extension stays cited/open. Display marker stays [O].

Externalization package (hand-off for outside specialists in conformal nets / subfactors / VOA). In the same six-part format as the `SEAM.EQUIV.01` externalization contract below: **(1) minimal theorem statement.** Let $\mathcal{A}$ be the $\mathfrak{so}(16)_1$ fermion-bilinear net on the seam circle (the half covered by the cited scaling-limit theorem, [verification/v973_seam_route_narrowing.py]), and let σ be the simple-current sector of the index-4 μ_4 Q-system (finite side explicit and machine-verified, v125/v154; net-level locality integer $h_s = 16/16 = 1$, v469; the generator is meanwhile *identified* at the lattice level as $\Psi_\lambda = Y(e^{(\omega_s,\omega_f)}, z)$ with $h_\lambda = 1$, [verification/v983_simple_current_generator.py]). *Claim:* there exists one normalized intertwiner Ψ_σ (a charged/spin field for σ — the identified Ψ_λ) satisfying G1 convergence on the finite-particle core with polynomial energy bounds, G2 relative locality to $\mathcal{A}$, G3 the μ_4 braiding/monodromy relation, and G4 Q-system closure; then the extension generated by $\{\mathcal{A}, \Psi_\sigma\}$ is the $(E_8)_1$ net and the v973 residual N1 + N2 (and N4 via uniqueness) closes. **(2) typed hypothesis list.** PROVEN-FINITE: the index-4 Q-system (v125/v154), the crossed-product/locality package (v469), the coverage carve with the named N1–N4 residual and the measured collar rate $p = 3.48$ ([verification/v973_seam_route_narrowing.py]), the gapped collar model (v367/v368). CITED-CLASSICAL: the Osborne–Stottmeister scaling limit (arXiv:2107.13834); Longo–Rehren / Böckenhauer–Evans simple-current extension theory; KLM μ-index classification. OPEN: C7 modular invariance of the completed μ_4/GSO extension, and G2–G4 as *net* statements (relative locality, braiding, Q-system closure at the continuum net). G1/C6 is no longer a blank: the quasi-free convergence theorem is written out in `articles/2026-08-28/psi_lambda_convergence_theorem_en.tex` (analytic log coefficient $2/\pi^2 = 0.20264$ vs measured 0.203) — that document *is* the hand-off. **(3) runnable finite models:** `verification/v125`, `v154`, `v469`, `v973` (embedded probes with pinned SPEC hashes), `v983` (the identified glue vector and its lattice shadow), `v988` (S3 exact + S1/S2 skeleton), `v994` (5/7 MMST identification criteria, including the $(E_8)_1$ vacuum character $[1, 0, 248, 0, 4124, 0, 34752]$). **(4) honest Lean surface:** none for G1–G4; the finite Q-system arithmetic only. **(5) must-fail controls:** the same construction attempted on $SO(16)_1$ *without* the order-4 clock must *not* produce a holomorphic net ($\det K = 4$ rival); a non-order-4 clock must break the marks; the trivial $M{=}3$ collar must give no chiral limit. **(6) acceptance tests:** a solution counts iff it (a) proves G1–G4 for one explicit Ψ_σ, or (b) proves that no local normalized generator exists (the registered kill — then the seam route is dead and the ledger records it as exactly that).

Research Contract — TFPT4D.LATTICE.ACTION.01 (the explicit finite 4D lattice action and its seven finite gates)

Typing. A research contract [O] (ledger row TFPT4D.LATTICE.ACTION.01); route step (2) of the DIMENSION.UPLIFT.FIREWALL.01 ordering made concrete — the single object on which nearly all open 4D questions become simultaneously testable. **Demand.** Exhibit one fully explicit finite family of 4D Euclidean lattice theories

$$\mathcal{Z}_a[J] = \int \mathcal{D}U\, \mathcal{D}\psi\, \mathcal{D}\Phi\; e^{-S_a[U,\psi,\Phi]+J\mathcal{O}}, \qquad S_a = S_{\mathrm{Wilson}}[U] + \bar{\psi}\, D_{\mathrm{GW}}[U,\Phi]\, \psi + S_\Phi[U,\Phi] + S_{\mathrm{seam/top}},$$

where U are the fundamental holonomies (TFPT fixes representation content, charge lattice, coupling relations and boundary conditions), D_{GW} is a Ginsparg–Wilson/overlap operator for the TFPT chiral representation content (the CHIRAL4D.NOMIRROR.01 operator), Φ is the *only* allowed relevant scalar/link-mixing sector (EWSB downstream), and $S_{\mathrm{seam/top}}$ is the boundary/topological module transporting the seam data into the bulk — first at finite lattice spacing a, deliberately *without* continuum claims. **Finite gates** (all machine-checkable at finite size; the promotion criteria for any candidate): **(T1)** exact gauge invariance; **(T2)** a positive transfer matrix (RP at finite a); **(T3)** the seam boundary restriction reproduces the TFPT seam data *exactly*; **(T4)** anomaly bookkeeping including the global $SU(2)$ anomaly; **(T5)** the index/generation count; **(T6)** the mirror spectrum separated; **(T7)** at least one *nonvanishing* connected four-point function. **Kill tests.** No local reflection-positive action with the correct seam restriction exists (master route dead); every candidate is generalized-free (T7 unreachable); a hidden freely tunable dimensionless parameter is required (the strong parameter-freeness claim is refuted); the uniform twistor flat direction shifts *normalized* observables rather than only the vacuum normalization of $\mathcal{Z}$ (a genuine undetermined modulus). **Relation.** The continuum limit is the separate contract QFT4D.OS.RECON.01; the unitary dilation is DYN.UNITARY.DILATION.01; $d = 4$ is typed by DIMENSION.SELECTOR.4D.01 [C]; the generating-functional collapse is FTRANSFER.GENERATING.01. Chiral lattice state of the art is calibrated in CHIRAL4D.NOMIRROR.01 (Lüscher's exactly gauge-invariant abelian construction, arXiv:hep-lat/9811032; the nonabelian anomaly analysis, arXiv:hep-lat/9904009; reflection positivity proven for *free* overlap fermions, Kikukawa–Usui arXiv:1005.3751 — none of this is the interacting chiral Standard Model). Nothing in the corpus closes any gate. Status: TFPT4D.LATTICE.ACTION.01 [O].

Executed update (2026-08-28, promote round): the T1–T7 harness plus a named Euclidean T2 kill and a Hamiltonian-route T2 clearance ([verification/v989_lattice_gate_battery.py]**; no gate closed).** T4 exact SM anomaly bookkeeping (a lepton-drop mutant fires). T5/T6: overlap index equals flux q on an 8×8 torus; Ginsparg–Wilson circle to machine precision. The Euclidean overlap candidate is killed at T2: $|\det D_{\mathrm{ov}}|^2$ is PSD at $N_t = 2$ and indefinite at $N_t = 4$ (det-only $\lambda_{\min} \approx -0.249$); Wilson fermions (ultralocal in time) stay PSD — the break is pinned on overlap time nonlocality. The Hamiltonian route clears T2 by construction ($H = H^\dagger$ exact $\Rightarrow$ $T = e^{-aH}$ symmetric PD; interior Gauss exact including the seam-sector intertwiner; background-field μ_4 split with $k = 1, 3$ degenerate). Display marker stays [O].

Registered decision (2026-08-28, QFT4D.LATTICE.FUNDAMENTAL.01): the lattice-fundamental reading — a typed route/typing, *not* an [E] claim. The physical completeness criterion is this finite local unitary lattice theory (Hamiltonian route), not the continuum OS limit. That limit (QFT4D.OS.RECON.01) is retyped as mathematical reinforcement; Poincaré invariance is an IR fixed-point property. The display marker of this contract stays [O]; nothing closes. See the decision paragraph at QFT4D.OS.RECON.01.

Amendment (2026-08-29): a single finite box is not a world — the fundamental object is the quasilocal consistent family $\{H_\Lambda\}$ with $\tau_t = \lim e^{itH_\Lambda} A e^{-itH_\Lambda}$; continuum optional, thermodynamic limit and controlled IR universality mandatory. Full prose at QFT4D.OS.RECON.01.

Executed update (2026-08-30, review wave 7): 2+1D scaffold coherent ([verification/v1008_master_assembly_scaffold.py]**; Decision typing UNCHANGED).** Combinatorial $K = 0$, unique-

contractive $|R'| < 1$, QWZ wall + DET4 mirror, $\mathbb{Z}_6$ assembly $(1, 2, 3, 0)$. $L = 3$ spaces out of suite. No [E] claim, no marker move.

Executed update (2026-08-30, afternoon harvest; [verification/v1011_3p1d_ladder_battery.py]; Decision typing UNCHANGED). 3+1D minimal object viable; fully coupled link–wall at minimal scale; Kronecker limitation removed. No [E] claim, no marker move.

Executed update (2026-08-30, late-evening harvest; [verification/v1013_thermodynamic_dynamics.py]; Decision typing UNCHANGED). Mandatory-dynamics leg closed at class level (Lieb–Robinson + τ_t + $\mathcal{A}^G$ + state existence). Uniqueness / limit gap / IR universality remain. Full prose at `QFT4D.OS.RECON.01`. No [E] claim, no marker move.

Research Contract — `SEAM.DETLINE.UNIFICATION.01` (the determinant-line unification hypothesis)

Typing. A research *hypothesis* contract [O] (ledger row `SEAM.DETLINE.UNIFICATION.01`) — a precise conjecture, *not* a result. **Observation.** Three separately open structures may be projections of *one* geometric object: **(1)** the seam extension is described by a Q-system/simple current (`SEAM.SIMPLECURRENT.GENERATOR.01`); **(2)** the chiral fermion measure lives on a determinant line (`CHIRAL4D.NOMIRROR.01` demand C1); **(3)** anomalies and CP phases appear as holonomies of that determinant line — the *finite* Quillen identity is already two-sided ([verification/v974_alpha_faces_computed.py]: Berry integral = Poincaré–Lelong divisor integral = +1), and the continuum Bismut–Freed identification is the named missing step of `ALPHA.QUILLEN.EXACT.01`. **Demand (the compatibility theorem).**

$$\mathrm{Res}_{\mathrm{seam}} \det D_{4D} \;\cong\; \det D_{\mathrm{seam}}$$

as line bundles *with connection*, and the holonomy of the bulk determinant line equals the discrete TFPT orientation phase (the classically invisible $\operatorname{sgn} J$ orientation bit, now machine-checked at matrix level with the Wilson-plaquette interference readout as its minimal observable, [verification/v977_transfer_unistochastic_wilson.py]; the bulk-determinant identification demanded here stays open). If such a theorem exists, seam extension, chiral anomaly freedom, the generation index and the discrete CP orientation become different projections of the same object. **Kill tests.** The two determinant lines differ by a nontrivial local factor no counterterm removes; the bulk holonomy disagrees with the discrete orientation phase on a computable family. *Honest boundary:* the continuum Quillen/Bismut–Freed identification is exactly the critical missing step; nothing in the corpus proves any part of the 4D side. Status: `SEAM.DETLINE.UNIFICATION.01` [O].

Executed update (2026-08-28, promote round): a finite 2+1D/1D bulk–edge shadow is executed ([verification/v991_detline_bulk_edge.py]; continuum Bismut–Freed stays [O]). On the QWZ collar, bulk twist-torus curvature $2\pi \cdot (+1)$ equals the seam holonomy winding $+1$ (difference 5.6×10^{-16}); the $M = 3$ control is zero on both sides; a conjugation mutant flips both signs coherently. Honest: finite shadow [E], not the 4D Bismut–Freed identification. Display marker stays [O].

Executed update (2026-08-30, late-evening harvest; [verification/v1014_bridge_refinements.py]; the contract stays [O]). The finite restriction isomorphism $\mathrm{Res}_{\Sigma} L_{\mathrm{bulk}} \cong L_{\Sigma}$ is verified (c_{phase} constancy $3.7{\times}10^{-16}$, conjugation covariance exact, $C = W_{\mathrm{bulk}} = W_{\mathrm{seam}} = +1$ / mirror -1). Orientation fixes only the conjugate-pair branch ($2 \to 1$); a constant $U(1)$ anchor remains, fixed by the A_0 reference normalization of `ALPHA.QUILLEN.EXACT.01`. Continuum Bismut–Freed stays [O].

Research Contract — FTRANSFER.GENERATING.01 (four transfer bridges from one generating functional)

Typing. A research contract [O] (ledger row FTRANSFER.GENERATING.01); the *constructive successor* named by the FR.TRANSFER.01 typing guard. The four continuous-transfer interfaces (F_{pole}: source masses → pole, v183; $F_{\text{Boltzmann}}$: CP source → η_B, v169/v184; F_{relic}: $(f_a, m_a, \theta_i, N_{\text{DW}}) \to \Omega_a$, v185; F_{QCD}: $(\alpha_s, b_3, \Lambda_{\text{QCD}}, \text{lattice}) \to m_p/m_e$, v164) are currently four *separate* bridges. **Demand.** If the 4D theory $\mathcal{Z}_a[J]$ of TFPT4D.LATTICE.ACTION.01 exists with a controlled continuum limit (QFT4D.OS.RECON.01), the four bridges must arise from *one* generating functional $W[J] = \log Z[J]$: lepton and quark masses as poles of two-point functions (zeros of $\Gamma^{(2)}$); m_p/m_e from the baryon and electron correlator poles of the *same* theory; η_B from the real-time Schwinger–Keldysh dynamics of the *same* theory (CP violation, B violation, nonequilibrium); the axion relic from the cosmological solution of the *same* normalized action. F_{transfer} is then not another 3×3 matrix; it is the full $W[J]$. **Binding.** The FR.TRANSFER.01 guard stays in force: the four frontier observables remain [C]/[O]-typed until $W[J]$ exists and each bridge is *derived* from it — this contract permits no marker upgrade; it names the single object whose existence would collapse the four transfer problems into computations within one theory. **Kill test.** The four bridges provably require mutually inconsistent normalizations of the same $W[J]$. Status: FTRANSFER.GENERATING.01 [O].

Executed update (2026-08-28, promote round): a finite $W[J]$ is executed at 1+1D $\mathbb{Z}_6$ ([verification/v990_wj_transduction_mechanism.py]; the FR.TRANSFER.01 guard stays binding — no upgrade). Content shadow: the real $6Y$ table as a $\mathbb{Z}_6$ centre. Derivative identities: $\partial W/\partial J = \langle O \rangle$ (residual 6.3×10^{-11}), Hessian = Kubo (residual 1×10^{-7}). The four transfer bridges remain un-derived; this is the finite-level mechanism, not $W[J]$ of the 4D theory. Display marker stays [O].

Research Contract — GAUGE.DETLINE.FIXPOINT.01 (all three gauge couplings from one zeta-det functional)

Typing. A research contract [O] (ledger row GAUGE.DETLINE.FIXPOINT.01), registered 2026-08-28 from review wave 3. **Demand.** All three gauge couplings (g_1, g_2, g_3 / equivalently α, $\sin^2\theta_W$, α_s) arise as stationary points of *one* nonabelian zeta-det / inflow functional on the determinant line — the nonabelian upgrade of the $U(1)$ face ALPHA.QUILLEN.EXACT.01. **Acceptance.** Unique stationary points, positive Hessian, $\alpha_s(M_Z)$ without measured input, RG from the matching scale reproduces the data, mutants fail. **Honest bound.** $\alpha_s(M_Z)$ is currently an *external input* (the QCD matching / lattice input of F_{QCD}); this contract names the missing first-principles replacement, it does not claim the number is derived today. **Kill tests.** A unique stationary point fails to exist; Hessian not positive; mutants (wrong level, wrong inflow multiplicity) still pass; RG from the functional fails to land on the observed couplings. **Relation.** Generalizes ALPHA.QUILLEN.EXACT.01. Status: GAUGE.DETLINE.FIXPOINT.01 [O].

Executed update (2026-08-28, completeness wave): the naive grammar extension is a structural kill ([verification/v996_gauge_grammar_kill.py]; the contract stays [O]). Simplest-extension H0 of the $U(1)$ α-grammar to $SU(2)/SU(3)$ is structurally inapplicable: negative b_2/b_3 flip the stationary-root signs ($SU(2)$ inverse -151.2, $SU(3)$ -115.9); 64 conventions, 0 hits; the $U(1)$ control 137.0359992168 survives every mutant. Exponential detline-curvature convergence to 2π with rate $0.6936 \sim \ln 2$ is a numerical Bismut–Freed *shadow*, not the continuum theorem (canonical note on ALPHA.QUILLEN.EXACT.01). Naive census-only extension excluded [E]; genuinely nonabelian Casimir/instanton structure is required — typed. Honest: $\alpha_s(M_Z)$ remains an external input. Display marker stays [O].

Executed update (2026-08-29, review wave 5): fixed-volume mechanism executed, thermodynamic step-scaling divergent (exploration probes gauge_stiffness_fixpoint_probe.py + gauge_stiffness_stepscaling_probe.py; *not* a new module; the contract

stays [O]). At fixed volume the stiffness map has a unique fixed point with contraction -0.632, exact q^2 and $SU(2)$ Dynkin-ratio 4.0 content laws, Weyl exact. The preregistered thermodynamic step-scaling $L = 2..8$ (certified Lanczos) is *divergent*: $g_\star$ oscillates 0.65..1.69, $|R'| > 1$ at $L = 4, 5, 8$, and $1/L$ vs $1/L^2$ extrapolations are inconsistent. Typed hurdle: the 1+1D toy has no plaquette self-interaction; the real contract needs $\geq$ 2+1D. The v996 H0 grammar kill stands. Display marker stays [O].

Executed update (2026-08-30, review wave 7): the 2+1D plaquette scaffold lifts the 1+1D no-plaquette hurdle at scaffold level ([verification/v1008_master_assembly_scaffold.py]; the 4D fixpoint stays [O]). Combinatorial $K = 0$; frozen-link $q^2 \sim 4$ with $K(q = 0) = 0$; unique-contractive selector $|R'| < 1$ (L2 selector 0.578 $\to$ L3 0.936 harvest pins; $L = 3$ Lanczos out of suite). The thermodynamic 4D fixpoint and $\alpha_s(M_Z)$ as an external input remain open. Display marker stays [O].

Executed update (2026-08-30, afternoon harvest; [verification/v1011_3p1d_ladder_battery.py]; BINDING constraint recorded, display stays [O]). Holonomy stiffness collapses in the confined phase (coupled K dies above $g \sim 1.6$; charged spinor channels enhance $\sim 2.014\times$ at weak g but do not prevent the collapse at minimal volume). The matching point of the coupling functional *must* be defined in the deconfined/weak regime. Charge-0 control matches the uncharged periodic-face curve to 10^{-10}. Display marker stays [O].

Research Contract — GRAV.SPIN2.EMERGENCE.01 (massless transversal spin-2 from the same spectral determinant)

Typing. A research contract [O] (ledger row GRAV.SPIN2.EMERGENCE.01), registered 2026-08-28 from review wave 3. **Demand.** From the *same* spectral determinant that carries the gauge fixpoints, a massless transversal spin-2 pole with two helicities, universal $T_{\mu\nu}$ coupling, diffeomorphism Ward identities, and positive Hessian / reflection-positive transfer. **Relation.** The entanglement-equilibrium Einstein equation (v358/v359) stays a *downstream / conditional* readout — it presupposes a local QFT and can never be cited as producing one; GRAV.NONCIRCULAR.01 is *binding* (BW/CHM uses remain assumption-declared). **Kill tests.** A massive or longitudinal extra mode; non-universal coupling; failed diffeomorphism Ward; negative Hessian / RP violation on the spin-2 sector. Status: GRAV.SPIN2.EMERGENCE.01 [O].

Executed update (2026-08-28, completeness wave): quadratic witnesses, with the local truncation's ghost typed ([verification/v997_spin2_smg_witnesses.py]; the contract stays [O]). Exact Barnes–Rivers decomposition of the quadratic spectral action: $R{+}R^2$ is *clean* (one massless TT pole, two helicities, scalaron at $M_{\rm scal}$ exact). The local a_4 Weyl^2 truncation *necessarily* carries an opposite-residue spin-2 ghost — typed. The contract's content is pinned on the untruncated form factor $a(\Box) = e^{-\Box/M^2}$. Einstein equation stays downstream/conditional; GRAV.NONCIRCULAR.01 remains binding. Display marker stays [O].

Executed update (2026-08-29, review wave 4): quadratic Hamiltonian half executed ([verification/v1000_spin2_sk_witnesses.py]; the contract stays [O]). Positivity exact; 2 TT helicities at all 5852 nonzero momenta ($N = 6..12$); $\omega = |q|$ with $q_i = 2\sin(k_i/2)$ and axis expansion $\omega = k - k^3/24$ exact. Must-fail: a mass term gaps TT; a wrong-sign kinetic term yields two negative directions. Spectral-action Hessian identification, nonlinear Ward and universal coupling stay [O]. Display marker stays [O].

WAVE-6 firewall (2026-08-30): GRAV.WEINBERG.WITTEN.01 registered [O]; this contract's display stays [O]. The Weinberg–Witten theorem is the binding firewall *before any spin-2 marker move*: the microscopic Hamiltonian family must *not* possess a fundamental Lorentz-covariant local conserved $T_{\mu\nu}$; Lorentz symmetry, diffeomorphism Ward identities and the spin-2 mode must emerge in the *same* IR limit. Cross-link only (this row does not list GRAV.WEINBERG.WITTEN.01 as a dependency — that would cycle).

Executed update (2026-08-30, evening TOE-gate wave): interacting TT half-step ([verification/v1012_toe_gate_witnesses.py]; the contract stays [O]). Conserved interaction-complete lattice stress $\sim 10^{-15}$; TT spectral positivity exact; sum rules $\sim 10^{-15}$; the dominant positive-Z pole persists under ϕ^4 (gapped, as expected for massive generic matter); Z trend k_1 suppressed 0.958 / k_2 enhanced 1.033; free-limit analytic match. Missing: TFPT content, $k \to 0$ volume scaling, Ward identities. Display marker stays [O].

Executed update (2026-09-05, round-3 integration): gapped-parent/TT boundary (GRAV.PARENT.GAP.NO_GO.01; [verification/v1024_parent_internal_boundaries.py]; no parent marker move). A uniform full-parent gap $\Delta > 0$ excludes a massless pole with $\omega(k_{\min}) \to 0$. Away from $k = 0$ the symmetric-traceless target has five pointwise components and the transverse-traceless projector has rank two, but its direction-dependent $k \to 0$ limit prevents an exact strictly finite-range direct TT projector. A viable escape therefore requires a critical physical sequence together with local tensor gauge constraints whose Ward identities generate the TT sector only in the infrared. The finite representation count is not a derived TFPT stress operator, and no massless pole, universal coupling, diffeomorphism Ward identity or gravitational continuum theorem is constructed. GRAV.SPIN2.EMERGENCE.01, T7 and TOE remain [O].

Research Contract — FTRANSFER.SK.RHO0.01 (Schwinger–Keldysh completion: cosmological transfers need (S, ρ_0))

Typing. A research contract [O] (ledger row FTRANSFER.SK.RHO0.01), registered 2026-08-28 from review wave 3; mechanism executed 2026-08-29. **Demand.** Cosmological transfers ($F_{\rm relic}$, $F_{\rm Boltzmann}$, and any nonequilibrium 4D readout) need the pair (S, ρ_0), not $W[J]$ alone — the Euclidean-cap / KMS initial state. **Typed candidate.** The $\theta_i = 3\pi/5$ saddle $S_{\rm init} = -\kappa\cos(5\theta - 3\pi)$ as the selection. **Arithmetic.** Stationary points $\theta = (3 + k)\pi/5$; the μ_4 orientation picks $k = 0$. **Relation.** FTRANSFER.GENERATING.01 covers *equilibrium* correlators of $W[J]$ only; this contract is the Schwinger–Keldysh completion. **Kill tests.** The (S, ρ_0) pair is inconsistent with the $W[J]$ of TFPT4D.LATTICE.ACTION.01; the $3\pi/5$ saddle is not selected by μ_4 orientation (a different k is forced); the Euclidean cap fails KMS. Status: FTRANSFER.SK.RHO0.01 [O].

Executed update (2026-08-29, review wave 4): mechanism executed ([verification/v1000_spin2_sk_witnesses.py]; the contract stays [O]). KMS/FDT on the exact Bohr grid $\sim 10^{-17}$; ρ_0-dependence contrast 3.550×10^{-3} versus static 10^{-16}; saddle 5-fold degenerate with $3\pi/5$ unique in the chosen μ_4 lift [C]. No cosmological solve. Display marker stays [O].

Executed update (2026-08-30, review wave 7): finite uniqueness closed with two typed review corrections ([verification/v1009_rho0_minimizer.py]; the contract stays [O]). The constrained minimizer is uniquely the normalized KMS compression (Klein); Hessian spectrum strictly positive on $[3.76, 12.6]$; centre dimension 1; compression identity $\sim 8\times10^{-17}$. TYPED: V1000_CONSEQUENCE_MISMATCH — the compression kills the SK response (RMS $\sim 1.5\times10^{-31}$ versus frozen contrast 3.550×10^{-3}), so the proposed constraint set is too restrictive; and CENTER_FLATNESS_PREMISE_FALSE — Umegaki uniqueness holds with a nontrivial centre. The canonical formula needs a weaker admissible set. Display marker stays [O].

Executed update (2026-08-30, evening TOE-gate wave): selection-principle vacuity typed ([verification/v1012_toe_gate_witnesses.py]; the contract stays [O]). Eight unique candidates. DOUBLE DIAGNOSIS: hard Gauss / $\mu_4, +$ sectors kill the SK response (RMS $\sim 10^{-31}$); every response-alive set is exactly $\rho_{\rm KMS}$ even *without* μ_4 — the entropic-proximity selection principle is vacuous at finite level. Verdict NO_SET_WORKS. BINDING: the canonical state requires a different principle (e.g. Euclidean-cap / orientation-branch). Display marker stays [O].

Executed update (2026-08-31, Monday-morning harvest; [verification/v1016_state_gap_batteries.py]; the contract stays [O]). The R_4 seam-modular cap is the unique finite-level non-entropic selector ($D_{\rm tr}(\text{KMS}) = 0.1434$, SK response alive 3.038×10^{-3}, μ_4-invariant, conditional

$3\pi/5$ saddle preserved). R_1 vacuous, R_2 response-dead, orientation-free mutant loses saddles. Caveat: the difference lives in the neutral spectator factor — finite proxy. Display marker stays [O].

Executed update (2026-09-05, round-3 integration): parent-internal cap and enlargement boundaries (`FTRANSFER.PARENT.Z5CAP.01`; [`verification/v1024_parent_internal_boundaries.py`]; no parent marker move). The selected internal twelve-dimensional $\mathbb{Z}_5$ cap has a unique subspace ground state and gap 0.197423715893; its positive vacuum-preserving extension retains the checked finite parent gap and a nonzero mixed response. However, the five-cycle is chosen rather than derived, the cap breaks the checked D_4 action (residual 1.094700), and the extension vanishes on a large complement. It therefore does not select a global ρ_0. The exact Weyl-pair divisibility obstruction applies only to the unprojected parent dimension; the physical Gauss-projected dimension has not been counted. Likewise $M_n' \cap M_{nd} \cong I_n \otimes M_d$ is an irreducibility/prime-completion obstruction, not a general dynamical-coupling no-go. `FTRANSFER.SK.RHO0.01`, T8 and TOE remain [O].

Research Contract — `SEAM.MMST.TYPEIII.CHARGED.01` (one scaling-limit theorem)

Typing. A research contract [O] (ledger row `SEAM.MMST.TYPEIII.CHARGED.01`), registered 2026-08-29 from review wave 5. **Demand.** *One* scaling-limit theorem $\Psi_{\lambda,N} \to \Psi_\lambda$ strongly on the finite-energy core + crossed-product continuity, from which $B \cong (E_8)_1$, $\mu = 1$, type III_1, unique rotational KMS, unique conditional expectation, deck equidistribution and modular invariance all follow as corollaries. **Children** (listed here, not listing back): `SEAM.SIMPLECURRENT.GENERATOR.01`, the continuum leg of `SEAM.STATE.RPMIXING.01`, the rigidity premise of `ALPHA.QUILLEN.EXACT.01`, the P1 reading of `AX.P1.01`. **Cited document.** `articles/2026-08-29/holomorphic_kms_extension_en.tex` (13 pp, compiled): unconditional rotational-KMS uniqueness via KLM + Longo–Tanimoto *given* the identification; P1 = theorem-conditional on seam identification + $\beta_{\text{angle}} = 2\pi$. Axiom typing of `AX.P1.01` UNCHANGED. **Kill tests.** The scaling limit fails to exist; uniqueness of rotational KMS fails; type III_1 fails; a competing conditional expectation exists. Status: `SEAM.MMST.TYPEIII.CHARGED.01` [O].

Executed update (2026-08-30, review wave 7): five lemmas in-house; residual TEL-B-EXTERNAL + ALG-EXH boxed ([`verification/v1006_mmst_lemma_battery.py`]; the contract stays [O]). Cited `articles/2026-08-30/mmst_charged_scaling_limit_en.tex` (21 pp) and `articles/2026-08-30/externalization_mmst_handoff_v2_en.html`. In-house: L3 crossed-product; UGF $K_G = \pi^2/4$ (measured plateau ~ 1.023, max measured K 1.029399 at $N = 16$); TEL-a N^{-2} with exact isometries ($A_{\text{TEL}} = 5120(1 + \pi/2)$, measured envelope $3.420384 < 8$); integer D5+A3 pairings; $\mathbb{Z}_4$ outerness in the limit + finite-inner correction (256-dim gauge unitary is inner). Residual boxed: TEL-B-EXTERNAL (Hankel C_R; HS remainder sourced from the paper max 2.105079358 / fit 2.107565497 / v998 bound 2.11, not recomputed) + ALG-EXH (two Buchholz–Verch estimates). Display marker stays [O].

Executed update (2026-08-31, Monday-morning harvest; [`verification/v1016_state_gap_batteries.py`]; the contract stays [O]). The third (certified-tail) TEL-B route is BLOCKED: $N^2 D_N$ grows $530 \to 19032$, remainder UV-supported. Strictly smaller external target: the scalar dyadic increment $|\|R_{2N}\| - \|R_N\|| \le A_R/N$ with $A_R < 43$ sufficient for $C_R < 3$ (measured N-increments 0.120–0.123, factor ~ 350 headroom). N-uniform CAR nuclearity PROVED ($\nu_N \le e^{16 C_\beta}$; $C_\beta = (67.97, 29.73, 12.15)$ for $\beta = (0.5, 1, 2)$) as ALG specialist compactness input; ALG-EXH projector-identification UNCHANGED. Display marker stays [O].

TEL-B reduction (2026-09-04)

The residual `TEL-B-EXTERNAL` is restated below. The earlier scalar dyadic increment $|\|R_{2N}\| - \|R_N\|| \le A_R/N$ with $A_R < 43$ sufficient for $C_R < 3$ is superseded — A_R was measured 0.1238

(r645) but is no longer the route. Numerically closed ($1.7833 + 0.7863 = 2.57 < 3$; conditional analytic $2.9495 < 3$ with $c_E = 0.30$); round 3 closed C1, C2a and the interval-TV input. Round 4 below now closes CF, DG and the relaxed uniform TEL-B norm, including the finite-grid aliases: $\|R_N\|_{\mathrm{HS}} < 2.995906 < 3$ for all even $N \geq 16$. ALG-EXH and the charged MMST identification remain open. Not a marker move; display stays **[O]**. Evidence (experiments, not `veri`-citable): `experiments/tfpt-discovery/mmst_telb_trace_decomposition_probe.py`, `mmst_telb_cover_split_probe.py`, `mmst_telb_bound_a_probe.py` (9/9), `mmst_telb_bound_b_probe.py` (17/20, G18–G20 open by design). The historical v998 2.11 value is not promoted to an all-volume theorem; the new load-bearing certificate is [`verification/v1026_telb_round4_closure.py`].

Cover identity and sawtooth split. Corpus convention e^{-ipx}, $\alpha = \frac{1}{4}$, $M = 4N$, $d = x - y$. The four-fold cover identity is

$$(\Gamma_0 - \Gamma_\alpha)(x,y) = (1-i)\,C_M(d+N) + 2\,C_M(d+2N) + (1+i)\,C_M(d+3N).$$

The remainder splits as $R_N = R_N^{\mathrm{sm}} + M_N$ with $M_N = -(S_N + H_N) \otimes Q$. The scalar sawtooth response has the closed form

$$s_N(x,y) = \frac{-(1+t^2) + 3it - it^3}{4N(1-t^2)}, \qquad t = \tan\frac{\pi(x-y)}{4N}.$$

Triangle target: $\|R_N\| \leq \|R_N^{\mathrm{sm}}\| + \|M_N\| < 3$.

Bound (A) — native interval input closed (`SEAM.MMST.TELB.ROUND3.01`; [`verification/v1022_telb_round3_certificates.py`]). The continuous family $F = P - \mathrm{saw} \cdot dP$ has vanishing derivative jump ($\mathrm{rank}\, dF' = 0$, $\|dF'\| = 3.4\times10^{-10}$); the first discontinuity sits in F'' (rank 4, norm 2.000000029). The 2×2 crossing lemma supplies the removable crossing chart. The all-mode BV bound $|\hat{F}(m)| \leq \mathrm{TV}(F'')/(2\pi|m|)^3$ with 256-bit Arb/Acb arithmetic on 5812 rational cells gives

$$\mathrm{TV}_p(F'') \leq 15.8176959629889243576\ldots < 15.928674, \qquad K_3^{\mathrm{TV}} < 2.517464500835.$$

The cut contribution has exact Hilbert–Schmidt norm 2; the remaining cells use validated eigensystems, Taylor enclosures and an analytic ninth-derivative majorant. Thus the former float64/Richardson constant is retired. Round 4 additionally certifies the infinite Fourier core below 1.783260551 and the finite-grid alias contribution below 0.017333. The actual uniform smooth remainder bound is $\|R_N^{\mathrm{sm}}\|_{\mathrm{HS}} < 1.800594$; the older finite 1.7833 claim was not justified by the limiting core and is withdrawn.

Bound (B) — historical reduction; closed in round 4 below. Relaxed target $\|S_N + H_N\|^2 \leq 1.2167^2/2 = 0.740179$ (measured limit 0.5913). Write $S_N + H_N = A_\infty + \mathrm{Res}_N$ with $\|A_\infty\|^2 = \frac{5}{4} - 8/\pi^2$ exact. Lemma 6 cross term ≤ 0.3813 (all even $N \leq 512$, and 1024/2048/4096). Needed $\sum|\mathrm{Res}_N|^2 \leq 0.276873914$; measured max 0.154983791. Master identity (divided differences via the cotangent partial fraction; weight $\{(m+\alpha)/N\}$ is N-periodic; residual 2.3×10^{-15}). One-variable reduction

$$N \cdot \mathrm{Res}_N(k,l) = r(u,v) + A\frac{(1-\lambda)\,\Delta\eta - \lambda\,\Delta\varphi_N}{v-u}$$

(residual 1.16×10^{-14}), with $u = k/N$, $v = l/N$, $A = \sin^2(\pi\alpha)/\pi^2$, and

$$\begin{aligned}
\eta(k) &= -\psi(k+1-\alpha) + \log|k| + \pi\cot(\pi\alpha)\,1_{k<0},\\
\varphi_N(k) &= \Phi_N(k) - \log N - \gamma - \Phi_\star(k/N),\\
\Phi_\star(u) &= \log\Big|\frac{u}{\tan\pi u}\Big| - \pi\int_{u-1/2}^{u+1/2}\sigma\cot(\pi\sigma)\,d\sigma - \pi u\cot(\pi\alpha).
\end{aligned}$$

Closed bounds $\pi^2/3 \leq \Phi_\star'' \leq \pi^2 - 4$. The kernel r is singular only on $u = 0$, $v = 0$. The former offset midpoint values were $C_\star = 0.152044949/0.152045069$; the native certificate below replaces them as the load-bearing bound. Cell averages $(\mathcal{K}_N)_{kl} := N\iint_{Q_{kl}} r$ obey $\|\mathcal{K}_N\|^2 \leq C_\star$ exactly

(Cauchy–Schwarz on each cell). Measured $\sqrt{N}\,\|E_N\| = 0.1943\ldots0.2013$ ($N = 16\ldots512$). Required $c_E \le 0.542702$, margin 2.70. Assembled $(\sqrt{0.1525} + 0.2013/4)^2 = 0.1943 < 0.2769$.

C1 and C2a closed (2026-09-05; `SEAM.MMST.TELB.C1C2A.01`; `[verification/v1025_telb_c1_c2a_certificates.py]`). For C1, the exact simplification of the real kernel is integrated over 3888 analytic rectangles at 100-bit precision. Logarithmic charts remove the two axis singularities, a removable seam chart treats the opposite-sign corner, and the omitted axis union is bounded by $3.1570760454\times10^{-7}$. Parseval supplies the exact imaginary contribution $1/12$, giving the outward enclosure

$$C_\star \le 0.1520453048\ldots < 0.1525.$$

For C2a, exact cotangent reindexing gives the 3/4-point Peano identity

$$\varphi_N(k+1) - \varphi_N(k) = hf((k+3/4)h) - \int_{kh}^{(k+1)h} f(u)\,du, \qquad h = N^{-1}.$$

The partial-fraction formula $f'(u) = \sum_{m\in\mathbb{Z}\setminus\{0\}}(u-m)^{-2}$ implies $\pi^2/3 \le f' \le \pi^2 - 4$ and makes the increments monotone. Shifted Euler summation with its periodic B_3 remainder reduces the two endpoints to outward constant comparisons in v1025:

$$N \sup_k |\varphi_N(k)| \le 0.508228779140631\ldots < 0.509 < 0.51 \qquad (N \ge 16 \text{ even}).$$

C2b scalar reduction and the former remaining target (`[verification/v1022_telb_round3_certificates.py]`). The analytic one-dimensional reduction, with its final arithmetic enclosed by Arb, gives

$$\begin{aligned} c_\varepsilon &= 0.288547359238391\ldots, \\ c_\zeta &= 0.110677064900715\ldots, \\ c_{\text{potential}} &= 0.399224424139106\ldots < 0.400. \end{aligned}$$

The executable encloses $1 \le n \le 15$ and checks positive-coefficient rational tail comparisons for $n \ge 16$. The sign-controlled digamma and midpoint inequalities which justify those tails are analytic lemmas of this proof, not an independent formalization inside the executable. The former CF/DG targets were

$$\sqrt{N}\,\|(K'_N - K_N)_{\text{offdiag}}\|_{\text{HS}} \le 0.139, \qquad \sqrt{N}\,\|(\text{Res}_N - K_N)_{\text{diag}}\|_{\text{HS}} \le 0.060.$$

The diagonal is Hilbert–Schmidt orthogonal to the off-diagonal, and these two bounds suffice for the stated $c_E \le 0.542702$ budget. Round 4 proves the stronger constants 0.065697 and 0.042959, respectively, and gives $c_E < 0.466902$. The proof and interval arithmetic are in the round-4 supplement below and v1026; these are no longer open inputs.

ALG2 correction (binding). The former Vandermonde-to-generated-algebra implication is withdrawn. The exact one-mode countermodel has full CAR algebra M_2, while the even/code-generated subalgebra is diagonal; the annihilator stays at distance at least 1 from it. Fourier/Vandermonde inversion therefore supplies only full-CAR compression, not the microscopic generated algebra. ALG2 requires an independent FE-GEN compression-surjectivity theorem or a precise gauge/code-projected microscopic algebra argument. The positive ALG1 estimate is likewise conditional on the generated projection containing the microscopic low-energy sectors. The round-4 sector/frame lemma gives a sufficient conditional interface, not a microscopic FE-GEN proof. ALG-EXH and `SEAM.MMST.TYPEIII.CHARGED.01` remain [O].

Retired routes (do not retry). Absolute-value majorants ($3\sqrt{N}$); global BV majorant; per-cell log counterterms at $\pm\frac{1}{2}$ (nothing to counterterm); shifted-cell majorisation (ratio 0.73); Szegő/Widom split (Fisher–Hartwig $\pi^{-2}\log N$ per summand); trace decomposition (each term diverges $\propto \log N$, cancellation $2/\pi^2 + 2/\pi^2 - 4/\pi^2$).

Research Contract — `TFPT.TOE.COMPLETE.01` (the named AND of eight completeness gates)

Typing. A research contract [O] (ledger row `TFPT.TOE.COMPLETE.01`), registered 2026-08-29 from review wave 4. This row *names* the AND of eight gates; it does *not* close any of them and it does *not* upgrade a marker. Dual rest: the compiler residual Rest $= v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ sits beside Rest_{TOE}; this contract is the named AND, not a closure of either.

The eight gates. **(T1)** structure postulate → P1/P2/compiler (`AX.P1.01`, `AX.P2.01`, `TFPT.IRREDUCIBLE.01`, `DIMENSION.SELECTOR.4D.01`).
(T2) seam = $(E_8)_1$ proven (`SEAM.EQUIV.01`, `SEAM.EQUIV.MMST.01`, `SEAM.EQUIV.TWISTOR.01`, `SEAM.SIMPLECURRENT.GENERATOR.01`, `SEAM.MMST.TYPEIII.CHARGED.01`, `SEAM.DETLINE.UNIFICATION.01`, `SEAM.STATE.RPMIXING.01`).
(T3) quasilocal local unitary 3+1D Hamiltonian family (`QFT4D.LATTICE.FUNDAMENTAL.01`, `TFPT4D.LATTICE.ACTION.01`, `DYN.UNITARY.DILATION.01`, `DYN.MARKOV.EMBED.01`).
(T4) chiral SM with local measure + uniform mirror gap (`CHIRAL4D.NOMIRROR.01`, `CHIRAL4D.MIRROR.DET16.01`).
(T5) IR Lorentz + confinement + clustering + nontrivial scattering (`QFT4D.OS.RECON.01`, `SEAM.BULK4D.RECON.01`).
(T6) all three gauge couplings + neutrino texture internally fixed (`ALPHA.QUILLEN.EXACT.01`, `GAUGE.DETLINE.FIXPOINT.01`, `FLAV.NUSCALE.01`–`.06`, `FLAV.NU.TEXTURE.MECHANISM.01`).
(T7) quantum massless spin-2 with universal coupling (`GRAV.SPIN2.EMERGENCE.01`, `GRAV.NONCIRCULAR.01`, `GRAV.WEINBERG.WITTEN.01`).
(T8) unique initial state + one SK functional generating all readouts (`FTRANSFER.SK.RHO0.01`, `FTRANSFER.GENERATING.01`, `OBS.TRANSDUCTION.01`, `PRED.JOINTLIKELIHOOD.01`).

Separation (binding).

$$\text{TOE}_{\text{Complete}} := \text{AND}(\text{T1}, \ldots, \text{T8}), \qquad \text{TOE}_{\text{Validated}} := \text{Complete AND independent holdouts passed.}$$

Every existing 4D / dilation / detline / SK / flavor / gravity contract is a *child of one gate* (listed above); children do not list this row back. Status: `TFPT.TOE.COMPLETE.01` [O].

Executed update (2026-08-30, evening TOE-gate wave): four witnesses in one battery (`[verification/v1012_toe_gate_witnesses.py]`; the contract stays [O]). T5 four IR witnesses executed plus the propagating-mode precondition (common-c needs dynamical gauge excitations). T7 interacting stress mechanism executed. T8 selection-principle vacuity typed — a non-entropic selection is required. T6/nu scaffold-level triple bound complete. Display marker stays [O].

Executed update (2026-08-30, late-evening harvest; `[verification/v1013_thermodynamic_dynamics.py]`, `[verification/v1014_bridge_refinements.py]`; the contract stays [O]). T3 mandatory-dynamics leg CLOSED at Hamiltonian-class level (Lieb–Robinson + τ_t + $\mathcal{A}^G$ + state existence). Within the finite W-bridge/compiler lane considered here, the remaining identification was routed through the external MMST/Quillen leg (TEL-B/ALG-EXH/P1 + the character-blind determinant-response map + Bismut–Freed/A_0). This historical statement does not classify or close the independent TOE gates T3–T8. Display marker stays [O].

Executed update (2026-08-31, Monday-morning harvest; `[verification/v1015_axiom_core_closure.py]`, `[verification/v1016_state_gap_batteries.py]`; the contract stays [O]). T1 structure-postulate route is now fully premise-supported at finite level (both W-bridge premises derived; axiom-core remainder 0 modulo the externalized MMST identification). First viable canonical-state principle (R_4 seam-modular cap). Third-route volume theorem cited-verified for finite-group DET chains. The old phrase "every remaining fundamental gap" is scope-corrected: only the finite compiler/W-bridge lane was reduced to MMST plus the 4D functional; the independent T3–T8 mechanisms were not thereby classified. Display marker stays [O].

Executed update (2026-09-05, T1–T8 integration; upstream round-3 inputs `[verification/v1022_telb_round3_certificates.py]`, `[verification/v1023_derivative_filtered_`

wall.py], [verification/v1024_parent_internal_boundaries.py], [verification/v1025_telb_c1_c2a_certificates.py]). With the round-4 and round-7 modules and complete proof supplements below, the current gate matrix is:

- **T1:** the finite axiom-core/W-bridge premises have executable support, but their unconditional structure-postulate provenance and the dimension selector remain open.
- **T2:** the relaxed uniform TEL-B norm is closed below $2.995906 < 3$ at fixed $M = 1$, $N_y = 8$ for all even $N \geq 16$, including C2b and finite-grid aliases. Microscopic one-edge generation, FE-GEN/ALG-EXH and MMST remain open; the sector/frame bridge is conditional and the old Vandermonde implication remains refuted. Round 7 constructs charged corners only after adding an auxiliary local $\mathbb{Z}_4$ flux register. The simple eight-channel quarter twist has $h = 1/4$, not $h_\lambda = 1$; the required microscopic dressing and gauge/code quotient have not been supplied.
- **T3:** dynamics exists only at Hamiltonian-class level. No shared complete 3+1D parent or continuum theorem is constructed.
- **T4:** exact canonical DET excitations and a signed $z = 1$ wall exist under the stated singlet and fixed-classical-background premises. The source's factorized onsite model has gap at least $1 + \sqrt{3}/2$; this bound is not an interacting hopping theorem. An ordinary bond violates the proposed vacuum-relative form bound, not every possible interacting gap argument. No hard bare-mirror gap, chiral quantum-link realization, measure or 4D continuum theorem follows.
- **T5:** finite IR witnesses and the v1020 spectator obstruction make prime completion binding, but OS reconstruction, confinement, clustering and nontrivial scattering remain open.
- **T6:** full Gauss counts remove the blanket projected-dimension obstruction (one site 3 mod 5, cube 0 mod 5 with the specified nine-irrep link truncation and full 192-mode site character), but give no canonical compatible clock family or neutrino mass texture. The symmetric candidate forbids every Majorana bilinear.
- **T7:** the local first-class constraint target now has an explicit zero-mean Weyl/Fock quantization, a distributional continuum limit and a positive covariant local free curvature field of helicities ± 2. Its unstable periodic zero block must still be removed globally. Prescribed conserved sources obey the linear physical-Fourier Ward identities, but dynamical quantum sources and universal nonlinear coupling are not constructed. None of these target-field results supplies a common-parent TFPT embedding.
- **T8:** a symmetric cap on the selected internal 12D pair space has gap one, but the invariant space is two-dimensional; symmetry does not select its ray, and its vacuum-preserving extension selects no global ρ_0 or SK functional.

All T1–T8 gates and `TFPT.TOE.COMPLETE.01` remain [O]; no marker moves. A single shared 3+1D parent satisfying T3–T8 is still missing.

Round 4: proved subgates and binding residuals (2026-09-05)

The following results concern the explicitly stated regulators and algebras. They do not identify a single complete physical 3+1D parent. All T1–T8 and `TFPT.TOE.COMPLETE.01` remain [O]. The finite regressions are checks of the proofs below, not substitutes for their quantifiers.

T2: uniform relaxed TEL-B theorem. `SEAM.MMST.TELB.NORM.01`. For the frozen $M_{\mathrm{mass}} = 1$, $N_y = 8$ collar, the original remainder obeys

$$\boxed{\|R_N\|_{\mathrm{HS}} < 2.995906 < 3 \quad (N \geq 16 \text{ even}).}$$

The interval inputs are v1022 (TV and potential), v1025 (C1/C2a), and [verification/v1026_telb_round4_closure.py] (CF, DG, signed CROSS, validated Fourier core and all-volume aliases). This

is not a proof of the historical sharper 2.11 or Bound-B target 0.62. In particular a limiting Fourier-core bound is not a finite-grid remainder bound.

CF: cell covariance and the periodic corner. Put $h = N^{-1}$, $A = (2\pi^2)^{-1}$, $J = N/2$, $I_N = \{-J, \dots, J-1\}$ and let bars denote normalized means over $[kh, (k+1)h]$. Define

$$f(u) = -\log|u| - \pi \mathbf{1}_{u<0}, \qquad \Phi(u) = \log\frac{|u|}{2|\sin \pi u|} - \pi u, \qquad F = f + \Phi,$$
$$c(d) = d^{-1} - \pi\cot \pi d, \quad c(0) = 0, \qquad Q_g(u,v) = \frac{g(v) - g(u)}{v - u}.$$

Partial fractions give $\Phi' = c - \pi$, $L = \sup|\Phi'| = \pi + 2$, $m = \inf \Phi'' = \pi^2/3$, $M = \sup \Phi'' = \pi^2 - 4$ and $D = M - m = 2\pi^2/3 - 4$, with $\int |\Phi'''| = 2D$. The identities used are the standard cotangent and polygamma identities DLMF 4.22 and DLMF 5.15; no asymptotic truncation is needed. The real kernel cancellation is exact:

$$r(u,v) = A\{c(v-u)(F(v) - F(u)) - Q_\Phi(u,v) - i\pi(F(v) - F(u))\}.$$

For independent uniform cell points U, V, set $d_0 = (l-k)h$ and $\delta = V - U - d_0$. Then

$$(\mathcal{K}_N)_{kl} = h\mathbb{E}r(U,V), \quad (\mathcal{K}'_N)_{kl} = hA\left[c(d_0)(\bar F_l - \bar F_k) - \frac{\bar\Phi_l - \bar\Phi_k}{d_0} - i\pi(\bar F_l - \bar F_k)\right].$$

Thus the imaginary part freezes exactly. If $v_k = \operatorname{Var} F(U_k)$, $|a'| \le a_1$, $|a''| \le a_2$, the independent-copy variance identity, $\mathbb{E}\delta = 0$, and $\mathbb{E}\delta^2 = h^2/6$ give

$$|\mathbb{E}[(a(d_0) - a(V-U))(F(V) - F(U))]| \le \frac{a_1 h}{\sqrt{6}}\sqrt{v_k + v_l} + \frac{a_2 h^2}{12}|\bar F_l - \bar F_k|. \tag{R4.1}$$

For $|g''| \le B$, the two coordinate Lipschitz constants of Q_g are at most $B/2$, so $\operatorname{Var} Q_g \le B^2h^2/24$ and $|\mathbb{E}(\delta Q_g)| \le Bh^2/12$. The two singular log cells have variance one; all remaining log-cell variances are at most $1/(12n^2)$. Therefore

$$(\textstyle\sum_k v_k)^{1/2} \le V(h) := \sqrt{2 + \zeta(2)/6} + L\sqrt{h/12}, \quad \textstyle\sum_{k,l}|\bar F_l - \bar F_k|^2 \le 2N^2\zeta(2).$$

The last bound follows from $\operatorname{Var} F = \pi^2/6$ and contraction of cell conditional expectation. Split the nonzero lags into central ($|l-k| \le N/2$), middle ($N/2 < |l-k| \le 3N/4$), and far regions. For the first two regions (R4.1) gives

$$C_{\text{central}} \le \frac{Ac'(9/16)hV(h)}{\sqrt{3}} + \frac{Ac''(9/16)h^{3/2}\sqrt{2\zeta(2)}}{12} < 0.027267,$$
$$C_{\text{middle}} \le \frac{Ac'(13/16)h^{3/2}}{\sqrt{6}}\left(\sqrt{H_{J-1}/3} + L/\sqrt{24}\right) + \frac{Ac''(13/16)h^{3/2}\sqrt{2\zeta(2)}}{12} < 0.055905.$$

In the middle count, negative index $-n$ has $n-1$ partners and positive index s has s partners; the singular cells have none. Consequently the log-variance sum is at most $H_{J-1}/3$ and the smooth part at most $L^2/24$. The sequence H_{J-1}/J^3 decreases for $J \ge 8$, by its one-step recurrence, so the displayed envelopes are evaluated at $N = 16$.

For the far seam let $r = N - (l-k)$, $t = 1 - (V-U)$ and $t_0 = rh$. The short periodic continuation satisfies $F(V) - F(U) = -tQ$ with $|F'| \le L_F = 2\pi$ and $|F''| \le M_F = 2\pi^2$. In $c(d) = (1-d)^{-1} + a(d)$ the apparent pole cancels exactly:

$$\mathbb{E}[(t_0^{-1} - t^{-1})(F(V) - F(U))] = -t_0^{-1}\mathbb{E}[(t - t_0)Q].$$

There are r pairs per orientation. With $a_1 = (16/11)^2 + c'(5/16)$, $a_2 = 2(16/11)^3 + c''(5/16)$,

$$C_{\text{far}} \le AM_F h^{3/2}\sqrt{H_{\lfloor N/4\rfloor}/72} + Aa_1L_Fh^{3/2}/6 + Aa_2L_F(5/16)h^{3/2}/12 < 0.009280.$$

Here $H_{\lfloor N/4\rfloor}/N^3 \le H_4/16^3$ follows by grouping four successive sizes. The remaining smooth quotient contributes $C_\Phi \le AMh\sqrt{\zeta(2)/72} < 0.002810$. The three lag regions are HS-orthogonal; only the last contribution uses the triangle inequality:

$$\sqrt{N}\|(\mathcal{K}'_N - \mathcal{K}_N)_{\mathrm{off}}\|_{\mathrm{HS}} \le \sqrt{C^2_{\mathrm{central}} + C^2_{\mathrm{middle}} + C^2_{\mathrm{far}}} + C_\Phi < 0.065697. \tag{R4.2}$$

DG: original diagonal, not a replacement recurrence. The csc-squared finite product identity and trigamma recurrence for the original remainder give, for $R_k = (\mathrm{Res}_N)_{kk}$,

$$R_{k+1} - R_k = -Ah^2\Phi''((k+3/4)h), \qquad \sum_k R_k = A\pi + A\tau(J),$$

where $T(x) = (x-1)\psi_1(x) + \psi(x)$ and $\tau(J) = T(J+5/4) - T(J+3/4)$. Since $T'(x) = 2\sum_{n\ge0}(n+1)/(n+x)^3 \le 4/J$ for $x \ge J \ge 8$, $0 \le \tau(J) \le 2/J = 4h$. The 3/4-point Peano kernel and $h\sum_k \Phi'(kh) = -\pi - 2h$ imply

$$|NR_k + A\Phi'(kh)| \le Ah(2 + 3D/2). \tag{R4.3}$$

On each diagonal cell the imaginary mean vanishes, $|Q_\Phi - \Phi'(kh)| \le Mh$, and $|c(d)| \le C_s|d|$, $C_s = (\pi^2/3)/(1-h^2)$. The worst logarithmic integral occurs adjacent to zero and is exactly $\mathbb{E}[|V-U||\log|V/U||] \le h/2$; also $\mathbb{E}|V-U|^2 = h^2/6$. Thus

$$\sqrt{N}\|(\mathrm{Res}_N - \mathcal{K}_N)_{\mathrm{diag}}\|_{\mathrm{HS}} \le Ah(2+3D/2) + AMh + AC_s(h/2 + Lh^2/6) < 0.042959. \tag{R4.4}$$

All scalar envelopes increase with $h \le 1/16$. With the independently certified potential contribution 0.399224424139108, orthogonality gives $c_E < 0.466902$, below the required 0.542702.

CROSS: the sufficient one-sided theorem. Let W_∞ be the unitary fractional shift with entries $e^{-i\pi/4}\sin(\pi/4)/[\pi(k-m-1/4)]$, let P restrict to I_N, $w = PW_\infty e_0$, $B = W_\infty P_{(-\infty,-1]}W_\infty^*$, $t_N = \|(1-P)W_\infty e_0\|^2$ and $\mu_N = \sum k|w_k|^2$. Crucially $BW_\infty e_0 = 0$. For the finite unitary overlap W_N and $\rho_m = \{(m+1/4)/N\}$, the source identity is

$$\mathrm{Res}_N = -\operatorname{diag}(k)/N + W_N \operatorname{diag}(\rho_m) W_N^* - PBP.$$

Hence $w^*\mathrm{Res}_N w \ge -\mu_N/N + (1-t_N)/(4N) - t_N$. Pairing signed momenta and using convex reciprocal-square tail integrals yields

$$\mu_N \le 64/675, \qquad Nt_N \le T_* := \frac{4A}{1-(3/32)^2}.$$

For $A_N = |e_0\rangle\langle e_0|/2 - |w\rangle\langle w|$, (R4.3) at $k=0$ and the exact identity $2\,\mathrm{Re}\langle A_N, \mathrm{Res}_N\rangle = (\mathrm{Res}_N)_{00} - 2w^*\mathrm{Res}_N w$ imply

$$N\,2\,\mathrm{Re}\langle A_N, \mathrm{Res}_N\rangle \le A\pi + \tfrac{A}{16}(2+3D/2) - \tfrac{1}{2} + \tfrac{128}{675} + 2(1+1/64)T_* < 0.282637. \tag{R4.5}$$

This is not an absolute-value bound: the negative finite cross terms explicitly falsify the absolute $0.283/N$ mutant. Cell contraction and C1 give

$$\|S_N + H_N\|^2_{\mathrm{HS}} \le \tfrac{5}{4} - \tfrac{8}{\pi^2} + (\sqrt{0.1525} + 0.466902/4)^2 + 0.282637/16 < 0.714386. \tag{R4.6}$$

Validated Fourier core and the missing finite-grid aliases. Use the sigma-x basis $H(p) = \left(\begin{smallmatrix} \sin p\, I_8 & B_p \\ B_p^T & -\sin p\, I_8 \end{smallmatrix}\right)$, $B_p = (1-\cos p)I_8 - S$, $S_{y,y+1} = 1$. The jump is $dP = E_{8,8} - E_{7,7}$ (zero-based indices), and $F(p) = P(p) - (1/2 - p/(2\pi))dP$ for $0 < p < 2\pi$. At the 512 midpoint nodes, 256-bit Acb validated eigensystems strictly separate eight occupied and eight unoccupied eigenvalues. The outward matrix DFT A_m obeys the exact midpoint alias formula $A_m = \sum_{\ell\in\mathbb{Z}}(-1)^\ell C_{m+512\ell}$.

The v1022 bound $\|C_m\|_{\mathrm{HS}} \le K_3/|m|^3$, $K_3 = 2.517464500835$, controls the entire alias error by $K_3\zeta(3)[(512-m)^{-3}+512^{-3}]$. The certified rational caps for $m = 1,\dots,16$ are, in order,

$$.872617451262, .119561574970, .036384250090, .013594203794,$$
$$.005638853577, .002552800879, .001281003972, .000727535964,$$
$$.000465728162, .000325832279, .000241157627, .000184924134,$$
$$.000145312983, .000116348415, .000094610814, .000077965240.$$

Since $\sum_{m>16} m^{-5} \le 1/(4\cdot 16^4)$, these give $2\sqrt{\sum_{m\ge1} m\|C_m\|_{\mathrm{HS}}^2} < 1.783260551 < 1.783261$. This is a bound on a dominant Fourier matrix, not yet on R_N^{sm}. Absolute summability gives the exact four-cover expression

$$A_N(d) = \sum_{q\notin 4\mathbb{Z}} (1-i^q) C_{d+qN}, \qquad \|R_N^{\mathrm{sm}}\|_{\mathrm{HS}}^2 = \sum_{d=-(N-1)}^{N-1} (N-|d|)\|A_N(d)\|_{\mathrm{HS}}^2.$$

Choose $q = 1$ for negative $d = -N+m$ and $q = -1$ for positive d; at $d = 0$ choose $q = 1$ once. The resulting dominant norm squared is $4\sum_{m=1}^{N-1} m\|C_m\|^2 + 2N\|C_N\|^2$, bounded by the infinite core. The other aliases have envelope

$$R(t) = \sum_{q\notin 4\mathbb{Z},\ q\neq 1} \frac{|1-i^q|}{|q-1+t|^3}, \qquad 0 \le t \le 1.$$

This nonsingular series is convex. Sum $|q| \le 100$ and add the full tail $4/100^3 + 2/100^2$; its outward bounds at $0, 1/2, 1$ are denoted a, b, c (approximately $2.528674155, 1.333485721, 2.086708706$). Use their actual Arb enclosures, not those abbreviated printed decimals. The piecewise affine interpolant $L(t)$ majorizes $R(t)$. For $f(t) = tL(t)^2$ the composite trapezoid rule on each half interval is exact for the cubic, including the derivative jump. For every even N,

$$T_N = \frac{2}{N}\left(\sum_{m=1}^{N-1} f(m/N) + \frac{f(1)}{2}\right) = I + D_*/(6N^2),$$
$$I = a^2/24 + ab/12 + b^2/3 + bc/4 + 7c^2/24,$$
$$D_* = 4b^2 - 2ab - a^2 + 5c^2 - 6bc.$$

Thus the alias norm is at most $K_3\sqrt{I + |D_*|/(6\cdot 16^2)}/16^2 \le 0.017332234304270 < 0.017333$ and $\|R_N^{\mathrm{sm}}\|_{\mathrm{HS}} < 1.800594$. Finally $R_N = R_N^{\mathrm{sm}} - (S_N + H_N)\otimes Q$, with $\|Q\|_{\mathrm{HS}} = \sqrt{2}$ from two orthogonal rank-one projectors. Using the unrounded alias enclosure with (R4.6) gives

$$\|R_N\|_{\mathrm{HS}} < 1.783261 + 0.017332234304270 + \sqrt{2(0.714386)} < 2.995906 < 3.$$

The fixed-width full collar still has two gapless edges. One-edge microscopic identification, FE-GEN, ALG-EXH and charged MMST do not follow from this norm theorem.

T3/T4: canonical DET excitations and the signed wall. `CHIRAL4D.MIRROR.SIGNED.CAR.01`. [`verification/v1027_signed_det_car_wall.py`] checks the following algebra and its finite countermodels. The old Hubbard operators $b_i = |\Omega\rangle\langle e_i|$ satisfy $\{b_i, b_j^*\} = \delta_{ij}P_\Omega + |e_j\rangle\langle e_i|$, not CAR, and $b_i^* b_j^* = 0$. For even $n \ge 4$ (in particular $n = 32$), write $F = |\mathrm{full}\rangle\langle 0|$ and define

$$U = I + (2^{-1/2}-1)(P_0 + P_F) + (F - F^*)/\sqrt{2}, \qquad a_i = Uc_iU^*.$$

This is a two-dimensional rotation and identity elsewhere. It is even; under the inherited full-determinant singlet premise it is onsite gauge invariant. It maps $|0\rangle$ to $\Omega = (|0\rangle + |\mathrm{full}\rangle)/\sqrt{2}$, so

the a_i satisfy CAR on the entire 2^n-dimensional Fock space and $a_i^*\Omega = e_i$. Products of even onsite rotations preserve cross-site CAR. The actual new parent is additive:

$$N_a = UNU^* = N + \tfrac{n}{2}(P_0 - P_F - F - F^*), \qquad Q \le N_a \le nQ, \quad Q = I - P_\Omega.$$

Thus ΔN_a has the same unique ground state and gap Δ as ΔQ, but is not the old flat parent. The high-degree local rotation is not asserted to be a linear Bogoliubov map or an efficient Fock simulator.

Let $A = A^*$ be a fixed c-number, gauge-covariant quasilocal one-particle operator, $\|A\| \le L$. Define

$$H = p^*Ap + \Delta a^*a + \lambda(a + gAp)^*(a + gAp), \quad M = \Delta + \lambda, \quad \beta = \lambda g^2, \quad \eta = \lambda g, \quad c = \lambda g^2\Delta/M.$$

If $cL < 1$ and $B = L + \beta L^2 < M$, the spectral block

$$h(s) = \begin{pmatrix} s + \beta s^2 & \eta s \\ \eta s & M \end{pmatrix}, \quad \det h(s) = Ms(1 + cs)$$

has precisely the original kernel, no extra zero, high band $\mu_+ \ge M$, and $|\mu_-| \le B$. Expansion gives $\mu_-(s) = s + cs^2 - \eta^2 s^3/M^2 + O(s^4)$, hence $z = 1$ when A is linear. For $(\Delta, \lambda, g, L) = (3, 1, 1/2, 2)$, $M = 4$, $B = 3$, $cL = 3/8$. Filling the negative band and subtracting its energy gives a nonnegative full-Fock Hamiltonian for each fixed classical background; $E_0 \ge -Lr|\Lambda|$ with r physical modes per site. The bare mirror operator nevertheless has low-band weight of order s^2. There is no hard bare-mirror response gap, nor an isolated entire many-body mirror continuum. Operator-valued quantum-link diagonalization does not automatically lift to a canonical Fock transformation (the explicit SWAP mutant fails CAR).

For a free illustration only, $\gamma_i = \tau_1 \otimes \sigma_i$, $\gamma_5 = \tau_3 \otimes I$, $w = \sum_i (1 - \cos k_i) - 1$, $R^2 = 1 + 2\sum_{i<j}(1 - \cos k_i)(1 - \cos k_j) \in [1, 25]$ and $D = I + (w + i\gamma \cdot \sin k)/R$ give $A = \gamma_5 D$, $A^2 = 2 + 2w/R$ and $\|A\| \le 2$, with one Dirac node at zero. The inverse-square-root series has norm ratio $12/13$ and exponential length $2/\log(13/12)$. This four-component regulator contains opposite Weyl sectors: it is not the demanded two-component chiral SM construction. The common-parent dynamical gauge, measure, anomaly/index and continuum obligations remain open.

T6/T8: full Gauss counts, internal cap and selection rules. `PARENT.GAUSS.D4.ROUND4.01`. [`verification/v1028_gauss_d4_cap.py`] uses the full exterior character, not a low-occupation truncation. Let W_{16} consist of $Q : (\mathbf{3}, \mathbf{2})_1$, $u^c : (\bar{\mathbf{3}}, \mathbf{1})_{-4}$, $d^c : (\bar{\mathbf{3}}, \mathbf{1})_2$, $L : (\mathbf{1}, \mathbf{2})_{-3}$, $e^c : (\mathbf{1}, \mathbf{1})_6$, $\nu^c : (\mathbf{1}, \mathbf{1})_0$, with charges $6Y$. For color weights $(1, 0), (0, 1), (-1, -1)$ and weak weights ± 1, set

$$P(z) = \prod_{w \in W_{16}} (1 + z^w), \quad h = 1 + (u + u^{-1})v^3, \quad \Delta_W = (1 - x/y)(1 - x^2y)(1 - xy^2)(1 - u^2).$$

There are twelve SM16 copies per site (192 fermion modes), a three-state Higgs space, and seam dimension four. The singlet multiplicity of a Weyl-invariant character is $\mathrm{CT}(f\Delta_W)$, not its zero-weight multiplicity. All representations descend to the $\mathbb{Z}_6$ quotient. In characteristic five, $P^{12} = P^2(z)P^2(z^5)$, so for $A = P^2$,

$$\mathrm{CT}(P^{12}z^r) = \sum_{a+5b+r=0} A_a A_b \pmod 5.$$

This finite sparse calculation has 20771 integer two-copy coefficients (17151 nonzero modulo five); it never enumerates 2^{192} states. The full one-site Gauss dimension is $4\,\mathrm{CT}(P^{12}h\Delta_W) \equiv 3 \pmod 5$. A neutral Majorana bijects the parity sectors, each of which has dimension $4 \pmod 5$. For the nine link irreps $0, Q, u^c, d^c, L, e^c, H, \mathrm{Adj}_3, \mathrm{Adj}_2$, whose squared dimensions sum to 137, the exact transfer matrix is $T_{RS} = \mathrm{CT}(P^{12}h\chi_R\bar{\chi}_S\Delta_W)$. An n-vertex open path has dimension $4^n(T^n)_{00} \pmod 5$. For $n = 1, \ldots, 8$ this is $3, 4, 3, 0, 0, 1, 4, 0$. The full open $2 \times 2 \times 2$ cube within this specified nine-irrep

link truncation (eight sites, twelve links) has dimension 0 (mod 5). Two pairwise contraction orders give reduced integer 48544377696195; this is not the Hilbert dimension. The bound $9^{12}4^8 < 2^{63}$ precludes contraction overflow. Independent exact raising-operator ranks check 56 singlets without Higgs ($556 - 500$) and 120 with Higgs ($1376 - 1256$) for one copy; the two-copy character result with Higgs is 561848. Thus the universal projected non-divisibility obstruction is false. An abstract Weyl pair exists when five divides the dimension, but locality, covariance, canonical choice, energy compatibility and compatibility between volumes are additional demands.

On the selected even 12-dimensional pair space $|k, g\rangle$, use $C = \mathrm{diag}(1, -i, -1, i) \otimes \mathrm{diag}(-i, -1, i)$ and $J|k, g\rangle = |-k \bmod 4, 4 - g\rangle$. Then

$$\psi = (|1, 3\rangle + |2, 2\rangle + |3, 1\rangle)/\sqrt{3}, \qquad h_{\mathrm{sym}} = I - |\psi\rangle\langle\psi|$$

is D_4 invariant, has unique ground vector and gap one, Schmidt rank three, and nonadditive HS norm $\sqrt{8/9}$. This is ordinary D_4 only on that selected even space, not the whole projective fermion lift. The invariant-vector space has dimension two, so symmetry does not select this ray. Extending the cap by zero outside this subspace leaves the old vacuum and does not select the full-parent state. The unitary $X = P_\psi + e^{2\pi i/5}(I - P_\psi)$ has order five but only two eigenvalues; it is not a Weyl clock, and its chosen order is not derived.

For the actual one-fermion phases $u_g = e^{-i\pi g/4}$, invariance of a Majorana bilinear requires $(u_g u_h - 1)M_{gh} = 0$. Since $2 \le g + h \le 6$, $M = 0$ for this symmetric candidate. A Hermitian pair kernel instead has commutant $\mathrm{diag}(a, b, a)$; the two transformation laws differ. The specified squared seam-pair coupling conserves $Q = N_\nu + 2Q_{\mathrm{seam}}$ and cannot generate anomalous pairing in its unique $Q = 0$ vacuum. These symmetries have not been established for every term of an eventual full parent. No physical neutrino texture is obtained.

T7: local constraints and zero-mode removal. `GRAV.SPIN2.LOCAL.CONSTRAINT.01.` `[verification/v1029_local_tensor_constraints.py]` implements a quadratic target, not a TFPT embedding. For six orthonormal symmetric tensor coordinates, let $t = (1, 1, 1, 0, 0, 0)^T$, $B(k)q = q_{ij}k_j$, $v = B^T k$, $s = v - |k|^2 t$. Set

$$K_p = I - tt^T/2, \quad K_q = |k|^2(I - tt^T) - 2B^T B + tv^T + vt^T, \quad H = \tfrac{1}{2}p^T K_p p + \tfrac{1}{2}q^T K_q q,$$

with constraints $C_0 = s^T q = 0$, $C = Bp = 0$. Direct polynomial algebra gives

$$BB^T = (|k|^2 I + kk^T)/2, \quad Bs = 0, \quad K_q B^T = 0, \quad K_p s = B^T k, \quad s^T s = 2|k|^4.$$

The four constraints commute and $\dot{C}_0 = k^T C$, $\dot{C} = 0$. For $k \neq 0$ they are independent first-class constraints, leaving $12 - 2 \cdot 4 = 4$ phase dimensions. Rotation to $k \parallel e_3$ and constraint/gauge reduction leaves the two normalized TT tensors, on which $H = \frac{1}{2}\sum_{a=+,\times}(p_a^2 + |k|^2 q_a^2)$. Local constraints therefore avoid putting the nonlocal TT projector into the Hamiltonian. This is the standard linearized-gravity mechanism, not a novel derivation of general relativity (see Green–Kiriushcheva–Kuzmin).

A finite-stencil regulator puts diagonal components on sites, off-diagonal ij components at offsets $(e_i + e_j)/2$, and vector parameters on links. Use $\kappa_i = 2\sin(ak_i/2)/a$ in every formula. Under a Brillouin shift, the sign flip U_i on vectors induces D_i on symmetric tensors, and $B(U_i\kappa) = U_i B(\kappa) D_i$, $s(U_i\kappa) = D_i s(\kappa)$, $K_q(U_i\kappa) = D_i K_q(\kappa) D_i$, $K_p = D_i K_p D_i$. Thus the half-grid symbol is consistently glued; its only zero is $k = 0$, unlike the eight-corner $\sin k_i$ mutant. At $k = 0$, however, all constraints vanish and $p_0 = rt$ gives $H_0 = -3r^2/4$: the unmodified periodic model is unbounded below. A positive radiative target requires removal of the entire global canonical block (q_0, p_0), an extra global zero-mean prescription, not a local constraint or a TFPT state-selection theorem. Furthermore finite-dimensional exact CCR are impossible by the trace: $[q_d, p_d] = i(I - dP_{d-1})$ for the explicit truncation. Collective low-energy emergence is not excluded, but needs a controlled microscopic sequence. No critical embedding, nonlinear constraints, universal coupling or global gravitational state is established. The distinction between a tensor target and controlled emergence also matters in Gu–Wen; neither cited paper fills the TFPT embedding gap.

ALG2: sector generation and conditional uniform section. `SEAM.MMST.ALG2.FRAME.01`. [`verification/v1030_alg2_sector_frame.py`] does not restore the refuted Vandermonde implication. Suppose, explicitly, $H = \bigoplus_r H_r$ and the actual neutral algebra is $\mathcal{A}_0 = \bigoplus_r \operatorname{End} H_r$. Join r, s if $P_s T P_r \neq 0$. Then $W^*(\mathcal{A}_0, T) = \operatorname{End} H$ precisely when this undirected sector graph is connected: the neutral commutant is $\bigoplus_r c_r I_r$, and each nonzero corner forces $c_r = c_s$. The finite bicommutant theorem proves the result. Genuine superselection components must be treated separately, not bridged by forbidden odd fields.

For a fixed D-dimensional low-energy space choose actual local words $Y_{j,N}$, $j = 1, \ldots, D^2$, with $\|Y_{j,N}\| \leq M$. Let Φ_N have columns $\operatorname{vec}(P_N Y_{j,N} P_N)$. If $\sigma_{\min}(\Phi_\infty) = \sigma > 0$ and $\|\Phi_N - \Phi_\infty\|_2 \leq \sigma/2$, then solving $\Phi_N a = \operatorname{vec} X$ gives, for $\|X\| \leq 1$,

$$P_N y_N P_N = X, \quad y_N = \sum_j a_j Y_{j,N}, \quad \|a\|_2 \leq 2\sqrt{D}/\sigma, \quad \|y_N\| \leq 2MD^{3/2}/\sigma.$$

This supplies a uniform section only under the stated microscopic word, norm and noncollapsing-frame hypotheses. Compression does not preserve products ($PABP$ need not equal $PAPBP$), and double-cutoff convergence does not control $|N\rangle\langle 0|$ on a fixed input cutoff. For ALG2 one needs a joint real-linear frame for $L_N(y) = (P_F y P_E, P_F y^* P_E)$ and uniform high-output tails of both x_N, y_N and their adjoints. If the joint error and each pair of tails are at most ε, the desired one-sided sum is at most 3ε. Neutral generation, charged corners, actual local words, their scaling limits, and these tails are not yet proved for the one-edge TFPT microscopic algebra. They, together with the common-parent gates, are the precise remaining research obligations.

Rounds 6 and 7: free quantum curvature, covariant extension, charged disorder, mirror parent, and first-order matter coupling

Exact scope and evidence ledger

This supplement records five written proofs and their machine regressions:

$$69 + 34 + 35 + 36 + 59 = 233$$

passing checks. The Round 6 prerequisite contributes 69 checks (53 exact and 16 floating regressions) [`verification/v1031_quantum_tensor_curvature.py`]. Round 7 contributes 164 checks: 130 exact, 13 source-contract, and 21 finite or floating regressions [`verification/v1032_covariant_curvature.py`] [`verification/v1033_charged_disorder.py`] [`verification/v1034_parent_mirror.py`] [`verification/v1035_matter_coupling.py`]. The finite and floating checks are implementation regressions, never substitutes for the quantified arguments below.

The Round 6/Round 7 Lorentz-covariant helicity-± 2 sector is a declared free regulator/Fock construction. It is not an emergent TFPT parent, a nonlinear completion, or a proof of universal matter coupling. The factorized DET result proves the gap of the momentum-cell parent actually specified there; it does not prove a gap for ordinary real-space hopping. The charged construction distinguishes the common quarter disorder $h_{\mathrm{deck}} = 1/4$ from the sourced field $h_\lambda = 1$. The matter construction retains the full Fourier factors $c_0 = -s^T q$, $\boldsymbol{c} = \mathrm{i}Bp$ and the paired $+k/-k$ real Hamiltonian; its sign is not stripped. All physical completeness gates T1–T8 and `TFPT.TOE.COMPLETE.01` remain [O]. **NO RH CLAIM.**

Round 6 prerequisite: a free quantum curvature field on the staggered tensor target

The displayed arguments below are the proof-bearing source. The current repository regression is [`verification/v1031_quantum_tensor_curvature.py`]; its finite checks audit the implementation and do not replace the universal steps in the proof.

Exact scope. This note quantizes only the free, quadratic, zero-mean curvature target whose classical constraint quotient was established at repository commit `4417f1ba8da3d5dc622c942efcbfc46ebaae546d`. It constructs neither a microscopic TFPT parent nor finite-dimensional quantum cells. It proves no nonlinear constraint algebra, state selection mechanism, universal matter coupling, interacting graviton theory, T7 closure, TOE claim, or RH statement. Strict lattice light-cone support remains false. The microcausality theorem below is a distributional statement about the $a \to 0$ limit of this free target.

Assumptions and preregistered obligations. Let a periodic cubic lattice have $V = L^3$ cells and spacing $a > 0$. The six orthonormal symmetric-tensor coordinates are ordered as $(11, 22, 33, 23, 13, 12)$, with the off-diagonal basis tensors normalized by $1/\sqrt{2}$. Component A lives at $x + am_A/2$, where

$$m_A = (000, 000, 000, 011, 101, 110).$$

We use the physical-position Fourier phase $\exp[-\mathrm{i}k \cdot (x + am_A/2)]$. The staggered derivative symbol is

$$\mathrm{i}\kappa_i^{(a)}(k), \qquad \kappa_i^{(a)}(k) = \frac{2}{a} \sin \frac{ak_i}{2}, \qquad r_a(k) = |\kappa^{(a)}(k)|.$$

It vanishes only at the homogeneous momentum in the first Brillouin zone. The complete homogeneous canonical block is removed as an additional zero-mean prescription. It is not declared gauge.

The proof was preregistered against the following obligations.

O1. Prove, for every nonzero real momentum and not from samples, that the curvature commutator operator is $r^4 P_{\mathrm{TT}}$, is positive, has rank two, and is transverse and traceless.

O2. Construct the Weyl quotient, prove the Heisenberg inequality by an explicit factorization, and identify the finite-volume GNS representation and ground state. A finite matrix approximation to the CCR is inadmissible.

O3. Treat k and $-k$, self-conjugate momenta, and half-grid Bloch signs without choosing a globally inconsistent polarization frame.

O4. Prove convergence of local states as $L \to \infty$ at fixed a.

O5. Either prove an $a \to 0$ tempered-distribution limit and its curvature microcausality, or leave that claim open. Finite grids may only be regression tests for the analytic proof.

The input from the classical target is the canonical reduced Hamiltonian and the already-proved staggered identities. In one momentum fibre set

$$Bq = q\kappa, \quad t = (1, 1, 1, 0, 0, 0)^T, \quad v = B^T\kappa, \quad s = v - r^2 t,$$

$$K_p = I - \frac{1}{2} tt^T, \qquad K_q = r^2(I - tt^T) - 2B^T B + tv^T + vt^T. \tag{1}$$

The four local constraints are the scalar $s^T q = 0$ and the three momentum constraints $Bp = 0$. For $r > 0$ they are independent and first class. The curvature variables are

$$E = K_q q, \qquad F = K_q K_p p, \tag{2}$$

and the classical result supplies

$$K_q B^T = 0, \quad K_p s = B^T \kappa, \quad K_q t = s,$$

$$K_q^2 = r^2 K_q + ss^T, \qquad K_q K_p K_q = r^2 K_q + \frac{1}{2} ss^T. \tag{3}$$

All matrices in (1)–(3) are real in the physical-position trivialization.

The exact spin-two commutator projector. For $\kappa \neq 0$, let $P = I_3 - \kappa\kappa^T/r^2$. The orthogonal spin-two projector on symmetric tensors is

$$(P_{\mathrm{TT}}h)_{ij} = P_{ia}P_{jb}h_{ab} - \frac{1}{2}P_{ij}P_{ab}h_{ab}. \tag{4}$$

It has rank two, $P_{\mathrm{TT}}^2 = P_{\mathrm{TT}} = P_{\mathrm{TT}}^T$, and its range is $\{h : h\kappa = 0,\ \mathrm{Tr}\, h = 0\}$.

Theorem 2 (Polynomial curvature projector)**.** *For every $\kappa \neq 0$,*

$$C := K_q K_p K_q = r^4 P_{\mathrm{TT}}. \tag{5}$$

Consequently

$$C^T = C \geq 0, \quad C^2 = r^4 C, \quad \mathrm{Tr}\, C = 2r^4, \quad \mathrm{rank}\, C = 2, \quad CB^T = 0, \quad Ct = 0. \tag{6}$$

Although (4) has inverse powers of r, C itself is a homogeneous degree-four polynomial in κ.

Proof. The TT space is $\mathrm{Ker}\, B \cap t^\perp$. If h is TT, (1) gives $K_q h = r^2 h$ and $K_p h = h$, hence $Ch = r^4 h$. Equations (3) give $CB^T = 0$. They also give $Ct = K_q K_p s = K_q B^T \kappa = 0$. The scalar and momentum constraints are independent at $r > 0$, so $\mathrm{Ran}\, B^T + \mathrm{span}\{t\}$ is four-dimensional and is the orthogonal complement of the two-dimensional TT space. Thus C is r^4 on TT and zero on its orthogonal complement, proving (5). The properties in (6) follow immediately. Finally, $C = K_q K_p K_q$ is polynomial; equivalently the two factors of r^{-2} implicit in (4) cancel against r^4. A direct component multiplication of (4) is included in the executable checker. □

This theorem sharpens the off-shell identity in (3): the term $r^2 K_q + ss^T/2$ is exactly the polynomial numerator of the TT projector. No nonlocal TT projector is inserted into the lattice observable.

CCR quotient and finite-volume vacuum. Let $\mathcal{S}_L$ be the real finite-volume smearing space of curvature pairs $X = (f, g)$, with the homogeneous fibre absent and with the same reality/Bloch conditions as the fields. Define

$$\Phi(f, g) = E(f) + F(g), \qquad \sigma_L(X, Y) = \langle f, Ch_g\rangle - \langle h_f, Cg\rangle, \tag{7}$$

where $Y = (h_f, h_g)$ and the brackets denote the real lattice inner product. Canonical quantization means

$$[\Phi(X), \Phi(Y)] = \mathrm{i}\sigma_L(X, Y)I, \qquad W(X)W(Y) = e^{-\mathrm{i}\sigma_L(X,Y)/2}W(X + Y). \tag{8}$$

The null space of (7) is $\mathrm{Ker}\, C \oplus \mathrm{Ker}\, C$. We therefore define the physical Weyl algebra on

$$\mathcal{S}_L^{\mathrm{phys}} = \mathcal{S}_L/(\mathrm{Ker}\, C \oplus \mathrm{Ker}\, C), \tag{9}$$

not on the unreduced six-component smearing space. By (5), (9) retains exactly the two TT canonical pairs in each nonzero real momentum component.

Let $R = (-\Delta_a)^{1/2}$. It has fibre eigenvalue $r_a(k)$ and is strictly positive after removal of the homogeneous block. It commutes with C. Define the real covariance

$$\mu_L(X, Y) = \frac{1}{2}\langle f, CR^{-1}h_f\rangle + \frac{1}{2}\langle g, RCh_g\rangle. \tag{10}$$

Thus, fibre by fibre,

$$\frac{1}{2}\langle\{E, E\}\rangle = \frac{C}{2r}, \qquad \frac{1}{2}\langle\{F, F\}\rangle = \frac{rC}{2}, \qquad \frac{1}{2}\langle\{E, F\}\rangle = 0. \tag{11}$$

The unsymmetrized mixed covariances are $\langle EF\rangle = \mathrm{i}C/2$ and $\langle FE\rangle = -\mathrm{i}C/2$.

Theorem 3 (Proposition: Exact uncertainty factorization)**.** *The covariance (10) satisfies the CCR positivity condition, and it saturates it on every physical oscillator. If $C^{1/2} = r^2 P_{\mathrm{TT}}$ fibrewise, then*

$$\mu_L + \frac{\mathrm{i}}{2}\sigma_L = \frac{1}{2}\begin{pmatrix} CR^{-1} & \mathrm{i}C \\ -\mathrm{i}C & RC \end{pmatrix} = \frac{1}{2}\begin{pmatrix} C^{1/2}R^{-1/2} \\ -\mathrm{i}C^{1/2}R^{1/2} \end{pmatrix}\begin{pmatrix} R^{-1/2}C^{1/2} & \mathrm{i}R^{1/2}C^{1/2} \end{pmatrix} \geq 0. \tag{12}$$

Equivalently,

$$J(f,g) = (-Rg, R^{-1}f), \qquad J^2 = -I, \qquad \mu_L(X,Y) = \frac{1}{2}\sigma_L(X, JY). \tag{13}$$

Hence $\omega_L(W(X)) = \exp[-\mu_L(X,X)/2]$ is a pure regular quasifree state on the physical Weyl algebra.

Proof. All factors in (12) commute in momentum space, and multiplication gives the middle matrix exactly. This proves the uncertainty inequality rather than testing principal minors. On the quotient, J is well defined, compatible with σ_L, and satisfies (13).

For existence and purity, no abstract reconstruction theorem is needed. Choose a real orthonormal spectral basis common to R and P_{TT}. In the αth physical plane represent Q_α by multiplication and $P_\alpha = -\mathrm{i}\,\partial/\partial q_\alpha$ on $L^2(\mathbb{R}, \mathrm{d}q_\alpha)$, and put

$$E_\alpha = r_\alpha^2 Q_\alpha, \qquad F_\alpha = r_\alpha^2 P_\alpha, \qquad \Omega_\alpha(q) = \left(\frac{r_\alpha}{\pi}\right)^{1/4} e^{-r_\alpha q^2/2}. \tag{13a}$$

Direct Gaussian integration gives $\langle Q_\alpha^2\rangle = 1/(2r_\alpha)$, $\langle P_\alpha^2\rangle = r_\alpha/2$, and $\langle\{Q_\alpha, P_\alpha\}\rangle/2 = 0$. It also gives

$$\left\langle \Omega_\alpha, e^{\mathrm{i}(fE_\alpha + gF_\alpha)}\Omega_\alpha \right\rangle = \exp\left[-\frac{1}{4}\left(r_\alpha^3 f^2 + r_\alpha^5 g^2\right)\right], \tag{13b}$$

which is exactly $\exp[-\mu_L(X,X)/2]$ on that plane. The finite tensor product over all physical planes therefore constructs ω_L directly and makes regularity explicit. Multiplications and translations on $L^2(\mathbb{R}^N)$ have only scalars in their joint commutant (use the Fourier transform for the translation part), so this Schrödinger representation is irreducible. Its normalized Gaussian vector state is consequently pure. □

The state is the oscillator ground state, not just a formal Gaussian. In a real spectral basis of R and P_{TT} the reduced Hamiltonian is

$$H_L = \frac{1}{2}\sum_\alpha (p_\alpha^2 + r_\alpha^2 q_\alpha^2), \qquad a_\alpha = \frac{r_\alpha^{1/2} q_\alpha + \mathrm{i} r_\alpha^{-1/2} p_\alpha}{\sqrt{2}}. \tag{14}$$

Its GNS representation is the bosonic Fock representation, with unique vacuum $a_\alpha \Omega_L = 0$. Normal ordering gives $H_L^{\mathrm{N}} = \mathrm{d}\Gamma(R)$; without normal ordering the finite vacuum energy is $\frac{1}{2}\sum_\alpha r_\alpha$. There are exactly $2(V-1)$ real oscillators. Each oscillator requires an infinite-dimensional Hilbert space: no finite $d \times d$ matrices can satisfy $[Q,P] = \mathrm{i}I_d$, because taking the trace would give $0 = \mathrm{i}d$. Thus "finite-volume" never means a finite-dimensional CCR representation.

Here uniqueness means uniqueness of the regular finite-volume oscillator ground state on the zero-mean physical Weyl algebra. It is not a claim that all infinite-volume states or phases are unique. In particular, the omitted $k = 0$ fibre is not present in this algebra, so there is no zero-frequency coherent sector to translate. Restoring that fibre would remove the positive frequency hypothesis used in (14) and would require an additional state choice; the present theorem says nothing about such an extension.

The induced dynamics is

$$E(t) = \cos(Rt)E + R^{-1}\sin(Rt)F, \qquad F(t) = -R\sin(Rt)E + \cos(Rt)F, \tag{15}$$

so (10) is invariant and is a ground state for the positive frequencies.

Reality and Bloch gluing. Write $G_i = \mathrm{diag}((-1)^{(m_A)_i})$. Physical-position Fourier coefficients obey

$$\widehat{q}(k + 2\pi e_i/a) = G_i \widehat{q}(k), \qquad \widehat{q}(-k) = \overline{\widehat{q}(k)}, \tag{16}$$

and similarly for p, E, F. Changing k_i by $2\pi/a$ reverses $\kappa_i^{(a)}$ and gives

$$K_q(k + 2\pi e_i/a) = G_i K_q(k) G_i, \qquad C(k + 2\pi e_i/a) = G_i C(k) G_i. \tag{17}$$

Also $C(-k) = C(k)$. Therefore (7), (10), and (15) descend to the Bloch bundle and respect the real involution. No polarization vectors are chosen, so there is no hidden global-frame obstruction.

For a non-self-conjugate pair $\{k, -k\}$, the two complex TT amplitudes at k determine those at $-k$ and give four real oscillator coordinates. At a self-conjugate boundary momentum, (16) becomes $z = G_n \overline{z}$; the fixed real form of the complex two-dimensional TT space has real dimension two. If N_{sc} is the number of nonzero self-conjugate momenta, then

$$2N_{\mathrm{sc}} + 4\frac{V - 1 - N_{\mathrm{sc}}}{2} = 2(V - 1). \tag{18}$$

For odd L, $N_{\mathrm{sc}} = 0$; for even L, $N_{\mathrm{sc}} = 7$. This establishes the real mode count including Brillouin-zone corners.

Fixed-spacing thermodynamic state. At fixed a, extend the covariance symbols by zero at $k = 0$. The estimates

$$\|C(k)\| = r_a(k)^4, \qquad \|C(k)/(2r_a(k))\| = \tfrac{1}{2} r_a(k)^3, \qquad \|r_a(k) C(k)/2\| = \tfrac{1}{2} r_a(k)^5 \tag{19}$$

show that all symbols are continuous on the compact Brillouin torus. A finite-support smearing has a trigonometric-polynomial Fourier transform. Consequently every entry of (7) and (10) is a Riemann sum of a continuous periodic function. As $L \to \infty$ it converges to the corresponding Brillouin-zone integral.

Theorem 4 (Local thermodynamic state)**.** *For every finite collection of finite-support curvature smearings, the finite-volume Weyl expectations converge as $L \to \infty$ at fixed a. The limits define a translation-invariant quasifree state $\omega_{\infty,a}$ on the inductive local curvature Weyl algebra over $\mathbb{Z}^3$. Its equal-time CCR kernel is the same finite stencil C as in finite volume, while its vacuum covariance generally has infinite spatial range.*

Proof. Riemann-sum convergence and (19) give convergence of σ_L and μ_L. The global summand contributes zero, so deleting it causes no residual $1/V$ term. Pointwise limits of the positive characteristic functionals in (12) remain positive and normalized, and consistency under inclusion of supports gives a state on the inductive algebra. Translation invariance is manifest in the Fourier multiplier. Since C is polynomial in shifts, its CCR kernel is finite range; the fractional powers in (10) need not be. □

The same argument applies to (15) for each fixed time. It does not produce a strict lattice light cone. The analytic tails of the lattice wave propagator found in the classical construction remain present in quantum commutators.

Continuum distributions and microcausality. Take the thermodynamic limit first. For p in $\mathrm{BZ}_a = [-\pi/a, \pi/a]^3$, let

$$\kappa_{a,i}(p) = \frac{2}{a} \sin \frac{a p_i}{2}, \quad \omega_a(p) = |\kappa_a(p)|, \quad C_a(p) = C(\kappa_a(p)). \tag{20}$$

The band-limited interpolation of a lattice kernel is the tempered distribution obtained by integrating its multiplier over BZ_a. Equivalently, lattice fields with $[q_x, p_y] = \mathrm{i}\delta_{xy}$ are scaled as $a^{-3/2}$ distributions, so their smearings are $a^{3/2} \sum_x f(x + a m_A/2) E_A(x)$. For Schwartz test functions,

Poisson summation shows that direct sampling and the band-limited interpolation have the same limit. Here is the required uniform estimate. With

$$\widehat{f}_{a,A}(p) = a^3 \sum_{x \in a\mathbb{Z}^3} f_A(x + am_A/2) e^{-\mathrm{i}p\cdot(x+am_A/2)},$$

Poisson summation writes $\widehat{f}_{a,A}(p)$ as $\widehat{f}_A(p)$ plus the nonzero aliases $\widehat{f}_A(p + 2\pi n/a)$, each multiplied by a stagger phase of modulus one. For $p \in \mathrm{BZ}_a$ and $n \neq 0$,

$$|p + 2\pi n/a| \geq \pi |n|_\infty / a.$$

Every Schwartz seminorm is finite, so for every $M > 3$,

$$\sup_{p \in \mathrm{BZ}_a} |\widehat{f}_{a,A}(p) - \widehat{f}_A(p)| \leq C_{f,M} a^M \sum_{n \neq 0} |n|_\infty^{-M}. \tag{20a}$$

The largest covariance or commutator multiplier used below is $O(a^{-5})$, and $\mathrm{vol}(\mathrm{BZ}_a) = O(a^{-3})$. The difference between a sampled pairing and its band-limited pairing is therefore $O(a^{M-8})$; choosing any $M > 8$ makes it vanish. For spacetime Schwartz tests, the same estimate holds after spatial Fourier transformation with a coefficient rapidly decreasing in t, so its time integral obeys the same bound. Thus no unproved no-alias assumption enters the continuum limit.

Let $C_0(p) = C(p) = |p|^4 P_{\mathrm{TT}}(p)$ for $p \neq 0$, with its polynomial value $C_0(0) = 0$. Since $|\kappa_a(p)| \leq |p|$ on BZ_a, the covariance multipliers satisfy the a-independent polynomial bounds

$$\|C_a/(2\omega_a)\| \leq \tfrac{1}{2}|p|^3, \qquad \|\omega_a C_a/2\| \leq \tfrac{1}{2}|p|^5. \tag{21}$$

They converge pointwise to $C_0/(2|p|)$ and $|p|C_0/2$. Dominated convergence against the rapidly decreasing Fourier transform of any Schwartz test proves convergence in $\mathcal{S}'(\mathbb{R}^3)$ of the equal-time vacuum covariances. The limiting quasifree functional remains positive by (12).

For the spacetime commutators define the scalar lattice Pauli–Jordan multiplier

$$d_a(t,p) = \frac{\sin(\omega_a(p)t)}{\omega_a(p)}, \qquad d_a(t,0) = t. \tag{22}$$

Equation (15) gives

$$\begin{aligned} [E(t),E(0)] &= -\mathrm{i}C_a d_a, & [E(t),F(0)] &= \mathrm{i}C_a \partial_t d_a, \\ [F(t),E(0)] &= -\mathrm{i}C_a \partial_t d_a, & [F(t),F(0)] &= \mathrm{i}C_a \partial_t^2 d_a. \end{aligned} \tag{23}$$

For spacetime Schwartz tests, $|d_a(t,p)| \leq |t|$, $|\partial_t d_a| \leq 1$, and $|\partial_t^2 d_a| \leq |t||p|^2$. Together with $\|C_a\| \leq |p|^4$, these are integrable polynomial dominators after spatial Fourier transformation. Therefore (23) converges in $\mathcal{S}'(\mathbb{R}^{1+3})$ to the same formulas with

$$d_0(t,p) = \frac{\sin(|p|t)}{|p|}. \tag{24}$$

Theorem 5 (Continuum curvature microcausality)**.** *The limiting free curvature commutators vanish for spacelike-separated test supports. More precisely, all four matrix-valued distributions in (23) have support inside the Minkowski light cone.*

Proof. The inverse spatial Fourier transform of (24) is, up to the Fourier sign convention,

$$D_0(t,x) = \frac{1}{4\pi|x|}(\delta(t-|x|) - \delta(t+|x|)), \tag{25}$$

so $\operatorname{supp} D_0 \subseteq \{|x| = |t|\}$. The polynomial $C_0(p)$ is a constant-coefficient fourth-order spatial differential operator $C_0(-\mathrm{i}\nabla)$ in position space. The other factors in (23) are time derivatives of D_0. Distributional differentiation does not enlarge support, proving the claim. □

This closes O5 for the two-step limit $L \to \infty$ followed by $a \to 0$. It does not retroactively give strict support to (22) at any $a > 0$, and it does not prove that a TFPT microscopic dynamics has this scaling limit.

Executable certificate, costs, and remaining limits. The repository certificate [`verification/v1031_quantum_tensor_curvature.py`] is standalone and imports no proof text. It is a separate implementation check, not an external independent review. Its roles were fixed before execution:

- exact SymPy polynomial algebra checks (3), (5), (6), and the separate index formula (4);
- exact rational-momentum fibres, characteristic polynomials, TT action, and the factorization (12);
- exact Bloch transformations, boundary real forms, real mode counts, and the finite-matrix CCR trace no-go;
- finite floating-point checks of complete small-volume fibre spectra and two convergence diagnostics. These last checks are regressions only; they do not establish O1–O5.

The universal claims rest on the displayed polynomial identities, positivity factorization, Riemann-sum theorem, dominated-convergence bounds, and support argument. A failed finite regression would reject the implementation; a passed regression cannot replace any of those arguments.

The reported total of 69 checks is evidence-typed as follows: 53 are exact symbolic, rational, integer-counting, or finite-matrix no-go checks; 16 are floating-point regressions. The latter comprise 12 complete small-volume fibre-spectrum/uncertainty/gap checks and four thermodynamic or continuum diagnostics. No floating-point result is used in a theorem proof.

The exact core uses fixed 6×6 matrices of polynomial degree at most eight. Rational checks use four fibres. Boundary checks use the seven even-L self-conjugate corners. The numerical cost is $O(\sum L^3)$ over the listed small boxes plus at most a 96^3 scalar Riemann grid; memory is $O(96^3)$ floating-point numbers. No repository suite is needed for this isolated artifact.

The construction is ultraviolet-distributional. The continuum covariance grows as $|p|^3$ and $|p|^5$, so point fields and coincident products require smearing and, for composite observables, renormalization not supplied here. The un-normal-ordered vacuum energy diverges in thermodynamic/continuum limits. The curvature state is not claimed to have finite correlation range. The Hamiltonian written only in (E, F) contains inverse powers of R even though the CCR and equations are local. None of these limitations alters the finite-volume Fock construction or the stated distributional limits, but each blocks a stronger physical interpretation.

Finally, (E, F) is the spatial curvature/curvature-velocity pair of the linear target. This note does not reconstruct a complete Lorentz-covariant four-dimensional Riemann or Weyl tensor, construct boost generators, or prove that the regulator arises from a covariant microscopic theory.

Round 7: a Lorentz-covariant free curvature field on the Round 6 Hilbert space

The displayed arguments below are the proof-bearing source. The current repository regression is [`verification/v1032_covariant_curvature.py`]; its finite checks audit the implementation and do not replace the universal steps in the proof.

Scope and exact claim boundary

The Round 6 target gave positive quantum electric-curvature variables E, F and a microcausal continuum limit, but did not construct the full four-dimensional Riemann field or its Lorentz boosts. We close that specific free-field gap. A polynomial covariant curvature kernel equals a positive two-polarization kernel on the null cone. It defines a positive Gaussian Wightman field, the vacuum linearized Riemann (equivalently Weyl) tensor, with a strongly continuous positive-energy Poincare representation and helicities $+2, -2$. Its electric part is exactly $E/2$ and its time derivative is $F/2$; it adds no one-particle states. The construction is a free continuum extension of the target, not the missing derivation from the TFPT parent. It proves neither nonlinear gravity, local-net equality with the smaller E, F system, a local lattice realization of every curvature component, nor a complete T7 or TOE. No RH statement is involved.

Conventions and the algebraic bridge. Use $\eta = \mathrm{diag}(-1,1,1,1)$ and a covariant null momentum

$$k_\mu = (-r, p_1, p_2, p_3), \qquad r = |p| > 0, \qquad e^{ik_\mu x^\mu} = e^{-irt + ip\cdot\mathbf{x}}.$$

All tensor components in this note carry lower indices. For a symmetric metric amplitude h, define the linear curvature map

$$(A(k)h)_{\mu\nu\rho\sigma} = \frac{1}{2}\left(k_\sigma k_\nu h_{\mu\rho} + k_\rho k_\mu h_{\nu\sigma} - k_\rho k_\nu h_{\mu\sigma} - k_\sigma k_\mu h_{\nu\rho}\right). \tag{1}$$

It is the Fourier transform of

$$R_{\mu\nu\rho\sigma} = \tfrac{1}{2}(\partial_\rho\partial_\nu h_{\mu\sigma} + \partial_\sigma\partial_\mu h_{\nu\rho} - \partial_\sigma\partial_\nu h_{\mu\rho} - \partial_\rho\partial_\mu h_{\nu\sigma}).$$

No positive covariant quantum metric potential is assumed. Instead introduce the numerical tensor

$$\Pi_{\mu\nu,\alpha\beta} = \tfrac{1}{2}(\eta_{\mu\alpha}\eta_{\nu\beta} + \eta_{\mu\beta}\eta_{\nu\alpha} - \eta_{\mu\nu}\eta_{\alpha\beta}), \qquad K(k) = A(k)\Pi A(k)^T. \tag{2}$$

In a Euclidean orthonormal basis of symmetric 4×4 matrices, Π has six positive and four negative eigenvalues. It is not a physical Hilbert-space covariance by itself.

Let S embed a spatial symmetric matrix in the ij block of a spacetime matrix, with $h_{0\mu} = 0$. The orthogonal spatial transverse-traceless projector is

$$P_{ij} = \delta_{ij} - p_i p_j / r^2, \tag{3}$$

$$(P_{\mathrm{TT}}h)_{ij} = P_{ia}P_{jb}h_{ab} - \tfrac{1}{2}P_{ij}P_{ab}h_{ab}. \tag{4}$$

It is an orthogonal rank-two projector on $\mathrm{Sym}^2(\mathbb{R}^3)$.

Lemma 11 (Null-cone positive factorization)**.** *For every nonzero null $k = (-r, p)$,*

$$K(k) = A(k)SP_{\mathrm{TT}}(p)S^T A(k)^T = L(k)L(k)^T, \qquad L(k) = A(k)SP_{\mathrm{TT}}(p). \tag{5}$$

Consequently $K(k)$ is positive semidefinite of rank exactly two. It is nevertheless a polynomial of degree four in all four components of k.

Proof. Substitution in (1) gives $A(k)(k\xi^T + \xi k^T) = 0$ for every ξ. The identity (5) can be checked with $p = (0,0,r)$: the two surviving transverse orthonormal matrices are

$$h_+ = \mathrm{diag}(1,-1,0)/\sqrt{2}, \qquad h_\times = (e_1 e_2^T + e_2 e_1^T)/\sqrt{2},$$

and direct contraction gives $A\Pi A^T = (ASh_+)(ASh_+)^T + (ASh_\times)(ASh_\times)^T$. Spatial rotational covariance and degree-four homogeneity give the result for every nonzero p. The map is injective on the two-dimensional TT space because $(A(k)Sh)_{0i0j} = r^2 h_{ij}/2$. This proves rank two. Polynomiality follows directly from (2); no division by r occurs there.

For an independent algebraic certificate, the checker uses ten orthonormal metric components and 21 pair-symmetric bivector components. Write ω as an independent variable and $C(p) = r^4 P_{\mathrm{TT}}(p)$, the polynomial Round 6 bracket. It divides each of the 441 entries of

$$A(-\omega, p)(r^4\Pi - SC(p)S^T)A(-\omega,p)^T$$

by $\omega^2 - r^2$ and verifies an identically zero polynomial remainder. The off-shell identity and the identity without the trace-reversal term are separately required to fail. These are algebraic checks, not sampling of momentum directions or a numerical positivity test. □

The same map obeys both pair antisymmetries, pair interchange, the algebraic Bianchi identity $R_{\mu[\nu\rho\sigma]} = 0$, and the differential Bianchi identity $k_{[\tau}R_{|\mu\nu|\rho\sigma]} = 0$. On the TT null range the Ricci contraction vanishes, $\eta^{\mu\rho}R_{\mu\nu\rho\sigma} = 0$. For example, insert $h_{0\mu} = 0$, $p_i h_{ij} = 0$, $h_{ii} = 0$, and $k^2 = 0$ into (1); every contraction contains one of those four zero factors. Thus the constructed vacuum curvature is also its linearized Weyl tensor. These identities hold as operator-valued distributional identities after the null quotient below.

Positive field, local commutator, and boosts. Index the independent pair-symmetric bivector entries by I, J. Define the two-point distribution

$$W_{IJ}(x-y) = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3\, 2r}\, e^{-ir(t_x-t_y)+ip\cdot(\mathbf{x}-\mathbf{y})} K_{IJ}(-r,p). \tag{6}$$

The Bianchi identities make one of the 21 algebraic components redundant; retaining it with its null relation avoids an arbitrary choice of twenty coordinates.

Theorem 6 (Free covariant curvature field)**.** *Equation* (6)*, with Wick's rule for higher vacuum correlations, defines a positive free bosonic operator-valued tempered distribution on a symmetric Fock space. Its commutators vanish at spacelike separation. There is a strongly continuous unitary representation of the proper orthochronous Poincare group, with invariant cyclic Fock vacuum, spectrum in the closed forward cone, and covariant rank-four curvature transformation. Its only one-particle helicities are* $+2$ *and* -2*.*

Proof. For Schwartz test tensors f, factorization (5) gives

$$\|f\|_1^2 = \int \frac{d^3p}{(2\pi)^3\, 2r} \widehat{f}(k)^* K(k) \widehat{f}(k) = \int \frac{d^3p}{(2\pi)^3\, 2r} \|L(k)^T \widehat{f}(k)\|^2 \geq 0. \tag{7}$$

Quotient the zero-norm tensors and complete to $\mathcal{H}_1$. Write u_f for the one-particle image of the test tensor. With $a(u)$ antilinear in u, set $R(f) = a^\dagger(u_f) + a(u_{\bar{f}})$ on the finite-particle domain of $\Gamma_s(\mathcal{H}_1)$, using the Fourier convention implicit in (6). This is complex linear in f and obeys $R(f)^* = R(\bar{f})$ on that domain. It gives (6). The usual creation estimate $\|a^\dagger(u)\psi_n\| \leq \sqrt{n+1}\|u\|\|\psi_n\|$ and continuity of (7) make this an operator-valued tempered distribution. Wick positivity follows from this explicit Fock construction, not from assuming Π is positive.

Infrared convergence and temperedness are immediate: $K = O(r^4)$, so $K/(2r) = O(r^3)$ at zero and has only polynomial growth at infinity. Together with the d^3p measure the radial integrand is $O(r^5)$ near zero. Schwartz test functions control every ultraviolet moment.

Let

$$D_0(x) = \int \frac{d^3p}{(2\pi)^3\, 2r} (e^{-irt+ip\cdot\mathbf{x}} - e^{irt-ip\cdot\mathbf{x}})$$

be the scalar massless commutator distribution. Since $K(-k) = K(k)$, the curvature commutator is the constant-coefficient differential operator $K_{IJ}(-i\partial)D_0(x-y)$. The scalar commutator has light-cone support, and differentiation does not enlarge support. This proves locality of the curvature fields, despite the nonlocal spatial TT projector used to exhibit positivity.

All indices in (1)–(2) are tensorial and Π is built only from η. Therefore

$$K(\Lambda k) = D^{(4)}(\Lambda) K(k) D^{(4)}(\Lambda)^T \tag{8}$$

for the covector action of every proper orthochronous Lorentz matrix. The positive-null-cone measure is invariant. The induced contragredient test-tensor transformation, together with translations, consequently preserves (7) and its null space. It gives a unitary one-particle representation. The test-tensor action is continuous in the Schwartz topology, so the norm estimate gives strong continuity on a dense set and then on the completion. Second quantization gives the claimed Fock representation. Translations act by positive-energy null momenta; finite sums and closure keep the many-particle spectrum in the forward cone. The null cone tip has zero one-particle measure. Thus every nonvacuum n-particle wave function has strictly positive total energy almost everywhere, and the only zero-energy vector in this constructed Fock representation is the vacuum. This is not a theorem of uniqueness across all representations or all possible TFPT states.

For helicity, at $k = (-1,0,0,1)$ rotations about the third axis act on $(h_+, h_\times)$ by

$$\begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}.$$

Hence $h_+ \pm ih_\times$ have phases $e^{\mp 2i\theta}$. The translation part of the null little group is trivial on the curvature: put $q = (u^2+v^2)/2$ and

$$\Lambda(u,v) = \begin{pmatrix} 1+q & -u & -v & q \\ -u & 1 & 0 & -u \\ -v & 0 & 1 & -v \\ -q & u & v & 1-q \end{pmatrix}.$$

It preserves η and k, and sends the transverse covectors $e_1 \mapsto e_1 + uk$, $e_2 \mapsto e_2 + vk$. Each polarization change is therefore $k\xi^T + \xi k^T$, which $A(k)$ annihilates. This proves the two helicities without assuming away the noncompact little-group translations. The checker verifies these identities for symbolic u, v and all rotation angles through a rational parametrization. □

It is the same free target, with a larger covariant field family. The Round 6 continuum construction has, on the physical TT range,

$$E = r^2 q_{\mathrm{TT}}, \qquad F = r^2 p_{\mathrm{TT}}, \qquad \dot{q}_{\mathrm{TT}} = p_{\mathrm{TT}}, \qquad \dot{p}_{\mathrm{TT}} = -r^2 q_{\mathrm{TT}},$$

with $C = r^4 P_{\mathrm{TT}}$. In radiation gauge (1) gives exactly

$$R_{0i0j} = \frac{r^2}{2} q_{ij} = \frac{1}{2} E_{ij}, \qquad \partial_t R_{0i0j} = \frac{1}{2} F_{ij}. \tag{9}$$

Consequently the spectral electric-electric covariance is $C/(8r)$, electric-time-derivative covariance is $iC/8$ at equal time in the order $\langle R\dot{R}\rangle$, and derivative-derivative covariance is $rC/8$. These are precisely one quarter of the Round 6 EE, EF, FF covariances, respectively, including the nonsymmetrized commutator part.

For completeness this matching is more than equality of a few entries. Both one-particle spaces identify with the completion of spatial TT amplitudes in $L^2(d^3p/((2\pi)^3 2r))$. The map from curvature tests is $f \mapsto P_{\mathrm{TT}} S^T A^T \widehat{f}$. Electric tests alone have dense image: on every annulus $0 < \epsilon < r < R$ their map is multiplication by the invertible scalar $r^2/2$ followed by P_{TT}. Every smooth compactly supported TT amplitude on such an annulus can be extended to a spacetime Schwartz test in a tubular neighbourhood of the positive null cone. Cutting off at zero and infinity is dense in the one-particle norm. Thus (9) identifies the completed one-particle spaces, and the associated Fock spaces, with no extra physical oscillator or spectator.

There is an important locality qualification. At a fixed time,

$$R_{0ijk} = \tfrac{1}{2}(\partial_k p_{ij} - \partial_j p_{ik}) = \frac{1}{2r^2}(\partial_k F_{ij} - \partial_j F_{ik})$$

in Fourier notation; the spatial components similarly contain two spatial derivatives of $q = E/r^2$. Inverse powers of r^2 cancel in the covariant commutator kernel, but are still present in this reconstruction from E, F. The global Hilbert-space identification does not by itself prove equality of local observable nets, nor that every R component has a bounded-range lattice approximant in the previous E, F algebra. No such claim is used in the positivity, covariance, or locality theorem above.

Validation, provenance, and the remaining obligation. The repository certificate [`verification/v1032_covariant_curvature.py`] is standalone SymPy code. Its 34 exact checks include the all-momentum polynomial identity, gauge and Bianchi identities, null Ricci contraction, positivity factorizations, a nontrivial rational Lorentz boost, the full null little group, and the exact electric and mixed time-derivative covariance normalization. Four rational fibres supplement, but do not replace, the universal proof. No floating-point eigenvalue scan is used. The distributional/Fock completion rests on the written norm, continuity, density and support arguments, not on a finite matrix test pretending to construct an infinite-dimensional field.

The inputs are the declared Round 6 free target and ordinary linearized curvature algebra. No microscopic TFPT Hamiltonian is imported or derived by this checker; only the repository's reporting harness is shared. In particular this note does not manufacture the required shared parent: it does not prove that the microscopic seam, charged matter, mirror sector and gravity jointly flow to this field or to its nonlinear completion. A finite-spacing Lorentz-breaking regulator is not asserted to be exactly boost invariant. The vacuum homogeneous-block choice and the interacting constraint/renormalization problems remain separate.

For historical context, massless-spin field representations and the role of the non-semisimple little group are discussed in Weinberg's primary paper[1]. The result here is an explicit match of that free-field structure to the existing target, not a claim that free helicity-two fields are a new physical theory. T1–T8 retain their existing open status.

Round 7: a microscopic flux-string disorder operator for the frozen QWZ seam

The displayed arguments below are the proof-bearing source. The current repository regression is [verification/v1033_charged_disorder.py]; its finite checks audit the implementation and do not replace the universal steps in the proof.

Scope and exact claim boundary

There is a concrete microscopic boundary-condition-changing operator for the $M = 1$, $N_y = 8$ QWZ seam once one local four-level flux register is adjoined. Couple its clock Z directly to the seam hopping and let its shift X raise the flux. Dressing X by a quarter charge rotation on a fermion arc gives a disorder string D_A. It has nonzero $r \to r + 1$ corners, order four, commutes with canonical fermion parity, and its commutator with the actual QWZ Hamiltonian is supported exactly on the opposite endpoint bond. Thus this is not the earlier arbitrary block shift.

It is nevertheless not the TFPT λ field. The source's common quarter twist on eight complex edge channels has minimal charge $q = (1/4)^8$, hence $h_q = 1/4$, while the sourced $\lambda = (\omega_s, \omega_f)$ has $\|\lambda\|^2 = 2$ and $h_\lambda = 1$. Forcing weight one by an integer dressing inside the common scalar phase class requires odd fermion charge. The actual A_3 fundamental makes the missing datum explicit: it is a four-channel quarter disorder plus one endpoint fermion, combined with five half-twisted D_5 channels and a nontrivial cocycle/GSO identification. Those species-resolved couplings and that GSO map are not present in the audited QWZ path. A sharp endpoint insertion also has measured $O(1)$ lattice excitation energy; a smooth whole-circle dressing lowers this to $O(N^{-1})$ but loses sharp support. The result is therefore a positive microscopic deck-disorder construction and a narrow obstruction to identifying it with Ψ_λ, not a no-go for a future species-resolved extension. No RH claim is made.

Frozen source and target. The construction below uses the hopping matrix literally implemented in `verification/v988_psi_lambda_reduction.py`. With

$$T_x = \frac{\sigma_x}{2i} - \frac{\sigma_z}{2}, \qquad T_y = \frac{\sigma_y}{2i} - \frac{\sigma_z}{2},$$

the one-particle QWZ Hamiltonian h_r has mass $M = 1$, open y boundary, and on the bond $(N-1, y) \to (0, y)$ the forward hopping is $i^r T_x$. Every other longitudinal bond has T_x. The executable v988 campaign constant is $N_y = 6$, but its builder accepts N_y as an argument. This artifact calls that builder explicitly at the charged-limit manuscript value $N_y = 8$. The manuscript writes

$$p_n^{(r)} = \frac{2\pi}{N}\left(n + \frac{r}{4}\right). \tag{1.1}$$

That is its opposite-sign Bloch convention. With the checker convention $\psi_x = e^{ikx}$ and the executable forward seam amplitude i^r, the boundary eigencondition is

$$e^{i(r\pi/2 + kN)} = 1, \qquad k_n^{(r)} = \frac{2\pi n - r\pi/2}{N}. \tag{1.2}$$

[1] S. Weinberg, "Feynman Rules for Any Spin. II. Massless Particles", Phys. Rev. 134, B882 (1964), https://doi.org/10.1103/PhysRev.134.B882. The accessible abstract was used for context; no inaccessible theorem is imported as a proof.

The checker reconstructs the complete position-space matrix from these Bloch blocks and compares it entrywise with the imported v988 builder. Using the plus sign in this checker convention fails that matrix test. It also states that sixteen Majorana copies give eight complex edge bosons.

The target is not specified merely by the deck phase. The repository fixes

$$\omega_s = \left(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}\right), \qquad \omega_f = \left(\tfrac{3}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}\right), \qquad \lambda = (\omega_s, \omega_f), \tag{1.3}$$

so

$$\|\omega_s\|^2 = \frac{5}{4}, \quad \|\omega_f\|^2 = \frac{3}{4}, \quad \|\lambda\|^2 = 2, \quad h_\lambda = 1. \tag{1.4}$$

These exact claims occur in two frozen sources:

- `verification/v983_simple_current_generator.py`;
- `articles/2026-08-30/mmst_charged_scaling_limit_en.tex`.

The latter still lists a support-preserving microscopic realization and an independent FE-GEN or gauge/code-projected algebra theorem as open. We do not silently replace (1.3) by an arbitrary representative of a scalar $\mathbb{Z}_4$ phase.

A common microscopic Hilbert tied to the hopping. Let $\mathfrak{h}_N = \mathbb{C}^{2NN_y}$ and let $\mathcal{F}_N = \mathcal{F}(\mathfrak{h}_N)$ be the canonical number-conserving CAR Fock space. Adjoin one four-level link register $\mathcal{K} = \mathbb{C}^4$ at the chosen periodic seam. In the basis $\{|r\rangle : r \in \mathbb{Z}_4\}$ define

$$Z|r\rangle = i^r|r\rangle, \qquad X|r\rangle = |r+1\rangle, \qquad ZXZ^* = iX, \qquad X^4 = Z^4 = \mathbf{1}. \tag{2.1}$$

If $F = \sum_y |0,y\rangle T_x \langle N-1, y|$ is the forward seam hopping and $h_{\mathrm{cut}} = h_0 - F - F^*$, set

$$\widetilde{h} = \mathbf{1}_{\mathcal{K}} \otimes h_{\mathrm{cut}} + Z \otimes F + Z^* \otimes F^* = \bigoplus_{r=0}^{3} h_r. \tag{2.2}$$

Thus the register is not a decorative sector label: its clock occurs in the actual QWZ hopping term. On $\mathcal{K} \otimes \mathcal{F}_N$ take

$$\widetilde{H} = \sum_{r=0}^{3} |r\rangle\langle r| \otimes d\Gamma(h_r). \tag{2.3}$$

The four sectors now live in one canonical Hilbert space. This is an *auxiliary $\mathbb{Z}_4$ flux link*; no claim is made that the present TFPT code derives it.

The sharp line-ended operator. Choose an arc $A = \{0, 1, \ldots, \ell\}$ with $0 \le \ell < N-1$. Let u_A be the one-particle onsite gauge

$$u_A|x,y,a\rangle = \begin{cases} i|x,y,a\rangle, & x \in A, \\ |x,y,a\rangle, & x \notin A, \end{cases} \qquad \Gamma(u_A) = \exp\left(\frac{\pi i}{2} N_A\right). \tag{2.4}$$

Define

$$D_A = X \otimes \Gamma(u_A). \tag{2.5}$$

Theorem 7 (Microscopic QWZ disorder transporter)**.** *For every $r \in \mathbb{Z}_4$,*

$$P_{r+1} D_A P_r = |r+1\rangle\langle r| \otimes \Gamma(u_A) \neq 0, \qquad D_A^4 = \mathbf{1}. \tag{2.6}$$

Writing $\Pi = (-1)^{N_{\mathrm{F}}}$ for canonical total fermion parity,

$$[D_A, \mathbf{1} \otimes \Pi] = 0, \qquad (Z \otimes \mathbf{1}) D_A (Z^* \otimes \mathbf{1}) = i D_A. \tag{2.7}$$

Moreover

$$[\widetilde{H}, D_A] P_r = |r+1\rangle\langle r| \otimes d\Gamma(h_{r+1} - u_A h_r u_A^*)\, \Gamma(u_A), \tag{2.8}$$

and $h_{r+1} - u_A h_r u_A^*$ is supported only on the single longitudinal bond $(\ell, y) \leftrightarrow (\ell+1, y)$, for all $y = 0, \ldots, 7$. Its one-particle rank is $2N_y = 16$ and its norm is $\sqrt{2}$.

Proof. The corner and charge equations follow from (2.1). Since $u_A^4 = \mathbf{1}$ and X acts on a different tensor factor, $D_A^4 = \mathbf{1}$. A number rotation is an even CAR automorphism, proving the parity commutator.

The $(r+1, r)$ one-particle commutator block is $h_{r+1}u_A - u_A h_r = (h_{r+1} - u_A h_r u_A^*)u_A$; second quantization gives (2.8). Conjugation by u_A multiplies a forward hopping $x \to x+1$ by $u_{x+1}u_x^*$. At the original seam this factor is i, changing i^r to i^{r+1}. All bonds internal to A or its complement are unchanged. Only the other boundary of A has factor $-i$. Hence the original seam defect is transported to the far endpoint. Substitution of T_x gives rank 16 and norm $\sqrt{2}$; the companion probe checks the full matrices for all four sectors at $N_y = 8$. □

This is the promised positive result. A matter-only onsite gauge could not have done it: the product of link phases telescopes,

$$\prod_x a_x \frac{g_{x+1}}{g_x} = \prod_x a_x, \tag{2.9}$$

so it preserves the Wilson holonomy. The new register is exactly the extra degree of freedom that changes it. Unlike the Round-6 arbitrary J_r, the operator (2.5) is tied to (2.2), has a specified CAR action, and obeys the endpoint commutator identity (2.8).

Equation (2.7) proves compatibility with canonical fermion parity and with any GSO projector that is a function of that parity. It does *not* prove compatibility with the repository's character-level $O + S$ selector: the audited source supplies no microscopic projector identifying that selector with Π. This boundary is essential.

Why the constructed operator is not Ψ_λ. For a normalized complex chiral fermion, a twist by $e^{2\pi i q}$ has minimal bosonized weight $q^2/2$, with $q \in (-1/2, 1/2]$. The actual common phase i therefore fixes, for the eight complex channels in the source,

$$q_{\text{deck}} = \left(\tfrac{1}{4}, \ldots, \tfrac{1}{4}\right), \qquad \|q_{\text{deck}}\|^2 = \frac{1}{2}, \qquad h_{\text{deck}} = \frac{1}{4}. \tag{3.1}$$

It has order four but nontrivial spin $e^{2\pi i h} = i$. The target (1.3) instead has integer spin and weight one. Norm is invariant under an orthonormal change of current basis, so this is not repaired by relabeling the eight bosons.

The phase by itself does not determine an excited representative. Indeed

$$q' = \left(\tfrac{5}{4}, \tfrac{1}{4}, \ldots, \tfrac{1}{4}\right) \tag{3.2}$$

has the same scalar phase and $\|q'\|^2 = 2$. But $q' - q_{\text{deck}}$ is one unit of fermion charge. More generally, if $q_{\text{deck}} + n$ has norm two, then

$$2\left(\|q_{\text{deck}} + n\|^2 - \|q_{\text{deck}}\|^2\right) = 2\sum_j n_j^2 + \sum_j n_j = 3. \tag{3.3}$$

Modulo two, $\sum_j n_j$ is odd. Every such weight-one dressing therefore changes canonical fermion parity. Selecting (3.2) solely because it has the desired number would be precisely an arbitrary charge-basis shift, not a microscopic derivation of λ.

The $D_5 \oplus A_3$ coordinates show the missing operator content more constructively. The D_5 spinor part is a half disorder on five complex channels and contributes 5/8. For the A_3 part,

$$\left(\tfrac{3}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}\right) = \left(-\tfrac{1}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}\right) + (1, 0, 0, 0). \tag{3.4}$$

Thus it is a four-channel quarter disorder of weight 1/8 plus one endpoint fermion, giving the traceless A_3 fundamental weight 3/8. Combined with the five half twists, this gives $h_\lambda = 5/8 + 3/8 = 1$.

Equation (3.4) is the next microscopic specification. Indeed, $e^{2\pi i \omega_s} = (-1)^5$ componentwise and $e^{2\pi i \omega_f} = (-i)^4$ componentwise. A λ operator therefore needs species-resolved seam clocks Z^2 on the

five D_5 channels and Z^{-1} on the four A_3 fermions, the endpoint fermion, removal of the diagonal $U(1)$ in the A_3 realization, and the cocycle/GSO rule that makes the total field local and bosonic. The frozen QWZ code has one common scalar i^r phase and none of these four identification maps. The flux register solves the common-Hilbert and locality problem, but not this internal-charge problem. This redundant fermionic realization has nine complex fermions before the A_3 diagonal-$U(1)$ quotient; it is therefore not automatically an operator on the source's eight-channel Fock space. An explicit conformal embedding/isometry is part of the missing microscopic datum.

Energy and support diagnostics. There is a finite-lattice tradeoff that prevents another overclaim. *Portability amendment, 2026-09-06.* The $r = 0$ one-particle operator has an exactly degenerate zero-energy space. Filling the first half of a numerical eigenbasis silently selects a platform-dependent pure vacuum. Let P_- and P_0 denote its negative and zero spectral projectors, let $d = \operatorname{rank} P_0$, and let $m = NN_y - \operatorname{rank} P_-$. We instead declare the basis-independent one-body covariance

$$\overline{C} = P_- + \frac{m}{d} P_0 \quad (d > 0), \qquad \overline{C} = P_- \quad (d = 0).$$

It is the one-body covariance of the uniform fixed-particle mixture over rank-m zero-space fillings, not a uniquely selected pure Slater vacuum. The numerical zero cluster is required to be separated from all nonzero levels; this finite floating-point diagnostic is not an interval certificate. Applying the sharp D_A to that mixture and measuring above the $r = 1$ ground energy gives, for $N = 16, 24, 32, 48, 64$,

$$4.711531, \quad 4.689142, \quad 4.677637, \quad 4.665918, \quad 4.659977. \tag{4.1}$$

The measured cost is consistent with $O(1)$ at fixed $N_y = 8$ and the sharp endpoint commutator. This bare operator is not a low-energy weight-one state.

One can distribute the string by taking

$$u_N(x) = \exp\left[\frac{\pi i}{2}\left(1 - \frac{x}{N}\right)\right], \qquad D_N^{\mathrm{sm}} = X \otimes \Gamma(u_N). \tag{4.2}$$

Then the actual one-particle matrices give

$$N \left\| h_1 u_N - u_N h_0 \right\| \longrightarrow \frac{\pi}{2}, \qquad N\left(\operatorname{tr}\left[h_1 u_N \overline{C} u_N^*\right] - E_0(H_1)\right) \approx 6.904. \tag{4.3}$$

This does not discard other zero-space vacua. For either dressing u, set $A = P_0 u^* h_1 u P_0|_{\operatorname{ran} P_0}$, with eigenvalues $\alpha_1 \le \cdots \le \alpha_d$. Over every rank-$m$ zero-space filling, the exact variational extrema of the one-body excess are

$$\operatorname{tr}(P_- u^* h_1 u) - E_0(H_1) + \sum_{j=1}^{m} \alpha_j, \qquad \operatorname{tr}(P_- u^* h_1 u) - E_0(H_1) + \sum_{j=d-m+1}^{d} \alpha_j.$$

Indeed, in an eigenbasis of A, the diagonal entries of a rank-m orthogonal projector lie in $[0, 1]$ and sum to m; the minimum and maximum select the m smallest and largest eigenvalues. Both bounds are attained. At $N = 64$ the sharp interval is approximately $[4.644352, 4.675602]$, and the smooth N-scaled interval is $[5.333652, 8.474929]$. The previous 8.475 coefficient corresponds to one endpoint of this filling-dependent range; it was not a unique vacuum value. On all five sampled sizes, the sharp intervals lie in $(4, 5)$ and the smooth N-scaled intervals in $(5.2, 8.6)$. These are finite-grid diagnostics, not all-N asymptotic bounds. This is measured $O(N^{-1})$ energy, but u_N occupies the whole circumference and the coefficient contains full-band polarization. It is not a proof of conformal weight. Conversely, the sharp string has exact endpoint support but the $O(1)$ UV cost (4.1). Recovering both sharp local algebras and the weight-one edge primary requires the missing edge projection, renormalized endpoint field, species-resolved charge, and GSO cocycle.

As an independent finite diagnostic, momentum-block diagonalization gives

$$\frac{N}{2\pi}\left(E_0(h_1) - E_0(h_0)\right) \longrightarrow -\frac{3}{16} \tag{4.4}$$

for one two-edge QWZ copy. This is a sector-vacuum/Casimir shift, not the one-edge primary weight and not h_λ. No fit or finite sample in this section is promoted to an all-N theorem.

Executable certificate and claim boundary. The repository certificate [`verification/v1033_charged_disorder.py`] audits the frozen sources, proves the finite Weyl/parity/charge identities with rational and symbolic arithmetic, imports the executable v988 QWZ builder and compares it entrywise, verifies the endpoint commutator on the actual $M = 1, N_y = 8$ hops, and reports finite energy diagnostics.

Established: after adjoining one explicitly auxiliary local $\mathbb{Z}_4$ flux link, $D_A = X\Gamma(u_A)$ is a CAR-even, order-four, grade-one microscopic QWZ boundary-condition changer with an exact line-endpoint commutator.

Not established: that the present TFPT/GSO construction contains this register; that its $O + S$ selector is canonical parity; or that the common quarter disorder equals the $D_5 \oplus A_3$ field Ψ_λ. The latter identification is false for the minimal common twist because $h_{\mathrm{deck}} = 1/4 \neq 1 = h_\lambda$.

Round 7: the actual DET parent, its exact factorized gap, and the local-defect obstruction

The displayed arguments below are the proof-bearing source. The current repository regression is [`verification/v1034_parent_mirror.py`]; its finite checks audit the implementation and do not replace the universal steps in the proof.

Scope and exact claim boundary

The source audit separates three operators which had previously been discussed under one "mirror parent" label. The v1002 DET16 onsite term is the flat defect projector $Q = \mathbf{1} - P_\phi$. The v1008 wall scaffold instead tests the four-flavour momentum-cell Hamiltonian

$$h(d) = d(N-2) + 2Q, \qquad |d| \le 3/5.$$

The later v1027 identity constructs an additive dressed-number parent N_a; it is not the same operator as Q. For the actual v1008 momentum-factorized Hamiltonian we prove, without a grid scan, the volume-independent many-body and canonical-field threshold

$$\Delta_{\mathrm{fact}} \ge 1 + \frac{\sqrt{3}}{2} = 1.8660254037\ldots.$$

This is the exact infimum of the continuous dispersion envelope; the source's 129-point grid instead attains 1.8660698881694362. It is not a real-space locality theorem. Indeed, for two real-space DET cells, a newly chosen bare flavour-diagonal bond analogue T_{bare} satisfies (for even $n \ge 4$)

$$\|T_{\mathrm{bare}}|\Omega\Omega\rangle\|^2 = \frac{n}{2}|s|^2 > 0, \qquad (Q_x + Q_y)|\Omega\Omega\rangle = 0.$$

Consequently no finite κ can obey $T_{\mathrm{bare}} \ge -\kappa(Q_x + Q_y)$. An explicitly gauge-dressed operator, with the link shift applied as part of every hop, gives the same obstruction inside finite $\mathbb{Z}_2/\mathbb{Z}_4$ Gauss sectors. Thus the desired elementary relative-Q proof is impossible for the unmodified local parent; a genuine small-perturbation theorem or a changed local bond is needed. An explicit coupling-fixed bond repair is given, but is clearly marked as a new, unlanded interaction. No T3, T4, TOE, or RH claim is made.

Source contract and nomenclature. The certificate reads, but does not import, the following landed files at the audited repository tip:

source	exact object used here	status relevant to this note
v1002	$Q = \mathbf{1} - P_\phi$ on 16 complex modes	onsite projector only
v1008	$h(d) = d(N - n/2) + JQ$, $n = 4, J = 2$	momentum-cell scaffold
v1024	positive seam/number-square parent	separate construction
v1027	$N_a = \sum_i a_i^\dagger a_i$, $Q \le N_a \le nQ$	algebraic identity

In particular, "DET32" is sometimes used physically for 32 real components, whereas the repository's operator is named DET16 because it has 16 complex fermion modes. We use n for the number of complex modes and retain the repository's DET16 name.

Let

$$|\Omega_\phi\rangle = \frac{|0\rangle + e^{i\phi}|F\rangle}{\sqrt{2}}, \qquad P_\phi = |\Omega_\phi\rangle\langle\Omega_\phi|, \qquad Q = \mathbf{1} - P_\phi. \tag{10}$$

Here $|F\rangle = c_1^\dagger \cdots c_n^\dagger |0\rangle$ and n is even. Both $|0\rangle$ and $|F\rangle$ are assumed to be gauge singlets; this is the inherited full-DET-singlet premise, not a conclusion of this note. The common rotation $e^{-i\phi N/n}$ removes ϕ from (10) and commutes with N and ordinary flavour-diagonal hopping, so every spectrum below is phase independent.

The elementary algebra already confirms the proposed diagnosis:

$$[Q, (-1)^N] = 0, \qquad [Q, N] \neq 0, \qquad \|[Q, N]\|_{\mathrm{F}}^2 = \frac{n^2}{2}. \tag{11}$$

Thus the flat projector preserves parity, not naive mirror number. For the v1008 analogue $n = 4$, the last value is 8; for DET16 it is 128.

Exact theorem for the parent that v1008 actually computes. The v1008 routine `edge_cluster_hamiltonian` is

$$h(d) = d\left(N - \frac{n}{2}\right) + JQ, \qquad n = 4, \qquad J = 2, \qquad |d| \le t_{\mathrm{src}} = \frac{3}{5}. \tag{12}$$

The scan uses $d = \pm t_{\mathrm{src}} \sin k$. This is a separate Fock cell at every sampled momentum, not a real-space hopping Hamiltonian. In particular, a standard symmetric nearest-neighbour term $s\sum_x (c_x^\dagger c_{x+1} + \mathrm{h.c.})$ has dispersion magnitude $2|s|$ (up to a momentum/phase convention). Matching the v1008 dispersion amplitude would therefore give $|s| = t_{\mathrm{src}}/2 = 3/10$, not $3/5$.

Lemma 1 (complete one-cell spectrum). For $1 \le r \le n-1$, every occupation-r state has energy

$$E_r(d) = J + d\left(r - \frac{n}{2}\right). \tag{13}$$

On $\mathrm{span}\{|0\rangle, |F\rangle\}$, after removing ϕ, the matrix is

$$\begin{pmatrix} -nd/2 + J/2 & -J/2 \\ -J/2 & nd/2 + J/2 \end{pmatrix}, \tag{14}$$

with eigenvalues

$$E_\pm(d) = \frac{J}{2} \pm \sqrt{\frac{J^2}{4} + \frac{n^2 d^2}{4}}. \tag{15}$$

Proof. P_ϕ vanishes on occupation sectors $1, \ldots, n-1$, which gives (13). Its restriction to the empty/full doublet is the rank-one matrix specified by (10); the displayed characteristic polynomial gives (15). □

For the source values, $E_-(d) \le 0$, while the least intermediate energy is $2 - 3/5 = 7/5 > 0$. The ground state is therefore unique throughout the whole declared dispersion interval. A canonical c_i or $c_i^\dagger$ maps this ground doublet only to the $r = 1$ or $r = n-1$ sectors, with nonzero amplitude. The exact canonical spectral threshold at $x = |d|$ is consequently

$$g(x) = \frac{J}{2} - \left(\frac{n}{2} - 1\right)x + \sqrt{\frac{J^2}{4} + \frac{n^2 x^2}{4}}. \tag{16}$$

Theorem 1 (source-fixed all-volume factorized gap). Let K be any finite set of momentum cells, with arbitrary dispersions $|d_k| \le 3/5$, and define the graded tensor-product Hamiltonian $H_K = \sum_{k\in K} h(d_k)$. Then H_K has a unique product ground state and

$$\operatorname{gap}(H_K) \ge \inf_{0\le x\le 3/5} g(x) = 1 + \frac{\sqrt{3}}{2}. \tag{17}$$

The same number is the uniform lower edge of canonical one-fermion spectral support.

Proof. For $n = 4, J = 2$, differentiation of (16) gives the unique interior minimum

$$x_* = \frac{J(n-2)}{2n\sqrt{n-1}} = \frac{1}{2\sqrt{3}} < \frac{3}{5}, \tag{18}$$

and substitution gives $g(x_*) = 1 + \sqrt{3}/2$. The other empty/full eigenstate costs at least $J = 2$, and the $r = 2$ sector also costs at least 2, so neither undercuts (17). The even cell Hamiltonians commute on the graded tensor product and energies add. Any state orthogonal to the product ground excites at least one cell, proving the same bound independently of $|K|$. □

The analytic infimum $1.8660254037844386\ldots$ explains the lower envelope of v1008's reported probe minimum; it is not the exact minimum of the finite scan. The source grid $k_j = -\pi + j\pi/64$, $j = 0, \ldots, 128$, attains 1.8660698881694362. The locality limit is equally exact: a projector local in momentum is not a bounded-range real-space interaction. Equation (17) therefore does not prove the missing local, operator-valued TFPT parent theorem.

Why the desired relative-defect bound fails in real space. Put two DET cells on a bond and let

$$D = J(Q_x + Q_y), \qquad T_{\text{bare}} = s\sum_{i=1}^{n}(c^\dagger_{xi}c_{yi} + c^\dagger_{yi}c_{xi}). \tag{19}$$

This is a bare-bond theorem. The numerical choice $s = 3/5$ below is a newly chosen stress analogue, not a real-space coefficient derived from v1008. For even $n \ge 4$, the full-Fock CAR calculation contains only $2n$ nonzero output monomials and does not require construction of the 2^{2n} matrix.

Lemma 2 (minimal kernel incompatibility). For every even $n \ge 4$ and $s \ne 0$,

$$D|\Omega\Omega\rangle = 0, \qquad \langle\Omega\Omega|T_{\text{bare}}|\Omega\Omega\rangle = 0, \tag{20}$$

$$\|T_{\text{bare}}|\Omega\Omega\rangle\|^2 = \frac{n}{2}|s|^2 > 0, \qquad (Q_x + Q_y)T_{\text{bare}}|\Omega\Omega\rangle = 2T_{\text{bare}}|\Omega\Omega\rangle. \tag{21}$$

Proof. Only the $|F, 0\rangle$ and $|0, F\rangle$ components of the product DET vacuum can hop. Each has amplitude $1/2$. The two hopping orientations and n flavours produce $2n$ mutually orthogonal states, each of modulus $|s|/2$, which proves the norm in (21). Both sites have occupation 1 or $n-1$ after the hop, hence both local projectors P_x, P_y annihilate the image. □

The restriction $n \ge 4$ is necessary for that counting statement. At $n = 2$ the two orientations land on the same two basis states and add constructively. Direct CAR evaluation gives $\|T_{\text{bare}}|\Omega\Omega\rangle\|^2 = 2|s|^2$, rather than the former formula's $|s|^2$. The image is still nonzero, so the no-go below survives, but the claimed universal norm formula did not.

Theorem 2 (no finite relative-defect constant). There is no finite κ such that

$$T_{\text{bare}} \ge -\kappa(Q_x + Q_y). \tag{22}$$

Proof. If (22) held, the positive operator $B = T_{\mathrm{bare}} + \kappa(Q_x + Q_y)$ would have zero expectation in $|\Omega\Omega\rangle$ by (20). Positivity then implies $B|\Omega\Omega\rangle = 0$. Since $(Q_x + Q_y)|\Omega\Omega\rangle = 0$, this would say $T_{\mathrm{bare}}|\Omega\Omega\rangle = 0$, contradicting (21). □

For the newly chosen bare-bond analogue $n = 4, J = 2, s = 3/5$,

$$\|T_{\mathrm{bare}}|\Omega\Omega\rangle\|^2 = \frac{18}{25}. \tag{23}$$

Let $b = \sqrt{18/25}$ and normalize $|w\rangle = T_{\mathrm{bare}}|\Omega\Omega\rangle/b$. Direct CAR evaluation gives $\langle w|T_{\mathrm{bare}}|w\rangle = 0$ and $\langle w|D|w\rangle = 2J$. The compression of $D + T_{\mathrm{bare}}$ to $\mathrm{span}\{|\Omega\Omega\rangle, |w\rangle\}$ is

$$\begin{pmatrix} 0 & b \\ b & 2J \end{pmatrix}. \tag{24}$$

Thus the variational principle gives the exact strict bound

$$E_0(D + T_{\mathrm{bare}}) \le 2 - \frac{\sqrt{118}}{5} = -0.172556\ldots < 0. \tag{25}$$

The old product vacuum is not frustration free once ordinary hopping is on. This does not prove gap closure; it proves that the proposed elementary relative-Q route cannot establish a gap.

Gauge-dressed theorem and actual link action. The bare theorem above makes no Gauss-sector claim. For the $n = 4$ analogue and $q = 2, 4$, now adjoin an oriented link basis $\{|e\rangle : e \in \mathbb{Z}_q\}$ and define

$$U|e\rangle = |e + 1 \bmod q\rangle, \qquad T_U = s\sum_{i=1}^{n}\left(c_{xi}^\dagger U c_{yi} + c_{yi}^\dagger U^\dagger c_{xi}\right). \tag{26}$$

This operator is implementable on matter Fock space tensored with the finite link space; it is fermion-even because each summand is a matter bilinear. Take the endpoint Gauss constraints to be $n_x - e = 0$ and $n_y + e = 0 \pmod q$. The starting vector $|\Omega\Omega\rangle|0\rangle_E$ is physical because its site occupations are 0 or 4. Applying (26), including the link shift rather than assigning a flux afterwards, produces only

$$(n_x, n_y, e) = (3, 1, 3 \bmod q) \quad \text{or} \quad (1, 3, 1 \bmod q),$$

and in both cases $n_x - e = 0$ and $n_y + e = 0 \pmod q$. Hence the nonzero hopping image is a physical flux excitation. The checker applies the CAR and link actions term by term for both $q = 2$ and $q = 4$ and verifies every nonzero output, so Gauss preservation is not a post-hoc flux assignment.

Let $E^2|\pm 1\rangle = |\pm 1\rangle$ and

$$H_0 = J(Q_x + Q_y) + h_E E^2, \qquad h_E \ge 0. \tag{27}$$

Both terms annihilate the physical starting vector, whereas T_U does not. The same positivity-kernel argument therefore rules out any finite relative bound of T_U by this complete positive local parent. Moreover, with $|w\rangle = T_U|\Omega\Omega, 0_E\rangle/b$, the exact Ritz compression is

$$\begin{pmatrix} 0 & b \\ b & 2J + h_E \end{pmatrix}, \qquad E_0(H_0 + T_U) \le \frac{2J + h_E}{2} - \sqrt{\frac{(2J + h_E)^2}{4} + b^2}. \tag{28}$$

Thus the electric cost is part of the excited diagonal entry. For the same newly chosen bond analogue $J = 2, s = 3/5$ and the illustrative choice $h_E = 1$, $b^2 = 18/25$ and (28) is $(25 - \sqrt{697})/10 = -0.140075756489\ldots < 0$.

An exact local repair boundary. The following repair is included to make the result actionable, not to present a new source claim. On a bond let

$$R_{xy} = \mathbf{1} - P_x P_y = Q_x + Q_y - Q_x Q_y. \tag{29}$$

It obeys $0 \le R_{xy} \le Q_x + Q_y$. For the flavour-diagonal hopping K_{xy} with unit coefficient, $\|K_{xy}\| = n$ exactly. Therefore

$$R_{xy} K_{xy} R_{xy} \ge -n R_{xy} \ge -n(Q_x + Q_y). \tag{30}$$

On a graph of maximum degree z, replacing ordinary hopping by its projected version yields

$$J \sum_x Q_x + s \sum_{\langle xy \rangle} R_{xy} K_{xy} R_{xy} \ge (J - zn|s|) \sum_x Q_x. \tag{31}$$

For the newly chosen periodic-1D bond analogue $J = 2, z = 2, n = 4, s = 3/5$, the margin is $-14/5$: projection alone is not enough. This number is not source-derived. Mode-normalizing K/n would give the positive margin $2 - 2(3/5) = 4/5$, but that is another changed normalization.

There is also a coupling-fixed positive repair with no new tunable parameter:

$$B_{xy} = |s| \left[n R_{xy} + \mathrm{sgn}(s) R_{xy} K_{xy} R_{xy} \right] \ge 0. \tag{32}$$

For the newly chosen $s = 3/5$ bond analogue its added coefficient is $|s|n = 12/5$ per bond; neither value is inferred from v1008. The Hamiltonian $J \sum_x Q_x + \sum B_{xy}$ has the common product DET vacuum and is bounded below by $J \sum_x Q_x$, hence has gap at least J. If P_x and K_{xy} are gauge invariant/covariant as assumed, so are R_{xy} and B_{xy}. Equation (32) is nevertheless a new many-body bond counterterm and may alter the desired continuum dynamics. It is an explicit design option, not evidence that TFPT already contains it.

Keeping the ordinary hopping instead requires a real local-stability theorem. The existing repository audit correctly notes that such theorems yield a volume-independent window only under verified tensor-product/local-gap and small-coupling premises, and that their unevaluated constants do not certify the newly chosen $s = 0.6$ bond point. The exact obstruction above explains why those premises cannot be replaced by the wished-for one-line inequality.

Reproducibility and claim boundary. The repository certificate `[verification/v1034_parent_mirror.py]` is read-only. Its JSON separates 29 exact algebra checks, six exact source-contract checks, and one floating-grid regression. The printed tag identifies each class. The checker performs an AST source-contract audit, exact 16×16 onsite algebra, exact symbolic minimization, and a sparse CAR-monomial two-cell calculation. It does not build the 256×256 two-cell matrix; for general n the obstruction uses only $2n$ output monomials when even $n \ge 4$. It also retains the exceptional $n = 2$ collision as a regression and applies the $\mathbb{Z}_2$ and $\mathbb{Z}_4$ link shifts explicitly. The reported decimal radical values are labelled evaluations; the separately labelled 129-point value is a floating reproduction of the actual finite-grid minimum, not an exact trigonometric certificate.

What is finished is precise: v1008's factorized parent has an exact all-volume gap, while ordinary real-space hopping admits no relative bound against the flat local DET defects. What remains open is equally precise: no local operator-valued TFPT Hamiltonian at a source-derived real-space coupling has acquired a proved volume-uniform mirror gap, and no net-chiral T3/T4 construction follows. **NO RH CLAIM.**

Round 7: first-order matter coupling of the staggered tensor target

The displayed arguments below are the proof-bearing source. The current repository regression is `[verification/v1035_matter_coupling.py]`; its finite checks audit the implementation and do not replace the universal steps in the proof.

Result and boundary. Conditional on the declared staggered Fourier dictionary, every nonzero momentum admits an exact affine coupling to a prescribed classical source obeying the two Ward equations. Source energy and momentum must deform the scalar and vector constraints; with those terms, the fibre constraints propagate and the curvature has the classical retarded response displayed below. The executable certificate does not import repository code, but it does instantiate the declared tensor/vector masks and checks their three reciprocal-axis Bloch transitions. Dynamical matter is tested only in collinear one-dimensional sectors and only through first order in the coupling. A free scalar and a fermion with a nearest-neighbour one-particle Hamiltonian pass those tests after their composite lattice vertices are repaired. A naive scalar stress fails, and an onsite ϕ^4 interaction has an explicit homogeneous momentum defect that no periodic stress divergence can remove. The deleted homogeneous gravity block cannot absorb nonzero total matter energy or momentum. No full three-dimensional matter coupling, quantum Kubo calculation, nonlinear constraint algebra, interacting quantum gravity, Lorentz-covariant lattice Riemann tensor, TFPT parent, T7 closure, TOE claim, or RH claim is proved.

Fixed regulator and conventions. We state the spatial regulator of the preceding classical and quantum artifacts as the intended real-space interpretation. The checker below instantiates these declared masks algebraically, but does not import a repository implementation. On a periodic cubic lattice of spacing a, a component with mask $m \in \{0,1\}^3$ lives at $x + am/2$, and

$$D_i^{(m)} f(x) = \frac{1}{a} \begin{cases} f(x + ae_i) - f(x), & m_i = 0, \\ f(x) - f(x - ae_i), & m_i = 1. \end{cases} \tag{1}$$

Thus $D_i^{(m \operatorname{xor} e_i)} = -(D_i^{(m)})^T$ under the periodic lattice sum. Physical-position Fourier phases give symbol $\mathrm{i}\kappa_i$; the factor i is retained below, where

$$\kappa_i = \frac{2}{a} \sin \frac{a k_i}{2}, \qquad r^2 = \kappa^T \kappa. \tag{2}$$

The diagonal tensor components have mask zero, q_{ij} with $i \neq j$ has mask $e_i \operatorname{xor} e_j$, and a vector component j_i has mask e_i. Products below are placed on the indicated cells or links; no unmentioned continuum Leibniz rule is used.

In the six orthonormal symmetric-tensor coordinates $(11, 22, 33, 23, 13, 12)$ define the following real matrices; they do not include the Fourier factor i:

$$(Bh)_i = h_{ij}\kappa_j, \quad t = (1,1,1,0,0,0)^T, \quad v = B^T\kappa, \quad s = v - r^2 t, \tag{3}$$

$$K_p = I - \frac{1}{2} t t^T, \qquad K_q = r^2 (I - t t^T) - 2B^T B + t v^T + v t^T. \tag{4}$$

With the common Fourier-volume normalization suppressed, the real canonical Hamiltonian sums the paired complex fibres; the vacuum constraints at each nonzero momentum are

$$\begin{gathered} H_0 = \frac{1}{2} \sum_{k \neq 0} \left[p(-k)^T K_p(k) p(k) + q(-k)^T K_q(k) q(k) \right], \\ c_0 = -s^T q, \qquad \boldsymbol{c} = \mathrm{i} B p. \end{gathered} \tag{5}$$

Relative to the Round-6 real matrices, these are only nonzero sign/phase changes of the vacuum constraints. Their zero surface, rank, and the reduced E, F observables are therefore unchanged. The inherited exact identities are

$$K_q B^T = 0, \quad K_p s = B^T \kappa, \quad K_q t = s, \quad K_q K_p K_q = r^2 K_q + \frac{1}{2} s s^T = r^4 P_{\mathrm{TT}}. \tag{6}$$

Here c_0 is the Fourier transform of $D_i D_j q_{ij} - \Delta \operatorname{tr} q$, and c_i is the transform of $D_j p_{ij}$. Thus $\dot{c}_0 = \mathrm{i}\kappa_i c_i = -\kappa^T B p$ and $\dot{\boldsymbol{c}} = 0$. Dropping i and restoring it only after the algebra gives the wrong scalar sign, as the plane-wave mutant below demonstrates.

The coupling that actually preserves the constraints. Let (ρ, j_i, τ_{ij}) be a symmetric c-number source on the scalar, vector, and tensor staggered cells. Its two Ward equations are

$$\dot{\rho} + D_i j_i = 0, \qquad \dot{j}_i + D_j \tau_{ij} = 0. \tag{7}$$

At $k \neq 0$, add the spatial metric coupling and deform both constraints:

$$H_g(t) = H_0 + g\langle q, \tau(t)\rangle, \qquad \Psi_0 = -s^T q + g\rho, \qquad \boldsymbol{\Psi} = \mathrm{i}Bp - g\boldsymbol{j}. \tag{8}$$

The lapse and shift multipliers multiply Ψ_0 and $\boldsymbol{\Psi}$ in the total Hamiltonian. Thus (8) is not merely the isolated $q_{ij}\tau_{ij}$ vertex: the ρ and j_i couplings are present as constraint terms. Periodic summation by parts fixes all signs in (8).

Theorem 8 (Conditional affine source consistency in the complex Fourier fibre)**.** *Assume that (3)–(6) are the fibre dictionary of the intended staggered placement and that a prescribed c-number source satisfies (7). Then the four constraints in (8) are first class within this fibre algebra and propagate as*

$$\dot{\Psi}_0 = D_i \Psi_i, \qquad \dot{\Psi}_i = 0. \tag{9}$$

If one instead retains the vacuum vector constraint, its time derivative is

$$\dot{c}_i = -gD_j\tau_{ij} = g\dot{j}_i, \tag{10}$$

which is generically nonzero. Consequently "couple q_{ij} to stress" without deforming the constraints is inconsistent already at order g. Moreover, within the linear ansatz

$$H_{\mathrm{int}} = \gamma g\langle q, \tau\rangle, \quad \Psi_0 = c_0 + \alpha g\rho, \quad \boldsymbol{\Psi} = \boldsymbol{c} + \beta g\boldsymbol{j},$$

Ward propagation uniquely fixes $\alpha = \gamma$ and $\beta = -\gamma$. Thus (8), up to the overall normalization of g, is the smallest affine constraint repair of the spatial stress vertex.

Proof. Hamilton's equations give $\dot{q} = K_p p$ and $\dot{p} = -K_q q - g\tau$. At fixed momentum, (7) reads $\dot{\rho} = -\mathrm{i}\kappa^T j$ and $\dot{j} = -\mathrm{i}B\tau$. Therefore

$$\dot{\Psi}_0 = -\kappa^T Bp - \mathrm{i}g\kappa^T j = \mathrm{i}\kappa^T(\mathrm{i}Bp - gj) = \mathrm{i}\kappa^T\boldsymbol{\Psi},$$

$$\dot{\boldsymbol{\Psi}} = -\mathrm{i}gB\tau - g\dot{j} = 0.$$

This is (9) with no stripped-i intermediate convention. Since the sources are c-numbers, their Poisson brackets vanish and the affine shifts do not change the vacuum first-class brackets. Dropping $-g\boldsymbol{j}$ leaves (10). For the general coefficients, the vector residual is $-\mathrm{i}(\gamma+\beta)gB\tau$, while the scalar residual relative to $\mathrm{i}\kappa^T\boldsymbol{\Psi}$ is $-\mathrm{i}(\alpha+\beta)g\kappa^T j$; both vanish only for the stated coefficients. □

The sign is falsifiable on one mode. Take the plane wave $e^{-\mathrm{i}t+\mathrm{i}z}$ with $\kappa = e_3$ and amplitudes $\rho = j_3 = \tau_{33} = g = 1$. Both Ward equations hold. Choose $Bp = -\mathrm{i}e_3$ and $s^T q = 1$, so the repaired constraints vanish. Directly, $\dot{\Psi}_0 = \dot{\boldsymbol{\Psi}} = 0$. If one instead keeps the old stripped-i choice $\Psi_0 = +s^T q + g\rho$ while restoring $\boldsymbol{\Psi} = \mathrm{i}Bp - gj$, the propagation residual is $\dot{\Psi}_0 - \mathrm{i}\kappa^T\boldsymbol{\Psi} = -2\mathrm{i}$.

The source need not be positive, local in time, or generated by matter for this theorem. Fourier reality is explicit: $B(-\kappa) = -B(\kappa)$ while s, K_p, K_q are even. Hence $\Psi_0(-k) = \overline{\Psi_0(k)}$ and $\boldsymbol{\Psi}(-k) = \overline{\boldsymbol{\Psi}(k)}$ whenever the fields at $-k$ are the conjugates of those at k and g is real. For an unordered non-self-conjugate pair, the actual real Hamiltonian contribution is

$$H_{\{k,-k\}} = p(-k)^T K_p\, p(k) + q(-k)^T K_q\, q(k) + g\big[q(-k)^T\tau(k) + q(k)^T\tau(-k)\big],$$

up to the common Fourier-volume normalization. It equals its complex conjugate. To lift the result from the complex fibre calculation to the actual staggered lattice it must also obey the mask placements and Bloch gluing.

That gluing is checked explicitly for the declared regulator. Under the reciprocal shift $k \mapsto k + 2\pi e_a/a$, let $R_V^{(a)}$ change the sign of vector component a, and let $G_T^{(a)}$ change the sign of an off-diagonal tensor component precisely when its mask contains e_a. Diagonal tensor components are unchanged. Then

$$\kappa' = R_V^{(a)}\kappa, \quad B'G_T^{(a)} = R_V^{(a)}B, \quad s' = G_T^{(a)}s, \quad K_q' = G_T^{(a)}K_qG_T^{(a)}.$$

Since K_p commutes with $G_T^{(a)}$, the source gluing $\tau' = G_T^{(a)}\tau$, $j' = R_V^{(a)}j$, and $\rho' = \rho$ gives

$$\mathcal{J}' = G_T^{(a)}\mathcal{J}, \qquad \Psi_0' = \Psi_0, \qquad \mathbf{\Psi'} = R_V^{(a)}\mathbf{\Psi}, \qquad (q')^T\tau' = q^T\tau.$$

Here $\mathcal{J}$ is the response drive defined in (12). The checker verifies all these matrix identities for $a = 1, 2, 3$. Reality is most cleanly imposed on extended-zone labels $(-k, k)$; if $-k$ is folded into the first Brillouin zone by a reciprocal vector, the displayed transition matrices supply the additional mask signs. What remains unverified is that a particular repository implementation uses exactly these declared conventions.

Gauge-invariant retarded curvature response. The local curvature variables remain

$$E = K_q q, \qquad F = K_qK_p p. \tag{11}$$

They commute strongly with the gravitational parts of all four constraints: $K_qB^T = 0$ and $K_qK_ps = 0$. The c-number shifts in (8) have zero bracket with them, so E, F remain gauge invariant. They obey $\dot{E} = F$ and, on $\Psi_0 = 0$,

$$\ddot{E} + r^2E = -g\mathcal{J}, \qquad \mathcal{J} := K_qK_p\tau + \frac{1}{2}s\rho. \tag{12}$$

The retarded solution with vanishing past data is therefore

$$\boxed{\delta E(t,k) = -g\int_{-\infty}^{t} \frac{\sin(r(t-t'))}{r}\,\mathcal{J}(t',k)\,\mathrm{d}t', \qquad \delta F = \partial_t\delta E.} \tag{13}$$

This is the classical Hamiltonian source response in the complex Fourier dictionary, not a spectral-positivity surrogate. Identifying it with a quantum Kubo response requires importing the Round-6 CCR/Fock representation and evaluating the corresponding commutator; that calculation is not executed in this artifact.

Theorem 9 (Proposition: Ward compatibility of the response)**.** *The source in (12) is transverse and has the trace required by the sourced scalar constraint:*

$$B\mathcal{J} = 0, \qquad t^T\mathcal{J} = -(\ddot{\rho} + r^2\rho). \tag{14}$$

Consequently (13), with constraint-compatible initial data, satisfies

$$BE = 0, \qquad t^TE = g\rho, \qquad t^TF = g\dot{\rho}. \tag{15}$$

For a TT step source $\tau(t) = \Theta(t)e_+$ at $\kappa = (0, 0, r)$ and $\rho = j = 0$,

$$\delta E(t) = -g(1 - \cos rt)e_+, \qquad t \geq 0, \tag{16}$$

where $e_+ = (1, -1, 0, 0, 0, 0)^T/\sqrt{2}$.

Proof. Both $BK_q = 0$ and $Bs = BK_qt = 0$, proving transversality. From (6),

$$t^TK_qK_p\tau = s^TK_p\tau = \kappa^TB\tau = -\ddot{\rho}, \qquad t^Ts = -2r^2,$$

where the second Ward equation followed by the time derivative of the first was used. This proves (14). Taking the trace of (12) shows that $t^TE - g\rho$ obeys the homogeneous wave equation and hence vanishes for compatible initial data. Equation (16) follows from $K_qK_pe_+ = r^2e_+$. □

Notice that sourced E is not generally TT: it is transverse, while its trace is fixed by the energy density. Only its radiative part is TT. Calling all of E a free graviton after matter is added would erase the scalar constraint.

Dynamical matter: what is and is not proved. Let matter have canonical variables z and an uncoupled Hamiltonian H_m. If its zeroth-order densities obey (7), then

$$H = H_0 + H_m + g\langle q, \tau^{(0)}(z)\rangle, \qquad \Psi_0 = -s^T q + g\rho^{(0)}(z), \quad \boldsymbol{\Psi} = \mathrm{i}Bp - g\boldsymbol{j}^{(0)}(z) \tag{17}$$

obeys (9) modulo $\mathcal{O}(g^2)$. This is the precise first-order feedback result. It is not exact nonlinear closure. Indeed, even before computing the g-correction to the matter stress,

$$\{\Psi_0(x), \Psi_i(y)\} = -g^2\{\rho^{(0)}(x), j_i^{(0)}(y)\}_m, \tag{18}$$

plus analogous current brackets. These terms are generally nonzero and must be absorbed by second-order constraint and Hamiltonian corrections if a nonlinear theory exists. The checker gives a canonical scalar datum for which the matter bracket on the right of (18) equals 1/2 exactly.

The distinction between a canonical/Noether stress and the metric-response stress matters here, especially for spinors. Their equivalence requires hypotheses and, for spinors, a tetrad formulation; see Leclerc's primary analysis [1]. We therefore test only explicit lattice Ward complexes below and do not rename them a full Hilbert stress tensor.

Free scalar: a local repair, then a sharp ϕ^4 obstruction. It suffices to examine a collinear sector; failure there is failure of a purported universal cubic-lattice identity. Set $a = 1$ on a periodic one-dimensional row and write

$$g_x = \phi_{x+1} - \phi_x, \qquad \Delta\phi_x = g_x - g_{x-1}, \qquad \dot{\phi}_x = \pi_x, \quad \dot{\pi}_x = \Delta\phi_x - m^2\phi_x. \tag{19}$$

The site energy and link flux

$$\rho_x = \frac{1}{2}\pi_x^2 + \frac{1}{2}m^2\phi_x^2 + \frac{1}{4}(g_x^2 + g_{x-1}^2), \qquad j_x = -\frac{1}{2}(\pi_x + \pi_{x+1})g_x \tag{20}$$

obey the exact energy equation

$$\dot{\rho}_x + j_x - j_{x-1} = 0. \tag{21}$$

The continuum-looking averaged-gradient stress

$$\tau_x^{\mathrm{naive}} = \frac{1}{2}\pi_x^2 + \frac{1}{4}(g_x^2 + g_{x-1}^2) - \frac{1}{2}m^2\phi_x^2 \tag{22}$$

does not obey momentum continuity. An exact same-range correction is

$$\boxed{\tau_x^{\mathrm{free}} = \frac{1}{2}\pi_x^2 + \frac{1}{2}g_x g_{x-1} - \frac{1}{2}m^2\phi_x^2 = \tau_x^{\mathrm{naive}} - \frac{1}{4}(\Delta\phi_x)^2.} \tag{23}$$

Direct substitution gives

$$\dot{j}_x + \tau_{x+1}^{\mathrm{free}} - \tau_x^{\mathrm{free}} = 0. \tag{24}$$

The correction in (23) is exactly the finite-difference product-rule term; it vanishes at leading continuum order but is mandatory at finite spacing.

For the rational four-site datum $(\phi_0, \phi_1, \phi_2, \phi_3) = (0, 1, 2, 4)$, the four residuals obtained from (22) are

$$\left(-\frac{25}{4}, \frac{1}{4}, \frac{35}{4}, -\frac{11}{4}\right), \tag{25}$$

so this is an analytic mutant, not a rounding effect.

Now add $V_{\mathrm{int}} = \lambda\phi^4/24$ to ρ_x, the force $-\lambda\phi_x^3/6$, and the continuum-like pressure $-\lambda\phi_x^4/24$ to (23). Energy continuity (21) remains exact, but the momentum residual is

$$R_x := \dot{j}_x + \tau_{x+1} - \tau_x = \frac{\lambda}{24}(\phi_x + \phi_{x+1})(\phi_{x+1} - \phi_x)^3. \tag{26}$$

For the same datum,

$$\sum_{x=0}^{3} R_x = -\frac{17}{2}\lambda \neq 0. \tag{27}$$

Every periodic stress divergence sums to zero. Therefore no redefinition of τ_x alone—local or nonlocal—can repair (26) on this matter phase space while leaving j_x and its dynamics fixed. This is the first precise interacting obstruction, and it appears in constraint propagation at order $g\lambda$, not only in a higher-order gravity bracket.

There is a limited repair for the already-selected nonzero gravity modes. Let P_0 be the spatial mean and let $D\tau = \tau_{x+1} - \tau_x$. Then

$$\delta\tau = -D^{+}(I - P_0)R, \qquad D\delta\tau = -(I - P_0)R, \tag{28}$$

where D^{+} is any zero-mean inverse of D. This makes the nonzero-mode Ward identity exact and is constructed explicitly by cumulative summation in the checker. It is nonlocal, leaves the homogeneous momentum defect P_0R, and does not constitute a local interacting matter coupling. A full repair must instead alter the matter discretization/current or add degrees of freedom that carry the missing lattice momentum. The broader lattice literature likewise defines and renormalizes energy-momentum tensors by enforcing Ward identities, rather than assuming the continuum expression survives unchanged [2, 3].

Free fermion: exact collinear divided-difference vertices. A second collinear test uses a free chiral fermion whose *one-particle Hamiltonian* is nearest-neighbour, with dispersion $\epsilon(p) = \sin p$. This is a finite-spacing Ward test, not a claim of a doubler-free $3+1$ dimensional chiral theory. For a bilinear transferring momentum K, set

$$\kappa(K) = 2\sin\frac{K}{2}, \quad \epsilon_- = \sin p, \quad \epsilon_+ = \sin(p+K), \quad \bar{\epsilon} = \frac{1}{2}(\epsilon_+ + \epsilon_-). \tag{29}$$

The exact divided-difference identity is

$$\epsilon_+ - \epsilon_- = \kappa(K)\cos\left(p + \frac{K}{2}\right). \tag{30}$$

Consequently the energy, current, and stress vertices

$$\rho = \bar{\epsilon}, \quad j = \cos\left(p + \frac{K}{2}\right)\bar{\epsilon}, \quad \tau = \cos^2\left(p + \frac{K}{2}\right)\bar{\epsilon} \tag{31}$$

satisfy both one-particle Ward identities

$$(\epsilon_+ - \epsilon_-)\rho = \kappa j, \qquad (\epsilon_+ - \epsilon_-)j = \kappa\tau. \tag{32}$$

In real space, ρ, j, and τ in (31) have maximal hopping ranges one, two, and three, respectively: only the one-particle Hamiltonian is nearest-neighbour. The average factors are the fermionic analogue of the scalar repair in (23). Replacing j by the continuum vertex ρ fails: at $p = \pi/6$, $K = \pi/3$, one has $(\rho, j, \tau) = (3/4, 3/8, 3/16)$ and the first Ward residual is $-3/8$. Thus both bosonic and fermionic tests reject naive continuum vertices, while both free tests admit explicit finite-spacing corrections.

The homogeneous obstruction. The preceding free tensor construction removed the complete $k = 0$ canonical block to eliminate its negative trace direction. At $k = 0$, $B = s = K_q = 0$, so its vacuum local constraints vanish identically. Equation (8) would reduce to

$$\Psi_0(0) = g\rho(0), \qquad \boldsymbol{\Psi}(0) = -g\boldsymbol{j}(0). \tag{33}$$

A positive-energy matter state normally has $\rho(0) \neq 0$, and a state may also have $\boldsymbol{j}(0) \neq 0$. For $g \neq 0$, imposing (33) would delete either such component rather than source gravity. The conditional coupling must therefore be read in one of two narrow ways:

(a) couple only $(I - P_0)(\rho, j, \tau)$, equivalently fluctuations around a specified homogeneous background; or

(b) restore and consistently stabilize a homogeneous/global geometry sector.

The first option is sufficient for (9)–(16) at nonzero momentum but does not describe the gravitational response to total energy. The second is an open construction. Simply calling the removed block "gauge" is not a repair.

Quantum meaning and remaining obligation. Equation (13) is an exact classical Green-function solution of the conditional staggered Fourier equations and is gauge invariant and Ward compatible there. A smooth compact-time c-number source is expected to define linear forcing of the already-constructed oscillator Fock space, but the CCR import and Kubo commutator needed to prove that quantum statement are not executed here. Thus the generic prescribed-source fibre algebra is closed; its full staggered real-space and quantum realization remain open checks.

For dynamical scalar or fermion matter, (17) is an interacting Hamiltonian on the tensor–matter tensor product only as a formal first-order deformation. Equations (18), (26), and (33) are three distinct unclosed edges:

- g^2 current brackets require genuinely deformed constraints and gravity self-couplings;
- finite-spacing interacting matter may fail the spatial Ward identity already at order $g\lambda$;
- total energy has no receiving homogeneous gravity mode.

An exact graph energy current such as $J_{x\to y} = \mathrm{i}[h_x, h_y]$ addresses energy continuity only. It is not by itself a symmetric stress satisfying the vector Ward identity, and it does not close the Hamiltonian constraint algebra. Conversely, positivity of a TT stress spectral measure says nothing about these three equations.

Executable certificate and evidence typing. The repository certificate `[verification/v1035_matter_coupling.py]` imports no proof text. It checks the generic prescribed-source polynomial identities with their full complex Fourier signs, a conserved plane-wave witness, the mandatory failure of the old restored-i sign, paired Fourier reality, all three declared Bloch transitions, the classical retarded TT step solution, the two collinear Ward complexes and mutants, the ϕ^4 obstruction and projected repair, the nonzero matter Poisson bracket, and the fermion divided differences. It does not instantiate a full three-dimensional matter complex or execute a quantum Kubo calculation. Its successful run reports 55 exact symbolic/rational checks and four floating-point regressions. The floating checks sample generic fibres and random scalar/fermion data; they establish no universal statement.

The exact cost is fixed-size symbolic algebra plus a four-site matter system; the numerical cost is constant and negligible. The certificate is read-only.

References for the matter-coupling proof

[1] M. Leclerc, "Canonical and gravitational stress-energy tensors," arXiv:gr-qc/0510044 (2006).

[2] A. Buonanno, G. Cella, and G. Curci, "Lattice energy-momentum tensor with Symanzik improved actions," arXiv:hep-lat/9506008 (1995).

[3] L. Giusti and M. Pepe, "Energy-momentum tensor on the lattice: non-perturbative renormalization in Yang–Mills theory," arXiv:1503.07042 (2015).

Non-RH research through Round 27: a full-reference control theorem

Progress in a specified model, not physical TOE completion

The previously local research contracts through Round 27 are now shipped with their proof sources, source-pinned checkers, negative controls and saved validation records under `experiments/theory-contracts/`. They extend the positive matter/constraint constructions to an interacting scalar/charge model, local charge transport, exact scalar elimination and controlled signed history sums. Additional off-shell fields, selections and limiting hypotheses remain explicit. These constructions are not a derived common microscopic TFPT parent. All T1–T8 gates and `TFPT.TOE.COMPLETE.01` remain [O].

What the intermediate contracts establish. The geometry branch realizes the proposed elliptic boundary data but exposes its unselected markings and symmetry enlargement; it supplies neither physical dimension selection nor a selected chiral measure. The local positive and constraint-domain branches distinguish the separated completion from the first-vertex-preserving completion, including their different off-shell terms and form domains. The charge branch retains the actual eight-channel Gram form and cocycle. Round 19 couples charge to the scalar mass in a positive local construction rather than merely adding independent tensor factors; its added frame/lock laws are not derived microscopically. Rounds 20–24 introduce local charge transport and identify the changed charge carrier, density assumptions and finite-volume limits. They do not turn the original global charged operation into a local operation by relabeling it. Rounds 25–27 below control the explicitly specified $g = 0$ scalar/charge model at a finite Euclidean regulator. The separate proof files retain the hypotheses of each branch; results from incompatible completions are not combined into a single theorem.

The common reference. Fix $L \geq 3$, $N = L^3$, $a, m > 0$, $u, \nu \geq 0$, $J \geq 0$, a time step $\delta > 0$, $T \geq 3$ and $\beta = T\delta$. Every site's charge remains in $\mathbb{Z}^8$. The original signed hopping and potential are

$$\begin{aligned} H_{\mathrm{hop}} &= \kappa I - J A_{\mathrm{sign}}, & \kappa &= 48NJ, \\ D(n) &= \frac{1}{N}\sum_x e(n_x) + \nu \sum_{\langle xy \rangle} e(n_x - n_y), & e(p) &= \tfrac{1}{2} p^T G p, \\ V(n,\phi) &= D(n) + \tfrac{1}{2}\phi^T A(n)\phi, & A(n) &= -\Delta_a + m^2 + 2u\ \mathrm{diag}\, e(n_x). \end{aligned}$$

The $48N$ directed elementary shifts include their original phases. Thus $0 \leq H_{\mathrm{hop}} \leq 2\kappa I$. With $K_\phi = \sum_x \pi_x^2/2$, write the same symmetric primitive transfer as

$$S = e^{-\delta K_\phi/2} e^{-\delta V/2}, \qquad Q = S^* e^{-\delta H_{\mathrm{hop}}} S, \qquad Q_D = S^* S, \qquad C = S S^*,$$
$$Z = \mathrm{Tr}\, Q^T, \qquad Z_D = \mathrm{Tr}\, Q_D^T = \mathrm{Tr}\, C^T, \qquad \theta = \beta\kappa.$$

All traces in the next theorem may be restricted to any fixed total-charge sector. Scalar mass and charge coercivity give $0 < Z_D < \infty$. Crucially, Z_D includes *all* constant charge profiles and their charge-dependent scalar determinants, not just the zero profile.

Full-reference theorem and proof. Let Z_{abs} be the sum of moduli before cancellation, resolved into elementary hopping words, initial charge profiles and scalar integrals. Then

$$\boxed{e^{-\theta} Z_D \ \leq\ Z \ \leq\ Z_{\mathrm{abs}} \ \leq\ Z_D, \qquad \frac{Z_{\mathrm{abs}}}{Z} \leq e^{\theta}.} \tag{33}$$

Every elementary hop changes the complete profile, hence $\langle n | A_{\mathrm{sign}} | n \rangle = 0$. Scalar spectral Jensen gives $p_n = \langle n | e^{-\delta H_{\mathrm{hop}}} | n \rangle \geq e^{-\delta\kappa}$. Pinching Q into its charge blocks $P_n Q P_n = p_n Q_D(n)$ contracts the Schatten T norm, so

$$\mathrm{Tr}\, Q^T \geq \sum_n \mathrm{Tr}(P_n Q P_n)^T \geq e^{-\theta} \sum_n \mathrm{Tr}\, Q_D(n)^T.$$

For a finite partition, pinching is an average of unitary conjugations; triangle inequality proves contraction. For the countably infinite carrier, use a finite set of charge blocks and its complement, discard the latter's nonnegative trace contribution, then increase the set monotonically. This is a trace-power statement, not the generally false operator inequality $e^{-\delta\kappa}Q_D \le Q$.

For the absolute bound replace phases by bare translations only in the majorant. Each unsigned slice word U_j is a sector-preserving unitary, and C has nonnegative scalar integral kernels. Schatten Hölder yields

$$0 \le \mathrm{Tr}(U_1 C \cdots U_T C) \le \prod_{j=1}^{T} \|U_j C\|_T = Z_D.$$

The left side equals the sum of moduli of the corresponding original profile-resolved weights, not the modulus of an already cancelled trace. At total order k, the slice allocations and channel choices have combined weight $e^{-\theta}\theta^k/k!$. Their positive sum is finite and equals one, proving the absolute bound and justifying Tonelli followed by the signed rearrangement. As a second upper comparison, the unsigned hopping transfer is a positive contraction, so its partition is at most Z_D. These arguments concern the finite regulator, not a limit interchange.

The norm tools are standard; see Perez-Garcia, Wolf, Petz and Ruskai and Hansen and Pedersen. The full-reference application above removes the extra charge heat-trace factor of Round 26 without changing the original model. It still permits exponential growth $e^{48\beta NJ}$ and is not an efficient sign cure.

Three explicit error budgets. The exact Gram inverse satisfies $\|G^{-1}\|_\infty = 31$, so $D(n) \ge a_c\|n\|^2$, $a_c = 1/(62N)$. On the neutral sector define

$$\Theta(s) = \mathrm{Tr}_{\mathrm{neutral}}\, e^{-sD} \le \left(1 + \sqrt{\frac{\pi}{s a_c}}\right)^{8N}.$$

Let K bound all charge coordinates at *sampled slice boundaries* and P bound total elementary hopping order. If $P + 2 > \theta$,

$$\rho_{\mathrm{charge}} \le T e^{\theta} e^{-\delta a_c (K+1)^2/2} \Theta(\beta/2)^{1/T},$$
$$\rho_{\mathrm{hops}} \le \frac{\theta^{P+1}}{(P+1)!} \frac{1}{1 - \theta/(P+2)}.$$

For the first estimate, insert the boundary projection R_K in the joint trace. It commutes with C, not generally with hopping. Hölder gives $Z_D p_D(K)^{1/T}$, with $p_D = \mathrm{Tr}(R_K C^T)/Z_D$. In the neutral sector $Z_D \ge Z_{\phi,0}$ and scalar Gaussian comparison gives $p_D \le e^{-\beta a_c (K+1)^2/2}\Theta(\beta/2)$; then take the Tth root and the union over boundaries. No intermediate state inside a retained hopping word is discarded using this boundary estimate.

For a quadratic memory truncation with operator error ϵ_R, retain the full eliminated determinant. If $\eta_R = \epsilon_R/(\delta m^2) < 1$, Round 25/26 comparison gives

$$B_R = \frac{\epsilon_R |S_{\mathrm{ret}}| T g_m(0)}{2(1 - \eta_R)}, \qquad \rho_{\mathrm{memory}} \le e^{\theta}(e^{B_R} - 1).$$

Here $g_m(0)$ is the exact diagonal inverse of the temporal mass operator, or a certified upper bound. Restrict histories first and approximate the retained scalar factors second. Then $\rho = \rho_{\mathrm{charge}} + \rho_{\mathrm{hops}} + \rho_{\mathrm{memory}}$ bounds the relative partition error. If $\rho < 1$, the approximate partition is positive; bounded one-time selfadjoint charge observables of norm at most one have normalized error at most $2\rho/(1-\rho)$. Arbitrary signed multi-time insertions and reflection positivity of truncated weights are not covered.

Reproducible certificates and actual remaining cost. The planner uses integers and fractions, with upward roots, $e^\theta \le 3^{\lceil\theta\rceil}$, $e^{-d} \le 2^{-\lfloor d\rfloor}$ and $e^B - 1 \le B/(1-B)$ for $0 \le B < 1$. At $N = 27, T = 3, \beta = 1, J = 1/10$, full memory and a 1% omission target, the published Round 26 $K = 4096, P = 962$ improve to $K = 2670, P = 354$. Applying the same first-passing-integer search to the old inequalities gives $K = 4035, P = 956$: the gain is not just a finer search grid.

A genuine nonzero-memory example uses $L = 9$, cells of side 3, $a = m = 1$, arbitrary $u, \nu \ge 0$, $T = 33$, $\delta = 3/4$, $\beta = 99/4$ and $J = 1/866052$, hence $\theta = 1$. There are 513 retained sites and 27 interiors of 8 sites. The cell gap is 4, giving exactly $q_b = 1/4$. A temporal comparison mass squared $8/9 \le 1$ gives exactly $q_0 = 1/2$. The cyclic tail certificate at $R = 12$ retains 25 of the 33 quadratic time offsets and bounds the summed memory error by 0.2045%. With $K = 12235, P = 5$ the three omission budgets sum to less than 1%. The full determinant remains a full-history function. This is a declared weak-total-hopping example, not a fitted coupling.

No full interacting partition is numerically evaluated. The naive enumeration envelopes still have decimal-digit upper bounds 1912 and 25624 in the two examples, respectively. They are upper counts of one scheme, not measured runtimes or complexity lower bounds. Future numerical evaluation must additionally budget Gaussian and transcendental errors. The two new checkers have 23 and 25 exact groups with 18 source pins each; 37 isolated fixtures pass normally and under optimization, and the aggregate reruns 29 prerequisites. These counts are regressions, not numbers of independent physical theorems or proof-assistant certificates. See `experiments/theory-contracts/ROUND27_STATUS.md` and `experiments/theory-contracts/run_round27.py` for the reproducible entry.

The original sign obstruction and physical frontier remain. For overlapping original hops $A = W_{xy,e_2}$ and $B = W_{yz,e_3}$, the word ABA^*B^* closes with phase -1 because $e_2^T G e_3 = 1$. Its actual hopping matrix-element product is $-J^4$ for $J > 0$. Diagonal configuration rephasing telescopes around this loop and cannot remove it. This excludes that particular stoquastic rephasing, not all bases, regroupings or simulation methods. High-dimensional charge sums, full-history determinants, volume-uniform interacting continuum and real-time reconstruction remain open, as do a derived common parent, chiral completion, gravity and physical initial-state selection. No RH or empirical claim is made.

Research consolidation: source fields, dynamics and the remaining bridge (2026-09-09)

Status and provenance. The research archive after Round 27 is now integrated, including the Round 28–50 development chain and the subsequent named branches. These are written, model-specific arguments accompanied by executable controls, not newly formalized or independently reviewed theorems. All physical T1–T8 gates and `TFPT.TOE.COMPLETE.01` remain [O]. The source map, prerequisites and reproduction entry points are collected in `experiments/theory-contracts/RESEARCH_2026-09-09.md`. Historical records retain their original local/uncommitted labels and hashes; the integration does not retroactively change their provenance.

1. A charged field from the original filled microscopic sea

Fix the original complex, number-conserving QWZ strip at width eight, mass one and holonomy sector $r = 1$. Its one-particle space has dimension $16N$ over $\mathbb{C}$; it is not the compiler's sixteen real Majoranas. Write P_N for its negative-energy projector of rank $8N$ and Ω_N for the actually filled sea. Set

$$D_N = \frac{N}{2\pi} h_N, \qquad \mathcal{H}_N = d\Gamma(D_N) - \mathrm{Tr}(P_N D_N), \qquad Q_N = \mathcal{N} - 8N.$$

Here $\mathcal{H}_N \geq 0$ on the full finite Fock space, including charged sectors. For a smooth periodic function g use the physical top row, not an auxiliary charge register:

$$(R_N g)(x,y) = N^{-1/2} e^{-i\pi x/(2N)} g(x/N)\delta_{y,7}\chi_+, \qquad \Psi_N(g) = a_N(R_N g), \quad \chi_+ = 2^{-1/2}(1,1)^T.$$

These fields obey exact CAR, have charge -1, and are supported on the sampled support of g. Their adjoints have charge $+1$. For $k = j - \frac{1}{4}$, $p = 2\pi k/N$, $\rho = 1 - \cos p$ and the normalized top-row mode $r_{N,j}$, the original strip action gives

$$\|D_N r_{N,j}\| \leq |k|, \qquad \|(D_N + k) r_{N,j}\| \leq \frac{\pi k^2}{N} + \frac{2\pi^2 |k|^3}{3N^2}.$$

Indeed $h(p)\chi_+$ on the top row has the two orthogonal components $-\sin p\,\chi_+$ and $\rho\chi_-$; its norm is $2|\sin(p/2)|$. This is a bound on the actual microscopic action, not a fitted edge dispersion.

The geometric edge quasi-mode has amplitude proportional to $\rho^{7-y}\chi_+$ and residual ρ^8. For $|j| \leq M_N = \min(\lfloor N/8 \rfloor, \lfloor \sqrt{N} \rfloor)$, projection onto P_N for $j \geq 1$ and onto $1 - P_N$ for $j \leq 0$ produces an isometry J_N with $P_N J_N = J_N P_{\mathrm{cur}}$ exactly and $\|J_N e_j - r_{N,j}\| \leq 4\rho_j$. The discarded occupation of the raw top-row vector is at most $9\rho_j^2$, hence $O(N^{-4})$ for each fixed label. If $g = \sum_j f_j e^{2\pi i j x}$, the locality-to-polarization comparison is

$$\|R_N g - J_N \Pi_{M_N} f\| \leq 4\rho_* \|\Pi_{M_N} f\|_2 + \sum_{|j| > M_N} |f_j| \longrightarrow 0.$$

The last term is an ℓ^1 alias tail; replacing it by an unqualified ℓ^2 tail would not justify sampled locality.

Second quantization relative to the filled sea maps the target vacuum, particles and holes into the same source Fock space. With $a_j = b_j$ for $j \leq 0$ and $a_j = d_j^*$ for $j \geq 1$, the target is

$$H = \sum_j |j - \tfrac{1}{4}|\, n_j^{\mathrm{exc}}, \qquad Q = \sum_{j \leq 0} n_j^b - \sum_{j \geq 1} n_j^d.$$

The resulting contractions intertwine both field adjoints and charge in the limit on the common finite-energy core. Orthogonality of momentum residuals bounds the k-excitation generator error by k times the one-particle error, not by the $8N$ sea occupation. Since each excitation costs at least $1/4$, $k \leq \lfloor 4E \rfloor$ on an energy-E core. For $S_m = \sum_j |j - \frac{1}{4}|^m |f_j|$ one also obtains the uniform bounds

$$\|\Psi_N(g)\| \leq S_0, \quad \|[\mathcal{H}_N, \Psi_N(g)]\| \leq S_1, \quad \text{time-transport error} \leq |t| \left(\frac{\pi S_2}{N} + \frac{2\pi^2 S_3}{3N^2} \right),$$

with transported test function $g_t(x) = e^{it/4} g(x - t/(2\pi))$ in the source convention. All fixed smooth finite field words and their vacuum moments follow by bounded-product convergence. The top-edge charge-sector minima are

$$E_{\mathrm{top}}(q) = q^2/2 - q/4, \qquad q \in \mathbb{Z}.$$

They are not minima over both edges of the whole cylinder. The microscopic holonomy term has not been removed to fit a target. Full proof and controls: `microscopic-charged-car-limit/` in the archive.

2. Neutral multipoint limits, without changing the source

For the same strip and filled projector, the alternating four-word $U_a^* U_b U_c^* U_d$ cancels the outer history ramps. Normal ordering is the actual finite-sea trace subtraction. Positive heat regularization, Jensen and Cauchy–Schwarz give uniform L^p bounds for $1 < p < 2$ on the full four-endpoint torus, including collisions. They promote the separated-point limit to an L^1 limit. At physical $\delta = 4N^{-3/4}$ an auxiliary Fourier window of order $N^{1/4} \log N$ improves the normalized grid error to $O(N^{-3/4} \log^3 N)$; the interpolation contribution is $O(N^{-5/8})$. This changes only

the comparison window, not the physical source. Ordered analytic branches retain, for example, the quarter-circle phase $e^{i\pi/8}$. Pair-Gram positivity uses the corresponding reversed pair ordering; it is not a proof of Osterwalder–Schrader positivity. The even-word grid estimate for length $L < 28$ has bound $O(N^{(L-28)/32}\log^3 N)$; higher collision control is not thereby proved. See `microscopic-fourpoint-limit/` and `current-fourpoint-limit/`.

3. Full lattice dynamics and a physical neutral ground representation

A different, explicitly declared compact-$U(1)$ rotor/CAR test parent keeps $a = 1/12$, $\eta = 1/2$, $\beta = 1/4$, $\kappa = 1/100$, $M = 4$ and all integer electric fluxes. It is not selected by the TFPT compiler. The all-electric source majorant has contraction

$$\rho(T) = \frac{77T}{96} + \frac{53T^2}{14400}, \qquad \rho(1) = \frac{11603}{14400} < 1.$$

The subsequent all-observable argument counts all three vertices of each two-step hopping footprint. Its directed incidence is $J = 69/32$, and an s-cell observable has connected-series majorant $q^n(s/2)_n/n!$, $q = 4J|t|$. At $|t| = 1/16$, $q = 69/128 < 1$. Finite-volume maps therefore converge in norm on bounded local operators, uniformly on compact time intervals, and compose to all-time automorphisms of the full bounded quasi-local algebra. *Continuity qualification:* this group is not point-norm continuous on that full algebra. Unbounded electric flux gives an explicit rotor, and also neutral Wilson-loop, counterexample. Its maximal continuous subalgebra equals the free onsite continuous subalgebra and is proper, not norm dense in the full algebra.

Normalized periodic neutral ground-space projectors have locally trace-norm compact subsequences. Their limits are locally normal, stationary states with a strongly implemented dynamics and nonnegative generator on the neutral physical cyclic representation. This is not positive energy for arbitrary charged fields, nor uniqueness of the state. The bare all-low, zero-electric preparation is not a ground state. A hole/square-completion estimate improves the electric bound to

$$\sum_{a=1}^{3}\langle E_{x,a}^2\rangle \le C_* = \frac{1706590130}{3033378777} \simeq 0.562604, \qquad \langle n_{H,x}\rangle \le \frac{32}{9075}.$$

For the elementary neutral Wilson loop W and $S = \sum_{\ell\in\partial p} p_\ell E_\ell$, the full Hamiltonian obeys $W^*HW - H = \kappa(S + 2)$. Its limiting positive spectral measure has $m_1 = 1/50$ and $m_2 \le \kappa^2(16C_*/3 + 4)$, giving positive-energy weight at least 0.571383 and positive spectral infimum at most 0.035003 in the declared Hamiltonian units. This is neither a positive gap lower bound, a mass prediction nor a continuum limit. See `observable-dynamics/`, `neutral-ground-state/` and `ground-state-loop-response/`.

4. What the compiler, Clock and lattice audits actually decide

The charge-sign quotient $L/(1+J)L$ loses commutator information. The minimal additive refinement is the eight-bit symplectic space $L/2L$, with four-dimensional kernel $(1 + J)L/2L$. There are 64 quadratic-form-preserving sections, four compatible with the family marking; the deck exchanges complementary sections. The full target on $\ell^2(E_8)$ has charge shifts, cocycle and $H_0|\mu\rangle = |\mu|^2|\mu\rangle/2$, but is not a microscopic identification. The spinor charge maps to $s = (1/2, \ldots, 1/2)$, so $|s|^2/2 = 1$. Its order-four quotient grading must retain actual charge carry: $U_s^2 = -U_{2s}$ and $U_s^4 = U_{4s}$, not the identity.

For the compact parent, every site-fixed, rotor-independent diagonal phase symmetry of the actual low/high hoppings has $h_x = l_x = q_x$ and link phase $q_v - q_u$. Its implementation is $e^{-i\theta\sum_x q_x}e^{-i\theta\sum_x q_x G_x}$, hence only a scalar on $G_x = 0$. The distinct onsite energies reduce site-fixed internal $U(2)$ mixing to this diagonal class. In addition, paired low/high characters have even multiplicities, whereas the source Clock has multiplicities $(5, 1, 1, 1)$. There is no eight-mode intertwiner in that class. Spatial, flux-dressed or particle–hole actions are not excluded.

The compiler's scalar quadratic Majorana dynamics and Gaussian elimination do not produce a connected quartic vertex. In contrast, the actual rotor double commutator has a nonzero quartic component. That fact does not identify it with a selected effective compiler interaction. The frozen compact-parent matter symbol factors through $x(k) = \sum_i \cos k_i/6$; its Pauli Jacobian has rank at most one and its band separation is at least $55/16$. Retuning scalar functions of x cannot supply a three-direction Weyl node. See `charged-cocycle-lift/`, `clock-rotor-joint-charge/`, `clock-interaction-provenance/` and `parent-selection-audit/`.

5. The next closure test is one source-to-source bridge

The integer charged CAR field is substantial progress beyond neutral determinants, but is not yet the half-charged E_8 spin field. A sharp half-twist has a divergent polarization Hilbert–Schmidt sum $\sum_{k\geq 1} k/[\pi^2(k+1/2)^2]$. Its unrenormalized unitary limit is therefore not justified. The next construction must provide a renormalized, smeared inter-sector field, both adjoints, source energy bounds and charge/cocycle carry, and then identify the eight channels and their family/Clock marking from the same source. Appending eight independent copies or a formal lattice register would assume this bridge.

The accompanying double-cover, origin-cascade and prime-event studies are archived separately. Exact clock/interaction readouts are finite algebraic results; a sheet flip alone does not yield an Euler product, RH positivity or a factoring advantage. The old logarithmic cascade telescopes and retains its separate scale choice. No arithmetic or physical marker is promoted by those observations. T1 still requires principle and compiler selection; T2 the microscopic charged E_8 seam; T3 one selected local 3+1D parent; T4 a chiral anomaly-consistent gauge theory; T5 its physical continuum and scattering; T6 complete couplings and neutrino data; T7 universally coupled quantum spin two; and T8 a selected state and common source functional. Separate constructions are not their common-parent solution.

1 Universalraum synthesis: source selection, reconstruction and accessible dynamics

Scope and evidence, 12 September 2026. This section consolidates the local primitive-transfer investigation and the latest two available answers from each of four ChatGPT research tasks. It is not an independent rerun of all external certificates, an exhaustive conversation archive, or a derivation of a theory of everything. All T1–T8 and `TFPT.TOE.COMPLETE.01` remain open. The local checkout base is `66b91e40e245569f06ab440ead80f446c9be0ee5`; additional local files must not be interpreted as published results.

The central distinction is between *reconstructing a generator from specified data* and *deriving those data and the elementary generator class from the primitive source.* Several finite reconstruction steps are now sharply formulated. The second, forward implication remains open. No RH, general efficient factoring, P versus NP, or Hylæan capability claim is promoted.

1.1 Typed objects and prior local results

The integral compiler commutant, its positive cone, the sixteen-real-mode quadratic candidate, the seventy-state interacting chain, the filled QWZ strip and the compact-U(1) rotor parent are different objects. In particular, the quadratic candidate has eight complex modes and a 256-dimensional full Fock space. The designated H_{70} is a fixed-four-particle sector of an open four-site, eight-mode model; it is not an unbounded winding rotor or the internal Dirac operator.

The compiler-order audit distinguishes a unique minimal family-inner order from its unique maximal overorder, with integral indices four and four. These solve different selection problems.

The Hermitian 2×2 formula

$$X = \begin{pmatrix} t+z & x-iy \\ x+iy & t-z \end{pmatrix}, \qquad \det X = t^2 - x^2 - y^2 - z^2$$

provides an exact Lorentz-cone representation, not a derived spacetime metric, action or quantum-gravity measure. Under the matched-center and Clock-normalizer hypotheses, the actual eight-complex-mode source excludes a unital $M_2(\mathbb{C})$ identification carrying its whole Clock action. The active M_3 corner accommodates two distinct slow frequencies; a closed autonomous M_2 automorphism flow has only one positive Bohr frequency.

Other local controls delimit apparent simplifications. One-sided quaternion defects can be scalar, and Gram whitening can erase the relative rates. A unique Hilbert–Schmidt relational vector need not give a nontrivial physical reduced state. A four-point D_4-symmetric conservative real generator retains independent edge and diagonal weights. On the scalar circle, full rotation/reflection symmetry and differential order at most two imply $aD^2 + b$; fourfold symmetry alone does not.

The direct positive point-evaluation perturbation of $|D|$ is not a closed form in the audited setting; its positive Fourier-cutoff family has the free norm-resolvent limit. A negative renormalized coupling gives a nontrivial family with an unselected parameter. Under the direct rotor identification its selected vector need not have finite D^2 energy. Adding a positive ηD^2 term changes the form domain and introduces an unselected η. The free Dirichlet–Neumann operator $|D| = \sqrt{D^2}$ is an exact bridge, but not an identification of the two time generators.

Finally, for $u^* = -u$ and $u^2 = -I$,

$$(I + nu)/\sqrt{1+n^2} = \exp(\arctan(n)u).$$

An integral index $(1+n^2)^4$ of the unnormalized lattice action is not entropy production of this unitary, nor a derived physical clock.

1.2 Results reported by the four parallel tasks

The following are *attributed task reports*; their original finite-field and rank certificates have not all been independently rerun in this consolidation. The archived excerpts retain exact task identifiers and retrieval limitations.

1. **Vollständige Lösung entwerfen.** For the designated H_{70}, reported irreducibility over $\mathbb{Q}(\zeta_8)$ implies that linear zero conditions over this field annihilating the ground vector annihilate its entire 38-dimensional even reflection sector. Reductions at 449 and 977 are reported. This is a scoped obstruction to exact linear zero selection, not to all variational selection. The explicitly weighted two-state control

   $$Q = \begin{pmatrix} 2 & -2 \\ -2 & 4 \end{pmatrix}, \qquad \operatorname{spec} Q = \{3 - \sqrt{5}, 3 + \sqrt{5}\}$$

   has a simple positive minimum; changing relative weights changes its selected vector. A previous nearest-neighbor, traceless QWZ comparison reports rank 14 in a 15-parameter class, while allowing $h + \epsilon h^3$ preserves occupied projectors and changes relative energies.

2. **TFPT und Universalraum Ideen.** A declared neutral, finite-range quartic extension contains 110 nonzero expressions spanning 66 independent real-symmetric directions, including the original 17. A real even eigenstate can constrain at most 37 transverse directions in the 38-dimensional even sector, leaving at least 28 nonconstant eigenstate-compatible directions. A separate thermal reader is reported to have kernel $\operatorname{span}\{I, H_{70}\}$ for every finite $\beta > 0$, using a rank-64 commutator certificate and thermal positivity. Synthetic thermal numerical reconstructions are diagnostics, not independent primitive-state derivations.

3. **Untersuche das Gesamtbild.** A different product-enlarged class is reported to have dimension 133 and 95 nonconstant compatible directions; it must not be confused with the 66-dimensional class. The signed equal-weight Laplacian on the designated 70 configurations and 220 edges has a unique frustrated minimum but its state admits no nonconstant generator in the old comparison family. This rejects that particular selector, not TFPT generally. Two designated states share $C = I_8/2$ while their electric-form expectations differ, showing information loss in a one-body description.

4. **Thema vollständig lösen.** All seventy eigenvectors of the specified H_{70} are reported non-Slater, using Plücker-relation certificates in finite extensions at primes 347 and 23. Full two-particle data determine its nondegenerate ground state. Within the original family, sixteen nonconstant expectations and a positive supporting-operator problem determine the generator direction, conditional on the reported rigidity. Neither the state data nor the choice of family have thereby been obtained from P1/P2.

The findings agree on a structural requirement: the source must distinguish elementary generators from products of already allowed operations. Closing the observable algebra under composition is not a derivation of a list of independently weighted primitive energy terms.

1.3 Reconstruction contracts that must remain distinct

For a pure eigenstate and a specified Hermitian family O_i, the covariance form obeys

$$c^{\mathsf{T}}\Gamma c = \|(K - \langle K\rangle)\psi\|^2, \qquad K = \sum_i c_i O_i.$$

A one-dimensional nonconstant kernel identifies one generator direction, not the origin of the family or state. For a faithful mixed state, $\mathrm{Var}_\rho(K) = 0$ instead forces K to be scalar. Thermal reconstruction therefore needs its own stationarity-response contract. Adding separately measured eigenstate covariance matrices is not the same as mixing the eigenstates before computing one covariance matrix.

For a fixed-N system with a Hamiltonian of at most two-body order, the complete two-particle reduced density matrix determines its energy expectation. If the ground state ρ_0 is unique with gap $\Delta > 0$,

$$H - E_0 I \succeq \Delta(I - \rho_0)$$

implies uniqueness of the positive state completion having the same two-particle data. More generally,

$$1 - \mathrm{Tr}(\sigma\rho_0) \le \frac{\mathrm{Tr}(\sigma H) - E_0}{\Delta}.$$

This conditional argument is not a general uniqueness statement for arbitrary excited states. Likewise $\log((I - C)C^{-1})$ reconstructs the quadratic quasifree one-body reader, not every interacting state with that covariance.

A different sufficient finite contract starts with a fully specified selfadjoint strictly positive Hilbert-space operator T, a simple largest eigenvalue λ_0 and a positive step size δ:

$$H = -\delta^{-1}\log(T/\lambda_0).$$

It yields a unique ground ray and energy differences. It does not determine T or δ from symmetry or a common fixed vector. Computing $T = e^{-\delta H_{70}}$ first and then inverting it is circular as an origin argument. A completely positive channel is not this type of operator.

1.4 Exact access results in the local quadratic candidate

For the pinned real matrices,

$$D = \alpha J + gB, \quad [O, J] = [O, B] = [J, B] = 0, \quad P = \frac{1}{6}\sum_{k=0}^{5} O^k, \quad Q = I - P,$$

the real ranks are ten and six and $PJQ = PBQ = 0$. The original boundary cyclic space equals P. The induced Fock decomposition is 32×8; with the unchanged split dynamics, finite adaptive boundary protocols depend only on the accessible marginal. Hidden occupation cannot be read out by such protocols, even if other initial correlations are allowed.

The hidden complex energies are $\alpha - g, \alpha + \sqrt{3}g, \alpha - \sqrt{3}g$. The neutral slow transitions have magnitudes $(\sqrt{3} \pm 1)|g|$ and annihilate the empty vacuum at the audited point $(\alpha, g) = (1, 1/8)$. A conditional added mixed field exposes all sixteen linear directions. An even cross bilinear can generate all 120 quadratic directions with the full even boundary algebra and free derivation; this does not prove arbitrary many-body control.

The invariant hidden operator algebra is $M_2 \oplus M_2 \oplus \mathbb{C}^4$. Its frequencies are $0, \pm 2\alpha$; number conservation leaves only diagonal operators. This is a conditional restriction: the source has not proved that its Clock must be quotiented as gauge on every observable. The older $\mu_4/\mathbb{Z}_8$ seam/GSO discussion concerns a different Clock and cannot be transferred without an intertwiner.

The existing B decomposes exactly into fifteen channel-edge matrices. Each of the six triple-to-pair edges, if separately available together with D, opens all sixteen real linear directions. A modified fixed weight $B + xE_{14}$ also gives full access for every real $x \neq 0$: an eight-complex-column Krylov determinant is

$$-\frac{ix^3}{2^{41}}\left[x^4 + 7x^3 + (24 + 4i)x^2 + (32 + 26i)x + 24 + 36i\right].$$

The real and imaginary parts of the bracket have no common root. The cubic zero at $x = 0$ prohibits reading this exact-rank statement as a uniform sensitivity bound. Neither the weight nor edge selectability has been derived from the primitive source.

1.5 A covariant event channel with an actual boundary response

Let K be the Hermitian one-particle lift of the existing edge E_{14}. It obeys $K^3 = K$. Define

$$U(\theta) = I + (\cos\theta - 1)K^2 - i\sin\theta K, \qquad U_k = O^k U(\theta) O^{-k}, \qquad \Phi_\theta(\rho) = \frac{1}{6}\sum_{k=0}^{5} U_k \rho U_k^*.$$

This is a covariant, trace-preserving completely positive map. Averaging the Hamiltonians first instead preserves the P/Q split. The two operations must not be confused. For the invariant hidden one-particle state $Q/3$, writing $c = \cos\theta$,

$$\mathrm{Tr}(P\Phi_\theta(Q/3)) = \frac{35 - 34c - c^2}{108}.$$

At $\theta = \pi/2$ the value is $35/108$, without retaining an orientation record. The pulse itself acts only on carrier modes. With R_B the original boundary projector and h the unchanged source one-body Hamiltonian, $\rho' = \Phi_{\pi/2}(Q/3)$ obeys

$$R_B\rho' = 0, \qquad \mathrm{Tr}(R_B h \rho' h) = \frac{1}{24},$$

and hence

$$\mathrm{Tr}(R_B e^{-ihs}\rho' e^{ihs}) = s^2/24 + O(s^3).$$

The initial hidden state and the vacuum have identical accessible vacuum marginals, but this stipulated protocol separates them. The vacuum remains vacuum because the lifted operations preserve particle number. This is a conditional positive readout result, not a source-derived preparation or implementation. Covariance of density matrices also does not establish a stronger Gauss-law physical-state constraint.

1.6 New continuation: transfer succeeds but state selection fails

A possible, explicitly added continuous event model is

$$\mathcal{L}(X) = -i[h, X] + r(\Phi_{\pi/2}(X) - X), \qquad r > 0.$$

The event-only semigroup is the Poisson mixture of Φ^n; the rate r is unselected. With the Hilbert–Schmidt inner product,

$$\mathrm{Re}\langle X, (\Phi - I)X\rangle = -\frac{1}{12}\sum_k \|[U_k, X]\|^2_{\mathrm{HS}}.$$

The Hamiltonian part has zero real quadratic form. Consequently the fixed space of $\mathcal{L}$ is the common commutant of all U_k and h. The three distinct eigenvalues of each U_k make its commutant equal to that of K_k.

Exact linear equations give dimensions ten for the edge-only commutant and five after adding h. The joint algebra is

$$\mathbb{C}I_6 \oplus M_2(\mathbb{C}).$$

Indeed the boundary plane whose three coefficients sum to zero has a rank-two projector R with $K_k R = 0$ and $hR = R$. The two normalized positive states $R/2$ and $(I - R)/6$ are stationary and distinct. The explicit five fixed directions exhaust the computed commutant.

Thus the event mechanism transmits a signal but does not select a unique one-particle state. Full Fock number conservation adds further obstructions. Its 64-dimensional channel superoperator is not Hilbert–Schmidt-selfadjoint, so it is not the strictly positive selfadjoint T in the logarithmic transfer theorem. This is a typed exclusion, not a claim that no different source-derived transfer or detailed-balance representation can work.

1.7 Arithmetic bridges, acceptance gates and provenance

The archived 10 September rotor synthesis reports an exact change to residue/carry coordinates retaining the declared parent dynamics, arithmetic covering operations, conditional logarithmic weights, and finite-energy approximations with explicit error costs. Discarding the carry loses electric dynamics. Norm time, electric time and modular time are not interchangeable. A finite factoring example and static E8 counts do not supply an asymptotic factoring algorithm or a signed global RH proof. No fresh RH-index closure or Hylæan implementation audit was performed in this synthesis.

Gate	Remaining physical obligation
T1	Primitive choices, elementary operations and quantitative measure.
T2	Actual half-charged marked E8 seam field, energy/adjoints and scaling.
T3	One selected local or quasi-local unitary $3+1$D parent.
T4	Chiral SM gauge/Weyl construction, anomalies and mirror control.
T5	Interacting physical continuum and required Lorentz/scattering properties.
T6	Internal gauge couplings and complete neutrino texture and scale.
T7	Massless quantum spin two and universal coupling from the same parent.
T8	Selected physical state and a common source functional for readouts.

The next falsifiable source gate is to derive a weighted, phase-preserving grammar of elementary events, specify whether its output is a Hamiltonian, a positive transfer or a channel, and generate the appropriate state data without importing the desired H_{70}, its eigenstates or target couplings. Reconstruction, physical access, robustness and the downstream continuum bridges must then be checked separately.

The local verification results in normal and optimized modes are:

- `source_edge_access.py`: 44 exact checks;

- `covariant_edge_channel.py`: 20 exact checks;
- `edge_channel_stationary.py`: 15 exact checks.

They are not 79 closed physics claims. The complete local provenance and task report archive are indexed in `docs/TFPT_UNIVERSALRAUM_SYNTHESIS_2026-09-12.md`; the nontechnical version is `docs/TFPT_UNIVERSALRAUM_EINFACH_2026-09-12.md`. Earlier local branches retain their own hypotheses and verification records.

1.8 Proposed follow-ups and discriminating research directions

These are proposed investigations, not newly completed results or launched worker tasks. Their common acceptance rule is a forward derivation without importing the desired Hamiltonian, its eigenstates or target couplings. A genuinely independently derived primitive action may itself be the Hamiltonian; it need not be artificially required to be a different object.

F0: Independent reproduction. Obtain and pin the original 17- and 66-family certificates before using their strongest claims as local premises; then audit the non-Slater and cyclotomic irreducibility certificates. Keep the 133-dimensional product extension separate. Success means the same exact ranks, coefficient fields and class definitions, with numerical conditioning reported separately. Unavailable originals mean "not reproduced", not "false".

F1: Freeze the elementary grammar. For one original finite sector, map each primitive rule to its operator, phase, support, adjoint and composition. Distinguish physical operations from gauge identifications and elementary generators from composite words. A source-derived filtration or equivalent primitive rule must explain why composition does not automatically introduce independently weighted energy terms. Reject a cutoff chosen only to enforce the desired covariance rank.

F2: Derive process type and measure. Construct one source step or collar without the target generator as input. Declare whether the output is a Hamiltonian, a positive Hilbert-space transfer or a reduced quantum channel. Derive weights, step duration and any traced environment from the stated inputs. Test alternative weights with identical symmetry and fixed vector. A freely chosen metric in d^*Wd, pulse angle or event rate remains an additional datum. Channel positivity does not by itself license a Hamiltonian or Markov logarithm.

F3: State selection and correlation content. Compute the exact fixed algebra or ground space of the F2 candidate. Use eigenstate covariance, thermal stationarity or full two-particle data only under their respective contracts. Test identical one-body data with different four-point responses, and identical ground rays with different energy ratios. Repetition of the current channel and free flow cannot remove their common fixed algebra. Any operation removing that algebra must be independently supplied by the source, not by a target-state penalty.

F4: Operational response and costs. For a source-derived intervention, compare admissible preparations with the same initial accessible marginal at the original boundary. Bound the remainder in the conditional $s^2/24 + O(s^3)$ signal on an explicit finite time interval, and include preparation energy, precision and sampling costs. Preserve the no-intervention control. Exact cyclic rank alone is not successful physical control, and arbitrarily small edge asymmetry must not be compensated by unaccounted unlimited precision.

F5: Source-to-model maps. An explicit map between the successful candidate and H70, the rotor parent or the charged seam must preserve the relevant adjunction, charge/Clock action, states, dynamics and readouts on named domains. Matching spectra, dimensions or group names alone is

insufficient. An open chain must not be identified with a winding rotor, nor a free source with an interacting state without the additional operation being derived.

Four exploratory hypotheses, not established mechanisms.

1. *Positive frustrated selection:* derive the unavoidable positive minimum and relative defect weights from the source. The failed equal-weight H70 selector is not revived by adjusting its weights to the target.
2. *A genuine collar transfer:* derive a positive transfer directly from the actual boundary construction and its composition. Reflection positivity must not be inferred merely from a positive Gram matrix.
3. *An internal relational reference:* look for an already present source interaction coupling the untouched boundary plane while respecting the correct joint symmetry. Appending an arbitrary environment or projector does not establish source provenance.
4. *Thermal or multi-state readout:* exploit additional independently obtained state responses for robust reconstruction. Optimizing a synthetic temperature for conditioning does not select a cosmic temperature.

F6: Separate downstream acceptance. The half-charge, common $3+1$D parent, chiral, continuum, coupling and gravity obligations remain separate after any finite success. No new RH or factoring mechanism is proposed in this roadmap; future arithmetic work must first consult the existing source-bound proof and counterexample registers. A later Hylæan application could test order-independent consolidation on frozen inputs while preserving distinguishability and recall, but requires an actual implementation map and separate authorization.

F0 and F1 can proceed independently. F2 then supplies a candidate for both F3 and F4; F5 requires a successful typed source result. The immediate recommendation is to freeze one original-sector grammar and reproduce only the external certificates needed for its first forward test. Full acceptance and stop conditions are in `docs/TFPT_UNIVERSALRAUM_FOLLOWUPS_2026-09-12.md`.

Research Contract — `FLAV.NUSCALE.06` (pentagon-class misalignment candidate)

Typing. A candidate [C] (ledger row `FLAV.NUSCALE.06`); mechanism = the Q_+-to-flavor operator [O]. No seesaw closure; `FLAV.NUSCALE.05` unmoved. ([`verification/v1001_nu_pentagon_chain.py`].)

Executed content. $U_e = I$ inventory theorem; misalignment $U = U_{v9} R_{13}(\theta, \phi)$; pentagon double hit $\phi = 288^\circ = 4(2\pi/5)$ frozen, all three measured angles $\leq 0.56\sigma$ (honest max pull 0.557) at $\theta = 2\pi/35$, the unique LEE survivor of nine pre-declared candidates (LEE 2.7%); v3 chain SHA-16 `a4c28732fa687620` with $\Sigma = 0.0599$ eV, $m_\beta = 9.0$ meV, $m_{\beta\beta} \in [1.5, 3.8]$ meV, $\delta_{\rm CP} = 287.66^\circ$; v270–$\theta_{23}$ tension 1.85σ TYPED; census null 0/1607. **Kills.** DESI floor on Σ; DUNE $\delta_{\rm CP}$ discriminates 287.7 vs 240; JUNO. Status: Candidate [C]; no closure.

Research Contract — `FLAV.NU.TEXTURE.MECHANISM.01` (both structural nulls as binding constraints)

Typing. A research contract [O] (ledger row `FLAV.NU.TEXTURE.MECHANISM.01`), registered 2026-08-29 from review wave 5. ([`verification/v1003_nu_structural_nulls.py`].) **Executed content.** Charged-sector data $h = (1, 1, 1)$, $d = (64, 60, 64)$ (coset min-norms all 2) [E]. Schur-texture NULL [X]: all 8 preregistered (K, B) combinations fail; D4 reflection forces $\lambda_1 = \lambda_3$; hierarchy $(\varepsilon, 2\varepsilon, 3)$ structurally unreachable from D4-even seam data; required $K \sim [3536.998, 1768.499, 1/3]$ matches none of 9 comparators. Orientation-doubling structure theorem [E]: the $\operatorname{sgn} J$ odd sector

generically lifts the degeneracy (exact rational witness odd_scale $= 1/4$, $\gamma = 1/10$). Scale NULL [X]: 12/12 declared trials fail; required cancellation $1 - |c/a| = 1.885\times10^{-4}$ is not supplied by any frozen constant; η-branches are isospectral. **Consequence (binding).** The mechanism must be D4-odd *and* supply a $\sim 2\times10^{-4}$ even-odd cancellation; pure seam data excluded [X]-typed. `FLAV.NUSCALE.05/.06` unmoved; no seesaw closure. Status: `FLAV.NU.TEXTURE.MECHANISM.01` [O].

Executed update (2026-08-30, evening TOE-gate wave): κ six-fold NULL completes the scaffold bound ([verification/v1012_toe_gate_witnesses.py]; the contract stays [O]). Six pre-declared D4-odd candidates, all parity-checked, all NULL (five at 10^{-14}; $K_4 = 3.55\times10^{-3}$ is 18.8× too large). The 1.885×10^{-4} cancellation scale requires the true 4D functional — mechanism scale strictly beyond frozen finite objects. Display marker stays [O].

Executed update (2026-09-05, round-3 integration): parent-internal A_3–D_4 intertwiner (`FLAV.NU.PARENT.INTERTWINER.01`; [verification/v1024_parent_internal_boundaries.py]; no parent marker move). Using only the existing four-state seam clock and six ν^c modes, the finite module constructs a positive, gauge-neutral, parity-even, non-additive local term. Its checked gap is 0.25 conditional on the inherited full-parent bound, and the mixed resolvent is 0.104712041885. The selected pair-operator triplet carries ordinary D_4, whereas the full fermionic lift is projective:

$$C_f^8 = 1, \qquad C_f^4 = (-1)^{N_1+N_3} \neq (-1)^F, \qquad J_f C_f J_f = (-1)^F C_f^{-1}.$$

This supplies neither a neutrino mass matrix nor the required 2×10^{-4} cancellation and is not a 4D functional. `FLAV.NU.TEXTURE.MECHANISM.01`, T6 and TOE remain [O].

Research Contract — `SEAM.MILNOR.LOCALRING.01` (canonical identification of the seam collar with the reduced Milnor algebra)

Typing. A research contract [O] (ledger row `SEAM.MILNOR.LOCALRING.01`), registered 2026-08-30 from review wave 6. ([verification/v1004_milnor_frobenius_gray.py].) **Demand.** A *canonical* identification between the raw seam collar / E_8 transition bus and the reduced Milnor algebra $\mathbb{F}_2[y,z]/(y^2,z^4)$ preserving *simultaneously* the dual-number action, the Frobenius pairing, the Gray torsor, the deck action and CP. **Acceptance.** The identification must be geometric/mechanical (not basis-chosen) and must produce the rank clock and socle-CP as *outputs*. **Kill tests.** Any second inequivalent identification with the same invariants; failure of μ_4-equivariance. **Note.** Would reduce P2 / alignment-bit / Gauss-code / Brieskorn origin to one local ring. Currently [O] because the Milnor algebra of $x^2+y^3+z^5 \cong \mathbb{Q}[y,z]/(y^2,z^4)$ (dimension 8) and the spectral numbers $= \{1,7,11,13,17,19,23,29\}$ (the E_8 exponents) are *classical* (cite, no claim), and the corpus map is unproven.

Executed finite algebra (review wave 6; the contract stays [O]). New-in-corpus exact [E]-finite: rank clock $(2-a)(4-b)$, table sum $30 = h(E_8)$; $S = pqr\cdot\mu/8$; self-clock iff $\mu = 8$; $(2,3,5)$ unique among ordered spherical triples; Galois–Gray action (11 swaps sheets, 7 cycles the seam clock in binary Gray order $0\to1\to3\to2$, $-1 = (a,b)\mapsto(1-a,3-b)$); CP = socle/Frobenius duality (residue pairing nonzero iff $a+a'=1$, $b+b'=3$, $m+m'=30$); integral dual-number bridge $\mathbb{Z}[y,z]/(y^2,z^4)\otimes\mathbb{F}_2$ free of rank 4 over $D = \mathbb{F}_2[\varepsilon]/(\varepsilon^2)$, $\mathbb{Z}[i]/(2)\cong D$, basis-dependent. Interpretive [C]: $15 = 8_R + 7_{NS}$ as $PG(3,2) = AG(3,2)\sqcup PG(2,2)$. Must-fail: $(2,3,7)$ gives $S = 63 \neq 42$; A_8 self-clock passes but the exponent fingerprint fails; wrong socle rank 4. Status: `SEAM.MILNOR.LOCALRING.01` [O].

Executed update (2026-08-30, afternoon harvest; [verification/v1010_simplicity_bridge_census.py]; canonical status SHARPENED, display stays [O]). The strong all-structure identification is finitely obstructed: at $\mathbb{F}_2$, clock order 2 versus 4 and CP Jordan rank 2 versus 4 make the four-structure isomorphism set empty; at every upstairs home the same obstruction persists ($\mathbb{Z}/4$: $J^2 = -I$ versus Gray $P^2 \neq -I$, D_8 versus $C_4\times C_2$; Gaussian $k=2$ order collapse, $k=3$ rank/relation clash; affine conjugacy cannot repair the linear parts). Only a coarse

D-module+pairing bridge exists (witness constructed; $|\mathrm{Aut}(X)| = 36864$, $|\mathrm{Aut}(Y)| = 8$). The ring-internal identities of v1004 (rank clock, socle CP, self-clock) are unaffected. No marker move.

P2 rehabilitation note (not a third contract; [verification/v1005_relative_pencil_bridges.py], corrected by [verification/v1010_simplicity_bridge_census.py]). The v1005 circularity flag is corrected as forced provenance: the bit-selector ladder forces $c = 2$ and family $n = 3$ then forces Q column sums $(9, 5, 1)$; C's carrier-looking 5 is inherited from R column 3, not inserted. Unique primitive positive S_3-invariant covector $\varepsilon = (1, 1, 1)$. The winding bridge remains a *choice* that is conditionally unique: spectrum $\{5/2, 1, 1\}$ alone has 1078 solutions; democratic-image + carrier-anchor kernel leave exactly the corpus shift (count 1). Remaining obligation = seam-geometric derivation of those two directional premises. `AX.P2.01` typing UNCHANGED. Hankel $p_n p_{n+2} - p_{n+1}^2 = 2^{n+1}$ and winding spectrum $\{5/2, 1, 1\}$ remain [E]-finite (provenance v106/v119/v4). Klein-four SNF $(2, 2)$ only after $D = 2d$ is interpretive [C].

Executed update (2026-08-30, late-evening harvest; [verification/v1014_bridge_refinements.py]; `AX.P2.01` typing UNCHANGED). W-bridge P-anch is DERIVED: $e_1 = R^{-1}(1, 1, 2)^T$ uniquely via the winding/triple lock (not by g_{car} or by the repeated anchor entries). P-dem is PARTIAL: KMS equidistribution lives on the trivial character; the $\mathbb{Z}_4$ average annihilates the nontrivial character space. The missing object is the character-blind determinant-response map into the $v4$ row space — the same MMST/Quillen externalized leg as TEL-B/ALG-EXH/P1.

Executed update (2026-08-31, Monday-morning harvest; [verification/v1015_axiom_core_closure.py]; `AX.P2.01` typing UNCHANGED). W-bridge P-dem is now DERIVED: the abelian total-charge insertion is character-blind, $r = (1, 1, 1)$ exactly at collar sizes 12 and 16, democracy residual 0; mutants split $(1, 0, 0)$ and $(5/4, 1, 5/4)$. Together with P-anch (v1014) both W-bridge premises are derived at finite level; the axiom-core remainder is ZERO modulo the externalized MMST identification. Finite derivations are shadows; the continuum identification is the same external leg. Axiom typing UNCHANGED.

Research Contract — `GRAV.WEINBERG.WITTEN.01` (Weinberg–Witten firewall before any spin-2 marker move)

Typing. A research contract [O] (ledger row `GRAV.WEINBERG.WITTEN.01`), registered 2026-08-30 from review wave 6. Child of TOE gate T7; cross-link `GRAV.SPIN2.EMERGENCE.01` (listed here, not back — that would cycle). **Demand.** The microscopic Hamiltonian family must *not* possess a fundamental Lorentz-covariant local conserved $T_{\mu\nu}$; Lorentz symmetry, diffeomorphism Ward identities and the spin-2 mode must emerge in the *same* IR limit. **Acceptance.** Positive TT spectral density; exactly two helicities; massless pole; emergent diffeomorphism Ward identity; universal soft coupling; *and* an explicit statement of which Weinberg–Witten premise is evaded. **Kill tests.** A microscopic covariant local $T_{\mu\nu}$ together with a massless composite spin-2 claim. Status: `GRAV.WEINBERG.WITTEN.01` [O].

Externalization Contract — `SEAM.EQUIV.01` (the seam scaling-limit theorem, packaged for external specialists)

Typing. This section is a *hand-off document*: it packages the one remaining seam obligation as a self-contained problem statement for an outside specialist in conformal nets / vertex operator algebras / algebraic QFT. It *documents*, it does not close: the ledger row `SEAM.EQUIV.01` stays [O] (its conditional route `SEAM.EQUIV.MMST.01` is closed modulo cited theorems; the unconditional parent is the target below). Nothing in this section moves a marker, and no claim beyond the corpus is made. Everything an external solver needs — hypotheses, runnable finite models, the Lean axiom surface, the negative controls a valid solution must respect, and the exact acceptance criteria — is listed here with its verification module.

(1) Minimal theorem statement. *The scaling limit of the gapped μ_4-equivariant CAR collar exists in the local-net topology; the index-4 simple-current extension functor commutes with this limit; and the Osterwalder–Schrader reconstruction of the limit state is isomorphic to the holomorphic $(E_8)_1$ lattice conformal net.* Explicitly, with $\mathcal{A}_L$ the finite collar net ($D = 16$ Majoranas, gap $\Delta = 6\ln(3/2) > 0$, carrier clock μ_4) and α_{μ_4} the index-4 simple-current extension (the Longo Q-system of v125/v154), the target decomposes into four lemmata:

- **(L1) Uniform energy / nuclearity bounds.** The rescaled collar family satisfies uniform (in L) energy bounds and a nuclearity/compactness condition strong enough to extract a limit state on the quasi-local algebra. [O] — the finite side supplies the uniform gap (v302) and exact quasi-freeness (Wick/Pfaffian closure, v113/v155); N-uniform CAR nuclearity is now proved as compactness input ([verification/v1016_state_gap_batteries.py]: $\nu_N \le e^{16C_\beta}$) but does *not* identify P_{gen}/ALG-EXH; L1 stays [O].
- **(L2) Convergence of local algebras.** The local algebras of the rescaled collar converge (in the sense of the Osborne–Stottmeister scaling-limit framework) to the local algebras of a chiral CFT of central charge $c = 8$. The cited theorem covers free CAR lattice fermions in the range rank $\le c \le D$; the collar sits at $8 \le 8 \le 16$ (v356/v366). [O] as applied *with the μ_4-equivariant structure retained through the limit.* The coverage of the cited theorem is since carved exactly ([verification/v973_seam_route_narrowing.py]): *covered* is the $\mathfrak{so}(16)_1$ fermion-bilinear current algebra (strong convergence, Virasoro and correlator control, polynomial error bounds); *not covered* — the named residual — are N1 the μ_4/GSO extension $SO(16)_1 \to (E_8)_1$ as a net statement, N2 the twist/spin-field sector, N3 the one-sided chiral edge net, and N4 the OS/holomorphy selector; a measured collar convergence rate $p = 3.48$ lies inside the preregistered band (the trivial M=3 control does not converge). Completeness wave ([verification/v994_mmst_criteria_battery.py]): 5 of 7 MMST identification criteria are now executed at the finite level (measured $c = 8.005$, $(E_8)_1$ vacuum character $[1, 0, 248, 0, 4124, 0, 34752]$, $k = 1$, $\det D_8 = 4 \to \det E_8 = 1$); C6 is the written-out quasi-free theorem, C7 stays cited.
- **(L3) The crossed product commutes with the limit.** The index-4 simple-current extension (the μ_4 Q-system, equivalently the crossed product by the glue $\mathbb{Z}_4$, v469) commutes with the scaling limit: $\lim(\mathcal{A}_L \rtimes \mu_4) \cong (\lim \mathcal{A}_L) \rtimes \mu_4$. The finite Q-system is explicit and machine-verified (v125/v154); the commutation is [O].
- **(L4) Uniqueness of the holomorphic $c = 8$ reconstruction.** The OS reconstruction of the limit is a *holomorphic* (μ-index 1, one primary) $c = 8$ net, hence the $(E_8)_1$ lattice net — the same-central-charge rival $SO(16)_1$ ($\det K = 4$, four primaries) is excluded by the order-4 clock, *not* by $c = 8$ alone (v351/v378). The classification input is cited (KLM/Müger; lattice-net uniqueness); that the *reconstructed* net satisfies its hypotheses is [O].

Dimension typing (per `DIMENSION.UPLIFT.FIREWALL.01`): hypotheses *and* conclusion of L1–L4 live in 1+1D — the target is a chiral c=8 boundary object. No part of this contract, proved or not, closes a 3+1D claim; the seam→bulk uplift is the separate contract `SEAM.BULK4D.RECON.01`.

(2) Complete hypothesis list. Every input an external proof may use, typed. PROVEN-FINITE = machine-checked in the suite (module cited); CITED-CLASSICAL = a published theorem, used as stated, never re-proved; OPEN = the target itself.

Type	Hypothesis	Witness
PROVEN-FINITE	spectral gap $\Delta = 6\ln(3/2) > 0$ of the collar transfer	v302
PROVEN-FINITE	quasi-free CAR structure (one kernel, Wick/Pfaffian exact; boundary marginal of a Gaussian bulk is Gaussian)	v113/v155/v160
PROVEN-FINITE	MMST applicability arithmetic: $D = 2^{g_{\mathrm{car}}-1} = 16$, $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$, $8 \le 8 \le 16$	v356/v366
PROVEN-FINITE	explicit lattice realisation: p+ip collar, FHS Chern $\lvert C\rvert = 1$ at M=1 (control $C = 0$ at M=3); 16 Majoranas $\Rightarrow c_- = 8$	v367
PROVEN-FINITE	realised non-gappable chiral edge on the strip (bulk–edge, in-gap states, edge-localised)	v368
PROVEN-FINITE	strict locality topologically forbidden (Wannier winding $= \lvert C\rvert = 1 \neq 0$)	v461
PROVEN-FINITE	central charge $c = 8$ from Calabrese–Cardy entanglement scaling (fitted $c = 1.0000$ per mode, 16 Majoranas = 8 modes)	v376
PROVEN-FINITE	the $(E_8)_1$ character: $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$, one primary, 248 level-1 currents	v377
PROVEN-FINITE	torus (genus-1) discriminator: KLM μ-additivity $\mu = 16/4^2 = 1$ for the index-4 glue vs $16/2^2 = 4$ for index-2	v378
PROVEN-FINITE	reflection positivity of the collar measure by an exact positive-mixture Gram identity (symbolic PSD, no tolerance)	v379
PROVEN-FINITE	the index-4 Q-system exists explicitly (Longo axioms on $\mathbb{C}[\mathbb{Z}_4]$); pinned levels $(D_5)_1 \times (A_3)_1$, $c = 5 + 3$	v125/v154
PROVEN-FINITE	the order-4 clock is forced (four Gauss–Bonnet marks, $\lvert\mu_4\rvert = 4 = h(A_3)$); contract typing + import firewall (routes non-circular)	v216/v286
CITED-CLASSICAL	scaling limit of free lattice fermions → chiral CFT, rank $\le c \le D$ (Osborne–Stottmeister, CMP **398** (2023) 219, arXiv:2107.13834, on the MMST OAR framework arXiv:2010.11121)	v336
CITED-CLASSICAL	OS reconstruction / holomorphy selector (Adamo–Moriwaki–Tanimoto 2024)	v336
CITED-CLASSICAL	index-4 extension theory: Longo–Rehren Q-systems; Kawahigashi–Longo–Müger complete rationality; Xu orbifold heredity	v469
CITED-CLASSICAL	gapped free-fermion phases are invertible (Kitaev); nontrivial invertible phase ⇒ non-gappable edge (anomaly inflow)	v301
CITED-CLASSICAL	holomorphic $c = 8$ classification: #primaries $= \lvert\det K\rvert$; E_8 the unique even unimodular rank-8 lattice (Conway–Sloane)	v281
OPEN	**L1–L4 above** — the continuum existence, equivariant convergence, crossed-product commutation and holomorphic reconstruction, assembled	`SEAM.EQUIV.01`

(3) Available finite models (runnable). All models live in the repository and run standalone (Python 3, `mpmath`/`sympy`; `python verification/run_all.py` runs the whole suite):

- [`verification/v367_seam_s3_lattice.py`] — the collar lattice model (FHS Chern, controls)
- [`verification/v368_seam_s3_inflow.py`] — the strip edge / anomaly-inflow diagnostics
- [`verification/v376_seam_s3_centralcharge.py`] — entanglement central charge
- [`verification/v377_seam_s3_e8character.py`] — the exact $(E_8)_1$ character
- [`verification/v378_seam_s3_modular.py`] — torus degeneracy / modular data
- [`verification/v379_seam_s3_rp.py`] — reflection positivity
- [`verification/v125_glue_qsystem.py`] — the explicit index-4 Q-system
- [`verification/v154_simple_current_theorem.py`] — the simple-current extension theorem
- [`verification/v356_continuum_mmst_applicability.py`] — MMST applicability arithmetic

- [verification/v366_mmst_seam_collar.py] — the collar in the MMST class
- [verification/v286_seam_equivalence_contract.py] — contract typing and import firewall

The seam-relevant Lean formalisation lives under `experiments/lean4-carrier-rigidity/TfptCarrier/` (shipped as `lean4-carrier-rigidity/`; `lake build` clean).

(4) Lean axiom surface (honest). The Lean layer certifies the *logical composition* and the finite arithmetic — *not* the analysis. In `TfptCarrier/SeamScalingLimit.lean` (`FORM.SEAM.MMST.01`) the kernel proves, by `decide` with no axioms: the range $8 \le 8 \le 16$ (`mmst_range`), chirality $c_- = 8 \neq 0$ (`chiral`) and the holomorphy discriminator $\det K$: 1 vs 4 (`holomorphic_discriminator`). *Axiomatized* (named, cited, not re-proved): `mmst_scaling_limit`, `adamo_holomorphic_e8` and the structural input `collar_gapped`; the carriers `CollarGappedQuasifree`, `ChiralCFT_c8`, `SeamIsE8` are *opaque* `Prop`s. The same pattern holds in `SeamEquivChain.lean` (`kitaev_invertibility`, `mueger_klm_holomorphy`, `conway_sloane_e8` as axioms), `SeamApplicabilityLedger.lean` and `BWKeystone.lean`. No `sorry` in any seam module; `#print axioms` pins the residual to exactly the named cited theorems. **Caveat stated plainly:** no operator algebras, nets, or states are formalised — L1–L4 have *zero* Lean content beyond this composition audit, so the formalisation is thinner than a casual reading of "Lean-verified" would suggest.

(5) Must-fail controls. A submitted proof is accepted only if it *fails* on the following deformations (each already realised as a machine-checked negative control):

- **MF-S1 ($c = 8$ alone must not suffice).** Any argument that would equally conclude "$SO(16)_1$" is invalid: the proof must consume the order-4 clock / $\det K = 1$ discriminator, not just the central charge (the red-team point; v281/v378).
- **MF-S2 (wrong extension index breaks it).** Replacing the index-4 μ_4 glue by the index-2 $\mathbb{Z}_2$ glue must yield $SO(16)_1$ ($\mu = 4$), i.e. the E_8 conclusion must demonstrably fail (v154/v351).
- **MF-S3 (trivial collar).** At the trivial parameter $M = 3$ ($C = 0$, clean gap, no edge) the construction must produce *no* chiral limit (v367/v368).
- **MF-S4 (marks).** A non-order-4 clock, or a mark configuration away from the free μ_4 orbit, must break the $\tau = i$ pillowcase input (v216/v217).
- **MF-S5 (firewall).** No circular import: the proof must not assume the $(E_8)_1$ net on the seam at any intermediate step (the Route-i/Route-ii import firewall, v286).

(6) Acceptance tests. `SEAM.EQUIV.01` counts as solved exactly when: (i) L1–L4 are proved under the hypothesis list of (2), with any additional hypothesis flagged explicitly; (ii) the composed theorem outputs the $(E_8)_1$ net — one primary, 248 level-1 currents, torus degeneracy 1 — as a *consequence*, matching v377/v378; (iii) all five must-fail controls verifiably fail; (iv) no downstream TFPT readout (in particular no α data) enters as an input. Partial results that close a single lemma (e.g. L1 alone) narrow the target and are recorded as such; only the full chain moves the ledger — and any marker move happens in `status_ledger.csv` by the house process, never in this document.

Externalization Contract — `ALPHA.QUILLEN.EXACT.01` (the determinant-line variation theorem, packaged for external specialists)

Typing. The second hand-off document: the one remaining EM obligation, packaged for an outside specialist in determinant lines / η-invariants / index theory. It *documents*, it does not close: `ALPHA.QUILLEN.EXACT.01` stays [O] (external math, type A, v384), and it is a *face* of `SEAM.EQUIV.01` (v382) — not a second independent gate. The *value* α^{-1} is closed [E] regardless (unique root + ablation, v3); open is only "why *this* functional". Nothing here moves a marker.

Relation (2026-08-28). The nonabelian upgrade of this $U(1)$ face is the new contract `GAUGE.DETLINE.FIXPOINT.01` (all three gauge couplings as unique stationary points of one zeta-det/inflow functional; honest: $\alpha_s(M_Z)$ is currently an external input). This row stays the $U(1)$ face; that row is the three-coupling demand.

Relative-determinant note (2026-08-29, review wave 4). The correct object is the *relative* determinant $Q(A) = \log[\det' \Delta_A / \det' \Delta_{A_0}] + 8\, b_1\, c_3^6 \log \varphi(A)$; the BFK constant 2^{-4} is $A/\tau/\alpha$-independent at fixed cut number, so $\delta \log 2^{-4} = 0$ — no physical compensation needed (consistent with `COMPENSATION_INTERNAL` and the reviewer's correction). Absolute normalisation is fixed by the reference A_0.

(1) Minimal theorem statement. Two steps, both over the moduli of flat $U(1)$ connections on the μ_4 seam (the twist torus):

- **(T1) Occupied Berry line and continuum determinant-line comparison.** *Finite object.* Over the same twist torus B, specify the gapped collar Hamiltonians $H_L(b)$, their gauge identifications and $P_L(b) = \mathbf{1}_{(-\infty,0)}(H_L(b))$. The gap defines the smooth occupied bundle and its Berry line $L_{\mathrm{occ},L} = \det \mathrm{Ran}\, P_L$. Its FHS integer is $C_{\mathrm{FHS}} = +1$ within numerical tolerance at $L = 4, 6$ (v472). For $\nabla = d + A_{\mathrm{occ}}$ with $A_{\mathrm{occ}} = \mathrm{tr}(U^\dagger dU)$ and $c_1 = iF_{\mathrm{occ}}/(2\pi)$, the continuum Berry convention is $C_{\mathrm{Berry}} = \int_B F_{\mathrm{occ}}/(2\pi i)$, hence $c_1(L_{\mathrm{occ},L})[B] = -C_{\mathrm{Berry}}$. Identify the finite-mesh integer with this continuum value by a controlled mesh limit or an independent topological calculation; check duality before comparing its sign with a continuum theta-line convention. The gap check establishes no global nonvanishing section: such a section of this same line would force $c_1 = 0$, contradicting the nonzero Chern class. A global projector or ground-state ray is not a globally chosen ground-state vector.

  Continuum object. Define independently a chiral elliptic Fredholm family D_b^+, including domains, coupling data and gauge descent, with Quillen line $L_Q = (\det \ker D^+)^{-1} \otimes \det \mathrm{coker}\, D^+$, Quillen metric and Bismut–Freed connection. Fix the chirality/duality convention in the comparison. The null-mode locus $\Theta \subset B$ is allowed and must be specified; a globally invertible D^+ has a trivial Fredholm determinant line and cannot supply $|c_1[B]| = 1$.

  Comparison demand. Construct the comparison between $L_{\mathrm{occ},L}$ and L_Q (or its explicitly specified dual) over the common base and prove convergence at the level of Hermitian metrics and connections after a fixed relative renormalisation. Norm-resolvent convergence and a Fermi gap alone are insufficient: uniform control of the high-frequency/heat-kernel contributions and their parameter variations is required. Match the curvature forms and the holonomies on the two independent twist cycles; equal integrated Chern numbers do not determine the connection. The reference A_0 fixes the remaining constant normalisation. On $B \setminus \Theta$ the logarithmic determinant variation is ordinary; its extension across Θ must retain the kernel and divisor data.

  Dai–Freed data. A Dai–Freed vector requires an actual family of fillings and takes values in the pulled-back inverse boundary line. A global nonvanishing section may be asserted only on that pullback or on a specified anomaly-cancelled product, not on the uncompensated line with nonzero c_1. Alternatively, cylinders and mapping tori may define transport and holonomy without a global trivialisation. All continuum comparison statements remain **[O]**; the finite Berry/gap results of v472 are unchanged.

- **(T2) Variation = Bismut–Freed curvature = inflow response, with the exact corpus coefficients.** On the continuum line of T1,

$$\delta_\tau \Big(\log \det_\zeta \Delta_{U(1)} \;+\; 8\, b_1\, c_3^6\, \log \varphi_{\mathrm{seam}} \Big) = 0,$$

  i.e. the EM closure $F_{U(1)}(\alpha) = k_0\, \alpha^3 - 2\, c_3^3 \alpha^2 - 8\, b_1\, c_3^6 \ln(1/\varphi_{\mathrm{seam}}) = 0$ *is* the stationarity of the exact ζ-regularised Quillen functional, with every coefficient as it stands in the corpus: the Chern level $k_0 = |C| = 1$ (the computed bulk invariant of the S3 collar, normalised by the embedding index $k_Y = \mathrm{tr}(Y^2)/\,\mathrm{tr}(T_3^2) = 5/3$, v470); the Gauss–Bonnet boundary coefficient

$c_3 = 1/(8\pi)$ with heat-kernel powers c_3^3, c_3^6 (v216/v342); the $U(1)_Y$ a_4/index coefficient $b_1 = 41/10$ ($8b_1 = (4/5)\cdot 41$, v434); and the discriminant exponent $-\frac{5}{4} = -q(D_5)$ (v341). The derived stationarity must then have $\alpha^{-1} = 137.0359992168$ as its unique physical root (v3).

The open content splits into two named faces (one target, per v434/v435): the *analytic* face — T1 plus the identification of $\delta \log \det_\zeta$ with the seam Seeley–DeWitt data, i.e. the diagonal ζ-renormalisation at the marks and the multiplicity matching $41 = 10\, b_1$ (v484/v485) — and the *topological* face, the `EM1` lemma: the from-first-principles proof that $\partial_\tau \log Z_{\text{Maxwell}} = k_0\, \alpha^3$ with k_0 the bulk Chern level (v48/v435). *Both named faces now carry computed witnesses (2026-08-27,* ***`ALPHA.QUILLEN.FACES.01`*** *[E] new;* *`[verification/v974_alpha_faces_computed.py]`*; ***`ALPHA.QUILLEN.EXACT.01`*** *stays [O]):* the diagonal-ζ (analytic) face is executed as a BFK gluing — the gluing constant is exactly 2^{-N}, the jump matrix $16\, c_3\, \mathrm{circ}(2,-1,0,-1)$ *is* the inverse mark Green matrix, and the v485 scale channel is reproduced via the KMS unit, while against the v484 mark matrix the comparison is channel-wise exact but with the kernels in *swapped* μ_4 channels, so the analytic face is *sharpened* to the channel-swap question; the topological (`EM1`) face is computed — pumped charge = spectral flow = Středa = Chern = 1 on the v367 collar; and the finite Quillen identity is verified *two-sided* (Berry integral = Poincaré–Lelong divisor integral = +1, the difference an exact form). The remaining open content is typed: the rigidity premise, the channel-swap question, and the continuum Bismut–Freed identification; $\alpha^{-1} = 137.0359992168$ is re-anchored with split residual 6.4×10^{-58}. *Channel-swap update (2026-08-28, external master-object review wave 2;* *`[verification/v985_quillen_channel_swap.py]`*; *the contract stays [O]):* the swap question is *resolved at the finite level* by μ_4-character grading. Exactly: the jump pattern $\mathrm{circ}(2,-1,0,-1)$ has μ_4 eigenvalues $[0,2,4,2]$ (kernel in the k=0 channel), the mark pattern $\mathrm{circ}(0,1,2,1)$ has $[4,-2,0,-2]$ (kernel in k=2); the deck swap $r \to r{+}2$ ($P_0 \leftrightarrow P_2$, $P_1 \leftrightarrow P_3$) maps the jump |spectrum| onto the mark pattern *exactly* (the odd swap $r \to r{+}1$ fails — must-fail fires), so after the swap both kernels sit in the same channel and the graded determinant $\log \det'_{\text{gr}} = \sum_r (-1)^r \log \det'(\Delta|_{P_r})$ compares them well-definedly: the graded difference is c_3-*free*, $\log(\ln 2/4)$ exactly, and the channelwise unit ratio is uniform $= 4/\ln 2$ in every matched nonzero channel (the v974 doublet ratio extended to full uniformity). *Completeness-wave notes (2026-08-28):* the finite KMS rigidity premise is now executed ([`verification/v995_kms_subnet_rigidity.py`]: all 33 mixing directions killed; covariance redundant; 196608 → 0); type-III/continuum stays [O]. The finite-collar detline curvature converges exponentially to 2π with rate $\sim 0.6936 \sim \ln 2$ ([`verification/v996_gauge_grammar_kill.py`]) — a numerical Bismut–Freed shadow, not the continuum theorem. Still [O]: the continuum Bismut–Freed identification, type-III KMS uniqueness, and the 2^{-4}/Quillen-metric compensation.

Executed update (2026-08-30, late-evening harvest; [`verification/v1014_bridge_refinements.py`]**; the contract stays [O]).** Finite restriction isomorphism verified; orientation fixes the conjugate branch ($2 \to 1$); the residual constant phase is fixed by the A_0 reference normalization — recorded. Continuum Bismut–Freed stays [O].

(2) Complete hypothesis list.

Type	Hypothesis	Witness
PROVEN-FINITE	the Quillen split at the root: $\alpha^3 - 2c_3^3\alpha^2 = 8b_1c_3^6\ln(1/\varphi_{\text{seam}})$ at $\alpha^{-1} = 137.0359992$, residual $< 10^{-35}$	v341/v382
PROVEN-FINITE	unique root + ablation battery; CODATA deviation $\sim 2.9 \times 10^{-10}$ *as output, never input*	v3/v48
PROVEN-FINITE	b_1 is the $U(1)_Y$ a_4 coefficient: $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$, $(3/5)\cdot\frac{41}{6} = \frac{41}{10}$	v434 (v159/v246)
PROVEN-FINITE	the embedding index $k_Y = (5/6)/(1/2) = 5/3$; level-1 rigidity fixes the $U(1)_Y$ normalisation	v470
PROVEN-FINITE	the π-power ladder $\{0, 3, 6\}$: the cubic term is the unique metric-independent (topological) term	v435
PROVEN-FINITE	the quadratic (Calderón) term is the Gilkey a_4 gauge-curvature coefficient $\Omega^2/12$	v342
PROVEN-FINITE	bulk Chern invariant $C = 1/0/-1$ at $M = 1/3/-1$ (FHS, Bloch BZ)	v367/v470
PROVEN-FINITE	the finite Quillen object: twist-torus determinant-line curvature $= 2\pi \cdot 1$ exactly (10^{-9}, $L = 4, 6$); twist integer = Bloch integer (Niu–Thouless–Wu); Fermi gap open over the whole torus	v472
PROVEN-FINITE	the contact unit: "c_3 per insertion" = the KMS seam unit, derived for the finite cycle sector; closed-form mark determinant, $41 = Y^2$-weighted count of the 48 states	v484/v485
PROVEN-FINITE	solvable-model anchors: circle ζ-det variation in closed heat-kernel form; T^4 reaches the a_4 order; measure $(4\pi)^{-2} = 4c_3^2$	v391/v433
CITED-CLASSICAL	Quillen metric/curvature (Quillen 1985); Bismut–Freed curvature (CMP **106** (1986)); Dai–Freed section (JMP **35** (1994))	v472
CITED-CLASSICAL	η/CS reading of $\delta \log\det$ (APS 1975; Alvarez-Gaumé–Della Pietra–Moore)	v470
CITED-CLASSICAL	integer quantisation of the $U(1)$ response (TKNN; Avron–Seiler–Simon); boundary level = bulk response (Callan–Harvey)	v470
CITED-CLASSICAL	heat-kernel coefficients (Gilkey, *Invariance Theory*, Thm 4.8.16); the $\beta = a_4$ theorem (Vassilevich)	v342/v434
CITED-CLASSICAL	the embedding index $k_Y = 5/3$ (Ginsparg 1987)	v470
OPEN	**T1+T2 above**: the gap-stable continuum c_1 identification and the from-first-principles proof that $F_{U(1)}$ is the exact ζ-regularised Quillen functional (incl. the `EM1` lemma)	`ALPHA.QUILLEN.EXACT.01`

(3) Available finite models (runnable).

- [`verification/v472_quillen_detline_moduli.py`] — the finite Quillen object itself (real-space collar on an $L \times L$ torus with twisted boundary conditions; FHS curvature of the occupied-frame determinant line)
- [`verification/v470_alpha_inflow_level.py`] — the Bloch-level computation
- [`verification/v367_seam_s3_lattice.py`] — the underlying collar model
- [`verification/v3_em_alpha.py`] — the fixed point: unique root + ablation battery
- [`verification/v48_em_ward.py`] — the Ward decomposition
- [`verification/v341_alpha_quillen.py`] — the Quillen split and coefficient identities
- [`verification/v382_alpha_quillen_exact.py`] — the named target registration
- [`verification/v391_alpha_quillen_progress.py`] — circle ζ-det solvable model
- [`verification/v433_alpha_quillen_heatkernel.py`] — the T^4 a_4-order anchor
- [`verification/v434_alpha_quillen_betafunction.py`] — b_1 as the a_4 coefficient
- [`verification/v435_alpha_quillen_chernlevel.py`] — the π-power / level partition

- [verification/v484_seam_contact_unit.py] — the contact unit (c_3 per insertion)
- [verification/v485_contact_diagonal_closed.py] — the closed computable side

(4) Lean axiom surface (honest). There is *no* Lean formalisation of this target: no `TfptCarrier` module formalises the determinant line, the ζ-determinant, or the fixed-point functional. The only adjacent Lean content is the `SEAM.EQUIV.01` face (Sec. above — `SeamScalingLimit.lean` and companions, composition-level only) and kernel-checked edge/Chern arithmetic (`SeamEdgeChern.lean`). An external solver should treat the Lean layer as contributing *nothing* to this contract beyond the seam-face audit.

(5) Must-fail controls. A submitted derivation is accepted only if it respects the following (each realised in the suite):

- **MF-A1 (no experimental α).** The functional and its stationarity must be constructed from $(c_3, b_1, k_0, \varphi_{\text{seam}})$ alone; any appearance of the measured α (CODATA or otherwise) at *any* stage of the construction invalidates the derivation — the measured value is compared *after*, never inserted (v3).
- **MF-A2 (ablations destroy the hit).** Ablating the deck/carrier structure — neighbouring integer budgets for 41, dropping the μ_4 factor, or a Chern level $k_0 \neq 1$ — must move the root away from 137.0359992 beyond the corpus tolerance (the v3 ablation battery; v48).
- **MF-A3 (trivial/reversed collar).** The trivial collar $M = 3$ must give $C = 0$ (no level, no fixed point of this form); the orientation flip $M = -1$ must give $C = -1$ (v470/v472).
- **MF-A4 (the metric-independence partition).** Any proposed identification must reproduce the π-power ladder $\{0, 3, 6\}$: a derivation that attaches a $(4\pi)^{-d/2}$ heat-kernel measure to the cubic term (or strips it from the quadratic or transport terms) fails (v435).

(6) Acceptance tests (the external review's three, verbatim, plus one structural):

- **A1 — no experimental α input:** the derivation is checked to contain no use of the measured value anywhere; α^{-1} emerges as output only.
- **A2 — unique physical fixed point:** the *derived* variational equation has exactly one physical root, and it equals $137.0359992168\ldots$ to the corpus precision.
- **A3 — ablations destroy the hit:** MF-A2/MF-A3 verified to fail, machine-checkably.
- **A4 — structural:** T1 is accepted only with the uniform-gap hypothesis explicit (the Dai–Freed section nonvanishing over the whole moduli torus); T2 modulo T1 (or vice versa) narrows the target and is recorded as progress, but only the pair closes it. Any marker move happens in `status_ledger.csv` by the house process, never in this document.

Certifiability and recommended order

Building block	Machine	Hardness
W_{wall} enumeration ($C_U^{(1)}$)	Lean / Python / Wolfram	easy (*done*)
D_4 fixed-locus polynomial ($C_U^{(2)}$)	Sage / Mathematica	medium
selector uniqueness ($C_U^{(3)}$)	interval arithmetic	medium
SNF, determinants, spectra	Lean	easy–medium
principal-symbol ellipticity (G1)	xAct / Cadabra / Lean	medium
gap inequality (G5)	Lean	easy (*done*)
operator-norm bound (G5)	analysis + numerics	hard
projective limit (G6)	not primarily machine	very hard
Ward identities (G7)	BV/BRST formal + checks	hard

Recommended order.

Selector-triangle pairings → v_{geo} scale anchor → G_{net} inclusion theorem

The flavor interface (historically U_{wall}) is now reduced to the *selector-triangle pairings* (the dual normal pair (d, n) forcing R columnwise, v136/v139); it is finite, falsifiable and the cheapest visible upgrade. The v_{geo} scale anchor is the one dimensionful input (same nature as $1/G$). G_{net} — the index-4 seam-net inclusion certifying the metric sector — is more fundamental but a *programme*, not a single proof: heavy machinery, with the projective limit the genuine summit. F_{transfer} (source→pole / relic / cosmology) is the remaining downstream interface.

F_{transfer} **is a typed functor, not a bag of open topics.** It is standard physics fed TFPT *source* data, $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$, with exactly four interfaces:

Interface	Map	Status	Source data → observable
F_{pole}	source→pole masses	[E] / [C]	Koide $53/54 = a^T(R+Q)\mathbf{1}/(2\,\mathbf{1}^T R a)$ is an exact operator factor [E] (the missing Sheet-Diamond corner); the pole-transfer interpretation is [C] ([verification/v183_koide_f_corner_transfer.py])
$F_{\text{Boltzmann}}$	CP source→ η_B	[C]	$\tilde{m}_1{=}m_3/A_\Lambda$ anchors the washout, but $M_1{=}M_R\varphi_0^4$ only relocates the free input (M_R is the seesaw scale, not a compiler power), so η_B stays a *sharper scenario*, not a closure ([verification/v184_etaB_anchored_boltzmann.py])
F_{relic}	$(f_a, m_a, \theta_i, N_{\text{DW}}) \to \Omega_a$	[C]	the determinant-line axion can be all of DM, but only on the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — a viable scenario ([verification/v185_axion_relic_solver.py])
F_{QCD}	$(\alpha_s, b_3, \Lambda_{\text{QCD}}, \text{lattice}) \to m_p/m_e$	[O]	a QCD matching contract ($b_3{=}-7$ a carrier output), *not* a compiler number — the rejected 1836 near-formula is the discipline working

A machine guard enforces the typing: the four families must always carry a [C]/[O] marker and must never be promoted to a primitive [E] compiler prediction ([verification/v187_ftransfer_laws.py]). Exact *algebraic* sub-parts (the Koide factor, $b_3{=}-7$) may be [E]; the *physical* prediction never is.

The functor contract `CONTRACT.F.01`, four axioms ([verification/v213_ftransfer_functor.py]). Beyond the typing guard, the functor itself satisfies four checkable structural axioms, each instance a *discrete* compiler kernel [E] composed with an *external* continuous solver [C]: **(1) μ_4-deck equivariance** — the discrete kernels are carrier/deck invariants $\{|\mu_4|, N_{\text{fam}}, g_{\text{car}}, A_\Lambda\}$, and the Koide multiplier $\lambda_2{=}(2/3)^6{=}64/729$ *is* the μ_4-deck transfer eigenvalue (v82/v54); **(2) Plücker preservation** — the dimensionless readouts are Plücker readings ($53{=}a^T(R+Q)\mathbf{1}$, $\mathrm{Pl}(F){=}(1, 22, 30)$; $\|\mathrm{Pl}(K)\|_1{=}11$); **(3) positivity/stochasticity** — spec $T{=}\{1, (2/3)^6, (1/3)^6\}$ is positive (RP/OS), $\kappa_f{\in}(0, 1)$; **(4) external modules explicit** — the continuous content (pole transfer, the flavored Boltzmann network, the finite-T $\chi(T)$ ODE, the lattice C_N) is named and external, never a compiler number. So `CONTRACT.F.01` is the third research contract alongside (U_{wall}) and (G_{metric}): a typed functor, *not* a bag of open topics — but a consolidation, *not* a closure. *[2026-08-05 dated note — the jet route closes, the fibered refinement stands ([verification/v777_ftransfer_clock_jets.py]): the executed external-clock contract `FTRANSFER.CLOCKS.01` (below) fires its preregistered K1 kill sympy-exactly — the transported Schwarzians are $\{-\Delta^2/2, -\Delta^2/2, 0\}$, $\Delta = 6\log(3/2)$, so* no *common continuous clock exists for the four jets. The surviving architecture is measured: F_transfer is a* fibered *functor — one shared* discrete PGL_2 *base (the anchor cross-ratio of all four channels is* $4/3$ *exactly, symbolically* and *through the frozen* g_{*s} *table at dev* 6.5×10^{-4}*; the deck generator is the unit parabolic* σ*-translation,* $\mathrm{tr}^2 - 4\det = 0$ *— the v632 lattice-explicit discrete half) with channel-specific projective connections whose obstruction is the* constant *seam cocycle, values exactly* $\{0, \pm\Delta^2/2\}$ *(antisymmetry and the cocycle identity exact;* $d/d\sigma$ *identically zero); the classification matches the predeclared marker — thermal/proper-time clocks* $\{\tau, \beta\}$ *on the* $-\Delta^2/2$ *coset,*

the RG clock $\{\log\mu\}$ *on the* 0 *coset, the relic channel degenerate and honestly unclassified. The functor axioms above stand; the jet/continuous-clock unification demand is* replaced *by the fibered form, with the Schwarzian differences stored as the obstruction cocycle, not fitted away. The one named proof-grade next step — derive the coset assignment from KMS/modular clock typing — is executed one round later and is a* theorem *(*`[verification/v792_ftransfer_kms_coset.py]`*,* 18 *checks,* KMS-COSET-THEOREM*, registered as* `FTRANSFER.KMSCOSET.01`*):* $S=-(\ln\lambda)^2/2$ *with* λ *the* PGL_2 *multiplier of the flow's time-1 holonomy per clock e-fold; the classifier uses no Schwarzian and no derivatives (non-circularity gated on its own source), the blind assignment classifies the four deployed charts exactly as the fibered functor carries them, the unblinding is sympy-exact, and the controls fire (fake KMS caught; affine invariance; the* `v578` *Möbius clock as the positive case) — the fiber classification is derived, not case-inspected. The four observables stay [C]/[O]; no marker move.]*

A preregistered cross-ratio check for the four solvers — designed, and since executed on its discrete half. The design is machine-checked in `[verification/v566_parabolic_anchor_selfcode.py]`, the discrete-half execution in `[verification/v571_cr_discrete_corners.py]`. The disc -7 norm line $\alpha_{t+1}=\alpha_t+2$ (v533) equips the transfer path with a projective invariant: $\mathrm{CR}(\alpha_{-2},\alpha_{-1};\alpha_0,\alpha_1)=\frac{4}{3}$ *exactly* — and, honesty first, $\frac{4}{3}$ is the *universal* cross-ratio of any arithmetic progression, so the test content is not the number but the *discipline*: if the four solvers (J,K,C,F) export canonical one-dimensional projective states y_J,y_K,y_C,y_F — declared *before* any evaluation — then Möbius transport along the path forces $\mathrm{CR}(y_J,y_K;y_C,y_F)=\frac{4}{3}$, and the fourth state is determined fit-free by the first three (y_F reconstruction identity). Negative control with teeth: the raw determinant path $(2,4,14,32)$ gives $\mathrm{CR}=\frac{28}{25}\neq\frac{4}{3}$ — the already-known norm sequence cannot pass. *The discrete half is now executed*: the four corners are the in-repo kernels $J,K,C,F=M(1,t)=R+Q\,\mathrm{diag}(1,t,t)$, $t=-2..1$ (v224), and the corpus's one canonical declared solver reading — the Koide anchor response $\ell(M)=a^{\mathsf T}M\mathbf{1}$ (v183) — reads $(26,35,44,53)$ on them: $\mathrm{CR}=\frac{4}{3}$ exactly, and the fit-free reconstruction *forces* $y_F=53$ from $(26,35,44)$ alone — the Koide numerator is Möbius-determined by the three upstream corners (a new exact cross-corner identity); honesty theorem stated out loud: ℓ is affine on the path (step $9=N_{\mathrm{fam}}^2$), and *every* linear functional of the kernel passes — the check separates linear from nonlinear exports (the dominant-eigenvalue export fails at 1.3222, the determinants at $\frac{28}{25}$, both as they must) [E]. Status: the *continuous* half (the external solvers exporting their own states) stays open exactly as this contract fences it; contract markers unchanged [C]. *The frozen export protocol for the continuous half* (design [X], preregistered here): each instance must export, *before* evaluation, the canonical one-dimensional datum of its own native flow — the fixed-point multiplier of its contraction (cf. the native rates of v425: F_pole $(2/3)^6$ exact in-corpus, F_Boltzmann/F_relic/F_QCD external) — as a projective state on the path; only then is $\mathrm{CR}=\frac{4}{3}$ tested, with no renormalisation after the fact. *Executed* (`[verification/v575_cr_continuous_uniflow.py]`): the in-repo native exports form the quadruple $\{(2/3)^6,(2/3)^6,1,-7\}$ — the repetition is *forced* by the single-flow theorem (v425: F_pole and F_Boltzmann are literally the same seam contraction) — and a repeated quadruple gives $\mathrm{CR}\in\{0,1,\infty\}$ in every corner assignment (theorem, plus all 24 permutations): the continuous half can never pass *natively*; it closes onto the single-flow reduction, verdict UNIFLOW-DEGENERATE. The four independent external data are the dimensionful anchors (v_{geo}, M_R, C_p, θ_i) — [O] walls, not projective states — so a future continuous test needs genuinely *external* anchor-level exports; the meaningful cross-ratio content was and remains the discrete half. Both halves of the check are now executed [E]/[C]. *The jet-level successor is executed natively too* (`[verification/v578_native_jets.py]`): when point values degenerate, the Möbius-invariant *Schwarzian* of the exported trajectories is the right jet test — and natively, the exact v99 Koide trajectory has $\{q,t\}=-\Delta^2/2$ in wall time (constant, trajectory-independent; F_Boltzmann identical by the single-flow theorem) while being Möbius in its *own* exponential clock $u=e^{\Delta t}$ (Schwarzian 0), and the 1-loop QCD flow is Möbius in its own RG time already (Schwarzian 0): the native quadruple $\{-\Delta^2/2,-\Delta^2/2,0,\text{degenerate}\}$ shares *no* wall-time jet invariant — each instance is *projective in its own clock*, the obstruction is the clock, and the clocks are anchored

by the external data. Verdict MOEBIUS-IN-OWN-CLOCK; a future joint jet test needs external anchor-level jets; markers unchanged [E]/[C].

The discrete→dynamic lens: `F_transfer` is the readout end of the one principle (`[verification/v303_ftransfer_dynamics.py]`, `FR.DYNAMICS.01`). The typing above is sharpened by a *dynamical* reading. The main-branch mechanism is a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update $m \leftarrow (A+2I)/4\, m \to$ the E_8 marks, gap $6\ln(3/2)$). All four `F_transfer` instances share that *same shape*: [E] **F_pole** (Koide) *is* the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier $F'=(2/3)^6=\lambda_2$, the seam-clock subleading eigenvalue, so $|F'|<1 \Rightarrow$ a unique attractor at the *seam* rate (v82); [C] **F_Boltzmann** (η_B) is a gapped positive contraction (washout $\kappa_f \in (0,1)$ toward the unique balance, an H-theorem); [C] **F_relic** (axion) freezes at an adiabatic fixed point (the comoving number relaxes to a constant); [E] **F_QCD** (m_p/m_e) is the 1-loop RG flow with the carrier $b_3=-7$, whose Gaussian UV fixed point is the attractor (asymptotic freedom, $\Lambda_{\rm QCD}$ generated). *Honest boundary:* only F_pole has the seam rate $(2/3)^6$ exactly; the other three share the *shape* with external rates (thermal, cosmological, RG). So `F_transfer` is not a bolt-on but the downstream manifestation of the one discrete→dynamic principle — the compiler fixes the discrete seed + integer kernel, a gapped flow carries it to the observable — and the theory's simplicity is preserved precisely because the external rates are honestly fenced (v187), not because they reduce to the seam. (Even F_pole's flow time is non-integer, v93: the relaxation is the *continuous* semigroup $e^{t\log F}$ generated by the discrete operator — exactly the diffusion-semigroup reading of the main branch.)

The single-flow reduction (`DYN.TRANSFER.UNIVERSAL.01`). The dynamic lens is sharpened to a precise reduction (`[verification/v425_dyn_transfer_universal.py]`): the shared "shape" is literally the *one* native flow of the theory — the seam modular/recovery semigroup (v238/v240, gap $6\ln(3/2)$) — restricted to each sector, extending the static universal-gap principle (v383) to a dynamic one. [E] the v99 Koide generator $\dot q=(\Delta/N_{\rm fam})(q-2)(q-5)$ linearised at the attractor $q=2$ has rate $-\Delta$, so the time-1 contraction is $(2/3)^6$ (the recovery eigenvalue, $q=5$ unstable); [E] the Boltzmann washout is that same contraction, with the CP phase $4\pi/3$ native (v320); [E] the QCD rate is $b_3=-7$. So the transfer *dynamics* is native; only the *anchors* stay external — the IR pole scale and M_R are $v_{\rm geo}$-class, C_p is the lattice $O(1)$, and $F_{\rm relic}$'s cosmological initial condition θ_i is the one genuine wall. A reduction, *not* internalisation: no new premise, and the firewall (v187) holds — the observable stays [C]. *[2026-08-27 precision (binding; `DYN.MARKOV.EMBED.01`, `[verification/v971_markov_embedding_generator.py]`): the "one native flow" is a universal* contraction class *— shared gap and multiplier algebra — not one physical clock; the clocks are external and sector-specific (v723/v724/v777), and the continuous semigroup $e^{t\log T}$ is irreversible statistical dynamics, not unitary evolution — the (b)-vs-(c) firewall of the `DYN.MARKOV.EMBED.01`/`DYN.UNITARY.DILATION.01` contract pair above.]*

Both thermal-time routes are dead: the external clock contract stands (`FTRANSFER.STT.01` / `FTRANSFER.COCYCLE.01`). The radical counter-idea to the external-clock wall — Connes-style thermal time, "the four clocks = four modular flows of one state" — is now machine-killed on *both* checkable routes. [X] *Strong thermal time* (`[verification/v723_phys_modular_clock.py]`, 11/11, STT-KILLED + INTERNAL-PAIR-CONFIRMED): a modular generator is self-adjoint, hence diagonalizable — but the F_QCD 1-loop clock map $u \mapsto u+2b_0$ is *parabolic* and not diagonalizable (single Jordan block), and the symbolic lemma "exp of a real-diagonalizable matrix is diagonalizable" closes the door: no constant-rescaled modular flow produces the QCD clock; the pole multiplier is confirmed *exactly* $(2/3)^6 = 64/729$ (hyperbolic, dimensionless ⇒ internal; F_Boltzmann the same element by v425). [X] *Conditioned modular flows* (`[verification/v724_phys_t3b_modular_flows.py]`, 12/12, T3B-DEAD): with target-free graph projectors, *one* global KMS scale $\Delta = 6\log(3/2)$ and the Connes cocycle composition exact to 1.6×10^{-15} along J–K–C–F, the four conditioned flows come out *loxodromic/complex* — the frozen v578/v632 solver classes (hyperbolic / parabolic / degenerate) do *not* arise from conditioned modular flows of one internal mother state (the NEEDS-LINDBLAD escape is honestly refused). *Typed consequence:* the external clock contract is *confirmed* and sharpens to the anchor census {one unit = $\Lambda_{\rm QCD}$ ($v_{\rm geo}$-class by No-Unit v153), one

angle θ_i, one lattice $O(1)$ $C_p\}$ — T3's clock wall and T1's unit wall are *the same wall*, counted once. No marker moves; the functor typing above is unchanged.

The frozen external-clock contract (`FTRANSFER.CLOCKS.01`) — preregistered 2026-08-04, executed 2026-08-05. Since both thermal-time routes are dead, the clocks *must* be external — and without clocks fixed in advance every jet statement is arbitrarily reparametrizable. The contract therefore freezes them (design byte-frozen 2026-08-04 in `experiments/ftransfer-clocks/hypotheses/ftransfer_clocks_v1.yaml`; typed [X], a search-target design, *not* a claim). *The four frozen clocks, exactly one physical clock per solver:* F_pole = proper time τ (the on-shell/pole-mass phase clock; frozen gauge: one v99 seam unit = one proper-phase e-fold); F_Boltzmann = inverse temperature β (FIRAS-anchored adiabatic chart; one washout seam unit = one β e-fold; the z-linear alternative is recorded as *excluded*); F_QCD = RG time $t = \ln(\mu/M_Z)$ ($\overline{\text{MS}}$, 1-loop $b_0 = 7$ on the frozen window $[m_t, 1\,\text{TeV}]$; identity gauge); F_relic = cosmological e-folds $\ln a$ (Planck 2018 background; the flow is frozen, the jets *degenerate* — the channel is scored only on kills and dictionary validity). *The preregistered prediction* (FTC.P1): under the frozen affine dictionaries (slopes $-1, -1, +1, -1$ into the common chart $\sigma = \ln(\mu/M_Z)$) *one* constant-coefficient Riccati equation carries all non-degenerate transported jets — equivalently, the transported Schwarzians of F_pole/F_Boltzmann/F_QCD are *equal*; the seam channels transport to $-\Delta^2/2 = -2.959235170076978$ exactly, so the prediction forces the transported QCD Schwarzian onto that value. *Three kills:* K1 NO-COMMON-CONNECTION (symbolic inequality of the transported Schwarzians; on kill the executor must exhibit the forbidden reparametrization per failing pair); K2 FREE-REPARAM-NEEDED (any post-freeze change of a gauge or dictionary; the YAML sha256 is recorded by the executor); K3 SECOND-CLOCK-NEEDED (any channel export needing two time variables). *The honest prior, preregistered:* the *expected* outcome is the K1 kill — affine transport scales a Schwarzian by slope^2 and cannot move 0 (the QCD value) onto $-\Delta^2/2$ (the seam value); a K1 kill would finalize the v578 diagnosis at the *physical*-clock level and close the jet route of `CONTRACT.F.01`; ONE-CLASS would be the surprise. *Executed (2026-08-05, [verification/v777_ftransfer_clock_jets.py], one module from two probes, 24 + 14 checks, 0 failures):* the K1 kill fired *exactly* as the frozen prior expected — verdict NO-COMMON-CONNECTION. The freeze guards were byte-verified at run time (YAML sha256 `880224f7...`; the two data tables — Saikawa–Shirai 2018 g_{*s}, PDG 2024 $\alpha_s(Q)$ — were curated and SHA-frozen *before* the first run); all four dictionary-validity legs pass (Koide scheme check 571×; washout entropy-chart drift 1.15×10^{-4}; QCD overlay max dev 0.83%; relic onset slope dev 0.0602); the transported Schwarzians are sympy-exactly $\{-\Delta^2/2, -\Delta^2/2, 0\}$ — no tolerance involved — and the K1 duty is discharged: the pass-enabling reparametrization $h = \text{Möbius}(e^{\pm\Delta\sigma})$ is exhibited and is non-affine, exactly the freedom the freeze forbids; K2/K3 stay quiet. The same module carries the surviving architecture (the fibered refinement — see the `CONTRACT.F.01` dated note above). *[2026-08-06 completion (`FTRANSFER.KMSCOSET.01`, [verification/v792_ftransfer_kms_coset.py], 18 checks, KMS-COSET-THEOREM): the coset assignment of these clock charts is now a theorem from KMS/modular typing — $S = -(\ln\lambda)^2/2$ from the time-1 PGL_2 holonomy multiplier per clock e-fold, with a Schwarzian-free, derivative-free classifier (the detailed-balance multiplier cross-ratio as the discrete KMS signature), blind assignment and exact unblinding; the three-probe chain executor → fibered groupoid → coset theorem is whole.]* Nothing here upgrades `CONTRACT.F.01`: the four observables stay [C]/[O]; no marker move; no RH statement (unrelated surface; discipline note). So the visible residual is one geometric postulate (`QGEO.SYM.01`), one unit gauge (v_{geo}), and this one typed transfer functor — everything else is out-of-sample data.

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

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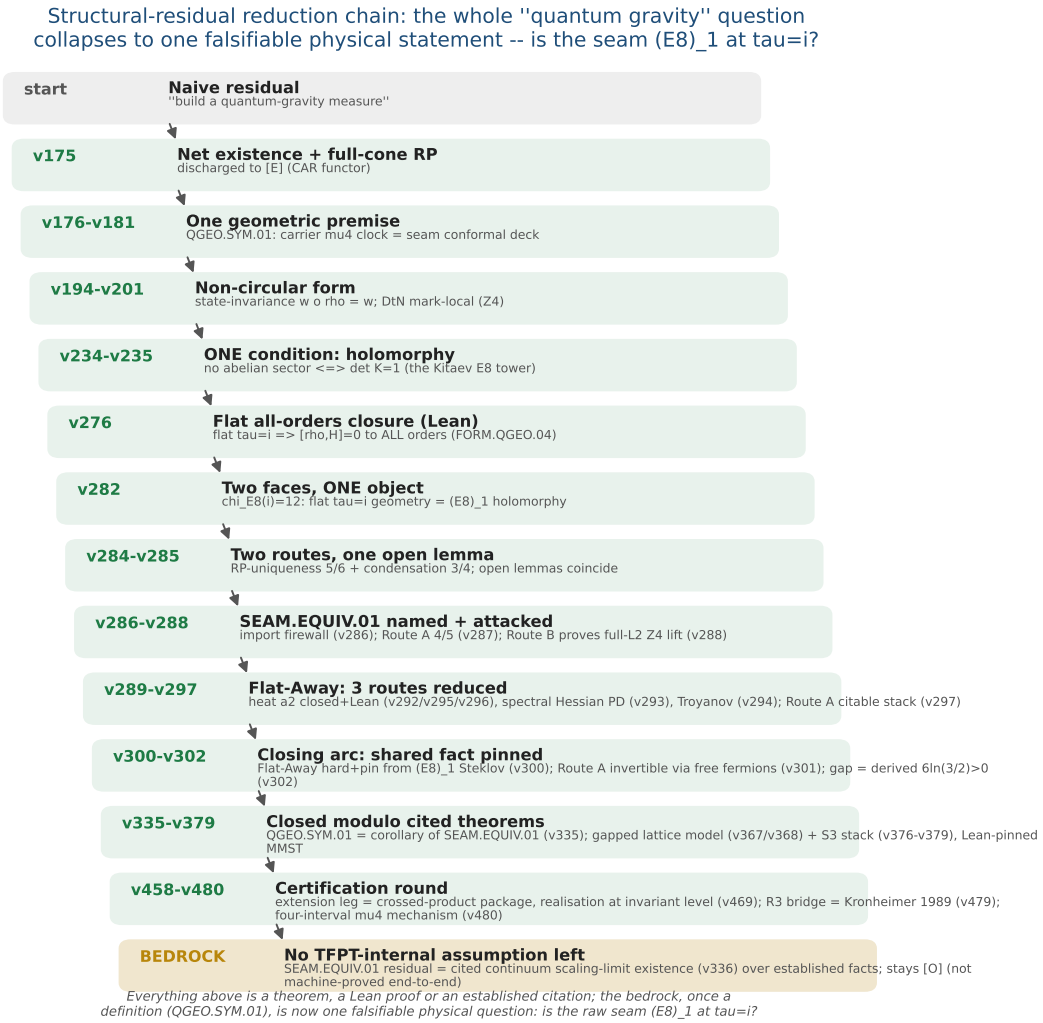


Figure 3: The structural-residual reduction chain (v175 → the certification round). The naive "build a quantum-gravity measure" shrinks, one machine-checked step at a time, from a vague measure to one falsifiable physical statement — *is the raw seam* $(E_8)_1$ *at* $\tau = i$*?* (`SEAM.EQUIV.01`; its conformal-deck face `QGEO.SYM.01` is a corollary, v335). The closing arc (v276–v302) pins the shared fact and derives the gap $6\ln(3/2) > 0$; the certification round discharges the extension leg to the crossed-product package (v469), the R3 graph→geometry bridge to Kronheimer 1989 (v479) and exhibits the four-interval μ_4 mechanism (v480). Every step above the bedrock is a theorem, a Lean proof or an established citation; the residual — the cited continuum scaling-limit existence (v336) — stays honestly [O].

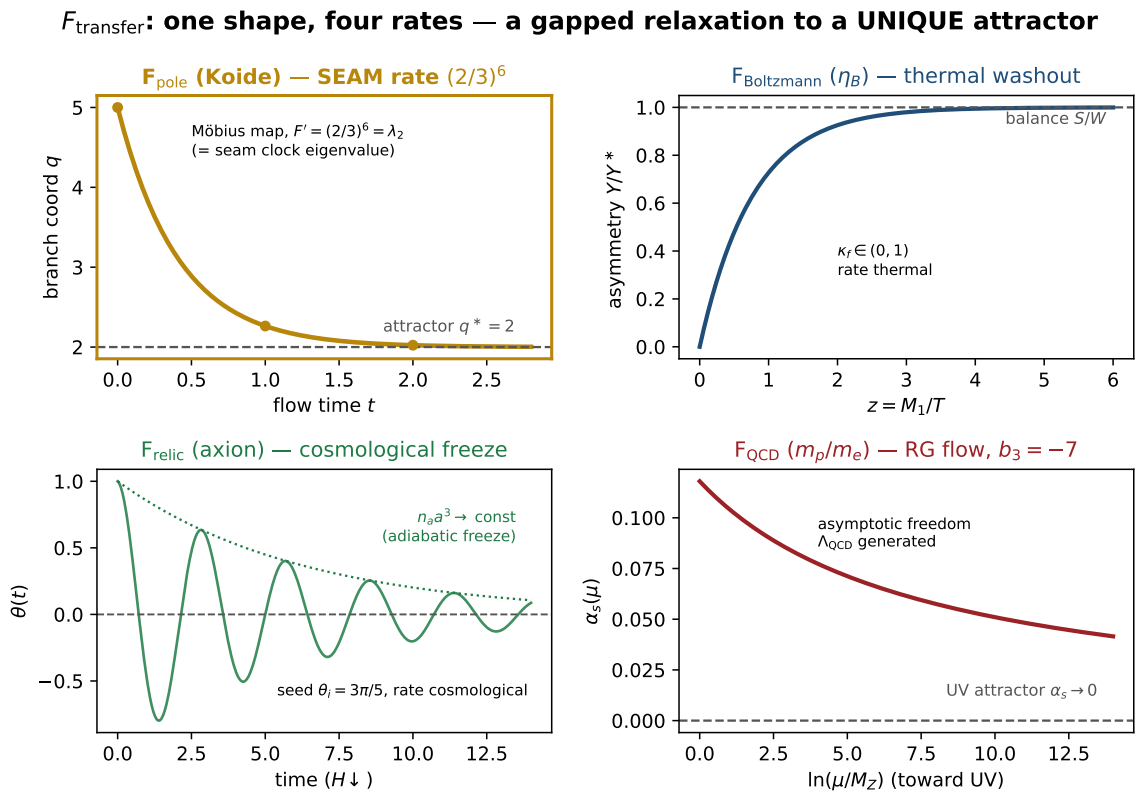


Figure 4: **The discrete→dynamic lens on `F_transfer` ([`verification/v303_ftransfer_dynamics.py`]).** All four transfer instances share *one* shape — a gapped, positivity-preserving relaxation to a *unique* attractor, the same Perron-Frobenius / H-theorem / RG-fixed-point structure as the main-branch update $m \leftarrow (A{+}2I)/4\,m \to$ the E_8 marks. *Only* $\mathrm{F}_{\mathrm{pole}}$ (Koide, gold) runs it at the *seam* rate: its Möbius multiplier is $F'{=}(2/3)^6{=}\lambda_2$, the seam-clock subleading eigenvalue. $\mathrm{F}_{\mathrm{Boltzmann}}$ (η_B, thermal washout), $\mathrm{F}_{\mathrm{relic}}$ (axion, cosmological freeze) and $\mathrm{F}_{\mathrm{QCD}}$ (m_p/m_e, the RG flow to the Gaussian UV fixed point with the carrier $b_3{=}-7$) share the *shape* with *external* rates. So `F_transfer` is the readout end of the one principle, not a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam.